have been, and still are, controversial. For example, reports such as that by Schleifer and Kilpper-Bälz (1984) have recommended the transfer of *S. faecalis* and *S. faecium* to the genus *Enterococcus.*

Although these terms sometimes are used interchangeably, in this text, both fecal streptococci and group D streptococci include the enterococcus group. Conversely, the enterococcus group does not include all of the organisms in either fecal streptococci or group D streptococci.

The use of enterococci or fecal streptococci as indices of fecal contamination, or as a means of determining the sanitary history of food products, their alleged association with foodborne illness, and their occurrence in human pathological conditions attests to their importance in public health.

Sources

Since enterococci are part of the larger group fecal streptococci, the sources of enterococci are also sources of fecal streptococci. However, if fecal streptococci are present, there could be organisms not included in the enterococcus group.

The primary source of the enterococci and fecal streptococci is the intestines of humans and warm-blooded animals. Levels in human feces may range from 10^4 to 10^9 per gram. The organisms are prevalent on the hide, hair, feathers, and hooves of animals and, upon slaughter, the processing equipment, the hands of the workers, and the food products become contaminated.

Animals excreting these organisms can cause widespread distribution of the enterococci. They have been reported in water, soil, vegetation, and insects (Mundt 1982). Due to this distribution, the organisms are found in a wide variety of foods at relatively low levels, and in some foods at rather high levels.

Methods

In order to be an acceptable indicator, relatively simple and standard methods should be available. No method for enterococci or fecal streptococci is listed in FDA (1978). Procedures for the KF streptococcal count and the GTC streptococcal count are described in APHA (1984). In a comparison of several media, M-Enterococcus agar yielded the best overall results (Pagel and Hardy 1980).

Since few media, if any, are specific for a particular organism or group of organisms, and because the species of *Streptococcus* may indicate the source of the organism, further biochemical testing of the organisms is warranted.

The tests include the finding of Gram-positive cocci that are catalase

negative and reduce 0.1 percent methylene blue. Also, lactose, sorbitol, glycerol, and arabinose fermentation, starch hydrolysis, gelatin hydrolysis, hemolysis and reduction of potassium tellurite and TTC can be determined for differentiation of the species. Facklam and Moody (1970) suggested using the bile-esculin test to segregate group D from other streptococci. Lee (1972) reported the use of a tyrosine decarboxylase medium. The enterococci are tyrosine decarboxylase positive.

Mundt (1973) suggested that the reaction in litmus milk and fermentation of melezitose and melibiose might be employed to distinguish between contamination representing recent pollution of human origin and the presence of *S. faecalis* as a member of plants with no sanitary significance.

Growth

The enterococci are able to grow at both 10° and 45°C. They can grow in broth with 6.5 percent NaCl and in media at pH 9.6. They can tolerate 40 percent bile.

The nutritional requirements for enterococci and some other fecal streptococci is such that they may fail to grow on simple culture media. However, most food products contain the nutrients needed for growth of these organisms.

Survival

The fecal streptococci survive outside the intestinal tract better than do the coliforms. *S. faecalis* remains viable longer than *E. coli* in acid foods. In frozen foods, the fecal streptococci remain viable for long periods. The enterococci survive spray-drying and storage in egg powder better than coliforms and *E. coli.* Cool, moist conditions prolong the survival of *S. faecalis* in soil (Kibbey, Hagedorn, and McCoy 1978).

The enterococci are thermoduric, withstanding heat that would destroy most nonsporeforming bacteria. The organisms can withstand 60° to 65°C for 30 min. *S. faecium* is the most heat-resistant species.

Enterococci are sensitive to chlorination. They are destroyed in 15 sec with 100 ppm chlorine at pH 8.4 and in 2 min with 10 ppm chlorine at pH 5.8 (Shannon, Clark, and Reinbold 1965).

Significance

The chief source of fecal streptococci is the intestinal tract of humans and animals. The occurrence of these bacteria in water or food implies either direct or indirect fecal contamination.

The occurrence of fecal streptococci in water generally suggests fecal

pollution. Their absence indicates little or no warm-blooded animal contamination. Fecal streptococci rarely multiply in polluted water; however, this may not be true for food products.

The presence of *S. bovis* or *S. equinus* indicates animal pollution rather than human contamination. Therefore, increased information is obtained if the species of fecal streptococci are identified.

Enterococci survive environmental conditions that destroy other microorganisms of sanitary significance. Due to their resistance to freezing, low pH, and moderate heat treatment, the enterococci have been suggested as an indicator in some types of food products.

The finding of enterococci or fecal streptococci in a food is not proof that enteric pathogens are present. Once the enterococci or fecal streptococci contaminate a food processing plant, they can become established and multiply on surfaces contaminated with organic material. From these sources, they can contaminate food products. Thus, the organisms are able to multiply in environments far removed from the original source of fecal contamination.

In this case, the presence of the organisms could be an indication of poor handling or sanitary practices. However, complete elimination of these organisms is extremely difficult, even under good sanitary conditions.

It has been stated that the enterococci cannot replace *E. coli* as an index for pollution. Also, the presence of the fecal streptococci, per se, gives little information, and either total counts or coliform counts are needed in conjunction with the fecal streptococcus or enterococcus count.

The use of these organisms in food products should be related to specific product practices, handling procedures, and the microenvironments of the food. In some products, their presence has some meaning as indicator organisms; in other products, their presence is meaningless.

OTHER INDICATORS

The staphylococci, or *S. aureus,* have been proposed as indicators. The presence of large numbers of *S. aureus* is an indication of a potential health hazard due to staphylococcal enterotoxin, as well as questionable sanitation, especially of processes involving handling of foods by humans.

Clostridia have been suggested, but they are not very specific for human feces. Since they are sporeformers, they may persist in foods when most enteric organisms are gone. However, two clostridia, *C. perfringens* and *C. botulinum,* are important in foodborne illnesses.

Spore counts of thermophiles have been used as an index of washing efficiency and cleanliness of certain vegetables.

Machinery mold (*Geotrichum candidum*) has been used as an indicator of sanitation of food-processing operations. It readily grows on food adhering to equipment surfaces and contaminates food that moves through the contamination. One study found little correlation between the aerobic plate count and incidence of the mold in frozen blanched vegetables (Splittstoesser et al. 1980).

The use of *Pseudomonas aeruginosa* as an indicator has been suggested. The presence of the organism in the intestinal tract of people, as well as its physiological characteristics, makes it a possible indicator of fecal contamination.

Bifidobacteria are present in the feces of humans and hogs (Resnick and Levin 1981) and only occasionally in the feces of cattle and sheep (Mara and Oragui 1983). Carrillo, Estrada, and Hazen (1985) reported that bifidobacteria show promise as a better indicator of recent fecal contamination in tropical freshwater than either *E. coli* or fecal coliforms.

The presence of certain viruses (phage) can be used as indicator systems. The correlation between coliforms and coliphages in natural water systems were highly significant (Wentsel, O'Neill, and Kitchens 1982). Stetler (1984) found that enteroviruses in water were better correlated with coliphages than with the bacterial indicator sytsems. Other researchers reported that bacterial indicators failed to predict the viral quality of marine water (Goyal et al. 1984). The presence of phage can be demonstrated in 6 to 16 hr, which is much more rapid than the present methods for bacterial indicators.

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8

Food Spoilage

As consumers, we depend upon our senses of sight, smell, taste, and touch to evaluate food quality. When based upon past experiences, these organoleptic evaluations are used to determine if food is spoiled. Since people have different past experiences and abilities, there are conflicting opinions concerning the point at which a food is no longer acceptable for consumption.

The deterioration of food can be caused by rodents, insects, tissue enzymes, nonenzymatic chemical changes, physical effects, and the action of microorganisms. We are concerned primarily with the deterioration of food by microorganisms and microbial enzymes. Tissue enzymes may aid in microbial deterioration of food by altering certain food components so they are more readily available for microbial use. The effect of tissue enzymes on deterioration is quite evident in overripe fruits.

The microbial deterioration of a food usually is manifested by alterations in the appearance, texture, odor, or flavor, or by slime formation. The appearance includes color changes, formation of pockets of gas or swelling, and microbial growth, especially that of molds. As some food products deteriorate, they tend to become soft or mushy. Degradation of food results in the formation of compounds that have odors and flavors different from those of the fresh food.

MICROBIAL FUNCTIONS

Like people, the microorganisms need food for energy, cell growth, maintenance, and reproduction. In the scheme of nature, one of the important roles of microorganisms is to degrade organic material that is no longer needed, so the elements can be reused. Unfortunately, the microorganisms attack our food resources with equal vigor.

The biochemical changes that occur in a food may be desirable or undesirable. Microorganisms change a food into another product, such as cabbage to sauerkraut, grape juice to wine, or alcohol to vinegar. Under controlled conditions, these reactions are desirable, but if we want grape juice, then the formation of alcohol and the production of wine are undesirable. The useful aspects of microorganisms in foods are dis-

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cussed in the next chapter. The basic reactions, whether desirable or undesirable, are essentially the same.

The system by which undesirable changes occur in a food is determined by the composition of the food (carbohydrate, fat, protein), the types and numbers of microorganisms associated with the food, the factors (intrinsic or extrinsic) that influence the growth of microorganisms, and changes in these factors that occur during processing or during deterioration.

The microorganisms present on food include those associated with the raw material, those acquired during harvesting, handling, and processing, and those surviving a preservation treatment and storage. The microorganisms may act singly or in groups to break down complex organic compounds. The active enzyme systems of the dominant microorganisms, as well as the natural enzymes of the food, will determine the course of degradation of a food.

FOOD COMPOSITION

Foods are composed primarily of water, carbohydrate, fat, and protein, as listed in Table 8.1. High-carbohydrate foods include fruits, vegetables, grains, honey, baked goods, and candy. Fats are prominent in

TABLE 8.1. APPROXIMATE RATIOS OF CARBOHYDRATES, FATS, AND PROTEINS IN VARIOUS FOODS

	Ratio of Components[a]		
Food	Carbohydrate	Fat	Protein
Fruit	88	7	5
Fruit juice	94	2	4
Jellies, jams	98	1	1
Vegetables	74	3	23
Cereals	85	3	12
Honey	99	0	1
Baked goods	75	15	10
Candy	83	11	6
Milk	42	29	29
Butter	1	98	1
Seafoods	12	8	80
Fish	0	18	82
Beef carcass, total edible	0	64	36
Pork carcass, total edible	0	84	16
Lamb carcass, total edible	0	58	42
Sausage products	6	62	32
Chicken carcass, total edible	0	21	79
Eggs	4	45	51

[a]Calculated from data of Watt and Merrill (1963).

foods such as butter, eggs, and red meats. Protein foods are considered to be animal products. High ratios of protein are in seafoods and poultry. The ratio of protein of vegetables and cereals is comparable to that of pork carcasses. In milk, there is less difference in the ratio of carbohydrates, proteins, and fats than in any other food listed in Table 8.1.

Carbohydrates are utilized more readily than are proteins or fats. The fermentation of carbohydrates produces acids and lowers the pH. The acidic condition tends to inhibit proteolytic organisms. The term *protein sparing* has been used to describe the metabolism of organisms in a food containing carbohydrates and proteins. Since very few foods are all carbohydrate, all fat, or all protein, more than one type of degradation may occur.

DEGRADATION OF COMPONENTS

Carbohydrates

The general formula of carbohydrates is $C_x(H_2O)_y$. The carbohydrates are divided into monosaccharides, disaccharides, and polysaccharides. The monosaccharides are polyhydroxy aldehydes (aldoses), or polyhydroxy ketones (ketoses). Some carbohydrates as well as some polyhydric alcohols are listed in Table 8.2.

There are some free monosaccharides, but most naturally occurring carbohydrates are disaccharides and polysaccharides. For utilization, bacteria first break down these complex carbohydrates into their constituent monosaccharides.

POLYSACCHARIDES. Plant cells and tissues have a fibrous substance embedded in an amorphous support matrix. The fibrous material resists tension, and the matrix resists compression. In plants, the primary fibrous substance is cellulose.

Plants have a rather wide range of matrix compounds. In higher plants, the matrix polysaccharides are classified into a group of acidic polysaccharides called pectins and a heterogeneous group of neutral polysaccharides called hemicelluloses. When stiffness is needed, a nonpolysaccharide, lignin, is incorporated. In algae (seaweeds), such things as carrageenans and agar are the matrix materials.

Pectin. Pectolytic enzymes are produced by a wide range of microorganisms, especially those causing soft rot. The pectic enzymes include pectin esterase (EC 3.1.1.11), polygalacturonases (EC 3.2.1.15; EC 3.2.1.67; EC 3.2.1.82), pectate transeliminases (EC 4.2.99.3), and lyases (EC 4.2.2.2; EC

TABLE 8.2. COMPOUNDS DEGRADED AS CARBOHYDRATES

Monosaccharides	Disaccharides	Polysaccharides
Hexoses	Maltose (glucose + glucose)	Dextrins
Aldoses	Lactose (glucose + galactose)	Starch
Glucose	Sucrose (glucose + fructose)	Glycogen
Mannose	Cellobiose (glucose + glucose)	Inulin
Galactose		Cellulose
Ketoses		Hemicellulose
Fructose		Galactan
Sorbose		Xylan
Polyhydric alcohols		Complexed polysaccharides
Mannitol		Glucosides
Glycerol		Salicin
Adonitol		Amygdalin
Dulcitol		Esculin
Sorbitol		Tannins
		Pectins
		Gums
		Mucilages

4.2.2.9; EC 4.2.2.10). The random splitting of glycosidic bonds results in softening and liquefaction.

The pectic enzymes are found in plant tissues as well as in many types of microorganisms. The plant enzymes act during the ripening of fruit to cause a softening effect. This is desirable to an acceptable level, but continued action by these enzymes causes excessive softness and a mushy product. When a microorganism is isolated from a softened fruit, the softening could have been caused by either the enzymes from the organism or the plant tissue. The degradation of pectin was reviewed by Chesson (1980). Perhaps this breakdown is the primary spoilage defect of plant tissue.

Cellulose. Only a few species of bacteria can decompose cellulose, but several filamentous fungi are capable of doing so. Cellulose is either hydrolyzed directly to glucose or to intermediates such as cellobiose.

The presence of readily metabolized substances such as glucose represses the production of the inducible cellulases. Hence, the breakdown of cellulose is of secondary importance in the deterioration of plant tissue.

Starch. This polysaccharide is an important component of many plant products. Several bacteria and molds possess an extracellular enzyme, diastase or amylase, which hydrolyze starch. Starch is converted either directly to glucose or through intermediates such as maltose.

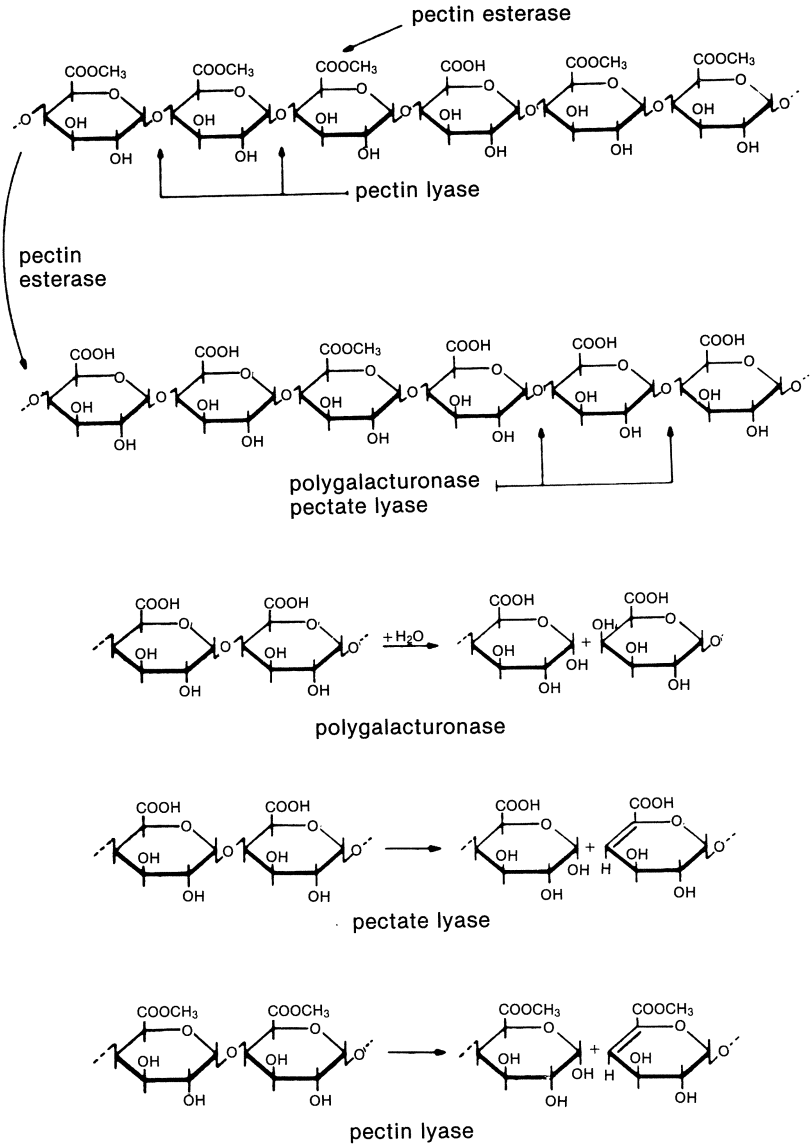


Figure 8.1. Breakdown of pectin.
Courtesy of Pilnik and Rombouts (1978).

MONOSACCHARIDES. Glucose is the main carbohydrate used as a carbon and energy source. The breakdown of these sugars can proceed by several pathways. Some of the reactions and intermediates of glycolysis are summarized in Figure 8.2. There are other systems for the metabolism of glucose such as the pentose-P-phosphoketolase and Entner-Doudoroff pathways.

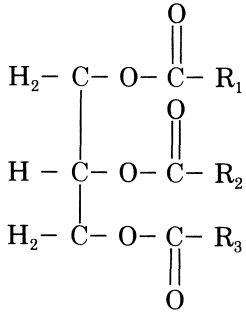
Pyruvic acid ($CH_3COCOOH$) is a key compound in the metabolism of glucose. Depending upon the metabolic pathways available to the organism and the environmental conditions, the pyruvate can be involved in many reactions, either fermentative or oxidative. Some of the metabolic products resulting from carbohydrates are listed in Table 8.3. The products include organic acids, alcohols, CO_2, H_2 and H_2O.

In aerobic respiration, pyruvate is converted into CO_2 and H_2O by means of the tricarboxylic acid (TCA) cycle, Krebs cycle, or citric acid cycle. To enter this system, the pyruvate is converted to acetate activated with coenzyme A. Only the aerobic and some facultatively anaerobic microorganisms possess an intact TCA cycle.

The pyruvic acid can be decarboxylated to form acetaldehyde (CH_3-CHO) and CO_2. The acetaldehyde can remain or be reduced to ethyl alcohol (CH_3CH_2OH), oxidized to acetic acid (CH_3COOH), or condensed to form acetoin ($CH_3COCHOHCH_3$) or acetylmethylcarbinol (AMC). The AMC can be oxidized to diacetyl ($CH_3COCOCH_3$), which has a butter flavor, or reduced to 2,3-butanediol ($CH_3CHOHCHOHCH_3$). Pyruvate can be aminated to form alanine.

Lipids

The principle lipids in foods are fats. Fats are esters of glycerol and fatty acids and are called glycerides. The generalized structure of triglycerides is



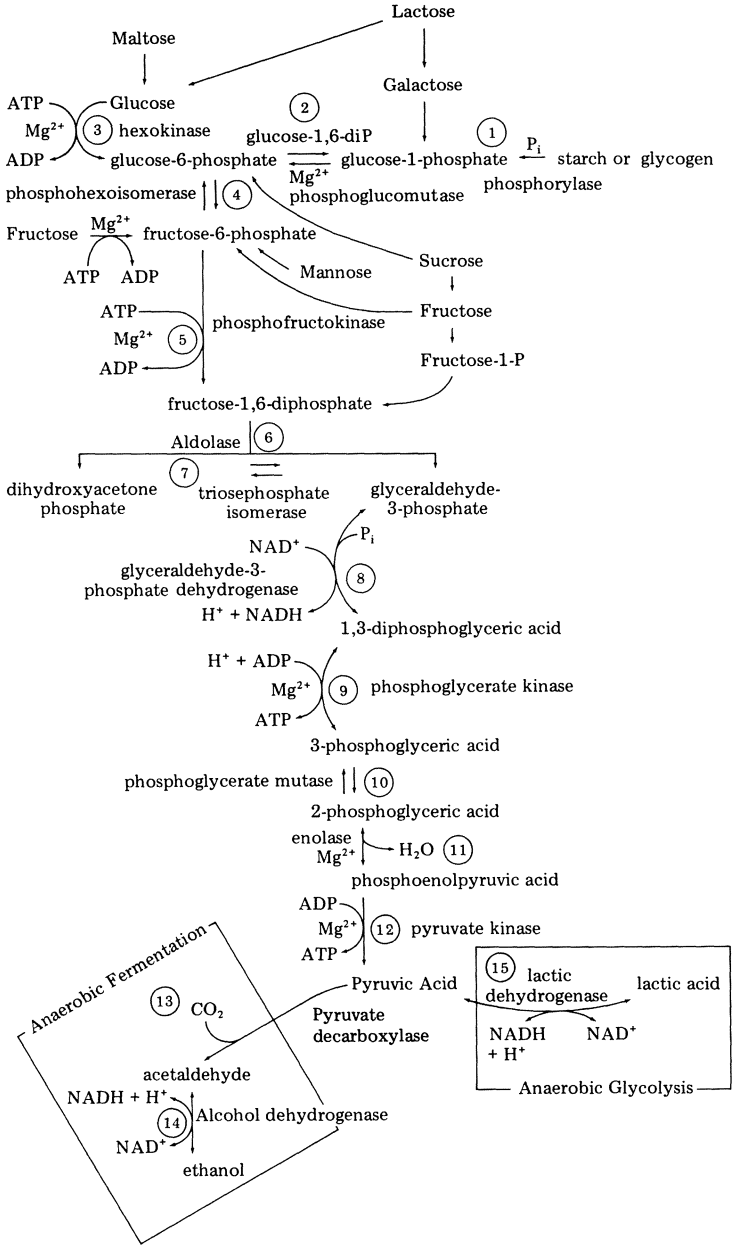


Figure 8.2. A glycolytic pathway for the degradation of various sugars.

Courtesy of Aurand and Woods (1973).

TABLE 8.3. SOME PRODUCTS OF CARBOHYDRATE METABOLISM

Organisms	Products
Leuconostoc mesenteroides	Lactic acid
	Ethyl alcohol
	CO_2
Leuconostoc cremoris	Acetic acid
	Acetoin
	Diacetyl
	CO_2
Saccharomyces (yeast)	Ethyl alcohol
	CO_2
Clostridium botulinum	Acetic acid
	Butyric acid
	Propionic acid
	Isobutyric acid
	Isovaleric acid
	Propyl alcohol
	Isobutyl alcohol
	Butyl alcohol
	Isoamyl alcohol
Propionibacterium	Propionic acid
	Acetic acid
	Isovaleric acid
	Formic acid
	Succinic acid
	Lactic acid
	CO_2
Escherichia coli	Lactic acid
	Acetic acid
	Formic acid
	CO_2
	H_2
Bacillus cereus	Acetoin
	Glycerol
	2,3-Butanediol
	Lactic acid
	Succinic acid
	Formic acid
	Acetic acid
	CO_2

The R-groups denote the carbon chain of the fatty acids. Some common fatty acids are listed in Table 4.9. The triglycerides may have the same fatty acid in all three positions, or two or three different fatty acids may be present.

Many foods contain fat that is susceptible to hydrolysis, oxidation, and other chemical processes that result in the formation of various compounds. Both desirable and undesirable flavor changes in foods are associated with these compounds.

The enzymes that hydrolyze triglycerides are called lipases. The desig-

nation of true lipase is given to enzymes that attack only water-insoluble substrates.

The oxidative deterioration of fats involves the reactions of unsaturated fatty acids with oxygen to give hydroperoxides. The hydroperoxides are not flavor compounds, but readily decompose to carbonyl compounds resulting in off-flavor or odor. The carbonyl compounds are mixtures of saturated and unsaturated aldehydes and ketones.

Deteriorated fat is called rancid. The simple release of free fatty acids is called hydrolytic rancidity, and oxidative deterioration is oxidative rancidity.

Hydrolytic rancidity is important in fats. Short-chain water-soluble fatty acids (butyric, caproic, and caprylic) cause obnoxious rancid flavors in milk.

Except for the release of volatile fatty acids, the main flavor changes are not due to hydrolysis, but to oxidation. The oxidation of the fat is more often due to tissue enzymes or autoxidation than to microbial activity. Autoxidation is accelerated by heat and light and is catalyzed by heavy metals and their salts, organometalic compounds, hemoglobin, hematin compounds, and photochemical pigments. Certain species of *Aspergillus* and *Penicillium* produce oxidative enzymes that catalyze the oxidation of free fatty acids to methyl ketones.

A pure fat is not attacked by microorganisms, since there must be a nutrient-containing aqueous phase in which the organism can grow. However, most of our fatty foods (butter, cream, margarine), contain an aqueous phase associated with the fat.

There are many lipolytic microorganisms. Some of the genera that contain lipolytic species or strains are listed in Table 8.4.

TABLE 8.4. SOME GENERA CONTAINING LIPOLYTIC SPECIES OR STRAINS

Bacteria	Fungi
Acinetobacter	*Absidia*
Aeromonas	*Alternaria*
Alcaligenes	*Aspergillus*
Bacillus	*Candida*
Chromobacterium	*Cladosporium*
Corynebacterium	*Endomyces*
Enterobacter	*Fusarium*
Flavobacterium	*Geotrichum*
Lactobacillus	*Mucor*
Micrococcus	*Neurospora*
Pseudomonas	*Penicillium*
Serratia	*Rhizopus*
Staphylococcus	*Torulopsis*
Streptomyces	

The lipolytic characteristics of microorganisms depend upon the fat in the substrate. Tom and Crisan (1975) showed that, although many organisms decomposed tributyrin or tween 80, only a few were able to decompose fish oil added to a medium.

With the breakdown of fats, various products are formed, including glycerol, free fatty acids, aldehydes, ketones, and alcohols.

Proteins

Proteins are composed of amino acids combined by peptide linkages. These linkages are broken by the addition of water catalyzed by enzymes. The proteins are degraded through proteoses, peptones, polypeptides, and dipeptides to amino acids. The enzymes that hydrolyze the peptide linkages of proteins are called proteases. These proteases convert the proteins into diffusible polypeptides and dipeptides, which can enter the bacterial cell. The polypeptides and dipeptides are without offensive odor.

The native proteins are resistant to attack by microorganisms. There are low-molecular-weight compounds, such as dipeptides and free amino acids in fresh meat, fish, and poultry tissue. These substances are readily used by the microbial flora. Spoilage of the protein-type foods may be evident before any significant amount of protein is degraded. In advanced spoilage, some protein is hydrolyzed by proteolytic enzymes, whether produced by the tissues or the microorganisms.

AMINO ACIDS. The degradation of amino acids is of primary importance in the spoilage of protein foods. The products that are formed depend upon (1) the type of microorganism; (2) the types of amino acids; (3) temperature; (4) the amount of available oxygen; and (5) the types of inhibitors that might be present.

There are many amino acids and hence, many products of amino acid dissimilation. Some products that are formed anaerobically are foul smelling. This form of degradation is called putrefaction. Aerobic degradation, with oxidation of the metabolic products, is called decay.

The two main reactions of microorganisms on amino acids are decarboxylation or deamination. Many of the reactions involving amino acids are shown in Figure 8.3. Other reactions of amino acids include the following:

Oxidation reduction

$$3CH_3CHNH_2\text{-}COOH + 2H_2O \rightarrow 2\ CH_3CH_2COOH + CH_3COOH + 3NH_3 + CO_2$$

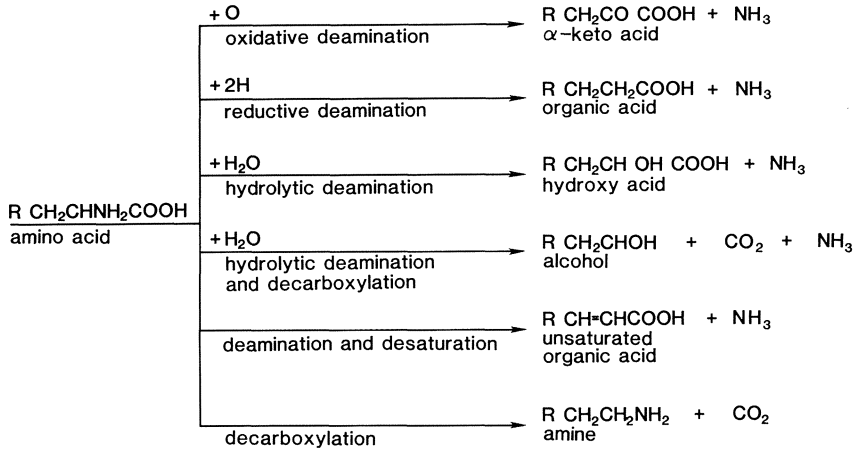


Figure 8.3. Reactions involving amino acids.

The overall reaction results in two organic acids (one with one less carbon atom), ammonia, and carbon dioxide.

Anaerobic degradation—release of hydrogen

$$5COOH\text{-}CH_2\text{-}CH_2\text{-}CHNH_2\text{-}COOH + 6H_2O \rightarrow 6CH_3COOH +$$
$$2CH_3CH_2CH_2COOH + 5CO_2 + 5NH_3 + H_2$$

The overall reaction involving, in this case, glutamic acid, yields acetic acid, butyric acid, carbon dioxide, ammonia, and hydrogen.

Transamination

Although the amino acid is not degraded in this reaction, transamination does alter the composition of the substrate, which will determine the final degradation product.

$$COOH\text{-}CH_2\text{-}CH_2\text{-}CHNH_2\text{-}COOH + COOH\text{-}CH_2\text{-}CO\text{-}COOH \rightarrow$$
$$COOH\text{-}CH_2\text{-}CH_2\text{-}CO\text{-}COOH + COOH\text{-}CH_2\text{-}CHNH_2\text{-}COOH$$

In this example, glutamic acid reacts with oxaloacetic acid to form α-keto glutaric acid and aspartic acid.

Mutual oxidation and reduction

Some amino acids (alanine, leucine, phenylalanine, and valine) can serve as hydrogen donators and are oxidized, while others (arginine, gly-

cine, hydroxyproline, ornithine, and proline) can serve as hydrogen acceptors and are reduced.

Tryptophan can enter reactions resulting in many products. One product, indole, is used in the identification of certain microorganisms. Another product is skatole. Small amounts of skatole are present in the perfume of the jasmine and orange blossoms. It is one of the two most important nitrogenous substances found in natural perfumes. In larger concentrations, skatole is a foul-smelling compound. It is found prominently in fecal material.

When sulfur-containing amino acids such as cystine or methionine are degraded, hydrogen sulfide as well as mercaptans, organic disulfides, and other sulfur-containing (thio) compounds may be formed.

The decarboxylation of the diamino acids, lysine and ornithine, yield cadaverine and putrescine accordingly:

$$NH_2\text{-}(CH_2)_4\text{-}CHNH_2\text{-}COOH \rightarrow NH_2\text{-}(CH_2)_5\text{-}NH_2 + CO_2$$

$$NH_2\text{-}(CH_2)_3\text{-}CHNH_2\text{-}COOH \rightarrow NH_2\text{-}(CH_2)_4\text{-}NH_2 + CO_2$$

Arginine is degraded through ornithine to putrescine.

By the dissimilation of amino acids, many diverse products, such as carbon dioxide, hydrogen, ammonia, hydrogen sulfide, organic acids, alcohols, amines, diamines, mercaptans, and organic disulfides may be formed. The production of ammonia and amines tends to cause an increase in the pH. Trimethylamine is more basic than is ammonia. An increase in the pH of a protein food indicates protein degradation, just as a decrease in pH results from the fermentation of carbohydrates.

OTHER DETERIORATION

Besides the changes caused by the degradation of carbohydrates, fats, and proteins, microorganisms can create unacceptable foods by changing the appearance of the food, either by mold growth or by altering the color of a pigment. Also, microorganisms can metabolize sugar and produce dextrans or levans that feel like slime on the surface of foods such as meat, poultry, or fish, or a condition called ropiness, in such foods as milk or bread.

Appearance

When microbial growth can be observed on a food, it usually is considered to be unacceptable. Pigmented bacteria can, at times, be observed on a food. Mold mycelia indicate potential spoilage; however,

mold is grown on certain types of cheeses, such as Roquefort and Camembert, to develop the distinctive flavors.

Various pigments (chlorophyll, carotene, myoglobin) are present in foods. The degradation of chlorophyll and carotene is related to the lipolytic breakdown of fats. A lipase in peas effectively degrades chlorophyll to pheophytin, and the degradation of both chlorophyll and pheophytin is related to an increase in fat peroxide. Carotene is fat soluble, and its destruction is related to fat decomposition.

The myoglobin pigment of meat muscle can be altered by oxidation-reduction reactions. Myoglobin is dark red to purple and is present in the interior part of meat. When the muscle tissue is cut and exposed to oxygen, the myoglobin is oxygenated to oxymyoglobin, which is bright red and is the usual pigment associated with meat. In myoglobin and oxymyoglobin, the iron is in ferrous form. When it becomes oxidized to the ferric form, metmyoglobin is formed. This pigment is dark brown.

Although these pigment changes do not necessarily make the food inedible, a consumer may consider the product unacceptable.

Slime Formation

Several bacteria produce microbial polysaccharides (dextrans, levans, or amyloses) from various disaccharides present in food. These polysaccharides form unpleasant slime in and on food, causing the food to be both unpalatable and unacceptable to the consumer. An example of this is the slimy or ropy texture of fruit concentrate or milk when infected with *Leuconostoc mesenteroides, Bacillus subtilis,* or *E. coli.* Sucrose and maltose are readily used by *E. coli, L. mesenteroides,* or *B. subtilis* to produce amyloses and dextrans. *B. megaterium* and *B. subtilis* produce levans, which are long-chain polysaccharides.

Dextran is a glucose polymer. Sometimes it is encountered as a slime in large globular masses during the processing of cane or beet sugar. It increases the viscosity of the sugar solution and retards filtration and crystallization. These globular masses are due primarily to activities of *Leuconostoc mesenteroides* and *L. dextranicum,* although *Streptococcus salivarius* and *S. bovis* are able to synthesize from sucrose and raffinose a dextran-type insoluble carbohydrate.

Slime on refrigerated fresh meat and poultry that appears soon after off-odor is caused by the growth of *Pseudomonas.*

MICROBIAL DEFECTS IN SPECIFIED FOODS

Presumably the enzymes produced by microorganisms, even if the microorganisms are not multiplying, could, over an extended storage time,

decompose the food and cause spoilage. However, spoilage is associated with large numbers of microorganisms. Therefore, the organisms that cause spoilage are those that can multiply and become dominant. The factors affecting growth and dominance are discussed in Chapter 4. These factors are considered in regard to the spoilage of specific types of foods.

Fruits

A large and varied population of microorganisms, including the spores of many types of fungi, contaminate the surface of fruits during the growing season. Relatively few of the fungi are capable of attacking the fruit before harvest. The ripening process increases the susceptibility of the fruit to invasion. Fruits harvested from the ground have a more varied microbial flora than those picked from the tree. Falling from the tree causes bruising, and contact with the grass and soil provides an added source of inoculum.

Fruits have a low pH that inhibits most bacteria. The acid-tolerant bacteria are mainly Gram-positive lactobacilli and leuconostocs. Usually fruits are spoiled by yeasts and molds that are acid tolerant.

The microbial defects of fruit and fruit products are listed in Table 8.5. *Penicillium* species are important spoilage organisms of most fruits. *P. digitatum* and *P. italicum* cause soft rot in citrus fruits, and *P. expansum* causes blue mold rot of deciduous fruits. Blue mold rot consists of soft, watery, tan to light-brown areas that are readily gouged out of the fruit flesh. The tissue has a moldy or musty odor and flavor. There are typical bluish green spores on the fruit.

Black rots in apples, citrus fruits, bananas, and pineapples are due to species of *Alternaria* as well as other molds. The tissue becomes soft and watery. For brown rot, the fruit has decayed areas, turning dark brown to black at the center. The mold spore masses are yellowish gray.

Anthracnose is a defect with scattered black or dark-brown sunken spots covering firm decayed tissue. Under moist conditions, there are pink spore masses on the spots.

Gray mold rot is evidenced by light-brown, fairly firm, watery decay, covered extensively with delicate dirty-white mycelium. Grayish brown velvety spots may be seen in advanced spoilage.

Sassa and Onuma (1983) isolated phytotoxic metabolites of *Macrophoma* which they called fruit rot toxins. These chemicals induce necrosis of pinhole-injured apples.

The loss of berries results from dehydration, discoloration, and overripe, mushy, damaged, moldy, or spoiled fruit. Berries are spoiled primarily by *Botrytis cinerea* and *Mucor mucedo*. However, certain fungi are prev-

TABLE 8.5. SOME MICROBIAL DEFECTS OF FRUITS AND THEIR PRODUCTS

Food	Defect	Organisms
Fresh fruit, general	Blue mold rot	*Penicillium expansum*
	Black mold rot	*Aspergillus niger*
	Souring, bitter flavor	*Streptococcus faecalis*
	Soft rot	*Byssochlamys fulva, Penicillium*
	Green mold rot	*Penicillium digitatum*
	Anthracnose	*Colletotrichum*
	Gray mold rot	*Botrytis cinerea*
Apples	Fungal	*Gloeosporium*
	Fermentation	*Torulopsis, Candida, Pichia*
	Moldy core	*Alternaria tenuis*
	Brown firm rot	*Trichoderma harzianum*
Bananas	Storage rote	*Colletotrichum, Gloeosporium, Botryodiplodia, Erwinia*
	Black rot	*Alternaria*
	Crown rot	*Colletotrichum, Fusarium, Verticillium*
Berries, general	Fungal rot	*Botrytis cinerea, Mucor mucedo, Epicoccum, Colletotrichum, Rhizopus, Alternaria*
Strawberries	Fermentation	*Kloeckera*
	Soft rot	*Rhizopus*
Blueberries	Black rot	*Alternaria*
	Soft rot	*Phomopsis*
Cherries	Brown rot	*Monilia, Sclerotinia*
Citrus fruits	Green mold	*P. digitatum*
	Blue mold	*P. italicum*
	Black rot	*Alternaria*
	Sour rot	*Geotrichum candidum*
	Stem-end rot	*Phomopsis, Diploida*
Dates	Fermentation	*Saccharomyces, Candida, Hansenias-pora, Torulopsis*
Figs	Souring	*Gluconobacter*
Olives	Softening (stem-end shrivel)	*Rhodotorula, Saccharomyces, Hansenula*
	Sloughing of skin	*Klebsiella, Enterobacter, Escherichia, Aeromonas liquefaciens*
Peaches	Brown rot	*Monilinia fructicola, Sclerotinia*
	Decay	*Rhizopus stolonifer*
Canned fruit	Butyric acid	*Clostridium*
	Soft rot	*B. fulva, B. nivea*
Apricots	Softening	*R. stolonifer, R. arrhizus*
Banana puree	Gas	*Bacillus licheniformis*
Grapefruit sections	Gas (CO_2)	*Lactobacillus brevis*
Mango juice	Gas	*Leuconostoc oenos*
Fruit juice	Souring, CO_2	*Lactobacillus,* yeast
	Acetification, vinegar	*Acetobacter*
	Moldy, surface	*Penicillium*
	Cloudy	Nonfermenting yeasts
	Cloudy, alcohol	Fermenting yeasts
	Buttermilk flavor	*Lactobacillus, Leuconostoc*
Jelly, jam, preserves	Fungal	*Xeromyces bisporus*
	Fermentation	Osmophilic yeasts
Wine	Acetification	*Acetobacter, Gluconobacter*
	Mousy odor, cloudy, slimy	Lactic acid bacteria
	Cloudy, gassy	Yeasts (*Saccharomyces, Pichia, Candida*)
	Acetaldehyde	*Saccharomyces oviformis, S. beticus*
	Flowers	*Candida vini*

alent on specific berries at various times during production, harvesting, and storage.

The species of several genera of yeasts (*Saccharomyces, Hanseniaspora, Hansenula, Pichia, Torulopsis, Candida, Debaryomyces,* and *Kloeckera*) can cause a fermentation defect in fruits.

The lactobacilli are important in the spoilage of fruit juice. Some strains are quite acid tolerant and can metabolize citric and malic acid. This reduces the acidity, resulting in a bland, rather flat flavor and a loss in astringency. The production of diacetyl by these organisms results in a buttermilk flavor in the fruit juice.

Leuconostoc mesenteroides produces dextrans, resulting in a slimy, unpleasant texture of fruit juices. Yeasts contaminate and ferment fruit juice, especially apple juice. If the sugar concentration is 10 to 30 percent, the osmophilic yeasts, *Saccharomyces rouxii* and *S. mellis,* can ferment the sugars to alcohol. *Acetobacter* can convert the alcohol to acetic acid, giving the fruit juice a vinegar flavor.

The most common defect of wine is acetification. This is caused by *Gluconobacter* and *Acetobacter.* All alcoholic beverages containing less than 15 percent w/v ethyl alcohol are subject to this type of spoilage.

Species of *Leuconostoc* (especially *L. mesenteroides*), *Streptococcus,* as well as some *Acetobacter,* can produce dextrans in sweet wines. In this case, a ropy texture develops which becomes slimy and viscous.

The mature spores of *Byssochlamys fulva* and *B. nivea* are thermoduric and may survive the heat treatment during fruit canning. These fungi can grow at rather low redox potentials, and their pectic enzymes cause spoilage, especially of canned berries. In some cases, such as *Rhizopus* spoilage of canned apricots, the mold infects the fresh fruit and produces heat-resistant enzymes. During storage of the processed fruit, the residual enzymes cause an unacceptable softening of the product. Ethiraj and Suresh (1985) isolated *L. oenos* from a blown can of mango juice.

Vegetables

Because vegetables are harvested from or near the soil, they are subjected to a heterogeneous flora of soil as well as to airborne microorganisms.

In general, the pH of vegetables is near neutrality, so that bacteria, as well as fungi, cause deterioration. Because tomatoes have a lower pH, the spoilage is similar to that of fruit, although bacterial spoilage also occurs. Microbial defects of vegetables are listed in Table 8.6. Black rot of sweet potato is shown in Figure 8.4.

Celery plants are infected by strains of *Sclerotinia sclerotiorum,* which produces a pink rot at the base of the plant. Carrots, cabbage, and other

TABLE 8.6. SOME MICROBIAL DEFECTS OF VEGETABLES AND THEIR PRODUCTS

Food	Defect	Organism
Fresh vegetables	Soft rot, mushy	*Erwinia carotovora, Pseudomonas fluorescens*
	Soft black rot	*Alternaria, Rhizopus nigricans*
	Black mold rot	*Aspergillus niger*
	Blue mold rot	*Penicillium*
Beans, snap	Anthracnose	*Colletotrichum*
	Blight	*Xanthomonas*
Cabbage	Leaf spot	*Alternaria*
	Gray mold	*B. cinerea*
Carrots	Soft rot	*E. carotovora, Rhizopus stolonifer*
	Fungal rot	*Fusarium*
	Decay, wet rot	*Sclerotinia sclerotiorum, Rhizoctonia carotae*
Celery	Fungal rot	*Mucor*
	Pink rot	*S. sclerotiorum*
Onions	Neck rot	*Botrytis allii*
	Rot	*Pseudomonas cepacia*
	Brown rot	*P. aeruginosa*
	Black mold	*Aspergillus niger*
Potatoes	Ring rot	*Corynebacterium*
	Dry rot	*Fusarium*
Tomatoes	Ferment	*Candida, Pichia, Hanseniaspora, Kloeckera*
	Fungal rot	*Alternaria, Aspergillus, Botrytis, Colletotrichum, Monilia, Penicillium, Rhizopus*
	Bacterial spot	*Xanthomonas*
	Soft rot	*Byssochlamys fulva*
Canned vegetables		
Corn, green beans, peas	Flat-sour	*Bacillus stearothermophilus*
	Sulfide stinker	*Desulfotomaculum nigrificans*
	Putrid swell	*Clostridium sporogenes*
	Hard swell	*Clostridium thermosaccharolyticum*
Tomato	Flat-sour	*Bacillus coagulans*
	Butyric fermentation	*Clostridium pasteurianum, C. butyricum*
Fermented vegetables		
Brine	Film forming	Yeasts (*Candida, Debaryomyces, Hansenula, Kloeckera, Pichia, Rhodotorula, Saccharomyces*)
Pickles	Soft	*Bacillus*
	Black	*Bacillus*
	Soft, mushy, slimy	*B. subtilis*
	Reduced acidity	Yeasts
Sauerkraut	Pink	*Rhodotorula*
Vegetable juice	Sour	*Lactobacillus, Acetobacter*

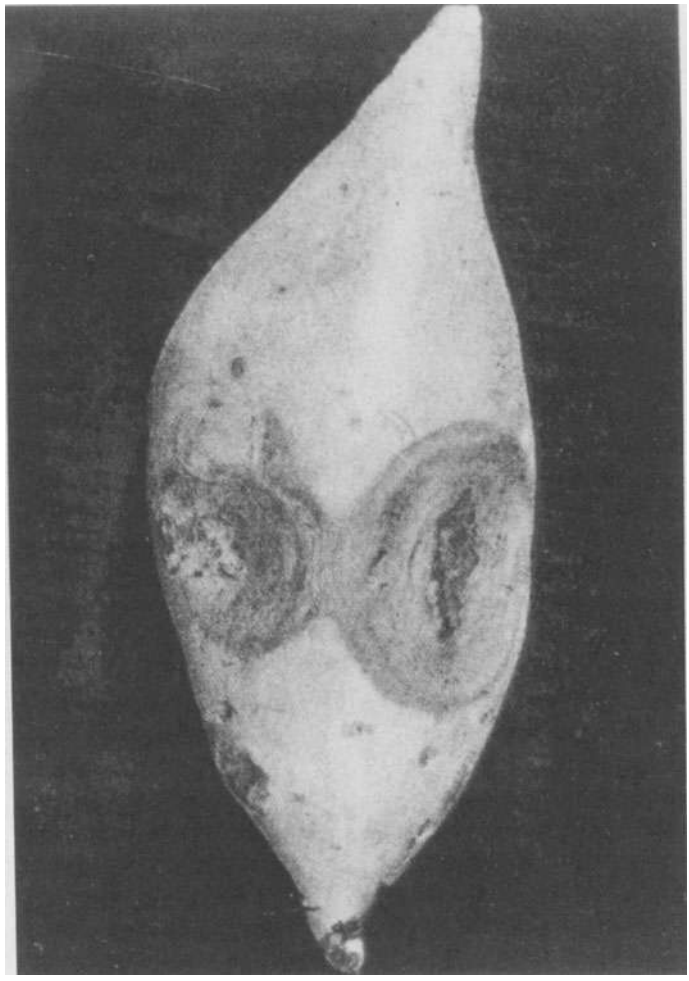


Figure 8.4. Black rot of sweet potato.
Courtesy of USDA.

vegetables are soft rot–spoiled by bacteria, such as *Erwinia carotovora,* as well as the fungi *Rhizopus, Aspergillus,* and *Alternaria.*

Canned tomato juice is subject to flat sour spoilage if not properly handled either during preparation or final heat treatment. Flat sour spoilage is attributed to the presence of and growth of *Bacillus coagulans,* which either survives normal heat processes or recontaminates the product.

Other Carbohydrate Foods

The microbial defects of these foods are listed in Table 8.7. Cereals, honey, molasses, syrup, biscuits, and candy ordinarily have water activities too low to support the growth of bacteria. However, due to storage at high relative humidity or production of water by metabolism, bacteria can be responsible for the spoilage of these products.

Fungi are the primary agents causing deterioration of damp grain. The alterations of grain caused by fungi include discoloration, decreased germination, loss in weight, formation of mycotoxins, mustiness, an increase in temperature, and a loss of nutrients.

Most raw sugars have an a_w between 0.65 and 0.75. Only osmophilic yeasts and some xerophilic fungi can grow on these raw sugars.

Soft-centered fondants and chocolates can be fermented by yeasts, which produce ethyl alcohol and CO_2. The gas buildup may cause the outer chocolate coating to burst. Foods such as molasses, honey, syrups

TABLE 8.7. SOME MICROBIAL DEFECTS OF MISCELLANEOUS CARBOHYDRATE FOODS

Food	Defect	Organism
Beer	Ropy	*Gluconobacter*
	Gas, slime	*Streptococcus lactis*
	Off-flavor	*Lactobacillus, Pediococcus, Brettanomyces, Candida, Pichia*
	Turbid	*Candida, Brettanomyces*
	Fruity	*Candida*
	Haze, diacetyl	Lactic acid bacteria, pediococci
Bread	Ropy, slime	*Bacillus subtilis*
	Black mold	*Rhizopus nigricans*
	Blue mold	*Penicillium*
	Pink mold	*Neurospora*
	Sour	Lactic acid or coliform bacteria
	Red	*Serratia marcescens*
Candy (chocolate cream)	Fermentation	Yeasts
Cereals, grains	Moldy	*Aspergillus, Penicillium*
	Discoloration	*Rhizopus nigricans*
Wheat	Pink	*Erwinia rhapontici*
Corn	Blue-eye	*Penicillium martensii*
Chocolate sauce	Cloudy	*Xeromyces bisporus*
Coconut	Rancidity	*Micrococcus luteus, Bacillus subtilis*
Dough "refrigerated"	Gas, slime, sour	*Lactobacillus, Leuconostoc, Streptococcus*
Honey	Fermented, yeasty	*Torulopsis,* osmophilic yeasts
Molasses	Gas, forthy	*Clostridium,* osmophilic yeasts
Peanuts	Moldy	*Fusarium, Penicillium, Aspergillus*
Soft drinks	Turbidity	Yeasts
Sugar solutions	Slime	*Leuconostoc mesenteroides*
	Ferment, yeasty	Osmophilic yeasts

(maple, corn, chocolate, soft drink), and candy are usually spoiled by osmophilic or osmotolerant yeasts, such as *Saccharomyces rouxii, S. bailii,* and *S. bisporus.*

Lactobacilli are hops tolerant. These organisms produce lactic acid, dextran hazes, and diacetyl in beer. A diacetyl flavor generally is not desirable in the light lager beers produced in the United States. Several yeasts, such as *Brettanomyces, Candida,* and *Pichia* can infect finished beer and, with secondary fermentations, cause defects (dextran hazes, films, off-flavors, and off-odors) in the beer.

The decarboxylation of hydroxycinnamic acids normally present in barley, by certain enterobacteria, such as *Hafnia,* result in a phenolic off-flavor in beer (Lindsay and Priest 1975). The genus name *Pectinatus* was suggested for an anaerobic bacterium that can grow in hopped beer, and cause turbidity with H_2S production (Lee et al. 1980).

Animal Products

Since fresh animal products are perishable, they are chilled and stored in ice or a refrigerator (0° to 4°C). This means that psychrotrophs become dominant and are the primary cause of spoilage. If these fresh products are mishandled and allowed to remain at room temperature (20°C), a more diverse flora may be present, since not only will many psychrotrophs grow, but also there will be some mesophilic strains.

The organisms most often involved with spoilage of refrigerated fresh meat, poultry, fish, and eggs are species of the genus *Pseudomonas. Pseudomonas putrefaciens* has been designated as being important in spoilage of meat and poultry. This organism does not conform to the given definition of *Pseudomonas,* so it was assigned the name *Alteromonas putrefaciens.* Since the DNA base composition is somewhat higher than the range for *Alteromonas,* the exact nomenclature should be delayed (Krieg and Holt 1984).

RED MEAT. In general, the muscle tissue of live, healthy animals is sterile. The organisms that are present in the carcass are concentrated in the lymph nodes. Researchers isolated organisms from 58 percent of 360 nodes they tested (Moo et al. 1980). Contaminated nodes are thought to be a source of bacteria for deep spoilage of postrigor meat. After slaughter, the defense mechanisms are slowed or halted. This enables bacteria to multiply and spread throughout the tissues.

The most common indications of spoilage are (1) off-odor and slime, usually due to the growth of aerobic bacteria on the cut surfaces of meat; (2) fungal growth, which is favored at water activities too low for bacterial growth; (3) bone taint, or deep spoilage, due to anaerobic or facultative

microorganisms; and (4) discoloration, primarily due to alterations of myoglobin, the muscle pigment.

There are some differences in the spoilage patterns of fresh and cured meat. There is only limited information on cooked meat spoilage. Generally, cooked meat is spoiled by the few types of organisms that can survive the cooking or those that gain access to the cooked product. Fresh, chilled meat spoilage is evidenced by off-odor and slime due to *Pseudomonas, Acinetobacter,* and *Alcaligenes.* Cured meats become sour due to the activity of *Micrococcus, Lactobacillus,* and *Microbacterium* species. Microbial defects of meat and meat products are listed in Table 8.8.

Fresh Meat. The rate of spoilage depends upon the numbers and types of organisms initially present, the conditions of storage (temperature and gaseous atmosphere), and characteristics (pH, a_w) of the meat.

In general, the whole carcass presents a surface of fat and connective tissue affording little opportunity for bacterial growth. The surfaces of meat cuts can support the growth of large numbers of bacteria, and ground meat offers not only ample and desirable surfaces, but a thorough inoculation of the meat during grinding.

To retard microbial growth and biochemical changes, fresh meat is stored at temperatures near 0°C. However, at times, the meat is, or may be, subjected to higher temperatures. The temperature of the animal is approximately 37°C. After slaughter, the temperature tends to increase and then change to that of the room in which the meat is held. During subsequent processing of the carcass, mishandling may occur and the temperature may rise. Furthermore, if a consumer purchases the meat and simply places it in a car trunk on a hot sunny day, the meat's temperature may be raised to a dangerously high level.

At room temperature (20°C), the dominant organisms on spoilage of sheep meat were *Escherichia* and *Acinetobacter*-like organisms (Rao and Sreenivasamurthy 1985). They stated that *Enterobacter, Pseudomonas,* and *Staphylococcus* also could form a major part of this flora.

To reduce the biochemical and microbial changes, fresh meat is stored at temperatures near 0°C. At low temperatures, spoilage is evidenced on the surface. In storage conditions in which a moist meat surface in maintained, spoilage is due to Gram-negative bacteria, primarily species of *Pseudomonas.* Species of *Aeromonas, Alcaligenes. Acinetobacter, Moraxella, Flavobacterium, Enterobacter, Microbacterium, Proteus,* and *Streptococcus* occasionally are found in the surface flora of spoiled meat.

The defects of the meat are off-odor, slime formation, and discoloration. The flavor would be unacceptable to most people. In one study, researchers could not detect proteolysis prior to the appearance of spoilage odor and slime (Dainty et al. 1975).

TABLE 8.8. MICROBIAL DEFECTS OF RED MEATS AND THEIR PRODUCTS

Product	Defect	Organism
Fresh, refrigerated (0°–5°C)	Off-odor, slime, discoloration	*Pseudomonas, Aeromonas, Alcaligenes, Acinetobacter, Microbacterium, Moraxella, Proteus, Flavobacterium, Alteromonas, Saccharomyces*
	Lipolysis, pungent odor	*Pseudomonas*, yeasts
	Moldy	*Penicillium*
	Whiskers	*Thamnidium*
	Black spot	*Cladosporium*
	White spot	*Sporotrichum*
Vaccum packaged	Acid, sweet, rancid	*Lactobacillus, Microbacterium, Enterobacter, Hafnia*
Cured meat		
Bacon	Cheesy, sour, rancid	*Micrococcus*
	Discoloration	Molds
	Slight souring	*Lactobacillus, Micrococcus, Vibrio, Alcaligenes, Corynebacterium*
	Putrefaction	*Clostridium sporogenes*
Vacuum packaged	Cabbage odor	*Proteus inconstans*
	Tainted	*Vibrio*
Brines	Turbid	*Debaryomyces, Kloeckera*
Ham	Surface slime	*Micrococcus, Microbacterium*, Yeasts
	Gassy or puffy	*Clostridium*
	Green discoloration	*Lactobacillus, Streptococcus, Leuconostoc*
	Bone and meat "sours"	*Clostridium*
	Surface growth (dry-cured)	Molds
Sausages	Slime on surface	*Micrococcus*, yeasts
	Gas production (vacuum packed)	*Lactobacillus*
	Greenish discoloration	*Lactobacillus viridescens, Leuconostoc*
Fermented sausage	Slime	Yeasts
	Spots	Molds
Canned		
Commercially sterile	Gas, putrefaction	Sporeformers (*Bacillus, Clostridium*)
Semi-preserved	Souring, discoloration	*Streptococcus*
	Putrefaction, gas	*Bacillus, Clostridium*

The important biochemical changes occur in the meat juices that contain free amino acids, nucleotides, and peptides. These nutrients are sufficient for microbial growth, and the metabolism of these compounds leads to the formation of H_2S, NH_3 indole, and various amines and diamines.

The pigment stability in meat involves factors such as oxygen penetration, bacterial growth, lipolysis, and the presence of flavin compounds. Exposure to light can enhance the destruction of oxymyoglobin on the

surface of beef. The H_2S produced during breakdown of sulfur-containing amino acids reacts with myoglobin at low oxygen levels, forming the green sulfmyoglobin.

Water activity can play a role in spoilage. Fresh meat will lose moisture if stored below 99 percent RH. Even at an RH of 99 percent, the growth of *Pseudomonas* is reduced. At an a_w of 0.96 or less, most of the usual microorganisms causing spoilage of fresh meat are inhibited. When the surface of meat is lower than a_w 0.96, the slower-growing fungi become evident. In this situation, *Thamnidium chaetocladioides* and *T. elegans* (cause of "whiskers"), *Cladosporium herbarum* ("black spot"), and *Sporotrichum carnis* ("white spot") growth can produce moldy meat. Gill, Lowry, and Dimenna (1981) reported that besides *C. herbarum,* black spot is produced by *C. cladosporioides, Penicillium hirsutum,* and *Aureobasidium pullulans.* White-spot colonies yielded *Chrysosporium pannorum* or *auremonium.* Besides *T. elegans,* whiskers colonies yielded *Mucor racemosus* (Lowry and Gill 1984).

Storage atmosphere can influence meat spoilage. When meat is vacuum packaged, the dominant flora tends to be *Lactobacillus* instead of pseudomonads. Other microorganisms include *Leuconostoc, Brochothrix, Enterobacter, Hafnia,* and *Altermonas.* Odors of stored, vacuum-packaged meat have been described as acidic, sour, sweet, cheesy, stale, or rancid.

The spoilage of meat under both aerobic and anaerobic conditions was reviewed by Gill (1983).

Cured Meat. Meat that is treated with salts such as NaCl, $NaNO_2$, and $NaNO_3$ differ from untreated fresh meat. The most apparent differences are in the color and flavor of fresh and cured meat. However, the intrinsic factors of cured meat, such as pH, a_w, and the presence of inhibitors, play an important role in the microbial ecology of these products. Besides ham, bacon, and corned beef, there are cured comminuted meats wieners, bolognas).

The a_w of cured meat (0.88 to 0.95) is lower than that of fresh meat (0.98–1.00) and the specific inhibitory effects of the added salts may account for the differences in the spoilage patterns of fresh and cured meats.

For bacon, the important types of organisms are *Micrococcus,* yeast, *Vibrio, Acinetobacter, Alcaligenes, Arthrobacter,* and *Corynebacterium* (Gardner 1971). It is generally believed that bacon in aerobic conditions is spoiled by micrococci. However, a significant portion of the flora on the lean portion of the bacon is Gram-negative rods. As the pH of bacon decreases from 6.0 to 5.5, the yeasts in the flora increase during spoilage. Many molds, such as *Alternaria, Aspergillus, Mucor,* and *Rhizopus* can cause surface discolorations.

According to Gardner (1971), chemical changes do not occur in bacon

until the total microbial count reaches 10^8/g or higher. The off-odors of spoiled bacon were described as cheesy, putrid, or sour by Dempster (1972), or as smoked fish and rancid cheesy by Ingram and Dainty (1971). The rancid cheesy flavor was thought to be due to lipolysis of fat by micrococci.

The predominant microflora of vacuum-packaged bacon is determined by the salt content. Species of *Micrococcus* give way to lactic acid bacteria in bacon with low salt (5 to 7 percent), while they remain dominant in bacon with higher salt (8 to 12 percent).

For ham, the important type of spoilage is "souring." This includes various spoilage problems from mild degradation to extreme putrefaction. Before the modern short-curing methods, clostridia caused bone sours prior to the meat being chilled or cured.

The short-cure commercial method of curing has greatly reduced the incidence of ham souring. However, gassy or puffy hams can result if the fresh meat is held too long at a moderate temperature prior to cure. This allows clostridia to grow. Proper slaughter and bleeding of hogs, prompt chilling of the meat, prompt handling, good sanitation, and use of pump and cover brines that have low microbial counts have aided in improving the microbial quality of ham.

Modern commercial ham has a milder cure (with less salt), milder smoking, a more moist surface, and vacuum packaging, than dry-cured hams. Slicing during packaging creates sources of contamination and more cut surfaces on which microorganisms can grow. Although the dry-salt–cured hams suffer from mold growth on the surface, the commercial hams allow the growth of micrococci, microbacteria, various lactic acid bacteria, and yeasts.

Comminuted and cured sausage products are subjected to various types of spoilage. Generally, these products are composed of mixtures of pork, beef, salt, sugar, sodium nitrite, and spices. Hence, the flora of these products is somewhat different from that of fresh meat. Bell and Gill (1982) found that the flora of freshly cooked luncheon meat consisted of about equal numbers of *Bacillus* and *Micrococcus*. These researchers reported that little microbial growth could be detected when this meat was held at 10°C for forty-two days. However, at 25°C, the *Bacillus* grew at the surface and reduced the nitrite level and lowered the pH from 6.8 to 6.2, as well as produced amylase enzymes. By the fourteenth day, a *Streptococcus* replaced *Bacillus* and continued lowering the pH to 5.2. A low pH, vacuum packaging and low-temperature storage help extend the shelf life of this type of meat product.

A surface slime of micrococci and yeasts can occur when sufficient moisture is present. As the surface dries, inhibiting the bacteria, mold growth can cause spoilage. When products such as luncheon meats are

vacuum packaged, aerobic growth is inhibited and lactic acid bacteria become dominant. Facultatively anaerobic yeasts may grow. The lactic acid bacteria produce CO_2 and can cause a swelling of the package, while the yeasts produce a surface slime. At 25°C, the shelf life of chub-packed luncheon meat was inversely proportional to the availability of oxygen (Bell and DeLacy 1982). At 25°C, the principal spoilage organisms of chub-packed luncheon meat were *Bacillus licheniformis* and *Streptococcus faecium. B. licheniformis* caused surface softening, discoloration, gas production, and product liquefaction, while the *S. faecium* produced lactic acid which caused souring of the meat product.

For bologna products, a casing is used. Moisture accumulates at the meat-casing interface, allowing the growth of bacteria. Micrococci can cause a slime layer during growth at this interface.

During storage of these meats, microorganisms of the genera (*Micrococcus, Lactobacillus,* and *Leuconostoc* can grow and cause souring of the product. The production of organic acids and reducing compounds by the bacteria can cause a fading of the pink color at the outer surface of the product, resulting in a ring of discoloration.

Green discolorations can occur in cured sausages. The greening may appear as rings, cores, or on the surface. *Lactobacillus viridescens* can grow in products with reduced oxygen tension and pH and is the organism primarily involved in greening of these sausages. Other *Lactobacillus* and *Leuconostoc* species sometimes are present. The bacteria produce peroxides that react with the cured meat pigment, causing the green discoloration. The first evidence is the appearance of small greenish spots on the damp surfaces of the cured sausage. These spots tend to spread and cover the sausage surface if favorable conditions exist for a sufficient time. In some cases, a slimy appearance may be noted.

Fermented Sausages. Some comminuted sausages, such as cervelat, Thuringer, and Lebanon bologna, are not only cured but also are fermented by lactic acid bacteria to produce a tangy flavor. Dependence upon the natural contaminants to ferment the product may cause spoilage, so a starter culture of homofermentative lactic acid bacteria is used. The lower pH of these fermented sausages helps control some types of bacteria, but they are susceptible to spoilage similar to cured sausage products. A surface slime due to yeasts, and discoloration due to mold, may occur.

Canned Meat. Canned meat products are subject to the same type of spoilage as other low-acid foods, if heat-resistance spores which survive the process can germinate and grow. In semipreserved or pasteurized cured products, such as canned ham, the curing salts and refrigeration are used to prevent spore germination and outgrowth. If not adequately processed, thermoduric cells, such as *Streptococcus faecium,* may survive and

cause souring. This organism may cause rapid discoloration after the product is removed from the can. If the spores of clostridia are able to germinate, and the resultant cells grow, gas may be formed along with extensive putrefaction. If held at a high temperature (55°C), *Bacillus* species (*B. stearothermophilus* or *B. coagulans*) may cause flat sour spoilage of canned meats and of other canned foods.

POULTRY. The spoilage of uneviscerated and eviscerated poultry is somewhat different from the spoilage of canned meat. However, because commercial poultry in the United States is eviscerated, only this type of poultry is considered in this section. Uneviscerated poultry is discussed by Barnes and Shrimpton (1957).

Immediately after processing, any of several hundred species of microorganisms might be found. However, as the poultry is chilled and held in cold storage, psychrotrophic microorganisms predominate and cause deterioration.

The main defects are off-odor, which appears at a bacterial load between 10^6 and $10^8/cm^2$, and slime formation, which occurs soon after off-odor is noted. As the number of bacteria increases, the flavor score decreases, species of *Pseudomonas* are the principal spoilage organisms. Besides *Pseudomonas*, other organisms, similar to those in fresh red meat spoilage, are found on spoiled poultry (Table 8.9). These include *Aeromonas, Moraxella, Alcaligenes, Flavobacterium,* and *Micrococcus.* When antibiotics were used on poultry, yeasts became the important spoilage microorgan-

TABLE 8.9. MICROBIAL DEFECTS OF POULTRY AND POULTRY PRODUCTS

Product	Defect	Microorganism
Poultry meat	Off-odor, slime	*Pseudomonas, Acinetobacter, Moraxella, Alcaligenes, Aeromonas, Alteromonas*
Shell eggs	Black rot	*Proteus, Aeromonas*
	White rot (colorless)	*Citrobacter, Alcaligenes*
	Sour	*Pseudomonas*
	Green white	*P. fluorescens*
	Musty	*Pseudomonas*
	Moldy	Many types of molds
	Red rot	*Serratia marcescens*
	Custard rot	*Citrobacter, Proteus, Enterobacter*
	Yellow and green rot	*Alcaligenes, Flavobacterium, Cytophaga*
Liquid whole egg	Fishy	*Pseudomonas, Flavobacterium, Chromobacterium*
	Off-odor, sour	*Proteus, Alcaligenes, Escherichia, Flavobacterium, Pseudomonas, Bacillus*
Liquid albumen	Off-odor	*Pseudomonas, Acinetobacter, Enterobacter*

isms. However, yeasts are not important on normally processed and eviscerated poultry.

Daud, McMeekin, and Thomas (1979) studied the flora of chicken meat held at 2°C. Before storage, the flora included species of *Micrococcus, Flavobacterium, Cytophaga, Acinetobacter, Moraxella,* and *Pseudomonas,* as well as enterics. During storage, the off-odor producers (principally *Pseudomonas* II) became dominant, accounting for 62 percent of the population after twelve days. At this time, species of *Pseudomonas* accounted for 98 percent of the bacterial population. Nonpigmented strains of pseudomonads produce more intense odors than do pigmented strains. Off-odors of poultry have been listed as sulfidelike, fruity, fishy, and like evaporated milk.

The chemicals produced by pseudomonads grown in chicken meat include acetone, 2-butanone, 2-pentanone, toluene, dimethyl benzenes, ethyl methyl disulfide, 2-heptanone, and dimethyl trisulfide (Pittard et al. 1982). Some of the more important volatile sulfur-containing compounds are dimethyl disulfide, dimethyl sulfide, methanethiol and propylene sulfide (Bowman et al. 1983).

Packaging poultry in oxygen-impermeable film or vacuum packaging it causes inhibition of psuedomonads so that the predominant flora at spoilage consists of lactic and enteric types of bacteria.

EGGS. It is generally accepted that, when an egg is laid, its contents are free from bacteria. But there are exceptions because of ovarian infections.

The egg contents are protected by the shell and associated membranes and chemical inhibitors in the egg albumen. This means that microorganisms must penetrate these barriers and then be able to grow inside the egg to cause spoilage. Penetration of eggs is aided by moisture on the shell. If the eggs are not properly stored or washed, penetration may be quite rapid, and spoilage can occur.

For egg products, the shell and membranes are removed. During processing, the liquid egg is subject to contamination from organisms on the egg shell, equipment, humans, added ingredients, and the final container.

Bacterial analysis of shell eggs during storage has revealed molds (*Penicillium, Aspergillus, Cladosporium, Rhizopus,* and *Mucor*), yeasts (*Rhodotorula*), and bacteria (*Pseudomonas, Micrococcus, Bacillus, Proteus, Alcaligenes, Flavobacterium, Citrobacter, Escherichia,* and *Enterobacter*).

As with other protein foods, species of *Pseudomonas* are the main spoilage organisms of shell eggs and egg products.

When bacteria grow within the egg, they decompose the contents and

form by-products. This results in characteristic odors, appearance, or colors from which the various rots acquire their name (Table 8.9).

The United States Department of Agriculture (USDA 1977) described a loss egg as "an egg that is inedible, smashed or broken so that the contents are leaking, frozen, contaminated, or containing bloody whites, large blood spots, largely unsightly meat spots, or other foreign material." Inedible eggs are due to nonmicrobial defects as well as microbial defects. An egg with a blood ring, embryo, or stuck yolk is inedible.

Besides these loss types, shell eggs will absorb odors from the storage atmosphere. Most vegetables and fruits will impart flavors and odors to shell eggs. If stored with apples, the eggs will be bitter and have a cardboard flavor and odor. If stored near gasoline or kerosene, the shell eggs will taste and smell like these compounds.

Eggs that become inedible due to microbial growth are listed by USDA (1977) as black rots, white rots, sour eggs, eggs with green whites, mixed rots, musty eggs, and moldy eggs.

Other designations for rotten eggs are fluorescent green, red, custard, colorless, green and yellow, mixed, and rusty red rots.

Black Rots. When viewed with a candling light, black rot eggs are virtually opaque. When broken out, the egg content has a muddy (dark brown) appearance and a repulsive putrid odor, and H_2S is evident. In many eggs, an internal gas pressure develops. The bacteria associated with this type of spoilage are species of *Proteus* and *Aeromonas.* Rots of this type are more likely to occur at room temperature (20°C) than in cold storage (4°C or less).

White Rot. Threadlike shadows may be seen in the thin white, and in later stages the yolk appears severely blemished when the shell egg is viewed with the candling light. When opened, the egg yolk shows a crusted appearance and frequently has a fruity odor.

This type of inedible egg sometimes is referred to as a colorless rot. Various organisms have been associated with this rot, including *Citrobacter, Salmonella,* and *Alcaligenes.*

Sour Eggs. These eggs are difficult to detect by ordinary candling, but they usually show a weak white and murky shadow around an off-center, swollen yolk. These eggs also are called fluorescent and are quite readily detected by observing with UV light. A green sheen is produced by species of *Pseudomonas.*

Since a green fluorescence is observed, these inedible eggs also have been termed fluorescent green rots.

Green Whites. This defect is caused mainly by *Pseudomonas fluorescens.* Green whites of broken-out eggs fluoresce when observed with UV light.

Eggs with green whites may or may not have a sour odor, since the green fluorescence can be observed long before any odor can be detected.

Musty Eggs. These frequently appear clear and free from foreign material when candled. The musty odor may be caused by odors in the atmosphere being absorbed by the egg contents. Also, some microorganisms occasionally invade shell eggs and produce a musty odor.

Moldy Eggs. Mold growth is visible as spots on the shell, in checked areas of the shell, or inside the egg.

The mold contamination of eggs seems to be due to the reuse of moldy packing materials. Several molds, such as *Penicillium, Alternaria,* and *Rhizopus* can grow on eggs. However, molds are not a prominent cause of egg spoilage.

Red Rot. These eggs are distinguished by a red discoloration of the albumen and the surface of the yolk. An ammoniacal to putrid odor may occur. *Serratia marcescens* has been considered as the cause of red rot.

Custard Rot. In this rot, the yolk is encrusted with custardlike material and occasionally flecked with olive-green pigment. The albumen becomes thin with an orange tint. There may be a slightly putrid to putrid odor. *Citrobacter* and *Proteus vulgaris* have been associated with this type of spoilage.

Mixed Rot. These addled eggs occur when the vitelline membrane of the yolk breaks and the yolk mixes with the white, resulting in a murkiness throughout the interior of the egg when viewed with a candling light. With no off-odor, these are referred to as odorless mixed rots.

Other Rots. Alcaligenes has been accused of causing both yellow rots and green rots. These rots are similar in odor and in the appearance of albumen. However, the yolk is dark yellow in the former and dark green to black in the latter case.

Rust red rot is associated with growth of *P. vulgaris.*

Egg Products. The spoilage of egg products is evidenced by off-odors described as sour, fecal, or fishy. Organic acids (lactic, succinic) are recognized as indices of decomposition of egg products.

SEAFOODS. Fish and other seafoods are subject to contamination of microorganisms in their marine environment, as well as those acquired during catching, handling, and processing. However, as with other foods, even though a mixed flora exists, certain types of organisms become dominant during storage of the product. The dominant types are those that can survive and multiply on or in the food at the temperature that prevails. The defects of seafoods are listed in Table 8.10.

TABLE 8.10. SOME MICROBIAL DEFECTS OF SEAFOODS

Product	Defect	Microorganism
Fish		
Fresh	Off-odor	*Pseudomonas, Alteromonas, Acinetobacter, Vibrio, Aeromonas, Moraxella, Proteus*
	Fruity	*Pseudomonas*
	Ammoniacal	*Pseudomonas, Alteromonas*
	H_2S odor	*Pseudomonas, Alteromonas*
Salted	Pink	*Halobacterium, Halococcus*
	Dun (red growth)	*Hemispora stellata, Sporendonema epizoum*
	Cheesy, putrefactive	Red halophilic bacteria
Crayfish	Sweet to foul odor	*Pseudomonas, Lactobacillus,* Coryneforms
Oysters	Pink	Yeasts (*Rhodotorula*)
Shrimp	Off-odor	*Pseudomonas*
Squid	Yellow discoloration	*P. putida*
	Red discoloration	*Serratia marcescens*

Fish. The spoilage pattern depends somewhat on the species and type of fish, the initial microbial flora, the area of catch, the method of catch, processing method, and method of storage.

The initial microbial flora is dependent upon the contamination of the water and bottom sediment from the area of catch. Fish caught on a line have lower counts than fish that are trawled by dragging a net along the bottom. The trawl net drags through the bottom sediment, which usually has high counts of microorganisms.

The temperature of the water affects the fish microflora. Fish from warm seas tend to have mesophilic strains; in cold seas, the organisms are primarily psychrotrophs.

The methods for handling the fish after catching influence the contamination. Fish do not have a sphincter muscle, so when any pressure is exerted, intestinal discharge occurs. This pressure can happen when the net is hauled aboard, when the fish are piled on the deck, or when whole fish are stored in ice.

The caught fish may be left whole, or they may be beheaded or gutted. A gutted fish may spoil somewhat differently than a whole fish, due to the intestinal flora.

The fish can pick up added contamination on shipboard if the holding areas are not maintained in a sanitary manner. The fish must be handled rapidly on the ship, and chilled. Usually the fish are mixed with ice or with chilled seawater for cooling; however, some ships have freezing facilities.

During processing, the organisms in the surface slime layer or the skin can be spread throughout the processing equipment, the workers, and onto the flesh of the fillet. Hence, the normal potentially sterile flesh

can be inoculated with millions of bacteria. With this contamination, the fish may be spoiled by the time they reach the consumer. The microbial flora on the fish leaving the processing plant may be different from that on the fish entering the plant.

Acinetobacter, Aeromonas, Alcaligenes, Bacillus, Clostridium, coryneforms, *Cytophaga, Flavobacterium, Micrococcus, Proteus, Psuedomonas,* and *Vibrio* have been isolated from fresh and spoiled fish. Members of the family Enterobacteriaceae rarely are found on fish but may be detected on fish harvested in polluted areas of both fresh and saltwater.

During storage of the fish at 0°C, the number of organisms increases after a one- to two-day lag period. During spoilage, the *Pseudomonas* and *Acinetobacter* become increasingly dominant, with *Flavobacterium* showing a transient increase. By ten to twelve days, *Pseudomonas* may comprise 90 percent of the population. The species of *Pseudomonas* tend to change as spoilage progresses. Regardless of which species are involved, organisms of the genus *Pseudomonas* are the most active spoilage organisms in fish stored at 0°C. The primary factor influencing the spoilage rate is temperature. As the temperature is increased from 0°C the growth rate of pseudomonads increases. However, as the temperature approaches 20°C, the mesophilic flora becomes evident.

Lipids in fish have a high content of polyunsaturated fatty acids. Hence, oxidative rancidity is more evident in fish than in most other animal tissue.

In marine fish, the trimethylamine oxide (TMAO) is reduced by bacterial and enzymatic action to trimethylamine (TMA), a spoilage product. The odor of trimethylamine at low levels is referred to as a stale fishy odor. Piperidine, δ-amino valeric acid, and δ-amino valeric aldehyde also have a fishy odor.

During spoilage, volatile bases, amines, and organic acids are formed by decarboxylation or by deamination of amino acids and organic bases. Hydrogen sulfide, mercaptans, and disulfides add to spoilage odors. Spoilage odors have been designated as fishy, stale, musty, rancid, sour, ammoniacal, yeasty, fruity, sweet, acid, and putrid.

Vacuum packaging of fish changes the spoilage flora in a manner similar to that of fresh meat. *Lactobacillus* and *Microbacterium* grow and cause souring, whereas the aerobic psuedomonads are inhibited. However, for vacuum-packaged sand flatheat fillets, McMeekin, Hulse, and Bremner (1982) reported *Alteromonas* species dominated the spoilage flora. Reducing the pH increased the number of Enterobacteriaceae. Packaging of swordfish steaks in 100 percent CO_2 resulted in lactobacilli becoming dominant (Lannelongue et al. 1982).

Shewan (1971) stated that bacterial spoilage can be prevented by storing fish below −10°C, so, from a practical viewpoint, the bacteriology of

spoilage is somewhat academic. This is also true for spoilage of meat and poultry products. It is unfortunate that so much of our meat, poultry, and fish is maintained in the so-called fresh condition, rather than being frozen.

Crayfish. The spoilage bacteria of freshwater crayfish was studied by Cox and Lovell (1973). They reported that spoilage of the tail occurred when the total aerobic count reached 10^9/g. In the fresh meat, *Micrococcus, Staphylococcus,* and *Alcaligenes* were the major genera, while in meat spoilage, at 0°C, *Pseudomonas,* and at 5°C, *Achromobacter* (*Acinetobacter, Moraxella*) predominated. These genera comprised the major bacterial flora of the spoiled meat at either temperature (0° or 5°C).

Clams. The microbiology of soft shell clams was reported by Cox (1965). The clams were judged to be unacceptable on the basis of black discoloration or a slimy appearance. The predominating spoilage bacteria are Gram-negative rods that produce sweet esterlike, musty, oniony, or cesspool-like odors on agar media. He suggested that the spoilage flora consisted mainly of *Pseudomonas* types.

Crabs. During processing of red crab, the aerobic plate count (35°C) increased from 170 CFU/g after cooking to 270,000 CFU/g in the finished product (Wentz et al. 1985). Loaharanu and Lopez (1970) studied the spoilage of pasteurized crab cake mix. When stored at 18° or 30°C *Bacillus* and *Micrococcus* species dominated, while at 2°C, the dominant genus was *Alcaligenes.*

Oysters. Frozen or fresh oysters spoil by either fermentative mechanisms (characterized by a sour odor) or by putrefactive mechanisms. Canned oyster decomposition is putrefactive. Oysters that are well cooked and then refrigerated spoil due to fat rancidity, rather than to microbial decomposition. Red or pink coloration of oysters may be due to *S. marcescens* or to pink yeasts, such as *Rhodotorula,* as well as to phytoplankton or eggs of the oyster crab. Various discolorations can occur in shellfish (Boon 1977).

Shrimp. These animals are caught at sea by towing a trawl net along the bottom. After one to three hours of trawling, the net is hauled aboard and emptied. Since the bottom sediment is more contaminated with bacteria than the upper water, one would expect harvested shrimp to have rather high bacterial counts. Novak (1973) reported counts of 31,000 to 1,200,000 bacteria per gram of whole shimp. He further reported that, although the head was 40 percent of the shrimp weight, it contained 75 percent of the bacteria. Therefore, heading the shrimp lowered the microbial count.

Bieler, Matthews, and Koburger (1973) reported that the heading of shrimp exposed tissues, allowed the gut to empty, and triggered enzymatic changes. Hence, they found that shrimp stored in ice maintained better quality with the head on than when headless.

The predominant organisms on iced raw shrimp are *Moraxella, Pseudomonas, Acinetobacter, Arthrobacter, Flavobacterium* and *Cytophaga* (Lee and Pfeifer 1977). They believed that the presence of *Arthrobacter* and *Acinetobacter* may indicate inadequate cleaning. The presence of *Moraxella, Flavobacterium,* and *Cytophaga* indicates the degree of secondary contamination, and *Pseudomonas* indicates the potential shelf life of processed shrimp. When shrimp was stored at 0°C for eight days, *Pseudomonas* species predominated (Matches 1982). Storage at 5.6 or 11°C resulted in the dominance of *Moraxella,* while at 16° or 22°C, *Proteus,* which produced indole, became dominant. Researchers reported that 42 of 1,647 isolates from putrid shrimp produced indole (Smith et al. 1984). The indole levels of severely decomposed shrimp were less than the 25 μg/100g of shrimp suggested as the defect action level by the FDA (Chang et al. 1983).

DAIRY PRODUCTS. Milk, when obtained from the cow, might not be sterile. During and after milking, the milk is subjected to organisms from various sources, although contaminated equipment used to handle, transport, store, and process the milk seems to be the main source of organisms. The genera found in cold-stored milk include *Pseudomonas, Acinetobacter, Aerobacter, Alcaligenes,* and *Flavobacterium* as well as members of the family Enterobacteriaceae. With extended storage of milk products at refrigerated temperatures, psychrophilic or psychrotrophic organisms are a cause of spoilage.

Spoilage of milk usually is caused by psychrotrophs that recontaminate the milk after pasteurization. Also involved are thermoduric psychrotrophs as well as heat-stable proteases produced before pasteurization.

The defects that can occur in milk due to microbial growth are (1) off-flavors; (2) lipolysis with development of rancidity; (3) gas production; (4) fermentation to lactic acid with souring; (5) coagulation of milk proteins; (6) viscous or ropy texture; and (7) discoloration. Some of the defects of milk and dairy products are listed in Table 8.11.

As with other animal products, species of *Pseudomonas* are prominent in producing defects in milk. They are associated with a bitter flavor and rancidity as well as a fruity odor. The fruity odor is due to a mixture of ethyl butyrate and ethyl hexanoate.

Certain lactic acid bacteria are involved with defects in fresh milk. Although the production of lactic acid and the resulting souring are desirable in fermented milk products, they are undesirable in fresh milk. Besides this type of defect, a strain of *Streptococcus lactis* designated as

TABLE 8.11. SOME MICROBIAL DEFECTS IN DAIRY PRODUCTS

Product	Defect	Microorganism
Milk		
Pasteurized, refrigerated	Rancid	*Pseudomonas, Alcaligenes, Staphylococcus*
	Ropy or slimy	Coliforms, *Pseudomonas, Alcaligenes, Micrococcus, Bacillus subtilis*
	Sour (acid, gas)	Lactic acid bacteria
	Discoloration	*Chromobacterium*
	Bitter, fruity	*Pseudomonas, Flavobacterium, Alcaligenes, Proteus, Acinetobacter*
Canned	Swelling, gas	*Clostridium sporogenes*
Cream	Foamy	*Candida, Torulopsis*
Butter	Surface taint	*Pseudomonas, Alteromonas*
Cheese	Gassy, butyric acid	*Clostridium tyrobutyricum*
	Gassy, floating, or split curd	*Leuconostoc, S. lactis* subsp. *diacetylactis*
	Moldy	*Penicillium, Scopulariopsis, Mucor,* other molds
Soft	Black mold (cat's fur)	*Mucor*
	Surface growth	*Torulopsis, Debaryomyces*
Cottage	Slimy curd, putrid odor	*Pseudomonas*
	Discoloration	*Flavobacterium,* yeasts, molds
	Slimy, gelatinous	*Pseudomonas, Alcaligenes, Flavobacterium,* coliforms
	Fruity	Yeasts
Cheddar	Sweet, yeasty, fruity	Yeasts
Swiss	Gassy, sweet	Yeasts (*Torulopsis*)
	Off-odor	*C. sporogenes*
Yogurt	Yeasty	Yeasts (*Torulopsis*)

variety *maltigenes* metabolizes leucine and produces 3-methylbutanol, which gives a malty flavor defect in dairy products. At the time the aroma first becomes detectable, counts range from 10^7 to 10^8 per milliliter. Ropy milk has been ascribed to several organisms, such as *Alcaligenes, Klebsiella, Pseudomonas,* and *Flavobacterium. Streptococcus lactis longi* and *Streptococcus cremoris longi* were isolated from ropy milk (Macura and Townsley 1984).

Butter. This is primarily fat with some moisture, carbohydrate, and protein. The main defect is due to rancidity either by autoxidation of fatty acids or by lipolysis due to microbial enzymes. The presence of short-chain acids (butyric and caproic) results in a rancid flavor. Molds are able to grow on the surface of butter. Putrid, proteolytic, and fruity flavors in butters are caused by psychrotrophic bacteria.

Surface taint and yeasty butter are other defects of this product caused by microorganisms. Butter is made from cream. If the cream has a defect, such as rancidity, it can affect the quality of the butter.

Cheese. Defects that can occur in milk may appear in other dairy products made from the defective milk.

In cottage cheese, the flavor defects of milk may be accompanied by a gelatinous or tapioca curd formation. Lipolytic and proteolytic organisms cause defects in cottage cheese.

P. fragi and *A. putrefaciens* have been associated with a slimy curd defect on the surface of cottage cheese. A fruity, putrid, or rancid odor and a fruity or bitter flavor may accompany this defect.

Surface discolorations may occur due to the growth of the pigmented *Flavobacterium. E. coli* can cause barny or unclean flavors and, if the cottage cheese is held at room temperature, the organism can cause a gassy defect.

The yeast *Rhodotorula* produces pink spots on the surface of the product. These pink spots eventually become a pink slime. *Torulopsis* also produces a slime, but it is yellow. *Geotrichum* produces off-white, tan, or yellow surface discolorations.

Molds multiply rather slowly. Cottage cheese may be spoiled by other organisms before mold growth is evident. However, there are occasions when growth of *Penicillium* and *Mucor* appears on the surface of the product.

A pink discoloration of Romano and other Italian cheese varieties was investigated by Shannon, Olson, and Vonelbe (1969). The pink discoloration occurred as a uniform band of color near the cheese surface, or as discoloration throughout the entire cheese. According to them, species of *Lactobacillus* (*L. helveticus* and *L. bulgaricus*) can cause this discoloration. Pink discolorations due to strains of *Propionibacterium,* certain reducing *Bacillus* species and reducing micrococci have been suggested as causing pink to red discolorations in various cheeses.

Early gas defects of semihard cheeses usually are caused by coliforms, while late gas defects are caused by fermentation by *Clostridium tyrobutyricum.*

The growth of molds such as *Penicillium, Mucor,* and *Aspergillus* can appear on the surface of cheese. Some of the molds growing on cheese can produce mycotoxins (see Chapter 6).

Fats and Oils

Rancidity, acidity, soapiness, and off-flavor defects can occur in fatty foods, such as butter, margarine, lard, and vegetable oils. The development of rancidity can be due to autoxidative deterioration, lipolysis by microbial lipases, or lipoxidation due to lipoxidase enzymes.

There are several lipolytic microorganisms (Table 8.4). Microorgan-

isms associated with the liberation of free fatty acids in butter and margarine include *Cladosporium, Candida, Pseudomonas, Micrococcus, Penicillium,* and *Geotrichum.* The hydrolysis of olive oil, coconut oil, and butterfat can be caused by species of *Micrococcus, Pseudomonas, Serratia,* and *Staphylococcus.*

The defect characterized by rancidity and acidity is due to the liberation of free fatty acids, especially butyric (C_4), caproic (C_6), caprylic (C_8), and capric (C_{10}), and the production of their corresponding methyl ketones. When lauric and myristic acids are liberated from triglycerides in coconut oil and butterfat, a soapy flavor defect can occur.

The spoilage of mayonnaise and salad dressings was investigated by Kurtzman, Rogers, and Hesseltine (1971). *Saccharomyces bailii* was the principal cause of mayonnaise spoilage, although *Lactobacillus fructivorans* also was involved. Species of *Bacillus* were isolated from several spoiled samples. The yeast *Zygosaccharomyces* was involved in spoilage of French salad dressing. Rather low numbers of yeast and bacteria were present in the spoiled product. They reported only five yeasts per gram in one spoiled mayonnaise sample. The highest yeast count they reported was 165,500 per gram.

Canned Food

Canned foods are discussed in the section on heat preservation, but some recognition of them should be made here.

Bacterial spoilage of canned food may be due to underprocessing, elevated storage temperature, or leakage of the container due to improper closures, rough handling, or defective cans.

Underprocessed cans of foods usually are spoiled by heat-resistant sporeformers (*Bacillus, Desulfotomaculum,* and *Clostridium*), which survive the process. Leakage can result in spoilage by nonsporeformers, which could not have survived the heat processing, but recontaminate the food after the heat treatment. Heat-resistant thermophilic organisms may survive the processing and, if the food is not cooled after heating, or if the cans are stored at elevated temperatures, these thermophiles can grow. There are four prominent thermophilic organisms causing spoilage of canned food. *B. stearothermophilus* causes flat sour spoilage of canned nonacid foods. It is facultatively anaerobic and can grow at 70°C. *B. coagulans* is less heat resistant but is more acid tolerant than *B. stearothermophilus.* It causes flat sour spoilage of canned tomatoes.

Clostridium thermosaccharolyticum forms large volumes of gas and resultant hard swells or blown cans. The spores are very heat resistant. Blown cans can be caused by various species of clostridia. *Desulfotomaculum nigri-*

cans produces hydrogen sulfide. This gas reacts with metallic ions from the container or food, forming sulfides, some of which are black.

Molds and yeasts are not significantly important in the spoilage of canned foods, except for mold spores such as *B. fulva,* which can survive fruit-processing treatments.

The movement of bacteria from contaminated cooling water into the processed canned food may result in various spoilage problems. Bean and Everton (1971) discussed a problem of pigmented bacteria surviving the chlorination treatment given the water used to cool heat-processed cans of product. No gas was noted (no swelling of the cans), but a canned pudding developed cheesy to sour flavors and there was separation of the product.

Bacillus subtilis and *Enterococcus faecium* were isolated from a can of coagulated evaporated milk by McKellar and Nichols-Nelson (1984). Neither organism alone caused the defect. They concluded that *B. subtilis* supplied low-molecular-weight nitrogenous compounds for the growth of *E. faecium,* which produced acid, causing the coagulation.

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9

Useful Microorganisms

Microorganisms are used in many facets of the food industry. Desired alterations of food by microorganisms are referred to as fermentations, regardless of the type of metabolism. By definition, fermentation is the anaerobic breakdown of an organic substance by an enzyme system, in which the final hydrogen acceptor is an organic compound. Hence, the aerobic oxidation of alcohol to acetic acid in vinegar production is not a true fermentation. Hence, for our purposes these alterations of foods are called food conversions. Since the enzyme systems of the microorganisms catalyze the changes in foods, for some reactions it is advantageous to use purified enzymes separated from the microbial cells.

Food conversions may reduce the food volume as well as aid in preservation. An example is the changing of milk to cheese. These changes can eliminate undesirable or toxic factors as well as enhance the nutrient value of foods. It is possible to convert substances presently considered wastes into useful products or foods.

Besides changing the form of food, microorganisms can be used to produce chemicals for food additives, such as acids, flavors, vitamins, amino acids, or protein. Systems using microorganisms or enzymes have been developed to assay for vitamins, amino acids, and other constituents of foods. These systems have certain advantages over normally used chemical systems for analysis.

FOOD CONVERSIONS

The conversion of one food to another may be called controlled degradation. Various types of food conversions are listed in Table 9.1. This table does not list all of the many food conversions. Other sources, such as Pederson (1979) and Reed (1982) should be consulted for more complete descriptions of food conversions than is possible in this text. In many countries, food fermentation is used as a system of preservation, whereas in the United States these foods are produced for their special flavors or other characteristics.

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TABLE 9.1. SOME MICROBIAL FOOD CONVERSIONS

Organisms	Substrates	Products
Lactic acid bacteria, species of *Leuconostoc, Lactobacillus, Pediococcus* and/or *Streptococcus*	Cabbage	Sauerkraut
	Cucumbers	Pickles
	Olives	Olives (green, ripe)
	Vanilla beans	Vanilla
	Red meat	Sausages, (salami, Thuringer, Lebanon bologna, cervelat, summer, pepperoni)
	Milk products	
	Cream	Sour cream, cultured butter, ghee
	Milk	Cultured milk, acidophilus, yogurt
	Milk	Cheese—unripened (cottage, pot, cream)
	Milk	Cheese—ripened (Cheddar, American, Edam, Cheshire)
Propionibacterium	Unripened cheese	Cheese (Swiss, Emmenthaler, Gruyere)
Brevibacterium linens	Unripened cheese	Cheese (limburger, brick, Trappist)
Penicillium roqueforti	Unripened cheese	Cheese (Roquefort, blue, Stilton, Gorgonzola)
P. camemberti	Unripened cheese	Camembert cheese
Lactic acid bacteria with yeasts	Taro	Poi
	Flour (dough)	Sour dough bread, pancakes
	Ginger plant	Ginger beer
	Beans	Vermicelli
	Rice, black gram	Idli
Saccharomyces, Aspergillus oryzae	Soybeans, wheat	Shogu (soy sauce), miso
A. flavus or *Mucor rouxii,* yeasts	Kaffir corn	Kaffir corn beer
Yeasts	Malt	Beer, ale, stout, lager, bock, porter, Pilsner
	Fruit	Wine, vermouth
	Wines	Brandy
	Molasses	Rum
	Grain mash	Whiskey
	Flour (dough)	Bread
	Fruit peel	Citron
Yeasts with *Acetobacter* or *Gluconobacter*	Sugar, fruit, potatoes, honey, malt, grain alcohol	Vinegar
Halophilic bacteria	Fish	Nuoc-mam-ngapi
Bacillus subtilis	Soybeans	Natto
Actinomucor elegans, Mucor	Soybean curd	Sufu (Chinese cheese, bean cake)
Rhizopus oligosporus	Soybeans	Tempeh, tempe
R. oligosporus	Coconut press cake	Tempeh bongkrek
Neurospora sitophila	Peanut press cake	Ontjom

Food Cultures

The food conversions or biotransformations are listed in Table 9.1 according to the microorganisms involved. The addition of previously fermented product or the selection of microflora naturally present have been used in traditionally fermented foods. However, to assure that proper reactions occur and to obtain larger yields of high quality product, cultures of selected and special organisms are added at the beginning of the food conversion. The culture may consist of one strain of one organism, or more likely, multiple or mixed strains of one or several organisms.

Microorganisms tend to be energy efficient, and they also have feedback mechanisms to prevent the accumulation of a product that is useless to them but may be useful to us. However, mutants of the organism may be found that do accumulate these products. This selection of exceptionally good organisms may require the screening of hundreds or thousands of organisms.

To improve the organisms in the culture, the manipulation of genetic information in the cells may be used. By genetic engineering, certain traits can be added to or subtracted from the cells. These manipulated cells may produce an improved product with a greater yield.

Since much work is needed to acquire an exceptional culture, it is important that the culture be properly maintained so that it does not lose its desirable traits. Quite often, the organisms contain desirable plasmids that can be lost after several transfers. Hence, after growth, the cultures are frozen or freeze-dried for storage. Frozen cultures are stored at $-20°$ to $-40°$C. How the culture is obtained and maintained depends upon the food substrate as well as on the company policy. There are companies that produce and sell commercial cultures for use by other companies. People in the dairy, meat, and baking industries tend to use commercial cultures, while beer and wine processors usually develop and maintain their own cultures.

Although some foods may be pasteurized or treated with chemicals, usually they are not sterile when subjected to fermentation. Hence, the inoculated culture must be sufficiently large and active so that it can outgrow the residual organisms. When an inoculum of 5 percent or more of the weight or volume of the food is needed, the amount of culture is increased by inoculation and incubation of increasing amounts of substrate.

In some cases, such as in the continuous production of alcohol, the immobilization of microbial cells may be advantageous. The immobilized cells are actually complex enzyme systems. The immobilization of cells was reviewed by Fukui and Tanaka (1982). The process is discussed fur-

ther in this chapter for some specific products as well as in the section concerning enzymes.

Lactic Acid Bacteria

The organisms referred to as lactic acid bacteria include certain species in the genera *Streptococcus, Pediococcus, Leuconostoc,* and *Lactobacillus.* The first two genera are homofermentative, the leuconostocs are heterofermentative, and the lactobacilli include both homofermentative and heterofermentative types. The homofermenters convert carbohydrates primarily to lactic acid, while the heterofermenters produce lactic acid and substances such as acetic acid, ethyl alcohol, and carbon dioxide. The homofermenters use the glycolytic (hexose diphosphate) pathway to degrade carbohydrates, while the heterofermenters use the phosphoketolase as well as alternative pathways. Some recent studies indicate that some homofermenters have the potential for heterofermentation. The genetic determinant for lactose metabolism by various streptococci is found in a plasmid, which can be lost or transferred (Fujita, Okamoto, and Irie 1984; Orberg and Sandine 1985; Scherwitz, Baldwin, and McKay 1983).

Many of the lactic streptococci cultures used in fermentations are susceptible to lytic bacteriophage. Lysis by the phage is believed to be the primary cause of slow acid production during the fermentation of dairy products (Teuber and Lembke 1983).

The lactic acid bacteria usually grow in a sequence in a food product. Generally, the leuconostocs or streptococci begin the fermentation and are followed by pediococci and lactobacilli. The production of lactic acid from sugar is used in vegetable, fruit, and dairy product fermentations. There are many similarities in the lactic acid fermentation, but because of the various substrates, there are some differences.

SAUERKRAUT. This means "acid cabbage." According to definition, sauerkraut is the clean, sound product of characteristic flavor, obtained by full fermentation, chiefly lactic, of properly prepared and shredded cabbage in the presence of not less than 2 percent nor more than 3 percent salt. Upon completion of the fermentation, it contains not less than 1½ percent of acid, expressed as lactic acid. Sauerkraut that has been rebrined in the process of canning or repacking contains not less than 1 percent of acid, expressed as lactic acid.

Sound heads of cabbage are prepared by washing and removing the outer leaves and any defective leaves (Fig. 9.1). The core is removed and the leaves are shredded. Shredding the leaves gives a larger total surface area and allows the extraction of juice. About 2.25 percent salt is added

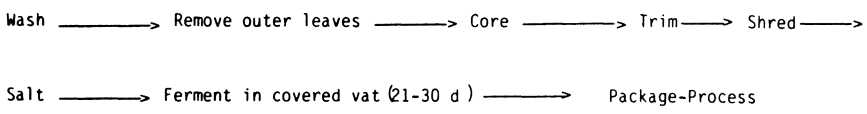

Figure 9.1. Flow sheet for sauerkraut.

uniformly to layers of shredded cabbage. The combination of salt and packing down of the cabbage expels the juice from the cabbage, and a brine results. The juice contains sugars and other nutrients from the cabbage. Depending upon the variety of cabbage and other factors, the sugar content is about 3 to 6 percent. The amount of sugar influences the fermentation and the final acidity.

When the vat or tank is essentially full, a plastic sheet is used as a cover to keep out dirt and air. The environmental conditions, numbers, and kinds of microorganisms, cleanliness of cabbage and vat, salt concentration and distribution, temperature, and covering influence the fermentation (Pederson 1979).

Many types of microorganisms are associated with raw cabbage. Most of these are not involved in the fermentation. Cabbage contains substances that are inhibitory to Gram-negative bacteria (Fig. 9.2). With the inhibitors, salt, and an anaerobic environment, the lactic acid bacteria tend to dominate. The fermentation is started by *Leuconostoc mesenteroides.* It converts the sugar to lactic acid, acetic acid, alcohol, CO_2, and other products that contribute to the flavor of sauerkraut. The CO_2 helps maintain anaerobic conditions in the fermenting cabbage.

As the acids accumulate, *L. mesenteroides* is inhibited, but the fermentation continues with *Lactobacillus brevis, Pediococcus cerevisiae,* and finally, *Lactobacillus plantarum.* The sequence of organisms and acid production during a typical sauerkraut fermentation are listed in Table 9.2.

The proper concentration of salt favors the growth of the lactic acid bacteria in the correct sequence. Too little salt results in poor flavor and soft kraut. Too much salt inhibits the lactic acid bacteria and may result in an acid flavor, darkening, and growth of yeasts (Pederson 1979).

Although salt influences certain organisms, the inhibition of undesirable bacteria is due primarily to the acid produced by the fermentation of sugars.

Keeping air from the fermentation is important in controlling molds and yeasts. Air allows the growth of these microorganisms on the cabbage surface and can result in softening, darkening, and the development of undesirable flavors. The product is not microbiologically stable, so it is important that it be refrigerated or heat processed after packaging.

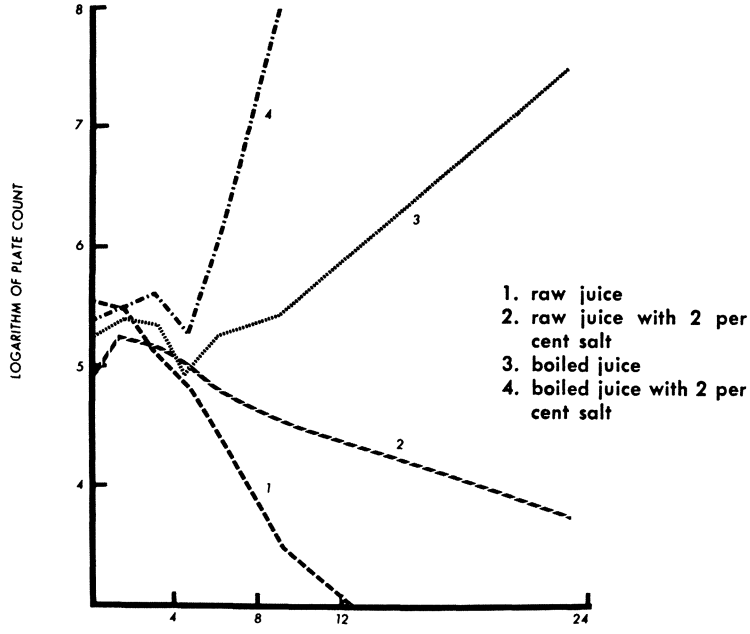

Figure 9.2. Bactericidal action of cabbage juice toward a Gram-negative bacterium obtained from a cabbage surface.
Courtesy of Pederson (1979).

PICKLES. There are many pickle products produced from cucumbers. Reportedly, three methods are available for pickle processing. Fresh-pack pickle products are directly acidified cucumbers (acetic acid is preferred) which are pasteurized. Various spices are added to give the characteristic flavor of the particular product. Researchers found that a combination of an acidity of 0.6 percent (acetic acid) and heating to an internal temperature of 71° to 77°C prevented spoilage of the product (Monroe et al. 1969).

For refrigerated pickle products, an acid brine is added to the jars containing washed, sorted cucumbers. Since they are not pasteurized, these pickles must be refrigerated.

The oldest method of producing pickles is by fermentation. Fermentation can be accomplished by the natural (traditional) method or by inoculation with pure cultures.

Natural Fermentation. Cucumbers from the field are washed and placed in a salt brine of about 25° salometer and allowed to ferment. The salt inhibits the growth of undesirable microorganisms and allows the salt-

TABLE 9.2. DEVELOPMENT OF ACID AND CHANGE IN BACTERIAL FLORA IN SAUERKRAUT FERMENTATION

Time (days)	Total Acid	pH	Total Plate Count × 10^5 per ml	Estimated Number of Each Type × 10^5 per ml					
				Aerobes	*Leuconostoc mesenteroides*	*Lactobacillus brevis*	*Lactobacillus plantarum*	*Pediococcus cerevisiae*	Yeasts
5/6	0.15	4.48	1,320	1	1,319				
1 1/6	0.30	4.23	4,400	—	4,400				
1 5/6	0.74	3.93	9,650	—	9,150		500		
2 1/6	0.77	4.00	4,660		4,660				
2 5/6	0.97	3.87	4,490		4,440	50			
4	1.18	3.67	1,250		687	62	313	188	
5	1.19	3.71	970			49	436	145	
6	1.16	3.63	1,410			70	1,270	70	
8	1.42	3.68	4,670			750	3,170	250	
10	1.57	3.59	2,300			1,035	690		
13	1.65	3.59	1,200			200	1,000		
17	1.61	3.58	546			200	1,000		
21	1.78	3.51	251				248		3

SOURCE: Courtesy of Pederson (1979).

tolerant lactic acid bacteria to ferment the sugars to lactic acid. Sugars and other nutrients diffuse from the cucumbers into the brine and are used by the microorganisms. The salt diffuses into the cucumbers during the holding period in the brine.

The formation of sufficient lactic acid is an important factor in the quality and preservation of the fermented pickle. The rate of acid production and the total acid produced depend on the variety and size of cucumber, initial salt concentration, the temperature, and the natural microflora of the cucumbers. During fermentation, the pH is lowered to about 3.5. During storage, salt is gradually added to a salometer level of 45° to 60°. This level of salt, with the low pH, halts enzymatic and bacterial activities and helps preserve the pickles. An important aspect of preservation is the removal of all fermentable carbohydrates from the pickles.

The natural microflora of the cucumbers is quite variable and includes bacteria, yeasts, and molds. The salt and a lowered redox potential favor the growth of facultatively anaerobic organisms. The vats are covered with plastic, and UV rays from sunlight or UV lamps are used to prevent surface growth of film-forming yeasts.

The initial microflora may contain molds primarily associated with parts of the flower still attached to the cucumbers. Molds do not survive due to the salt and to the low redox potential. However, the pectinolytic enzymes from molds have been involved with softening of brined cucumbers.

There is a miscellaneous group of yeasts at the beginning of the fermentation. Yeasts can affect the fermentation by utilizing sugars that would otherwise be metabolized to lactic acid by the lactic acid bacteria. Also, the yeasts can utilize the produced lactic acid, raise the pH, and allow other potential spoilage types of microorganisms to grow. The yeasts produce large amounts of gas. This is associated with bloater or hollow defect (Fig. 9.3). However, there are other causes for this defect.

The sugars that diffuse from the cucumbers are fermented sequentially by *Leuconostoc mesenteroides, Pediococcus cerevisiae, Lactobacillus brevis,* and *Lactobacillus plantarum.* Depending on the condition of fermentation, about 0.6 to 1.2 percent lactic acid is formed in about seven to fourteen days. As the pH is lowered to 3.2, the metabolism of *L. plantarum* is inhibited; and, in one study, about 0.25 percent sugar remained after lactic acid formation had ceased (Etchells et al. 1975).

Controlled Fermentation. The cucumbers are washed and sanitized with a chlorine solution that removes most of the undesirable microorganisms. After brining (20° to 25° salometer), the cover brine is acidified with acetic acid and buffered with sodium acetate or sodium hydroxide. The brine is purged with N_2 to remove dissolved CO_2. A pure culture of *Lactobacillus* is added for the fermentation. With controlled fermentation, the

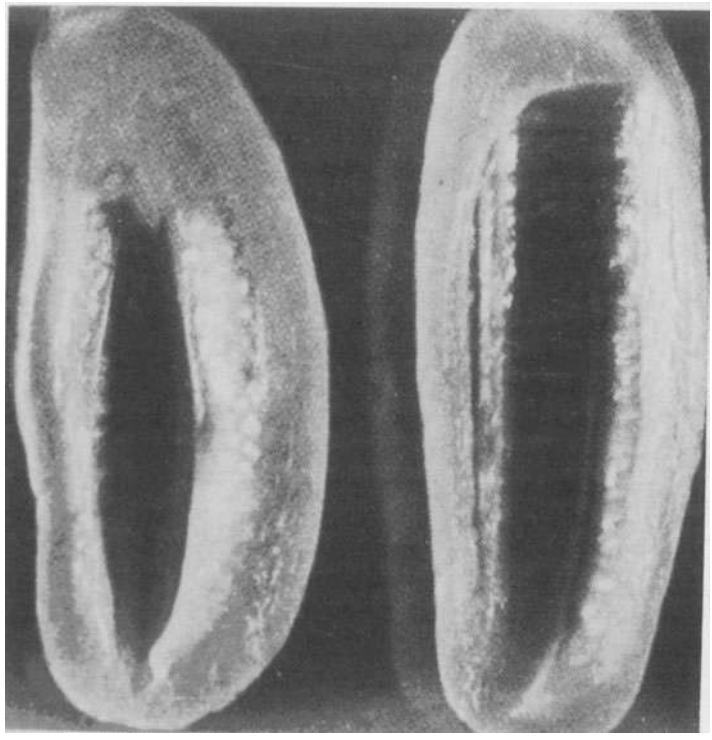

Figure 9.3. Severe bloater damage of pickles.

need to add more salt during storage is reduced. This is important because people are seeking low-salt diets and because the Environmental Protection Agency is phasing out the dumping of brine into streams.

The exposure of cucumbers to 100 percent oxygen prior to brining resulted in certain improvements during the early stages of brining (Fleming, Pharr, and Thompson 1980). However, the resultant pickles had serious bloater damage (Daeschel and Fleming 1981).

Defects. The two main defects of fermented pickles are bloaters and softening.

Bloaters are those fermented and cured pickles that float on the brine or are hollow or have large air spaces in the interior. Bloater formation is due to the accumulation of gas inside the cucumber during fermentation (Fig. 9.3). This can be caused by several factors. The growth of undesirable bacteria and yeasts can produce gases resulting in bloater formation. The respiration of cucumber tissue plus the fermentation by homofermentative *P. cerevisiae* or *L. plantarum* produces sufficient CO_2 to

cause bloater formation (Fleming et al. 1973; Etchells et al. 1975). The degradation of malic acid to lactic acid (malolactic reaction) was a major source of CO_2 when *L. plantarum* fermented cucumber juice (McFeeters, Fleming, and Thompson 1982). When a strain of *L. plantarum* that did not degrade malic acid was used, bloating did not occur (McFeeters, Fleming, and Daeschel 1984). Piercing of the fruit prior to brining controls this defect. Controlled fermentation, increasing the depth of the vat, and purging with N_2 reduce the incidence of bloaters (Fleming et al. 1977).

Bloaters are not a complete loss, since they can be used in cut pickle and relish products. However, their value is reduced by about 50 percent.

Softening of the salt-stock pickles is attributed to pectinolytic enzymes that degrade the cucumber tissue. One source of these enzymes is molds that enter the vat with the cucumbers, especially with portions of flowers that may remain attached to the cucumbers. Pectin-degrading enzymes are naturally present in the cucumber (fruit and seeds). The purging of brines at high rates of airflow results in increased softening (Costilow, Gates, and Lacy 1980). Gates and Costilow (1981) suggested that softening is caused by microorganisms growing in or on the cucumbers.

The control of softening involves the removal of any flowers attached to the cucumbers, by blanching at about 81°C, by adding $CaCl_2$, and by acidification (Hudson and Buescher 1986; McFeeters, Fleming, and Thompson 1985; Potts and Fleming 1982).

OLIVES. Olives are brined and fermented in a manner similar to cucumbers. Before brining, the olives are soaked in a 1.25 to 2.0 percent lye solution. This is necessary to hydrolyze oleuropein, a bitter factor in the olive. After treatment, the lye is removed by rinsing the olives in fresh water. Some nutrients are lost by this washing treatment. Therefore, excessive, unneeded washing is not desirable. If too much carbohydrate is lost, there will not be enough to develop sufficient acidity. To overcome this problem, reduced washing, addition of lactic acid to neutralize residual lye, or adding sugar to the brined olive for acid production has been suggested. The lye also may affect the microbial flora. If so, it is necessary to add cultures of desirable organisms.

After washing, the olives are brined. According to Pederson (1979), the brine concentration varies from 5 to 15 percent salt, depending upon the variety and size of the olives. As water diffuses out of the olives, and salt penetrates, the brine concentration is reduced. Additional salt is added to maintain the concentration. The vats, barrels, or other containers are covered to maintain a low redox level.

The entire fermentation process may take from two weeks to several

months. The same organisms active in the fermentation of cucumbers also are involved in olive fermentations. A level of at least 0.6 to 0.7 percent acid is needed for proper preservation and flavor of the product. Depending on the variety and type of treatment given the olives, the acidity varies from 0.18 to 1.27 percent (Samish, Cohen, and Ludin 1968).

Various factors, such as the origin, maturity, and variety of olives, treatment prior to brining, brine strength, sugar content, acidity, available desired microflora, and temperature influence the fermentation.

A defect known as sloughing spoilage includes severe softening, skin rupture, and flesh sloughing. The defect is caused by Gram-negative pectinolytic bacteria (Patel and Vaughn 1973) and occurs during the washing to remove the lye prior to brining. They described the bacteria involved as strains of *Xanthomonas* and various coliforms.

As with cucumber fermentations, yeasts can grow in or on the brine during fermentation of olives. Pink yeasts (*Rhodotorula*) and fermenting, pectinolytic yeasts (*Saccharomyces* and *Hansenula*) can cause softening of olives (Vaughn et al. 1969; Vaughn et al. 1972).

RED MEAT. Fermentation of sugars to lactic acid is utilized in the production of certain semidry and dry sausages. Various formulations of chopped meat are mixed with spices, sugar, salt, sodium nitrate, sodium nitrite, or a combination of these.

The mixture is held at a temperature that allows the desirable bacteria to produce lactic acid from the sugar. Although it is possible to use naturally occurring microorganisms, adding starter cultures of *Pediococcus* and *Lactobacillus* is preferred (Bacus 1984; Smith and Palumbo 1983). The culture may also contain one or more species of *Micrococcus,* although one team of researchers found that what were assumed to be micrococci in fermented meat were actually coagulase-negative staphylococci (Seager et al. 1986). Concentrated cultures are added at the rate of 40 g per 227 kg of meat. The use of starter cultures results in better color, aroma, flavor, and texture of the fermented product. Also, the processing time is reduced with a more rapid drop in pH and the yield is increased. Both frozen and dried cultures are available. The addition of starter cultures to the curing pickle of bacon helps prevent the formation of nitrosamines in fried bacon.

The fermented sausages are smoked and dried, during which the desirable characteristic flavors are developed.

POULTRY MEAT. Chicken and turkey meat, either alone or combined with beef, have been used in the production of dry fermented sausages. The fermentation is similar to that with red meat. Acton and Dick (1975)

reported that fermented and dried turkey sausage, on a fat-free basis, had a lactic acid content of 3.1 to 3.2 percent and a pH of 4.6.

DAIRY PRODUCTS. The production of lactic acid from the lactose in milk and the formation of flavor compounds are important in the manufacture of fermented dairy products (Fig. 9.6).

The main lactic acid formers are the homofermentative streptococci, *S. lactis,* and *S. cremoris* (Table 9.3). Strains of these organisms vary in the rate of acid production. Also, the rate is influenced by the temperature, pH, antibiotics, bacteriophage, stimulants, inhibitory compounds, milk composition, available nutrients, the condition of the culture, strain compatibility, and strain dominance.

Some strains of *S. cremoris* can degrade citric acid with the formation of diacetyl, an important flavor component of fermented milks. *S. lactis* subsp. *diacetylactis* produces not only considerable lactic acid, but also degrades citrate to diacetyl. Other flavor components include volatile acids, dimethyl sulfoxide, methyl ketones, lactones, acetaldehyde, and various esters.

The organisms that constitute the aroma and flavor producers are strains of *Leuconostoc* species (*L. cremoris, L. dextranicum,* and *L. mesenteroides*). *S. lactis* subsp. *diacetylactis* is included as a flavor producer. Some strains of *L. dextranicum* and *L. mesenteroides* produce lactic acid in milk, but rather slowly and in low amounts. *L. cremoris* produces diacetyl only in acidic substrates and is optimal at pH 4.3.

S. lactic subsp. *diacetylactis* degrades citric acid faster and produces more carbon dioxide than the leuconostocs. The excess gas gives fer-

TABLE 9.3. CULTURES AND CONDITIONS USED IN THE PRODUCTION OF SOME FERMENTED DAIRY PRODUCTS

Product	Cultures	Time (hr)	Temperature (°C)
Cultured buttermilk or cultured sour cream	*Streptococcus lactis, Streptococcus cremoris, S. lactis* subs. *diacetylactis*	18	22
Butter			
Acidophilus milk	*Lactobacillus acidophilus*	16–18	37–40
Yogurt	*Streptococcus thermophilus, Lactobacillus bulgaricus*	3	43–45
Cottage cheese	*S. lactis, S. cremoris*	18	22
		or	
		5	35
Cheddar cheese	*S. lactis, S. cremoris*		
Swiss cheese	*S. thermophilus, L. bulgaricus, Propionibacterium*		

mented milk drinks a desirable effervescence but is undesirable in cheese manufacture, since it may crack the cheese.

Besides streptococci and leuconostocs, certain species of lactobacilli are used in milk fermentations. Lactobacilli that have been found in fermented milk products include *L. bulgaricus, L. acidophilus, L. lactis, L. casei,* and *L. helveticus.* Some strains of lactobacilli produce diacetyl as well as lactic acid.

After pasteurization or heat treatment of milk, cultures containing the desired organisms are used to inoculate the milk for fermentation. A sufficient amount of culture must be present for the quantity of milk being fermented. Concentrated cultures that can be used to inoculate a vat of milk are available commercially. The propagation of cultures must be done under conditions that minimize contamination with undesirable bacteria or with phage that can infect the desirable bacteria (Fig. 9.4).

An important problem in the dairy industry is the susceptibility of lactic acid bacteria to phage. The rapid reproduction of phage can reduce or eliminate the streptococci in a fermenting milk so that an unacceptable product is produced. Phage resistance can be transferred between strains of streptococci (Sing and Klaenhammer 1986).

Both acid formers and flavor producers are important in the fermentation and hence, in the culture. If insufficient acid is produced, a product with poor keeping quality results. Without aroma and flavor producers, the fermented product is flat or acid and may have a metallic flavor.

The consumption of fermented milk rather than fresh milk has been suggested for people with an intolerance for lactose. There is less lactose in fermented milk, and the presence of viable lactic culture aids in the digestion of residual lactose.

The culture also produces chemicals besides acids that may inhibit spoilage or illness-producing organisms (Speck 1981).

Cultured Buttermilk. This can be made from whole milk, reconstituted nonfat dry milk, partially skim milk, or skim milk. From 0.5 to 2 percent fat tends to improve the consistency and flavor of cultured buttermilk (Fig. 9.5). The milk is heated to 85°C for 30 min or 88° to 91°C for 2.5 to 5 min. Heating destroys many bacteria, inactivates natural bacterial inhibitors, and helps prevent "wheying-off" of the product.

After heating, the milk is cooled to about 22°C and inoculated with a culture. Since good cultured buttermilk contains lactic acid and flavor compounds, the culture must contain a lactic acid producer (*S. lactis* or *S. cremoris*) and a flavor producer (a *Leuconostoc, S. lactis* subsp. *diacetylactis,* or both). The inoculated milk is incubated at 21° to 22°C until the titratable acidity (as lactic acid) reaches about 0.85 percent with a pH of 4.4 to 4.5. This requires about fourteen to sixteen hours. A desirable cultured

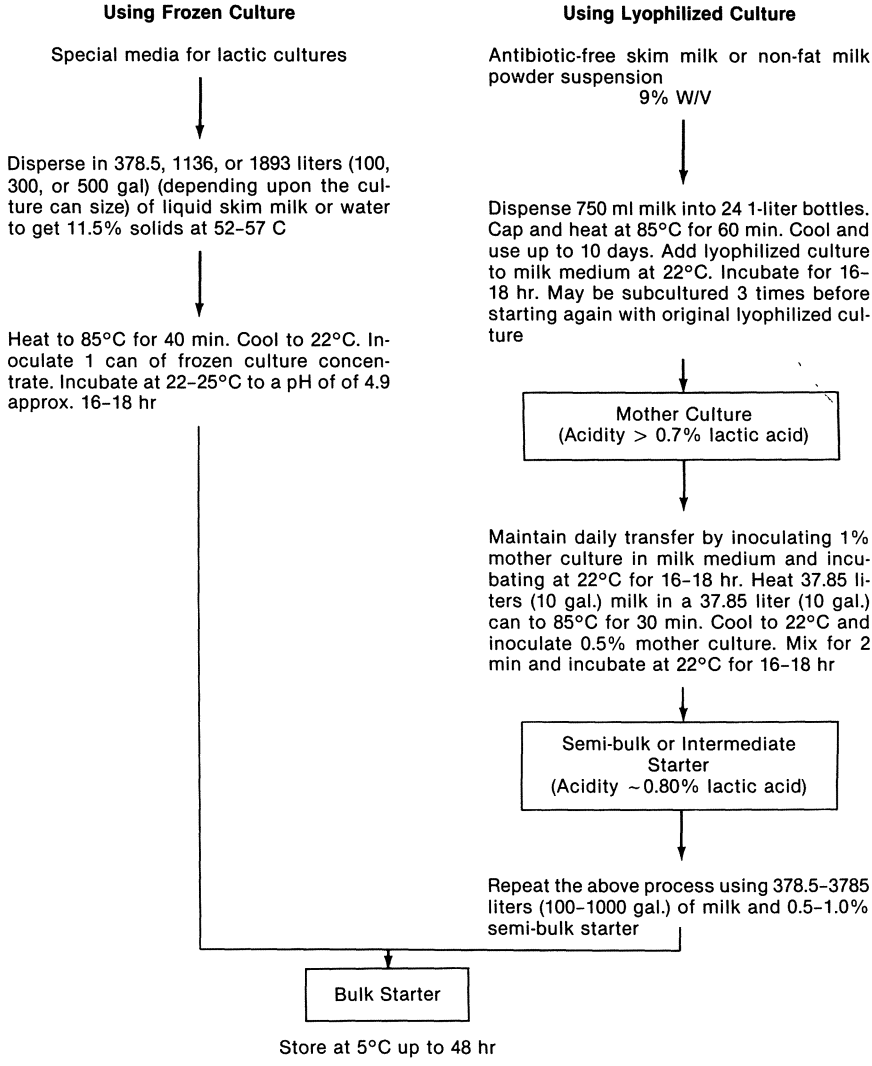

Figure 9.4. Flow diagram for production of bulk starter for certain dairy products. *Courtesy of Chandan (1982).*

milk should contain at least 2 mg diacetyl per liter. Hence, it is desirable to add citrate (about 0.25 percent) to the milk so that an acceptable level of flavor compounds is produced during fermentation. When the cultured buttermilk reaches the desirable stage, the fermentation is halted by cooling the product to 5°C for storge. A continuous fermentation process for buttermilk was discussed by Lelieveld (1984).

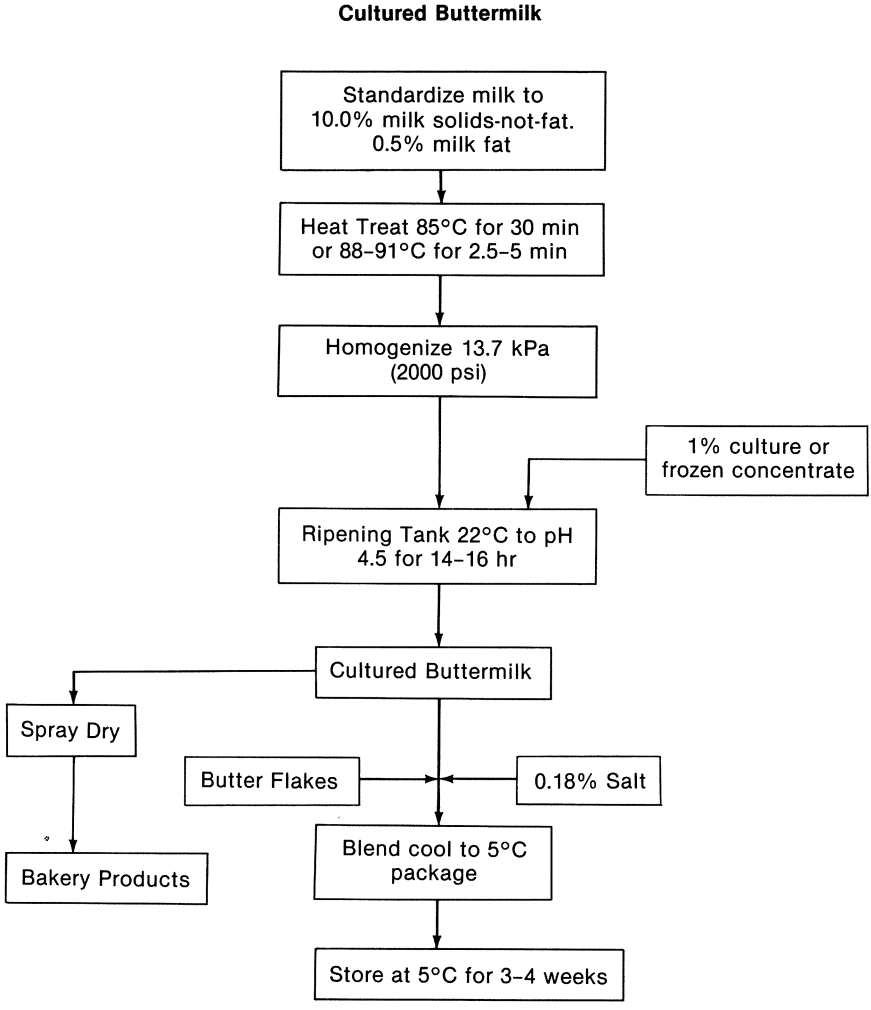

Figure 9.5. Flow sheet for the manufacture of cultured buttermilk.
Courtesy of Chandan (1982).

Acetaldehyde, produced by *S. lactis* subsp. *diacetylactis* and some other lactic acid formers, causes a defect in cultured milk that is called "green" or "yogurtlike" flavor. *L. cremoris* reduces the acetaldehyde to alcohol and prevents this defect. The chemical changes in milk during fermentation and during storage of the cultured buttermilk were reported by Marsili (1981). The formation of metabolites in milk by lactic cultures is shown in Figure 9.6.

Rather than fermenting milk to make cultured buttermilk, an acidi-

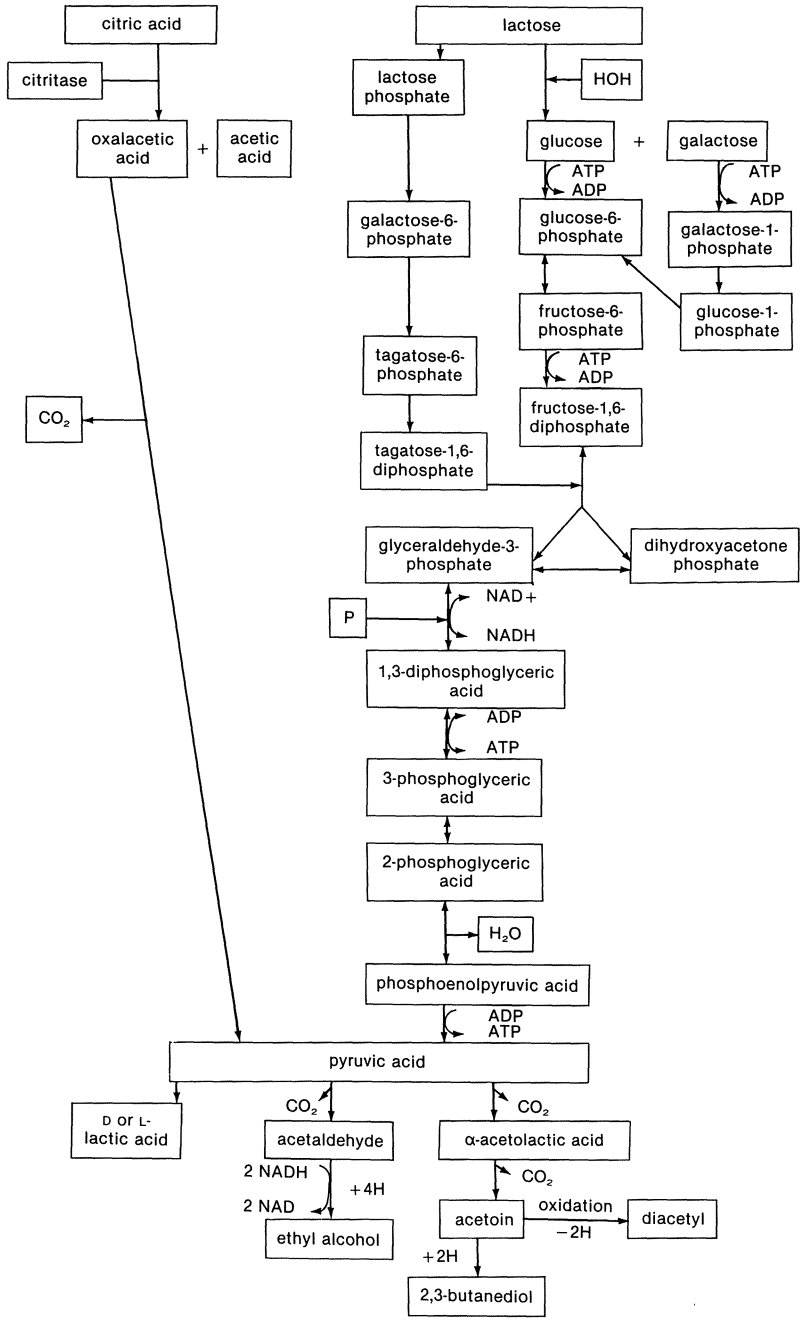

Figure 9.6. Formation of lactic acid and other metabolites in milk by lactic cultures. *Courtesy of Chandan (1982).*

fied buttermilk can be produced. Reportedly, the acidified, flavored product is consistently good, has a long shelf life, and can be made in less time, with less labor.

Cultured Sour Cream. This product is pasteurized and homogenized cream fermented in a manner similar to cultured buttermilk. It contains not less than 0.20 percent lactic acid and 18 percent butterfat.

Cultures in Butter. A lactic culture is added to cream and fermentation proceeds at 18° to 20°C. An acidity of 0.5 to 0.6 percent may be attained prior to churning. This butter is sold as the unsalted product. If cultured acid cream is used to manufacture salted butter, the product tends to deteriorate with a fishy flavor. Salted butter has better shelf life if the pH is near neutral.

For salted butter, a distillate from a culture can be added to give the desirable flavor.

Yogurt. In some countries, goat, ewe, mare, or cow milk is used to produce yogurt. In the United States either whole or skim milk from cows is used. The milk is standardized to 10.5 to 12.5 percent solids, heated to about 95°C for 30 min and, after homogenizing, cooled to about 43°C (Fig. 9.7). The frozen culture or bulk starter is a mixture of *Streptococcus thermophilus* and *Lactobacillus bulgaricus* in a 1:1 ratio. The combined action of these two organisms is needed to obtain the desired acid and flavor of the product. Yogurt is processed as either stirred or set, depending on whether the fermentation occurs before or after packaging.

The flavor depends to some extent on the production of acetaldehyde. Together, the cultures will produce about 25 mg per liter, but either organism alone produces about 8 mg per liter or less.

In a survey of 152 yogurt samples, Arnott, Duitschaever, and Bullock (1974) found that only 15.1 percent had a desired 1:1 ratio of *S. thermophilus* and *L. bulgaricus.* According to Moon and Reinbold (1976), *S. thermophilus* tends to outgrow *L. bulgaricus.* Symbiotic growth is inferred, but they found commensalism and competition between these organisms. This associated growth was reviewed by Radke-Mitchell and Sandine (1984). To halt the fermentation, the product is cooled to 5° to 10°C.

A continuous process for yogurt production using *S. thermophilus* and *L. bulgaricus* was described by Driesser, Ubbels, and Stadhouders (1977), MacBean, Hall, and Linklater (1979), and Prevost, Divies, and Rousseau (1985). The overall yogurt process was reviewed by Tamime and Deeth (1980).

Cottage Cheese. This is a soft, unripened cheese. Cottage cheese manufacture consists of coagulating the casein of skim milk by acid, cutting the coagulum into cubes, heating to reduce the moisture in the curd, washing

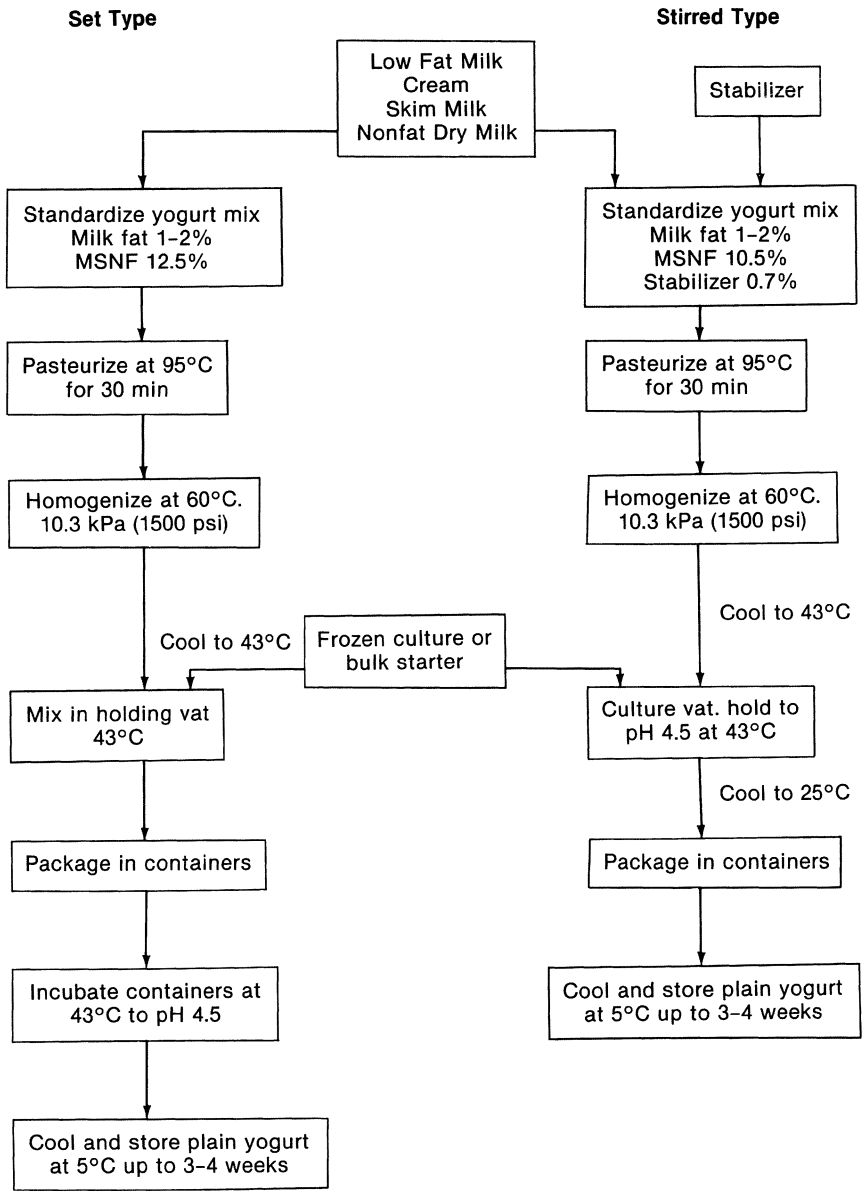

Figure 9.7. Flowsheet outline for the manufacture of set and stirred plain low-fat yogurts.
Courtesy of Chandan (1982).

to remove residual whey, and cooling the curd. A cream dressing may be added for texture and flavor. The product should not have more than 80 percent moisture.

The skim milk is pasteurized (62.8°C for 30 min or 71.7°C for 15 sec). Overheating can result in a soft coagulum and a lower yield. The inoculum should be able to produce lactic acid rapidly. *S. lactis, S. cremoris,* or a combination of these organisms is the culture of choice. *S. lactis* subsp. *diacetylactis* is not satisfactory since it produces large amounts of carbon dioxide. This gas can result in a floating curd defect. The culture is added at a level of 5 percent of the amount of milk.

The incubation temperature may vary from 20° to 33°C. For an 8-hr setting time, a temperature of 30° to 33°C is used, while for a 12-hr incubation, from 20° to 24°C is used.

Rather than using a lactic culture, the skim milk can be acidified by direct addition of lactic or other acid. The coagulation of the curd can be accomplished with acid, rennet, various commercial enzyme preparations, or a combination of acid and enzymes.

The curd is ready to cut when it is firm but not hard and brittle. The size of the cubes determines to some extent the size of the particles in the finished cheese. Cutting the curd is done when the pH is between 4.6 and 4.8. In this pH range, the curd will expel moisture most readily when stirred and heated, since pH range includes the isoelectric point of casein.

Heating of the curd halts the fermentation and aids in expelling water. When the curd has attained the proper firmness, the whey is drained and the curd is washed with cool tap water. A final washing is made with ice-cold water. When the curd is firm and dry, it is salted for flavoring.

The curd may be creamed to enhance the flavor. The flavor is obtained by adding starter distillate or cultured skim milk to the creaming mixture. As high a level of this mixture as possible is desirable, but a normal amount is 44 g of 14 percent fat dressing per 100 g of curd.

Skim milk concentrated by ultrafiltration is being used in the production of various cheeses. For cottage cheese, Kosikowski, Masters, and Mistry (1985) added skim milk retentate (the portion retained on the filter) to pasteurized milk to various protein levels. They reported that optimum flavor, body, texture, and appearance were obtained from retentate-supplemented skim milk at 1.7:1 total protein concentration.

The main considerations in producing a high-quality cottage cheese include controlling the properties of milk, using proper cultures, cutting the curd at the correct level of acidity and firmness, and cooking the curd to the desired firmness and solids content.

The cottage cheese can be contaminated by the wash water or by poor sanitation during creaming and packaging. With poor sanitation, the

shelf life may be only three to four days rather than three to four weeks. Surface spoilage (slimy defect) can be caused by *Pseudomonas* and *Alcaligenes.* Yeasts and molds indicate poor sanitation in the processing plant and can cause spoilage of cottage cheese with pH less than 5.0.

Cheddar Cheese. This cheese was first made in the village of Cheddar in Somersetshire, England. So much of this cheese is made in the United States that it is often called American cheese, or American Cheddar cheese. Cheddar is a firm, ripened cheese, ranging in color from nearly white to yellow-orange, depending upon the amount and type of coloring added.

In making Cheddar cheese, the lactic acid culture is added to a vat of pasteurized whole milk. The culture may be a pure culture of *S. lactis* or *S. cremoris,* but preferably a mixture of these organisms. *L. cremoris* may be added for flavor and its presence increases the rate of acid formation by lactic acid. A mixture of strains with different bacteriophage sensitivities is desirable to prevent potential lysis of the culture and failure of the starter to produce acid.

Kosikowski (1985) listed nine general steps used to produce cheese: prepare the milk; form the curd; cut the curd; cook; separate the curd and whey; salt; apply microorganisms for ripening; press; ripen the young cheese. These steps vary for different types of cheese. The process for Cheddar cheese is listed in Table 9.4.

Accelerated ripening of Cheddar cheese has been attempted by increasing the level of starter organisms, addition of lactase (β-D-galactosidase) to hydrolyze lactose prior to fermentation, by adding certain lipolytic or proteolytic enzymes, adding trace materials, or increasing the temperature of the aging process (Arbige et al. 1986; Aston et al. 1985; El Soda 1986; Law and Wigmore 1983; Ridha, Crawford, and Tamime 1984; Sood and Kosikowski 1979). The flavor of aged Cheddar cheese is regarded as a blend of fatty and organic acids, amino acids, carbonyl compounds, esters, alcohols, and sulfur compounds. Researchers reported the volatile flavor compounds to include ketones, aldehydes, alcohols, acids, esters, lactones, terpenes, alkanes, alkenes, alkylbenzenes, and chlorinated compounds (Liebich et al. 1970). However, McGugon, Emmons, and Larmond (1979) believed that the nonvolatile, water-extractable compounds, such as salts, amino acids, and peptides, make a significant contribution to the flavor of Cheddar cheese.

Ultrafiltered or reverse osmosis whole milk retentates have been used successfully in the production of Cheddar cheese (Bynum and Barbano 1985; Green 1985; Kealey and Kosikowski 1985). Although milk can be concentrated to 10:1 volume concentration ratio, the optimum concentration is 1.7:1 to 1.8:1 (Kosikowski 1986).

The defects in Cheddar cheese include bitter, fruity or acid flavors,

TABLE 9.4. TRADITIONAL PROCESSES OF CHEDDAR CHEESE MANUFACTURE

Process Step	Purpose	Remarks
Temperature adjustment	To bring temperature to optimum for starter organisms	Milk added to vat and adjusted to 30°–32°C
Starter addition	To develop about 0.01 to 0.02% acidity to assist rennet coagulation	Lactic starter used at about 1% for ½ to 1 hr
Settings	To coagulate the milk	Rennet, or suitable substitute, added at a rate of about 18 g/1,000 kg of milk to coagulate in 30 min.
Cutting	To promote whey removal	Curd is cut into 0.5- to 1-cm cubes
Cooking	To remove whey and firm curd	Temperature raised to 38° to 40°C in about 45 min and held about 45 min until firm.
Draining	To remove whey	Whey may be pumped through a separator to recover fat and then be stored for further processing
Cheddaring	To mat the curd and to develop the typical body of Cheddar curd	This is the characterizing step in Cheddar cheese manufacture; curd is cut into slabs, turned every 15 min, and piled every 30 min to give three to four high piles of matted curd. The curd is matted until the acidity reaches 0.5% (pH about 5.2)
Milling	To prepare curd for pressing	The curd is cut into pieces about 2.5 × 5 cm
Salting	To stop further acid production	Salt added to give 1.5 to 1.8% in finished curd. Salt is mixed and allowed to become absorbed into curd for 20 to 30 min
Pressing	To remove additional whey and arrive at final desired moisture content.	Cheese is hooped into forms or round or square hoops; usually 9 to 18 kg of finished cheese (9.7 to 19.5 kg of curd) Cheese is pressed at about 1.2 kg/cm^2 for 1 hr, then the cheese is removed from the press and dressed by placing a cloth around the curd to ensure a smooth-surface. The curd is then pressed for an additional 14 to 16 hr
Curing	To ripen the curd to give the characteristic body and flavor of Cheddar cheese	Cheese is placed in a 7° to 16°C curing room at 80% humidity for ripening. For a fully cured cheese, the curing requires 6 months to 2 years.

SOURCE: Adapted from Harper and Seiberling (1976).

or lack of flavor. Bitter-tasting peptides are a result of proteolysis of the casein and the inability of the culture to degrade the peptides to amino acids. Fast acid producers have a tendency to yield bitter-flavored cheese. Fruit-flavored cheese tends to contain high levels of ethyl butylate, ethyl hexanoate, and ethyl alcohol. Carbon dioxide produced by micoorganisms causes a slit-open defect in Cheddar cheese. This is the result of too many *S. lactis* subsp. *diacetylactis* in the starter culture or high levels of citrate-utilizing lactobacilli during ripening of the cheese.

Lactic Acid Bacteria and Other Bacteria

In some fermented products, not only is the production of lactic acid important; so are other compounds involved with flavor or other characteristics. Both dairy and vegetable products are manufactured with a combination of lactic acid bacteria and other microorganisms.

OTHER DAIRY PRODUCTS. As stated previously, lactic acid production is part of the process for all fermented dairy products. In some cases, acidulation with lactic acid can be used.

Swiss Cheese. This cheese is manufactured by using lactic acid bacteria and propionic acid bacteria. Swiss cheese is a hard cheese characterized by a sweet, nutty flavor, and by gas holes, or eyes, distributed throughout the cheese.

The organisms used in Swiss cheese manufacture are the lactic acid producers (*S. thermophilus* and *L. bulgaricus*) and a species of *Propionibacterium,* which produces propionic acid. *Propionibacterium freudenreichii* subsp. *shermani* is usually given credit as the species in Swiss cheese. However, *Propionibacterium freudenreichii* subsp. *freudenreichii* and *globosum,* as well as *P. acidi-propionici* and *P. jensenii,* may be found in Swiss cheese. The propionibacter are responsible for the characteristic flavor and eye formation in Swiss cheese.

After coagulation and separation of the whey, the curds are put into a cheese hoop and pressed lightly. The cheese is turned several times and is pressed more firmly each time. During the first two or three weeks of ripening, the cheese is salted and turned every one to three days.

Next, the cheese is moved to a prewarming cellar (17° to 20°C) for ten to fourteen days, and then to a fermentation room for six to eight weeks. The fermentation room is 22° to 23°C, with a relative humidity of 80 to 90 percent. During this time, the cheese is turned and washed with saltwater two to three times per week (Fig. 9.8).

The final process consists of curing at 7° to 10°C and high relative humidity. After four months, the cheese is mild and has the typical sweet

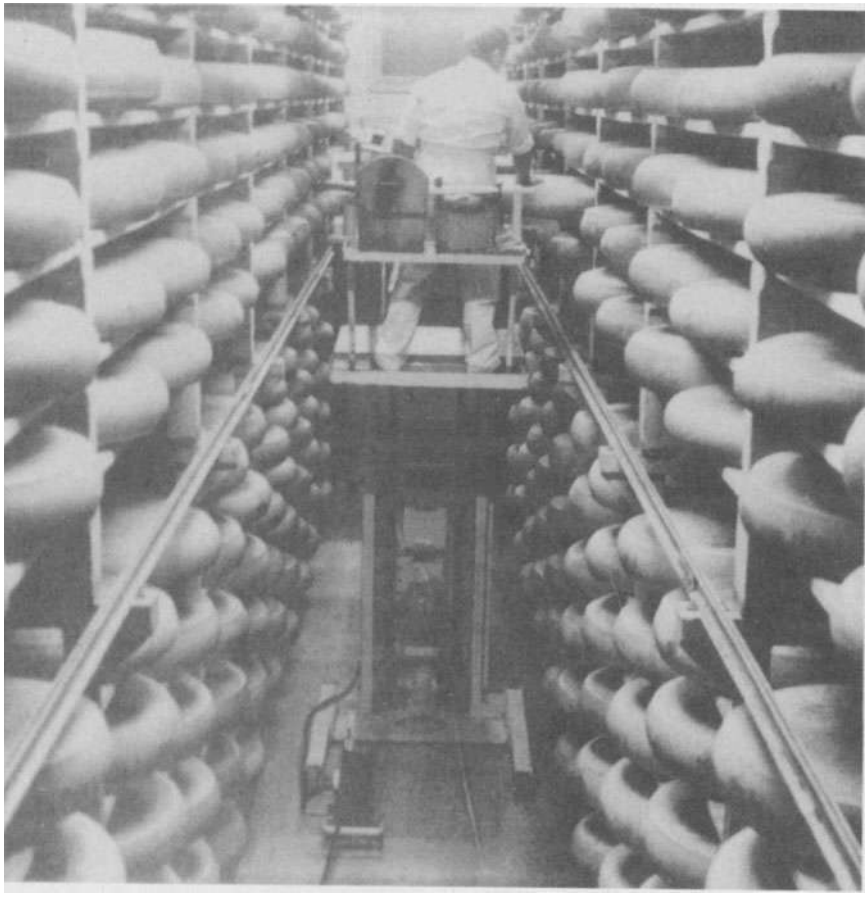

Figure 9.8. Swiss cheese in curing room.
Courtesy of Switzerland Cheese Association.

nutty flavor. Stronger flavors are developed by leaving the cheese in the curing room for eight to twelve months.

The propionibacter produce CO_2, which is not able to escape, so it concentrates in various places in the cheese. The pressure produced by this gas results in the eyes in Swiss cheese (Fig. 9.9). The turning, salting, washing, temperature, and humidity are essential to obtain uniformity of these holes. Undesirable organisms, such as butyric acid bacteria, produce off-flavors as well as gases such as hydrogen, which results in poor eye formation.

Further information on the production of Swiss cheese can be ob-

Figure 9.9. Typical eye formation in high-quality Swiss cheese. *Courtesy of Switzerland Cheese Association.*

tained from Auclair and Accolas (1983), Biede and Hammond (1979), Gilles, Turner, and Martley (1983), and Lawrence, Heap, and Gilles (1984).

Limburger. This is a semisoft surface-ripened cheese with a distinct odor and flavor. It is regarded as one of the most delicate and difficult cheeses to make. After the milk is coagulated, the curd is packed into rectangular forms, and residual whey is allowed to drain. When the cheese is firm enough to retain its shape, it is removed from the form and salted and turned frequently.

At the beginning of ripening, fungi predominate. They reduce the acidity so that bacteria can grow. After the fungi, micrococci may be present along with *Brevibacterium linens.* This organism (*B. linens*) is common in the slime of surface-ripened cheese. It is thought that proteolytic enzymes secreted by the organism diffuse into the cheese and cause the characteristic softening and flavor.

The flavor of Limburger cheese is probably due to sulfur compounds such as dimethyl disulfide, methanethiol, 2,3,4-trithiapentane, S-methylthioacetate, and dimethyl polysulfides as well as to phenol and indole (Cuer et al. 1979; Parliment, Kolor, and Rizzo 1982).

Although yeasts are listed as the initiators, these have been described as *Geotrichum candidum, Oidium lactis,* or *Oospora.* At the present time, *Oidium lactis* is *Geotrichum candidum,* which is a mold. Some *Oospora* species are now included in the yeast genus *Trichosporon.*

Lactic Acid Bacteria with Yeasts

Some fermented milks, such as kumiss and kefir, utilize a simultaneous production of lactic acid and alcohol. The alcohol is a product of yeast metabolism. The alcoholic content of kefir is about 0.3 to 1.0 percent, while that of kumiss is from 1 to 2 percent.

The lactic acid bacteria are *Streptococcus lactis* and *Lactobacillus bulgaricus.* The yeast is a lactose fermenter (perhaps *Kluyveromyces lactis*).

Other products in which lactic acid bacteria and yeasts are involved include sour dough, idli, and ogi.

SOURDOUGH. This type of dough was so prominent among prospectors of the Old West that they were named sourdoughs. San Francisco is still noted for sourdough bread.

The sourdough process uses a lactic acid bacterium for the souring and a yeast for leavening. Sugihara, Kline, and Miller (1971) screened 200 yeast isolates from sourdough. They reported two types of yeast, *Saccharomyces exiguus* and *Saccharomyces inusitatus.* The lactic acid bacteria isolated from sourdough were difficult to grow, and their characteristics did not fit any known species (Kline and Sugihara 1971). They suggested the name *Lactobacillus sanfrancisco* for this organism.

Lactic Acid Bacteria with Molds

Molds are involved in the aging and curing of several cheeses (Table 9.1). Two of the more familiar cheeses are Roquefort and Camembert.

ROQUEFORT CHEESE. Roquefort cheese is made from sheep's milk and is ripened in caves near Roquefort, France. A similar cheese that is common in the United States is made with cow's milk and is called blue cheese.

One of the characteristics of this cheese is the greenish-blue marbling of its soft, creamy-white interior. This marbled appearance is due to the growth of *Penicillium roqueforti,* a blue-green mold, throughout the cheese. So that oxygen is available to favor interior growth, the cheese is pierced with needles. *P. roqueforti* is able to grow at a lower redox potential than many other molds.

The characteristic sharp, biting flavor of blue cheese is generally attributed to hydrolysis of the cheese fat to fatty acids, and then the conversion of these fatty acids to methyl ketones. The development of blue cheese flavor was described by Coghill (1979), Jolly and Kosikowski (1975), and Kinsella and Hwang (1976).

CAMEMBERT CHEESE. This is a soft, surface-ripened cheese (Fig. 9.10). The coagulated curd is placed into hoops and allowed to settle for about two days. The cheese is then removed from the hoop, salted, and inoculated with *Penicillium camemberti.* The cheese is cured at about 12°C. Curing requires at least sixty days. The mold plus bacteria and yeasts ripens the cheese from the outside toward the center. During curing, a grayish-white feltlike growth of mold is followed by a secondary growth that produces a sliminess, and the surface becomes reddish to russet colored. The cheese has a mild to pungent flavor, probably due to carbonyl compounds (Karahadian, Josephson, and Lindsay 1985).

Milk concentrated by reverse osmosis or ultrafiltration can be used readily in the production of Camembert cheese (Honer and Horwich 1983; Rash and Kosikowski 1982).

Yeasts

The metabolism of sugar by yeast yields alcohol and carbon dioxide. The production of alcohol is important in various beverages (beer, wine, liquor), the CO_2 is used in the baking industry, and the yeast cells are a source of protein, lipid, and various useful chemicals and enzymes (Table 9.5). Although yeasts are the main organisms utilized in the production of these foods, other organisms may play a role.

ALCOHOL. Alcohol is produced during the metabolism of various sugars by yeasts as well as other microorganisms. Industrial alcohol,

Figure 9.10. Camembert cheese showing white surface mold during curing. *Courtesy of The Borden Co.*

TABLE 9.5. THE USE OF YEAST IN INDUSTRY

Fermentation Conditions	Product			
	Bread Dough	Lager[a]	Ale	Wine
Time of fermentation	1–3 hr	8–10 days	2–6 days	5–10 days
Temp. of fermentation	30°–35°C	10°C	20°C	15°–27°C
pH	5.2–4.7	5.2–4.2	5.2–4.0	3.5
Final EtOH conc. (%)	2–3	4	4	11–13
Cells,[b] start	275	6–10	6–10	5–10
end	300	30–70	30–70	50–150
Gas production rate[c]	18–35	16–25	16–25	25–32

[a]For primary fermentation.
[b]Expressed in million cells/g or /ml.
[c]Expressed as mM EtOH/hr/g yeast solids.
SOURCE: Ponte and Reed (1982).

which usually is produced chemically by the oxidation of ethylene, is used for many purposes. However, this discussion is limited to the alcoholic beverages beer and wine. Various strains of species of *Saccharomyces* are involved in the production of these beverages. Since the strains that are presently used cannot utilize starch or cellulose, these substances must be converted to glucose with enzymes or other microorganisms. However, yeasts are being isolated or developed through genetic engineering and tested for their ability and efficiency in converting starch or cellulose directly to alcohol (Amin et al. 1985; Calleja et al. 1982; Hawke et al. 1983; Laluce and Mattoon 1984; Russell et al. 1986). Perhaps in the future, new types of organisms will be used in food fermentations producing alcohol.

Beer. Beer is produced in several steps (Fig. 9.11). The first step is to make malt (Hudson 1986). Barley is cleaned, graded, washed, and steeped. It is then allowed to germinate for about five days. After germination, heat is used to stop the sprouting process and to dry the grain. The rootlets are screened from the resultant barley malt. Malting is used to develop enzymes such as amylases. The amylases degrade the starch in the grain to fermentable carbohydrates.

An adjunct, such as ground rice or corn, is mixed with the barley malt, wetted with clear, filtered water, and cooked. This mash is then strained. The filtered, clear, amber liquid, called wort, is obtained and pumped into the brew kettle. Here, hops, or hops extracts, the seasoning of beer, are added to the wort and the mixture is boiled to obtain the correct delicate hop flavor (Buckee, Malcolm, and Peppard 1982; Clarke 1986; Sharpe and Laws 1981; Verzele 1986). By this time the amylases have degraded the starch. The temperature attained during wort boiling halts the action of these enzymes.

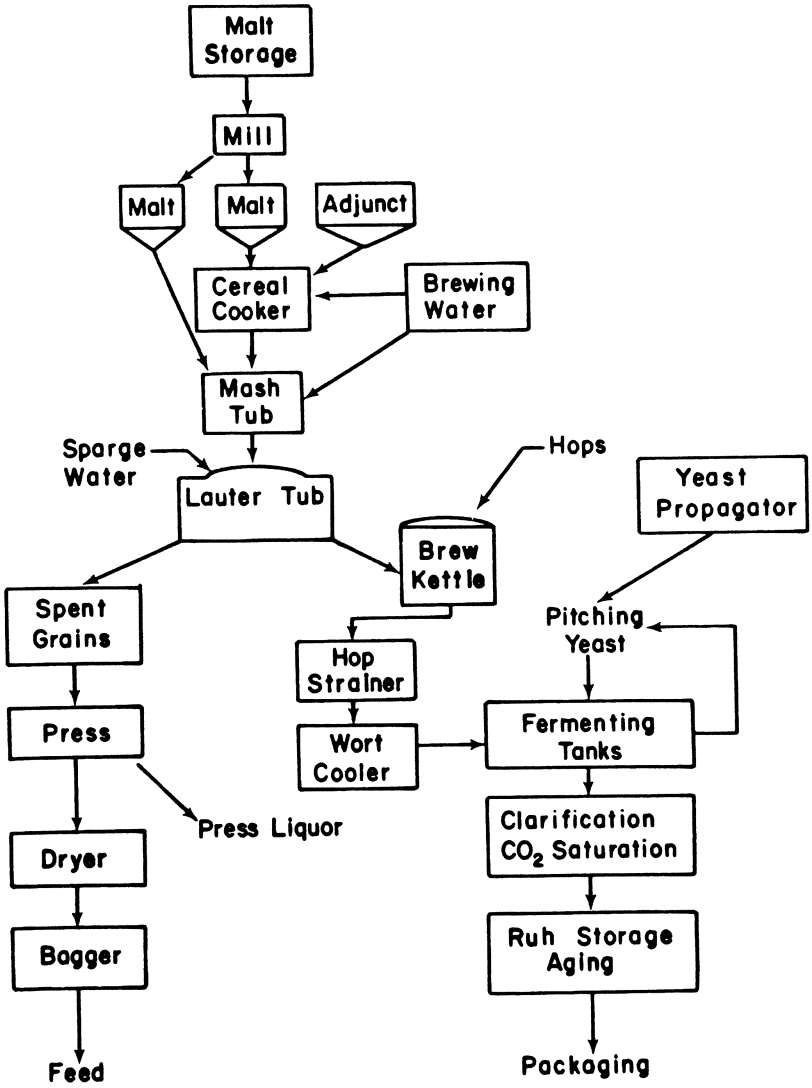

Figure 9.11. Processes of brewing.
Courtesy of Helbert (1982).

The hopped wort is strained, cooled, and then pumped to fermentation tanks. Here, yeast is added. The fermentation changes the sugars in the wort into alcohol and carbon dioxide.

For aging and natural carbonation, the beer is pumped into tanks that may contain substances such as beechwood chips to provide surfaces

for the yeast. Some freshly yeasted wort is added, and the beer is allowed to become naturally carbonated by a slow and quiet fermentation.

During this secondary fermentation, chill-proofing of beer can be accomplished by adding a proteolytic enzyme, such as papain. This hydrolyzes residual protein that would otherwise precipitate and cloud the beer when chilled. The addition of acetolactic decarboxylase helps remove acetolactic acid, the diacetyl precursor, and also aids in the maturation of beer (Godtfredsen et al. 1984).

The so-called light (lite) beers are made by including an enzyme, glucoamylase, before or during fermentation. This enzyme hydrolyzes most of the unfermentable dextrins in wort to glucose, which can be fermented by the yeast to alcohol. Hence, the beer contains a lower calorie level.

The beer is then filtered to remove yeast cells. The beer is either pasteurized (60°C for 15 to 20 min) after packaging, or it is packaged aseptically after bulk pasteurization or microfiltration. Bulk beer in kegs is not pasteurized, so it must be refrigerated.

Some yeasts settle to the bottom of the fermenting vat and are called bottom yeasts. These are strains of *Saccharomyces uvarum* and are used to produce lager beer. Other yeasts, strains of *S. cerevisiae,* tend to collect at the surface and are called top yeasts. The product of this fermentation is called ale. Each brewery has certain strains of these yeast species for its particular products. Not all strains will produce an acceptable beverage. Hence, when a good strain is obtained, it is handled very carefully. Through genetic alteration, new strains of yeast for brewing are being developed (Hammond and Eckersley 1984; Hinchliffe 1985; Hopwood 1981; Panchal et al. 1984).

Besides alcohol and carbon dioxide, there are various metabolic products that can affect the flavor of the beer. These compounds include ethyl acetate and other esters, fusel alcohols (pentanol, isopentanol, and isobutanol), diacetyl, 2,3-pentanedione, sulfur compounds (sulfites, sulfides, mercaptans, mercaptals, thioaldehydes, and thioketones), and leakage of amino acids and nucleotides from the yeast cells. The amount of these compounds depends upon the conditions of fermentation, wort composition, and yeast strain. Top-fermented ale usually contains more fusel alcohols than bottom-fermented lagers. The effect of the chemical composition on the flavor of beer was discussed by Meilgaard (1982).

There are factors in beer that limit the types of organisms that can grow. These factors include a low pH and redox potential, the isohumulones of hops that inhibit Gram-positive bacteria and the alcohol produced by the yeast. The contaminants are usually acetic acid bacteria, lactic acid bacteria, coliforms, and wild yeasts.

The acetic acid bacteria (*Acetobacter* and *Gluconobacter*) can oxidize

ethyl alcohol to acetic acid. In the anaerobic environment of active fermentation, this is not possible.

Some lactic acid bacteria (*Lactobacillus* and *Pediococcus*) have a tolerance to the isohumulones of hops. They are microaerophilic and tolerate acids. They produce lactic acid, diacetyl, off-flavors, turbidity, and ropiness in beer (Dolezil and Kirsop 1980; McMurrough and Palmer 1979).

The coliforms (*Klebsiella* and *Escherichia*) grow only when the pH is above 4.3. They impart various odors and flavors to the wort. If the wort is stored for future fermentation, coliforms can become a problem.

In addition to the so-called wild yeasts, the various strains of brewer's yeasts can cause contamination. Bottom yeasts may become contaminated with top yeasts, and vice versa. This will result in an alteration of the desired product.

A bacterium, *Zymomonas anaerobia,* has been found in beer and cider. It produces acetaldehydes and H_2S. The beer has a very unpleasant odor and flavor.

N-nitrosodimethylamine is a carcinogen and has been reported in beer (Scanlan et al. 1980; Spiegelhalder, Eisenbrand, and Preussmann 1979). The possible source of nitrosamines in beer as well as reduction or elimination were discussed by Wainwright (1986a, 1986b).

Wine. Wine is considered to be the oldest fermented alcoholic beverage. The term *wine* is applied to the product made by alcoholic fermentation by yeasts of grapes or grape juice, with an aging process. However, the products of fermentation of berries, fruit, and such things as honey, palm juice, rhubarb, and dandelion also are called wines. These are designated by the substance from which they were made (blueberry wine, dandelion wine).

The quality of the finished wine depends upon the grapes, fermentation techniques, aging, and blending. Anyone can make wine, but only the experts can produce high-quality wine.

Some of the factors influencing the quality of the grape are climate, soil conditions (temperature, fertility, type, and drainage), and variety of grape (Amerine, Berg, and Cruess 1972). A year with a good climate is known as a vintage year. If the climate is too cold, the grapes do not mature as well, producing less sugar and more acid.

The grapes are crushed and pressed to release the juice, which is called the must. The sugar content is determined with a hydrometer. The sugar content can be adjusted by adding sugar, mixing high and low musts together, or by adding water. Multiplying the hydrometer reading in degrees Brix by 0.55 to 0.56 generally gives an estimate of the percentage alcohol in the finished product. Jones and Ough (1985) reported some yields of 0.60 or higher.

The must contains many types of microorganisms. To control these contaminants, the must is treated with sulfur dioxide (SO_2), or potassium metabisulfite. The yeast primarily involved in the fermentation has been called *Saccharomyces ellipsoideus, S. vini,* or *S. cerevisiae* var *ellipsoideus.* However, Van der Walt (1970) classified the yeast as *Saccharomyces cerevisiae.* There are many strains of this yeast species. One study listed other *Saccharomyces* (*S. chevalieri, S. bayanus, S. italicus,* and *S. uvarum*) as so-called wine yeasts (Rosini et al 1982). Besides the various strains of *Saccharomyces,* Kunkee and Amerine (1970) listed about 160 species of yeasts that have been found on grapes, in musts, or in wine. These also may influence the overall fermentation of the must, especially if they are resistant to SO_2 treatment.

The added SO_2 not only inhibits unwanted organisms, but also stabilizes wine color. Due to health problems caused by the consumption of sulfites, there is a desire to eliminate this additive. Some strains of *S. cerevisiae* can stabilize wine color, apparently by producing sulfite. One strain produced 30 to 80 mg/L (ppm) (Suzzi, Romano, and Zambonelli 1985). Even if other yeasts are not inhibited, as the fermentation proceeds the alcohol-tolerant wine yeasts soon become dominant.

Thornton (1983) listed twelve desirable characteristics of a wine yeast:

1. Efficient conversion of grape sugar to alcohol
2. Rapid initiation of fermentation (48 hr)
3. SO_2 Tolerance
4. Ability to cause even fermentations
5. Ability to ferment at low temperatures
6. Ability to ferment to dryness (alcohol tolerant)
7. Good flocculation after fermentation to aid in removal
8. Production of a desirable bouquet
9. Low foaming
10. Low H_2S or mercaptan fermentation
11. For sensory quality of the wine, a relatively high glycerol production
12. Production of a relatively low amount of higher alcohols.

Although all of these characteristics are probably not possible to attain in any one yeast strain, improvements can be made through hybridization, mutagenesis, cell fusion, transformation, and genetic engineering (Eschenbruch et al. 1982; Romano et al. 1985; Thornton 1983, 1985). Killer wine yeasts that will inhibit (kill) unwanted yeasts, are being developed (Hara, Iimura, and Otsuka 1980; Seki, Choi, and Ryu 1985; Shimizu et al. 1985).

For red wine, the must is allowed to ferment with the skin and pulp of

the red grapes. The fermentation is allowed to proceed until the correct amount of color is extracted from the skin. During this early stage, aeration is used to promote the growth of the yeast.

After this initial fermentation, the must is drawn off in a process called racking. The must is put into closed tanks (Fig. 9.12) to provide anaerobic conditions for the main fermentation. The produced CO_2 tends to purge the must of oxygen and inhibits aerobic organisms, such as *Acetobacter*. The fermentation of white table wines is usually at 10° to 15°C, while red wine is at 25° to 30° C. It may require from one to four weeks to complete the fermentation, which is evidenced by the cessation of bubbling due to CO_2 production.

The wine is pumped into barrels, vats, or tanks for aging. The aging time varies with the type of wine. During aging, the bouquet and aroma of the wine are developed. Various compounds, such as esters, are formed. Most wines are blended to maintain flavor consistency. This is usually done during the aging period.

After aging, the wine may be pasteurized at 60°C for 30 min and

Figure 9.12. Stainless steel fermenting tanks. Note their location outdoors.
Courtesy of Wine Institute, Amerine, Berg, and Cruess (1972).

bottled or, in some cases, packed in kegs for shipment, and bottled elsewhere. When properly aged wine is bottled, it will continue to improve for several years.

There are special procedures for certain products, such as dessert wine and champagne. Dessert wines are sweet and have an alcohol content of about 20 percent. To attain these conditions, the fermentation is halted while there is sufficient sugar remaining by adding brandy. The higher alcoholic content of brandy increases the alcohol in the resultant wine.

For champagne and other sparkling wines, the wine is put through a secondary fermentation. New must, sugar, and yeast are added to the stock wine, which is then bottled or put into a closed tank. The CO_2 that is produced remains in the wine, which causes the effervescence.

Although acids in grape juice contribute to the organoleptic quality of wine, retard microbial spoilage, and stabilize color, too much acid is undesirable. Amelioration (adding a sugar solution) not only dilutes the acid but also dilutes other components, including flavor compounds.

A fermentation called the malolactic fermentation, can be used to reduce the acidity. Malic and tartaric acids account for about 90 percent of the acidity in grapes. Primarily three genera of lactic acid bacteria (*Lactobacillus, Leuconostoc, Pediococcus*) are involved in the reaction to convert malic acid to lactic acid and CO_2. This fermentation usually follows the alcoholic fermentation. The general equation of the reaction is

$$\underset{\text{Malic Acid}}{HOOC\text{-}CH_2\text{-}CHOH\text{-}COOH} \rightarrow \underset{\text{Lactic Acid}}{CH_3\text{-}CHOH\text{-}COOH} + CO_2$$

The reaction goes through an intermediate, pyruvic acid, which is then reduced to lactic acid. The monocarboxylic lactic acid is not as acid as the dicarboxylic malic acid. Hence, the acidity of the wine is reduced.

Not all lactic acid bacteria can tolerate the alcohol and low pH in order to convert malic acid to lactic acid. *Leuconostoc oenos* is usually the organism involved. It grew in wine with 14.2 percent alcohol and 11.2 ppm SO_2 at pH 3.2 or above (Davis et al. 1986). The organism did not grow or alter malic acid in wine with 11.9 percent alcohol and 72 ppm SO_2 at pH 4.0 or lower. It is evident that all conditions that may affect growth of an organism need to be considered. The use of immobilized cells of *L. oenos* for the malolactic fermentation was considered by McCord and Ryu (1985).

The yeast *Schizosaccharomyces pombe* utilized all of the malic acid when the must pH was over 3.0 (Yang 1973). Even at pH 2.5, about 70 percent of the malic acid was metabolized. With *S. pombe,* ethyl alcohol and carbon dioxide are the end products; however, the production of undesir-

able flavors and aromas has limited its usage. It may be possible to alter the genetic material in a desirable yeast to produce the enzyme systems to convert malic to lactic acid (Snow 1985; Williams et al. 1984).

Certain defects can occur in wine. The lower the sugar content, the less likely that spoilage is to occur. Thus, dry wines are more stable than other types of wine. Spoilage is evident by flavor or odor changes as well as haze or gas formation. Alterations may be due to molds (Daly, Lee, and Fleet 1984), lactic acid bacteria (Edinger and Splittstoesser 1986), acetic acid bacteria (Drysdale and Fleet 1985; Joyeux, Lafon-Lafourcade, and Ribéreau-Gayon 1984) and chemical reactions (Simpson, Bennett, and Miller 1983; Somers and Ziemelis 1985).

To prevent microbial spoilage of the finished wine, it is important to deactivate any residual microorganisms before or after bottling. This can be accomplished by pasteurization, addition of inhibitors such as SO_2, or filtration. The delicate flavors of some wines are harmed by heating or by adding SO_2. For these wines, filtration is the preferred method of removing microorganisms (Reeves 1983; Scott, Anders, and Hums 1981).

Baked Products. The leavening or raising of dough or batters is due to the incorporation of air into the product. This may be accomplished by whipping, such as egg whites in angel cakes, by adding chemical agents that react to produce gas, or by fermentation of sugars by microorganisms to produce carbon dioxide.

Although many microorganisms ferment sugars with the release of carbon dioxide, *Saccharomyces cerevisiae,* or baker's yeast, is best adapted for leavening of bakery products. Yeast fermentation has certain advantages over chemicals, since it can contribute a characteristic flavor and aroma to the product and gas evolution can continue over a longer time. The products of yeast fermentation also affect the texture of the dough.

Baker's yeast is used in the manufacture of bread, rolls, sweet dough, pretzels, and crackers. Although some yeasts can hydrolyze starch present in the flour, baker's yeast (*S. cerevisiae*) cannot. Enzymes (amylases) naturally present in the flour, or added enzymes (diastatic malt or amylase) hydrolyze the starch, making it available to the action of the yeast enzymes.

Baker's yeast is available as dried or compressed yeast. Commercial dried yeast has about 8 percent moisture and compressed yeast about 70 percent moisture. Trivedi, Cooper, and Bruinsma (1984) described the process of protoplast fusion used to combine the desired characteristics of two yeast strains into a hybrid yeast strain. The resultant commercial product of this new strain has been called quick-rising, highly active, and instant-active dried yeast.

Generally, baker's yeast is contaminated to some extent with bacteria, molds, and other yeasts. Proteolytic and lactic acid bacteria found in

yeast are potentially important in altering the pH and structure of cracker dough (Fields, Hoseney, and Varriano-Marston 1982).

A typical sponge dough process is shown in Figure 9.13. It involves blending a portion of the flour and water with yeast and yeast food and allowing this to ferment 3 to 5 hr. The remainder of flour and water plus salt, fat, milk product, and such things as dough improver, crumb softener, mold inhibitor, and enrichment (vitamins and minerals) are mixed with the fermented sponge (Dubois 1981; Fowler and Priestly 1980; Mann 1986).

In continuous bread-making processes, the yeast is grown and fermentation occurs in a preferment or brew, which contains little or no flour. The fermentation products (alcohol, acids, aldehydes, carbon dioxide) in the preferment contribute to the flavor of the finished product. The leavening action is accomplished during proofing (holding the mixed dough for about 45 min in pans before baking).

The heat to which the fermented dough is exposed during baking kills the yeast cells, drives off the alcohol, and sets the flour protein surrounding the entrapped carbon dioxide.

Yeasts with Acetic Acid Bacteria

This combination of organisms is used in the production of vinegar and in the fermentation of cacao beans and citron.

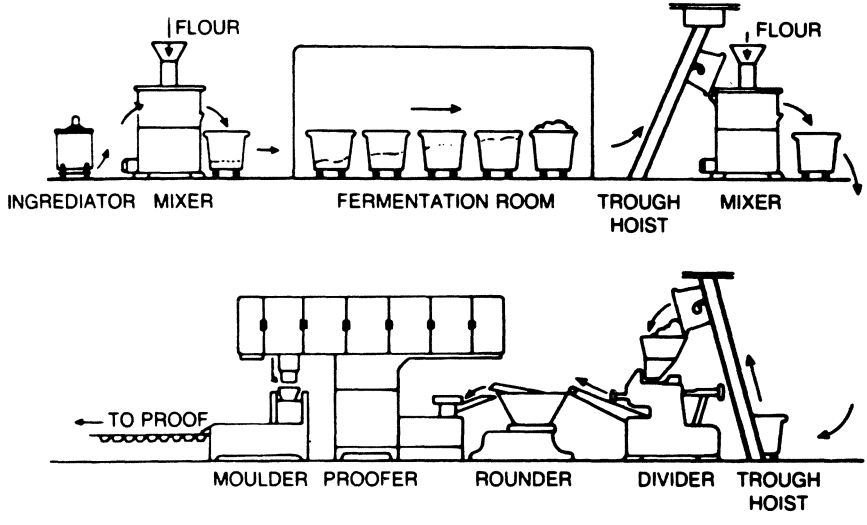

Figure 9.13. Schematic diagram of sponge dough process.
From Seiling (1969).

VINEGAR. Vinegar is produced by an alcoholic fermentation followed by oxidation of the alcohol to acetic acid. The vinegar must contain at least 4 g of acetic acid per 100 ml. The strength of vinegar is referred to in grains, with 10 grains equal to 1 percent. Hence, 4 g/100 ml is 4 percent or 40-grain vinegar.

Vinegar can be produced from any food product that can be fermented by yeast to ethyl alcohol. Among the materials used are fruits, fruit juices, tubers, cereal grains, molasses, honey, coconut, beets, malt, and refiners' syrup. Vinegars are classified according to the raw material from which they were made. In the United States, the term *vinegar* implies that it was made from apples. The names *cider vinegar* and *apple vinegar* also are used for this product. Wine vinegar is made from grapes. Distilled, grain or spirit vinegar is made from an alcoholic solution that has been distilled.

It is essential that alcoholic fermentation be complete before vinegar formation, since the acetic acid interferes with further alcohol production. The overall chemical changes can be represented by the reaction

$$C_2H_5OH + O_2 \xrightarrow[\textit{Acetobacter}]{} CH_3COOH + H_2O$$

Several side reactions occur which alter the final composition of the vinegar.

There are various systems for converting alcohol to vinegar. All of the methods provide a means of bringing together the alcohol, air, and *Acetobacter.* The air supplies the oxygen needed by the strictly aerobic *Acetobacter,* and for oxidizing the alcohol through acetaldehyde to acetic acid. The efficiency of aeration determines the rate of conversion of alcohol to acetic acid. There are three basic methods: (1) the slow, open-vat method; (2) the trickle, generator method; and (3) the submerged or bubble method.

Open-Vat. In the open-vat method, the substrate is placed into vats or barrels and inoculated with fresh vinegar or with mother of vinegar (a thick gelatinous skin formed during acetification). Since aeration is inefficient, the conversion may require several months. The system developed in Orleans, France, was a slow process adapted from the older methods and is known as the Orleans process.

Generator. In the quick process, the vinegar is manufactured in a generator in which the substrate is trickled over a packing material (beechwood shavings, coke, charcoal, ceramics, corn cobs) on which the *Acetobacter* is inoculated. The packing material allows a large surface area for the components to come together. The heat caused by the reaction results in

air moving from the bottom through the generator and out of the top. When excess heat is produced, cooling of the generator system is needed. One pass of the substrate through the generator is not sufficient to convert all the alcohol to acetic acid, so the product is recirculated to complete the conversion. Not only is the generator method quicker than the vat method, but also higher grain vinegars (over 100 grain) may be obtained. The amount of acetic acid in the vinegar depends on the alcohol content of the substrate. Theoretically, 1.0 percent alcohol should yield 1.3 percent acetic acid. However, actual yields vary from 0.8 percent to 1.0 percent, due to inefficiencies and residual alcohol in the vinegar.

Bubble. The submerged culture, or bubble method, was developed from the methods used for antibiotic production. In this system, air is pumped into the vat containing the stirred substrate (alcohol) and *Acetobacter.* Recent research has attempted to increase the productivity of this system (Ghommidh, Cutayar, and Navarro 1986; Okuhara 1985).

Besides acetic acid and water, vinegar contains residual ethanol, traces of ethyl acetate, fusel alcohols and their acids, and traces of other compounds.

Biologically produced acetic acid in vinegar can be distinguished from that produced chemically from petroleum or coal by measuring the ^{14}C content with a scintillation counter (Kaneko, Ohmori, and Masai 1973). The basis is that fossil fuels have a lower ^{14}C content than recently produced carbon.

Yeasts with Molds

In the production of sake, molds are used to hydrolyze the starch of rice to carbohydrates fermentable by yeasts to alcohol. Although many microorganisms may be present and affect the reactions and product, the principal organisms are *Aspergillus oryzae* and a strain of *Saccharomyces cerevisiae.* Kodama (1970) discussed the manufacture of sake and the various organisms that might be involved.

Molds

The molds are used in many fermentations, especially in Oriental countries. The most readily recognized mold-produced product is soy sauce.

SOY SAUCE. This is a dark-brown liquid that is an important flavoring material for the normally bland rice diet of the Far East. Also, it is used on vegetables, meats, poultry, and fish. Of all the fermented foods of

Asian countries, soy sauce has received the greatest acceptance in the United States. The soy-sauce production process is shown in Figure 9.14.

Soaked, steamed soybeans are mixed with roasted wheat in a ratio of about two to one. This is inoculated with a mold, *koji.* The koji is cooked rice that is inoculated with a species of *Aspergillus* and incubated for three to five days. The mold species that have been used are *A. oryzae, A. soyae,*

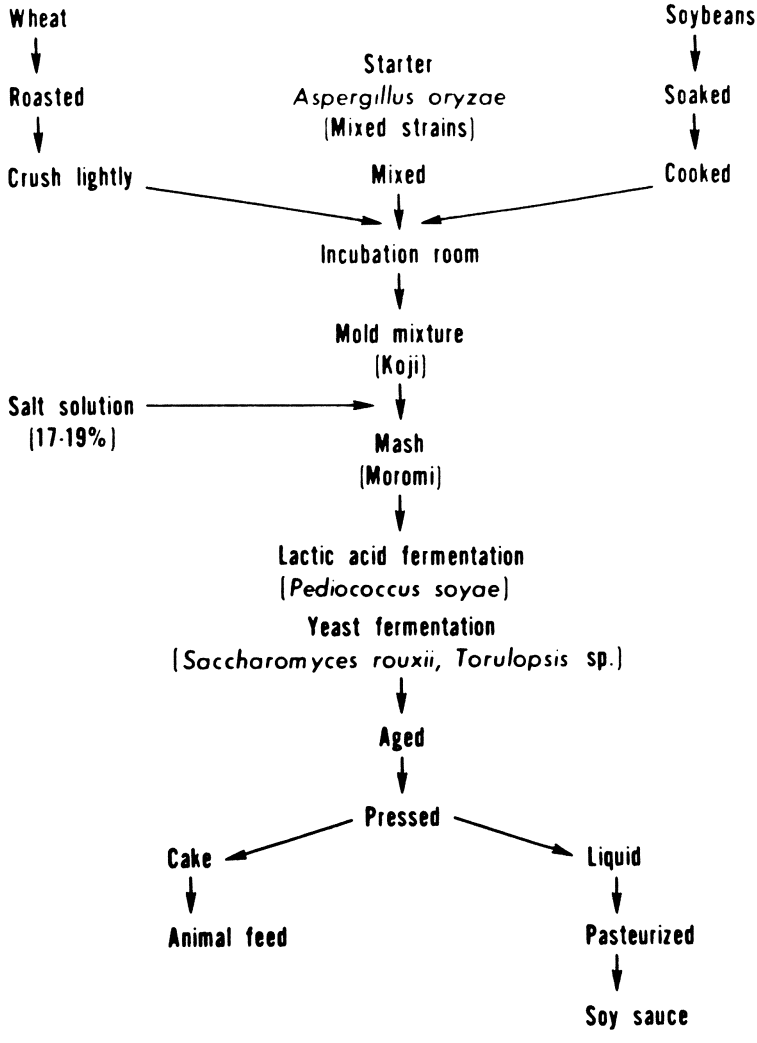

Figure 9.14. Flow sheet for production of soy sauce.
Courtesy of Wang and Hesseltine (1982).

or *A. japonicus.* This mixture is stored until the mold covers the surface of the soybean-wheat mixture. Brine (1 kg salt/4 liters of water) is added at the rate of about 2 liters/kg of soybeans. This blend, or moromi, is fermented for three to six months, first at 15°C, then at 25°C. The mold enzymes hydrolyze the carbohydrates and proteins. Bacteria appear in the early stages of the fermentation and lower the pH from 6.7 to 5.0. These bacteria are in the genera *Pediococcus* or *Lactobacillus.*

In the later stages, the osmophilic yeast *Saccharomyces rouxii* ferments part of the sugars (from the hydrolysis of carbohydrates) to produce alcohol (about 2.5 percent). Besides *Saccharomyces,* other yeasts that may be involved include *Hansenula* and *Torulopsis.*

In the final phase of the fermentation, the pH is lowered to about 4.5 to 4.8. At this time, the fermented mash is filtered, and the liquid obtained (soy sauce) is pasteurized at 80° to 85°C and bottled for use.

There are several variations in the procedure and several varieties of soy sauce in Asian countries (Fukushima 1985; Pederson 1979). The flavors in soy sauce were discussed by Nunomura, Sasaki, and Yokotsuka (1976a, 1976b, 1984).

A system using enzymatically hydrolyzed soybeans fermented by immobilized whole cells of bacterial and yeast cells has been described (Osaki et al. 1985).

TEMPEH. Tempeh is a fermented food characteristic of Indonesia. The product made from soybeans is called tempeh kedalee, while that made from coconut press cake is called tempeh bongkrek.

There are several methods for making tempeh. Briefly, for the soybean product, the beans are soaked and boiled and the hulls removed. The beans may be cooked, and the bean mash made into cakes, inoculated with tempeh mold, wrapped in banana leaves, and fermented. There are several species of *Rhizopus* that can ferment the beans to a desirable product, but *R. oligosporus* is considered to be the tempeh mold. At a temperature of 30° to 40°C, the tempeh is ready to eat in 24 to 48 hr.

Tempeh has been described as having a bland flavor, is quite nutritious, and has simple, low-cost processing techniques.

Miscellaneous Fermentations

There are certain fermentations in which individual or many microorganisms may be involved. Among these is the fermentation of glucose in egg white.

EGG WHITE. Egg white contains about 0.5 percent glucose, a reducing sugar. In commercially dried egg white (3 to 9 percent moisture), glucose

will combine with amino acids to form brown insoluble products, due to the Maillard reaction.

The simple solution to this problem is to remove the glucose. There are several systems that can be used for this purpose. Many microorganisms ferment or oxidize glucose so that it is not reactive with the amino acids. In a natural fermentation, the liquid albumen is allowed to remain at room temperature and the contaminants acquired during the breaking and separation of the egg white will eliminate the glucose. With this system, there may be undesirable organisms that degrade the protein, or those that are health hazards, such as the salmonellae, which also grow in the fermenting product.

Seeding of the albumen with specific organisms was used, so that there would be some semblance of a controlled fermentation. The organisms included coliforms (mainly *Escherichia coli* or *Aerobacter* [*Klebsiella*]), yeasts (species of *Saccharomyces* and *Candida*), and streptococci. Although these fermentations were acceptable, due to the lack of microbiologists in many egg products operations, control tended to suffer.

An enzyme isolated from *Aspergillus niger* is called glucose oxidase since, in its presence, glucose is oxidized to gluconic acid. The advantages of using this system are that the microbial load is not increased as in fermentation, and the glucose is not lost but remains in the product as gluconic acid. This means the product yield is greater. This enzyme system is discussed further in the section on enzymes.

ENZYME SYSTEMS

Enzymes are organic catalysts that allow reactions to occur under relatively mild conditions. Hence, they are well suited for use in the food industry. There are cases in which enzymes can be used more effectively than live microbial cells. When a series of reactions or several types of reactions are desired, such as in wine production, the intact microorganism is the system of choice, but if only one reaction is needed, separated enzymes may be beneficial. Also, some reactions or new products or processes may be possible only through enzyme activity.

Industrial enzymes are obtained from plant and animal tissues and from microorganisms. Since microorganisms can be grown in large amounts in controlled conditions, they offer an unlimited source of many enzyme systems. Of the thousands of enzymes, only a relative few have been developed for commercial use.

Microorganisms synthesize all enzymes intracellularly. They secrete certain hydrolases (carbohydrases and proteases) into the surrounding substrate. These enzymes hydrolyze large molecules into smaller ones, which can then be brought into the cells for further breakdown. All types

of enzymes remain inside the cell to conduct the normal cellular activities. It is evident that the extracellular hydrolases are easier to obtain than are the intracellular enzymes, since obtaining the latter requires breakage of the cell wall and membranes.

On a comparative basis, enzymes are rather expensive. If they are added directly to a food, they are difficult or impossible to recover, so they are used only once. Since enzymes are catalysts, they should be reusable. To accomplish this, processes have been developed for immobilizing enzymes on inert carriers. Immobilization may be accomplished by adsorption, chemical attachment (covalent bonding), microencapsulation (enclosing within a material), covalent cross-linking, entrapment, or copolymerization (Fig. 9.15). Various carriers or supports have been suggested for the immobilization process (Böing 1982; Fadda et al. 1984; Imai et al. 1986; Weetall and Pitcher 1986; Wongkhalaung et al. 1985). Since various enzymes and substrates differ in their properties, no one system is useful for all enzyme systems. Also, the process in which the enzyme is used will influence the type of carrier that is used. Each of the immobilization methods has certain advantages and disadvantages. The advantages are that the enzymes can be removed from the reaction and they can be reused. In some cases, an immobilized enzyme can be used at higher temperatures than can the corresponding free enzyme (it has a higher temperature of inactivation). Since the rate of chemical reactions is increased as the temperature is raised, this is an advantage to the processor. Some problems with immobilized enzymes include the reduction of enzyme activity upon immobilization, loss of activity after repeated usage, mechanical degradation of the enzyme or its support after usage, and microbial growth, with the enzyme acting as the substrate. Enzyme activity (enzyme decay) declines simply because of time.

Besides stability and reuse, other advantages of immobilization include using a continuous process rather than a batch operation, better control of the process, halting the reaction by removing the enzyme, and less product inhibition. One can conceivably use immobilized enzymes in a series so that several reactions can be obtained during the treatment.

The concept of enzyme immobilization has been applied to microbial cells by attaching or entrapping in various substances (Bisping and Rehm 1986; Fukui and Tanaka 1982; Johansen and Flink 1986). The advantages of cell immobilization over that of enzymes include the formation of multistep and cooperative enzyme systems and the elimination of the need to extract and purify enzymes from the cells. Also, the biological activities of immobilized cells are maintained much longer than those of immobilized enzymes. Some disadvantages are that the cells may contain unwanted enzymes, and there are permeability barriers of the cells and entrapment supports to the reactants.

Immobilized cells can be used advantageously in continuous pro-

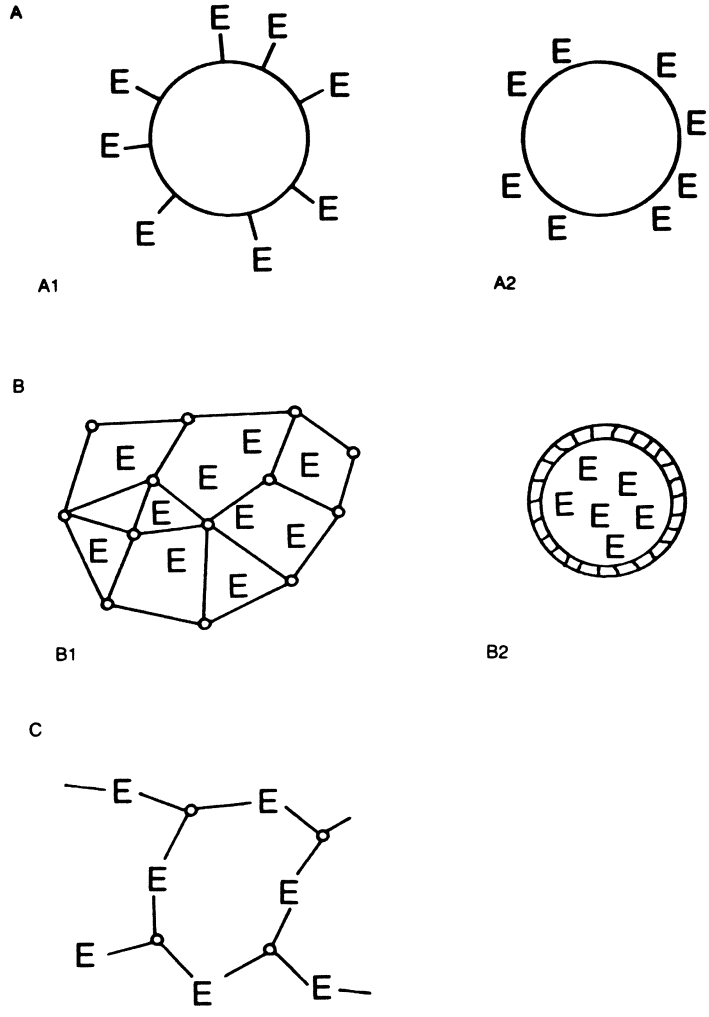

Figure 9.15. Methods of enzyme immobilization. (*a*) Bonding: (*1*) covalent bonding; (2) adsorption. (*b*) Inclusion; *(1)* entrapment; *(2) microencapsulation. (c) Cross-linking.*
From Böing (1982).

cesses. These cells can be removed from the product more readily than can the free cells, and they can be reused. Compared to free cells, the immobilized cells have a higher rate of product formation, and the yield is greater. This is due to an increased cell density of immobilized cells. With higher productivity, smaller reactors can be used, with a reduction in overall cost of the product.

Some microbial enzymes that are useful in the food industry are listed in Table 9.6.

Amylases

These enzymes hydrolyze starch. Most starches are composed of amylose and amylopectin. Amylose is an unbranched polysaccharide with glucose units joined by 1,4-α-glucosidic linkages. Amylopectin is a branched polysaccharide with the glucose units as in amylose and 1,6-α-glucosidic linkages at the points of branching. The products of hydrolysis of starch depend upon the specific amylase that is present.

Commercial amylases are not pure and contain small amounts of maltase and other carbohydrases. In general, amylases can hydrolyze gelatinized starch more readily than raw starch.

The dextrins and oligosaccharides released from starch by the amylases are important in the color and texture of crust and the shelf life of bread. These enzymes can replace the diastatic malt in doughs when the flavor of malt is not needed. Amylases help prevent the staling of bread. They are also used in the manufacture of syrup for use in confections, as well as in combination with glucose isomerase to produce high-fructose corn syrups.

ALPHA AMYLASE (3.2.1.1). These enzymes are obtained from *Aspergillus oryzae, Aspergillus niger, Rhizopus oryzae, Bacillus subtilis,* and *B. licheniformis.* They rapidly fragment starch to dextrins. They do not hydrolyze the 1,6-α-glucosidic linkages of amylopectin.

The enzyme facilitates filtration by converting gelatinous starch to soluble dextrins. It clarifies fruit juices which are turbid due to starch. It is useful in the baking, brewing, and alcohol industries, converting starch to fermentable components. By its action on cocoa starch, it helps ensure stability of chocolate syrup.

The thermostabilities and effective pH ranges vary among the fungal and bacterial amylases (Wasserman 1984). Hence, all of the characteristics of amylases or any other enzymes must be considered when an enzyme is selected for a particular application.

Beta Amylase (3.2.1.2)

These enzymes hydrolyze starch from the nonreducing ends of the glucosidic chains and release maltose. They do not attack the 1,6 linkages. In combination with alpha amylases, they are used in brewing, fermentation, and baking industries.

TABLE 9.6. SOME MICROBIAL ENZYMES USEFUL IN FOODS

Enzyme	Source	Application
Amylases	*Aspergillus, Rhizopus, Bacillus*	Conversion of starch to fermentable components in baking, brewing, and syrup manufacture Clarification of fruit juices Scrap candy recovery Vegetable canning
Catalase	*Micrococcus, Aspergillus*	Decomposition of H_2O_2 in dairy and egg products
Cellulase	*Trichoderma, Aspergillus*	Conversion of cellulose to fermentable components Clarification of fruit juices and wine
β-Galactosidase (lactase)	*Kluyveromyces marxianus, Aspergillus*	Conversion of lactose to glucose and galactose in dairy products
α-Glucosidase (maltase)	*Saccharomyces uvarum*	Conversion of maltose to glucose in brewing
Glucose isomerase	*Actinoplanes, Arthrobacter, Streptomyces*	Conversion of glucose to fructose in corn syrup
Glucose oxidase	*Aspergillus, Penicillium*	Conversion of glucose to gluconic acid in liquid eggs Remove oxygen from juices or from head space of containers
Invertase	*Saccharomyces cerevisiae, Candida utilis*	Conversion of sucrose to glucose and fructose Prevents granulation in soft-centered confections Used in production of artificial honey
Lipase	*Aspergillus, Rhizopus, Penicillium, Candida*	Conversion of fat to glycerol and fatty acids Flavor production in cheese Removal of egg yolk from egg white
Pectic enzymes (pectinases, pectin esterases, polygalacturonase, polymethylgalacturonase)	*Aspergillus, Rhizopus, Kluyveromyces*	Clarification of wine and fruit juices; release of juice from fruit for increased yield, removal of pectin for concentrated fruit juices
Proteases (peptidases, rennin)	*A. oryzae, A. nidulans, Penicillium chrysogenum, B. subtilis, B. stearothermophilus, Endothia, Mucor*	Conversion of proteins to peptides and amino acids (improves dough in baking, meat tenderizing, milk clotting, enzymes for cheese, chillproofing of beer)

GLUCOAMYLASE (3.2.1.3). This is a glucose-liberating amylase. Glucoamylase is used to convert partially hydrolyzed corn starch (treated with acid and alpha amylase) to corn syrup containing more than 90 percent glucose on a solids basis (Walton and Eastman 1973). An immobilized two-enzyme (glucoamylase and alpha amylase) system for this purpose was described by Hausser, Goldberg, and Mertens (1983).

The enzymes can be obtained from organisms such as *Aspergillus niger, A. oryzae, Rhizopus delemar, R. oryzae,* and *R. niveus.*

Catalase (1.11.1.6)

Catalase is produced by many organisms. Commercially, it is obtained from a strain of *Aspergillus niger* or *Micrococcus lysodeikticus.* This enzyme catalyzes the reaction in which hydrogen peroxide is decomposed to water and oxygen. When hydrogen peroxide is added to milk for cold sterilization, catalase is used to remove the excess hydrogen peroxide. Catalase is present in commercial glucose oxidase preparations to release O_2 from H_2O_2 for the oxidation of glucose to gluconic acid, such as in the process for liquid egg white.

Cellulase (3.2.1.4)

Cellulase catalyzes the hydrolysis of the 1,4 linkages of cellulose to form dextrins and usable sugars. Several fungi, including the thermophilic fungi, are sources of cellulase. The fungi include species or strains of *Myrothecium verrucaria, Stachybotrys atra, Trichoderma viride, Trichoderma reesii, Aspergillus niger, Chaetomium thermophile, Sporotrichum, Talaromyces, Thermoascus,* and *Humicola.*

Cellulase is used to remove the cellulose cloud and clarify citrus juices, to improve the body of beer, to aid in extraction of flavoring compounds (such as essential oils), to increase the digestibility and nutritive value of plant products, and to treat cellulosic wastes for potential use as feed, food, or as fermentable sugars (to alcohol). Some problems involved in cellulose conversions were discussed by Saddler (1986).

Beta Galactosidase (3.2.1.23)

The common name for this enzyme is lactase. It catalyzes the hydrolysis of lactose to glucose and galactose. This enzyme can remove the lactose from milk, making it acceptable to people with lactose intolerance. Lactose is less soluble and less sweet than the hydrolysis products glucose and galactose. Hence, hydrolysis of lactose improves the solubility of dried milk and the consistency of concentrated milk, ice cream bases,

and frozen milk. Lactose can crystallize in frozen products, which results in graininess or sandiness in these foods. Frozen desserts made with the hydrolyzed product requires the addition of fewer sweeteners, resulting in a lower-calorie product.

The high lactose content of cheese whey has made this by-product difficult to use. With hydrolysis of the lactose, a sweet whey is obtained that can be used in other products or as a substrate to produce yeast.

The enzymes from different sources vary in their physical properties. They have many microbial sources, including *Kluyveromyces marxianus, Aspergillus niger,* and *A. oryzae.*

Glucose Isomerase (5.3.1.5)

Glucose is obtained by the enzymatic (amylases) hydrolysis of starch, such as corn starch. Glucose isomerase catalyzes the reversible isomerization of glucose to fructose. This reaction is important because the relative sweetness of fructose is two and one-half times that of glucose. With cornstarch as the starting material, high-fructose corn syrup is obtained. This syrup contains about 42 percent fructose on a dry basis. With further processing, higher-level fructose products can be produced (55 percent or 90 percent).

The enzyme is produced by various bacteria, including species of *Streptomyces, Bacillus, Actinoplanes, Arthrobacter,* and *Microbacterium.*

Glucose Oxidase (1.1.3.4)

This enzyme catalyzes the oxidation of glucose to gluconic acid. The excess hydrogen peroxide is then catalyzed to water and oxygen by the presence of catalase. Hence, the enzyme system is called the glucose oxidase-catalase system.

Glucose oxidase is obtained from *Aspergillus niger,* although other organisms, such as *Aspergillus oryzae, Penicillium glaucum,* and *P. notatum* also produce this enzyme. Researchers reported that the best source of the enzyme was a strain of *Penicillium purpurogenum* (Nakamatsu et al. 1975).

This enzyme can be used to remove glucose from egg white, whole eggs, or pork prior to dehydration. This improves the quality and shelf life of the dried product.

Since it is specific for glucose, the enzyme can be used as an analytical tool to estimate the amount of glucose in foods. This is especially useful in the estimation of glucose formed by the hydrolysis of lactose or starch in the food industry. Also, the assay is used clinically for determining glucose in blood and urine (Jackson and Conrad 1985). The enzyme can also be used in immunohistochemical procedures (Rathlev et al 1981).

Since the enzyme uses oxygen to react with glucose, it can be employed as an oxygen scavenger for food systems that deteriorate in an oxygen atmosphere. The removal of oxygen by this system may increase the shelf life or aid in retaining the quality of foods. An increase in shelf life of fish treated with glucose oxidase was reported by Field et al. (1986).

Invertase (3.2.1.26)

Invertase catalyzes the hydrolysis of sucrose to fructose and glucose. The resultant invert sugar is sweeter than sucrose.

Although several microorganisms possess the ability to hydrolyze sucrose, the yeasts have been used as a source of invertase. The yeasts include *S. cerevisiae, S. rouxii, K. marxianus,* and *Candida utilis.*

Invertase can be used whenever it is desirable to have sucrose hydrolyzed. It is used in confections to convert solid sucrose mixtures to a fluid consistency after covering with a substance such as chocolate, to produce soft-centered confections.

Limonoate Dehydrogenases

Limonin is a bitter component in citrus products. If this bitterness could be controlled, there might be an increased demand for citrus products. Immobilized cells of *Arthrobacter globiformis* or *Corynebacterium fascians,* which produce the enzyme, were used to reduce the bitterness of citrus juices (Hasegawa, Patel, and Snyder 1982; Hasegawa et al. 1985).

Lipases (3.1.1.3)

Lipase acts on fats, hydrolyzing them to monoglycerides, diglycerides, or glycerine, and free fatty acids. Many microorganisms produce lipase enzymes. The commercial sources include *A. niger* and *A. oryzae* as well as *Candida lipolytica, C. cylindracea, Penicillium roqueforti, Mucor, Rhizopus, Pseudomonas,* and *Torulopsis ernobii.*

Lipases isolated from microorganisms have been used to develop flavor in cheese or cheeselike foods and for treating milk fat for use in ice cream, margarine, and butter (Kilara 1985a). It can be used to improve the flavor of bakery and other products. It can improve the whipping properties of egg white by removing any contaminating egg yolk fat. Since the reactions are reversible, it is thought that fats can be taken apart and other fatty acids replaced, thereby producing special modified fats for specific uses.

Pectic Enzymes

These enzymes hydrolyze pectin and pectic substances to lower-molecular-weight compounds. Various names have been used for these enzymes. The usual enzyme system is pectinase, which generally is a mixture of polygalacturonase (3.2.1.15) and pectic methylesterase (3.1.1.11). Pectic lyase (4.2.2.3) and pectin *trans*-eliminase (4.2.99) were suggested as clarifying agents for fruit juices (Ishii and Yokotsuka 1973).

Pectic methylesterase causes demethylation, while the polygalacturonase catalyzes the hydrolysis of α-1-4-galacturonide bonds of pectin. The release of methyl groups exposes carboxyl groups, which, in the presence of calcium or other multivalent ions, form insoluble salts that can be removed.

The commercial sources are *Aspergillus niger* and *Rhizopus oryzae;* however, several other fungi, such as *Aspergillus soyae, A. japonicus, A. wentii, Penicillium glaucum,* and *P. expansum,* may be used.

Pectin remaining in fruit juices tends to hold other substances in suspension as a colloidal system. By hydrolyzing the pectin, the protective colloidal action is destroyed, so suspended substances will settle out and can then be removed from the juice by filtration.

Not only do pectic enzymes act as clarifying agents, they also improve pressability and color extraction, prevent gelation of fruit-juice concentrates, speed filtration, and increase the yield of free-run juice. Pectinase is used to remove the gelatinous coating from coffee beans for fermentation and processing.

Proteases

This is a group of enzymes that catalyze the hydrolysis of peptide bonds in proteins and yield peptides of lower molecular weight. Most proteases split specific peptide linkages. Hence, it is necessary to select the appropriate protease or combination of enzymes so that the desired reaction is obtained.

Proteases can be obtained from any proteolytic organism. The commercial sources of proteases include *Aspergillus oryzae, A. saitori, Bacillus subtilis, B. licheniformis,* and *Streptomyces griseus.*

The applications of proteases include the chillproofing of beer, meat tenderization, the production of protein hydrolysates, mellowing of the dough in bakery products, the production of fish protein solubles, and the improvement of protein recovery from oilseed meals.

In baked products, the use of proteases in the sponge dough reduces mixing requirements. The treated doughs are more extensible and can be molded more easily than untreated dough. Proteases are of value in

controlling the pliability, eliminating buckiness, and assuring proper machinability of the dough, as well as increassing loaf volume.

Proteases are added to beer or ale during the finishing operation to prevent an undesirable haze when these beverages are cooled. The hydrolysis of high-molecular-weight proteins prevents the formation of the haze. The use of proteases in foods has been discussed by Kilara (1985b) and Löffler (1986).

Milk-Clotting Enzymes

These are proteolytic enzymes that have an important function in the manufacture of cheese from milk. During lactic acid fermentation of milk, rennet (rennin) is added to the vat to aid in the clotting process of casein to form a curd (McMahon and Brown 1984). Rennet is obtained from the fourth stomach (abomasum) of suckling calves. It is a mixture of enzymes, primarily chymosin, which has a high ratio of milk-clotting activity to proteolytic activity. Because fewer calves are being slaughtered, rennet substitutes have been investigated. Problems of using rennet substitutes can usually be traced to the extent and type of proteolytic activity relative to their ability to clot milk.

Although many microorganisms produce enzymes that clot milk, only three fungi have been used to any large extent for commercial enzyme production (*Endothia parasitica, Mucor miehei,* and *M. pusillus*).

CHEMICALS

Some organisms do not participate in food fermentations directly but are used to produce chemicals that are added to foods. Such chemicals as vitamins and amino acids upgrade the nutrient value of food, and gums and dextrins improve the physical characteristics of foods. Demain (1971) listed alcohols, amino acids, antibiotics, antioxidants, coloring agents, enzymes, nucleotides, organic acids, plant growth regulators, polyols, polysaccharides, protein, sugars, and vitamins as fermentation products used in the food or feed industries.

Amino Acids

Amino acids can be produced by chemical synthesis, but both optical isomers are obtained. In contrast, the synthesis by microorganisms results in the L-isomer which is the form used by biological systems. The microbial sources of certain amino acids are shown in Table 9.7.

Lysine and methionine are important as additives to plant proteins

TABLE 9.7. SOME MICROBIAL SOURCES OF AMINO ACIDS

Amino Acids	Microbial Sources
Alanine	*Corynebacterium, Brevibacterium*
Aspartic acid	*Escherichia, Serratia, Bacillus, Pseudomonas*
Glutamic acid	*Corynebacterium, Brevibacterium, Micrococcus, Microbacterium, Bacillus, Pseudomonas, Streptomyces, Aeromonas, Flavobacterium*
Lysine	*Micrococcus, Bacillus, Escherichia, Torulopsis, Saccharomyces, Proteus, Erwinia, Candida, Corynebacterium*
Methionine	*Pseudomonas, Rhodotorula, Micrococcus, Candida*
Ornithine	*Micrococcus*
Phenylalanine	*Escherichia, Micrococcus*
Threonine	*Escherichia*
Tyrosine	*Micrococcus*
Tryptophan	*Claviceps, Serratia, Micrococcus, Rhizopus, Hansenula, Candida, Escherichia*
Valine	*Enterobacter, Escherichia, Klebsiella, Micrococcus, Brevibacterium*

to increase the nutrient value. Glutamic acid is used in the production of monosodium glutamate, which is a flavor enhancer in many foods. Compared to free cells, immobilized cells produce increased amounts of amino acids (Umemura et al. 1984; Wada et al. 1980).

Other Organic Acids

Various organic acids are added to foods. The main supply of citric acid was lemons, but now it is obtained by fermentation of sugars with *Aspergillus niger.* Acetic and lactic acids are discussed in regard to vinegar production and vegetable and dairy fermentations. In lactic acid production, *Lactobacillus delbrueckii* is used because the reaction can be accomplished at 50°C. This high temperature suppresses many other organisms (psychrotrophs and mesophiles). Some microbial sources of organic acids are listed in Table 9.8.

TABLE 9.8. SOME MICROBIAL SOURCES OF ORGANIC ACIDS

Acid	Microorganism
Acetic	*Acetobacter*
Citric	*Aspergillus, Penicillium, Trichoderma*
Gluconic	*Penicillium*
Lactic	*Lactobacillus delbrueckii*
Propionic	*Propionibacterium*
Pyruvic	*Pseudomonas*
Succinic	*Brevibacterium*

Microbial Gums

Gums are used in the food industry in aqueous systems when there are problems involving suspension, thickening, emulsion stabilization, and rheological modifications. One of the sources of gums is microorganisms. Several microbial gums have been produced and tested. One that is available commercially is xanthan gum. It is an optional emulsifying and stabilizing ingredient in French dressing and a stabilizing ingredient in certain cheeses and cheese products. It has applications in canned foods, dry mixes, frozen foods, juice drinks, relish, sauces, gravies, syrups, bakery fillings, and bakery flavor emulsions.

In the production of xanthan gum, *Xanthomonas campestris* is grown in a well-aerated medium containing glucose, a nitrogen source, potassium phosphate, and trace elements. The gum is produced as an exocellular polysaccharide coat surrounding the cell wall.

Vitamins

Vitamins are added to various foods for enrichment. Only a few vitamins are produced by fermentation. Riboflavin, produced chemically, is used in food, while that produced by fermentation is used in feeds. Some microbial sources of vitamins are listed in Table 9.9.

Flavoring Compounds

Autolyzed yeast extract is used as a flavoring agent in many foods (soups, bouillons, sauces, gravies, entrees and side dishes, snack foods, canned meat products). An enzymatic system for autolysis of the yeast cells was described by Knorr et al. (1979).

MICROBIAL PROTEINS

Due to the potential increase in population and the limited supply of land for agriculture, alternative sources of food have been investi-

TABLE 9.9. SOME MICROBIAL SOURCES OF VITAMINS

Vitamin	Microorganism
Ascorbic acid	*Pseudomonas*
β-Carotene	*Rhodotorula*
B_{12}	*Bacillus, Propionibacterium, Streptomyces, Pseudomonas, Escherichia, Nocardia, Rhizobium*
Riboflavin	*Aspergillus niger, Eremothecium ashbyii, Ashbya gossypii, Candida, Streptomyces*

gated. One of these sources is microbial cells. Various names, such as novel protein, unconventional protein, minifoods, petroprotein, and single-cell protein (SCP), have been used for these products. Actually, the products are microbial cells or microbial proteins. Microbial protein (MP) is used in animal feed more than directly as human food.

Either live or dead microorganisms are present in nearly all of the food that we eat. In some products, such as yogurt, the presence of certain microorganisms is used as a sales gimmick. Thus, the use of microorganisms for food should not be objectionable if the microorganisms are not pathogenic and produce no harmful toxins.

Advantages

Microorganisms have many advantages over other types of food sources. Microorganisms do not require much space; they do not compete for the arable land that is available; they can be genetically manipulated to produce certain types of products; they have a rapid rate of growth and, since growing conditions can be easily controlled, there are no environmental problems as in normal agriculture, such as freezing, floods, or drought. Besides these advantages, microorganisms can be used to digest and convert wastes into useful products. This also reduces the problems of waste disposal.

Substrates

Various substrates ranging from CO_2 with sunlight for algae to complex substances for bacteria, yeasts, and molds have been proposed for the production of microbial protein. These substances are derived from petroleum, natural gas, industrial products, agricultural crops, and various wastes.

According to Calam and Russell (1973), to be economical, the minimum capacity of a processing plant should be 50,000 tons of SCP per year. This means that an adequate supply of substrate is needed for microbial growth and SCP production. Petroleum and natural gas products, such as kerosene, fuel oil, gas oil, *n*-alkanes, methanol, and ethanol were considered desirable because they were readily available in an apparently unlimited supply. However, there are problems in utilizing fuel oil, gas oil, and *n*-alkanes. Besides the real or imagined shortages and the increasing prices of petroleum, these substances are not miscible with water. Hence the mixture must be agitated constantly to keep the cells in contact with the substrate. A high volume of oxygen is needed, and ammonium salts or nitrates must be added as a nitrogen source for the

microbial cells. Only about one-half of the hydrocarbon is converted to cells, the rest being oxidized to carbon dioxide and water.

Other problems with petroleum usage include the need to remove the excess heat due to the oxidation of the hydrocarbons and also the separation of the cells from substrates, such as fuel oil or gas oil. There are potentially carcinogenic polycyclic hydrocarbons in oil-based fermentations.

Methanol is regarded as a very promising substrate. This alcohol can be produced from natural gas that would normally be burned at remote oil wells. Also, it can be produced from coal or wood. Methanol is relatively cheap, there is an ample supply, it is essentially pure (99.5 to 99.8 percent), it is miscible with water, less oxygen is required, less heat removal is needed, it is easily stored and handled, contamination is minimized due to restrictive use by only a few microorganisms, and the cells are readily separated from this substrate (Cooney and Makiguchi 1977; Dijkhuizen, Hansen, and Harder 1985; Jara, Allais, and Baratti 1983; King 1982).

Ethanol can be obtained from ethylene, a petroleum product, or by fermentation of carbohydrates. The use of ethanol as a substrate is similar to methanol (Laskin 1977).

Methane, derived from fossil fuels or by degradation of waste products, can be used as a substrate. The process was discussed by Wilkinson (1971). Another gas that has been considered as a substrate is carbon monoxide (Meyer 1980).

Some agricultural crops that are primarily carbohydrate could be used as substrates to produce SCP, and this protein could then be used to upgrade the nutritional value of the food crop.

The disposal of our wastes is a problem that is increasing every year. If our wastes could be used to produce a usable product such as SCP, we might be solving two problems, waste disposal and a food source.

For the United States, organic agricultural wastes have been estimated at over a billion metric tons annually (Bellamy 1974). The annual world production of straw is estimated at a billion tons. Generally only one-third to one-half of the agricultural crop is harvested and the remainder is usually left in the field. This residue is often a harbor for pests and plant pathogens. For some vegetables, such as peas, the entire plant is hauled to the processing plant and, after the peas are removed mechanically, the pea vines accumulate and may become a disposal problem.

Crop residues, such as corn stalks and cobs, straw, pea and bean vines, and other vegetable-processing wastes contain from 30 to 60 percent cellulose, 10 to 30 percent hemicelluloses, 5 to 20 percent ash, 4 to 18 percent lignin, and relatively small amounts of protein and fat.

Urban and industrial organic wastes accumulate at about 200 million metric tons annually. Some of these wastes are not biodegradable and others may contain toxic chemicals that make them unsuitable as substrates.

The suitability and economics of these wastes depend upon their availability (the amount and location), composition, biodegradability, treatment(s) needed, presence of toxic materials (including pesticides), and various costs (product gathering, transportation, and storage).

Some of the wastes that have been considered as potential substrates are animal wastes (Kargi et al. 1980), sewage, sugar cane bagasse (Molina et al. 1984), straw (Pavlostathis and Gossett 1985; Taniguchi et al. 1982). Various wastes from food-processing plants have been investigated as potential substrates for SCP production (Calleja et al. 1986; Clementi, Moresi, and Rossi 1985; Rale 1984; Shay and Wegner 1986). Some wastes are seasonal and diluted with water. Hence, there are costs to concentrate the dilute wastes and to accumulate enough to use over the entire yearly operation of a processor.

Some wastes, especially cellulosic wastes, need to be treated with acid or alkali so that they can be used by microorganisms. A source of nitrogen (ammonium salts or nitrates) and minerals such as calcium, iron, phosphorus and magnesium needs to be added for acceptable growth of microorganisms. Since these different types of wastes vary in their composition, it is difficult to establish a plant and procedure to utilize all of them. Only a few accumulate in a volume capable of supplying an SCP facility, and even then, the gathering and transportation costs must be considered.

Microorganisms

All types of microorganisms (bacteria, yeasts, molds, higher fungi, algae, and protozoa) have been suggested as potential sources of protein. For economic as well as other reasons, the organisms should be able to grow on inexpensive, simple media (preferably waste products); it is helpful if they can grow in a continuous culture; they should be easily separated at harvest; pure cultures with known genetic factors and means of altering these factors should be available; there should be no toxic or allergenic factors in the culture; the product should be palatable or readily disguisable, contain high-quality protein, and be easily packaged and stored; and the remaining effluent should be of a quality that is easily disposed of, with little or no further treatment.

There are certain advantages and disadvantages for using each type of microorganism in this process. Bacteria have the fastest growth rate, but being the smallest, they are more difficult to harvest. Molds are large

and are the easiest to harvest, but they have a slower growth rate than either bacteria or yeasts. The yeasts are easier to harvest than are the bacteria.

The type and species of microorganism used in the process depend upon the substrate, the conditions for growth, the equipment, and other factors.

YEASTS. As a food for humans, the yeasts have been more apparent than have the other microorganisms. Yeasts have been used as a food source since workers in Germany developed the process during World War I. The first successful commercial process of yeast production was from molasses and ammonia. Since this beginning, several types of substrates have been used to grow yeasts for feed and food. Yeast extracts are used in soups, sauces, relishes, and some processed meats.

Although some yeasts can multiply by a sexual process of spore fusion, under most circumstances they reproduce by budding. The time required for a mother cell to produce a daughter (the doubling time) for most yeasts with favorable growing conditions varies from one to three hours. Cooney, Levine, and Snedecor (1975) cited references showing that for three species of yeast on a methanol substrate, the doubling time varied from three to nine hours.

Various yeasts have been considered as potential food sources, including species of *Candida, Saccharomyces, Rhodotorula, Torulopsis, Hansenula, Kluyveromyces, Debaryomyces, Pichia,* and *Kloeckera.*

Yeasts can be used for producing vitamins, enzymes, high-quality protein, and other physiological essentials. Since yeasts can be bred for selective purposes, it is possible to obtain special yeasts that are richer than ordinary yeasts in certain nutritional components, such as high-protein yeasts.

In considering the nutritive value of yeast, the proximate analysis of the yeasts is usually published. The protein content can vary from 30 to 65 percent, depending upon the yeast and the substrate. The amino acid content of these proteins varies considerably. In general, the amino acid analysis of the yeast protein compares favorably with animal protein, except that the methionine content is low. According to the review of Reed (1981), the protein efficiency ratio (PER) of baker's yeast (2.02) can be increased to 2.77 by the addition of 0.5 percent methionine. Martini, Miller, and Martini (1979) calculated the PER of various yeasts and reported that they varied from 1.2 to 2.2. The replacement of 8 percent wheat flour with torula yeast flour increased the PER for rats fed the resultant bread from 1.31 to 2.28 (Lin, Chastain, and Strength 1986). According to Shacklady (1972), the biological value (BV) of dried whole egg is 90, while that of yeast protein varied from 46 to 61. When 0.3 percent

methionine was added to the yeast protein, the BV increased to values of 91 to 96, indicating that the supplemented yeast compares favorably to plain dried whole egg, or dried skim milk (BV of 87). The relatively high lysine content of yeast will improve the nutrient value if added to grains such as corn, wheat, or rice. It would not be as useful to add yeast to soybeans, since they too are low in methionine. Macmillan and Phaff (1973) listed *Rhodotorula gracilis* as a source of methionine.

One problem with SCP in general is the high nucleic acid content (6 to 16 percent) of microbial cells. The end product of purine metabolism is uric acid. Although we can tolerate low levels, when the daily nucleic acid intake is over 2 g, the uric acid, being insoluble at body pH levels, can cause conditions such as gout or kidney stones. Methods for reducing or removing the nucleic acids from yeast protein have been reported by Gierhart and Potter (1978), Sarwar et al. (1985), and Shetty and Kinsella (1979), and reviewed by Waslien and Steinkraus (1980).

In a review, Harrison (1970) reported that ingestion of live yeast cells is not desirable, because they can remove vitamins and other nutrients from the digestive tract. Whole dead cells are not easily digested because of the cell wall. With this inability to digest the cells, the protein is not usable. To make the protein available, the cells can be disintegrated mechanically or by enzymes.

DeGroot et al. (1975) fed yeasts to rats over a two-year period and found no adverse effect on mortality, rate of body weight gain, hematology, kidney function, fertility, histological changes, or incidence of tumors. Yeast SCP fed to rats yielded no toxicological effects and no abnormal reproductive performance in either male or female rats (Ashraf et al. 1981). Other researchers found an increased resistance to respiratory disease among monkeys fed brewer's yeast (Sinai et al. 1974); they thought that the yeast stimulated phagocytosis.

Chickens fed 5 to 10 percent yeast diets were significantly lighter than those on a control diet (Shannon, McNab, and Anderson 1976). This was thought to be due to less feed intake when yeast was incorporated.

Yeast, per se, has not replaced meat proteins in the human diet, primarily on the basis of flavor and other quality attributes. Dried yeast has very few functional properties, but yeast protein has several. The refined proteins can be spun or otherwise processed to function in various ways and can be added to foods in the same manner as vegetable proteins (Hayakawa and Nomura 1978; Huang and Rha 1978).

BACTERIA. The use of bacterial cells or protein as food lags behind yeasts. However, they have been consumed in fermented foods such as yogurt.

Generally, bacteria multiply faster than do yeasts. Most bacterial SCP

has a higher level of protein and a higher BV than yeast SCP. Although bacteria, being smaller, are more difficult to recover than are yeasts, there are methods using flocculation, filtration, and centrifugation that allow recovery, but at added expense.

The type of bacteria that is used depends upon the substrate. Species of *Pseudomonas* are grown on substrates of methanol, kerosene, fuel oil, or gas oil (Lipinsky and Litchfield 1974). Abbott, Laskin, and McCoy (1973, 1974) described the growth of *Acinetobacter calcoaceticus* on ethanol. They stated that the composition of an organism is strongly dependent upon its growth rate and environment. They obtained a higher protein content of the cells when the growth rate was increased. The fermentation of rice straw with a species of *Cellulomonas* and *Alcaligenes* was discussed by Han (1975) and Han and Callihan (1974). It was suggested that the microbial protein could be used for nonruminant feeding and the residue could be fed to ruminants.

Rhodopseudomonas gelatinosa is a photosynthetic bacterium utilizing CO_2 and sunlight. *Thermomonospora* and *Thermoactinomyces* are thermophilic bacteria that produce protein on cellulose substrates. *Methylococcus* and *Hyphomicrobium* species utilize methane. *Hydrogenomonas* can assimilate hydrogen and carbon dioxide for SCP production. These were discussed by Bellamy (1974) and Litchfield (1977, 1980). A system using immobilized *E. coli* to produce protein has also been described (Inloes et al. 1983).

The dry weight of bacterial cells is 47 to 86 percent protein. The nitrogen digestibility and weight gain of rats were improved when cells of *Pseudomonas* were homogenized before feeding (Yang, Yang, and Thayer 1977). At levels of 10 to 20 percent of the diet, methanol-grown bacterial cells resulted in reduced growth rates and other adverse effects on young chicks (D'Mello and Acamovic 1976).

Besides the problem of harvesting, the nucleic acid content of bacterial cells appears to be higher than that of yeast cells. Nucleic acid content up to 20 percent was suggested (Shacklady 1972). This can be reduced with known processing systems (Yang, Thayer, and Yang 1979). When fed to humans, bacterial cells have caused gastrointestinal problems and other adverse reactions (Litchfield 1980).

MOLDS. Molds are complex, multicellular, aerobic organisms. They grow over a wide range of pH, temperature, and substrates. They contain the vitamin B complex and from 13 to 60 percent crude protein. The methionine content of mold protein is rather low, as is true for yeast protein. The tryptophan and cystine contents are also low. For other amino acids, mold protein compares favorably with milk and fish meal proteins.

Molds have a slower growth rate with a lower nucleic acid content and are easier to separate from the substrate than either yeasts or bacteria. The mold filaments can be physically aligned into a fibrous texture resembling meat or poultry or even the flaky texture of fish. A fungal protein, mycoprotein, made from *Fusarium graminearum,* was to be market tested in the United Kingdom in 1981 (Anon. 1981). Reportedly, it is nutritious and safe to eat. However, there was a concern that it might be deficient in iron and zinc (Litchfield 1983).

The cellulolytic fungus *Trichoderma viride* and other *Trichoderma* have been grown on feedlot wastes, cereal straws, and other cellulosic wastes (Griffin, Sloneker, and Inglett 1974; Han and Anderson 1975; Peitersen 1975a). Other molds tested as potential sources of SCP include an acid-tolerant *Scytalidium acidophilum* grown in peat extract (Martin and White 1985), and a thermotolerant *Phanerochaete chrysosporium* grown on vinasse (Cardoso and Nicoli 1981a, 1981b).

MUSHROOMS. Higher fungi, or mushrooms, have been known as a source of food since the beginning of human history. Their efficiency of converting carbohydrates to protein is approximately 65 percent. As with other fungal protein, mushrooms have a low content of methionine and tryptophan. The nucleic acid content of mushrooms is lower than that of microorganisms (Fabregas and Herrero 1985). The use of mushrooms was reviewed by Chang (1980).

MIXED CULTURES. In nature, mixed cultures are present during the degradation of cellulosic and other wastes. In these wild, mixed cultures, the organisms consume each other, so that a large amount of protein does not accumulate. However, in a controlled mixed culture, two organisms may grow better and produce more protein.

Mixed cultures of *Trichoderma viride* and a yeast (*Candida utilis* or *Saccharomyces cerevisiae*) were used with alkali-treated barley straw (Peitersen 1975b). The overall rate of protein production was increased by use of the mixed culture.

ALGAE. The algae are the simplest plants that contain chlorophyll. Their use as a potential source for food should be readily apparent, since they begin the food chain for sea life (Ryther and Goldman 1975).

The algae vary in size from microscopic forms to the giant seaweed or kelp that may be 40 or more feet long. The familiar habitat of algae is any body of sunlit water, ponds, streams, lakes, or oceans. There are also terrestrial algae.

Since algae contain chlorophyll, they can utilize solar energy by photosynthesis, like the higher plants, to produce food from CO_2, H_2O, and

inorganic salts. They have many of the advantages listed for the other microorganisms. Generally, the algae have a slower growth rate than other microorganisms. The algae require light for growth, so either artificial light is needed, or growth is limited to areas with warm, sunny climates. When grown outdoors in ponds or on sewage, as has been suggested, there is the problem of contamination with undesirable bacteria. *Spirulina maxima* is grown successfully in the naturally alkaline water of Lake Texcoco in Mexico.

For centuries, algae have been harvested and consumed by people in the Lake Chad area of Africa, with no apparent ill effects. The red algae have been harvested for their agar and carrageenan. The brown algae, *Laminaria* and *Macrocystis,* are sources of algin. Agar is used for solidifying media, as well as in gels for dessert. Carrageenan and algin are added to food products to improve the texture. A red algae, *Porphyra,* has been used in Japan. It has been estimated that algae can produce from 30 to 80 tons of protein per hectare per year compared to conventional crops producing 0.4 to 2.5 tons per hectare per year (Grobbelaar 1979).

The protein concentration of dried algae varies from 5 to 64 percent. Cells of *Chlorella* contain about 60 percent protein (Endo, Nakajima, and Chino 1974). The different levels are due to differences in algae and in growth conditions. The PER of algal protein ranges from 1.25 to 2.6 and the BV from 54 to 72. The analysis of the proteins of algae shows that they are low in the sulfur-containing amino acids (cystine and methionine).

There have been toxic effects noted due to certain algae. The toxic effect of "red tide" on the killing of fish is well known. Some species of blue-green algae have caused the death of animals. Shilo (1967) reviewed algal toxins, but no toxins were discussed for the microalgae *Chlorella, Scenedesmus,* or *Spirulina.* Humans can tolerate some algal protein, but if the daily intake of algae exceeds 100 g per day, there may be gastrointestinal distress.

For good digestibility, the cells must be broken. For human acceptance, the protein will need to be purified and processed into other foods.

Due to the many difficulties in growing, processing, and use in human foods, other microorganisms show more promise as potential sources of protein than do algae.

The Potential of Microbial Protein

Many people are optimistic about the future prospects of microbial protein. With improved processing techniques, the potential will, no doubt, also increase. The main use of SCP is in animal feed, but some has been used in human food and this use should increase in the future.

The two main concerns are the safety and cost. As with any situation or product, there is always a risk factor. Feeding trials with rats and other animals indicate that the product is safe. However, for human consumption, it is necessary to reduce the nucleic acid content of the product. Gastrointestinal distrubances have occurred in humans fed certain SCP. Until problems such as these are solved, only low levels of SCP can be used as human food.

The cost of producing SCP is such that it may not be able to compete with other plant proteins, but it can compete with many animal proteins. The nutritive value of SCP is generally lower than animal proteins, but the SCP is improved by processing the cells, isolating the protein, and supplementing it with the limiting amino acids.

If wastes, which cost money for disposal, can be used to produce SCP, we might consider the savings in the disposal cost in the overall economics of the SCP. Since SCP can be spun into fibers like plant proteins, it could be used to make synthetic meats or added as extenders of meat products. By means of genetic engineering and biotechnology, better yields at lower costs may be achieved, resulting in advantages for the production and use of SCP.

With our present and future needs for protein, the use of microorganisms as a source of feed or food cannot be overlooked.

ASSAY

Microorganisms can be used to determine or assay the amount of vitamins and amino acids in food products.

The use of living cells for analytical purposes gives a high degree of sensitivity as well as biological specificity, because of the particular responses of the metabolic processes of the cell. The reagents are the test organism and the medium used for growth. The reaction is the metabolic response, or lack of response, by the cell. By using serial dilutions of the sample in the growth medium, inoculating with the test organism, incubating, and determining the response of the cell, we can determine the amount of substance being assayed. Automated systems for vitamin assay were described by Einarsson and Snygg (1986) and Guilarte (1983) and were reviewed by Gregory (1983). Enzymatic assays of amino acids have been developed (Roy 1979; Tonogai et al. 1983; Tuffnell and Payne 1985).

Some of the microorganisms and vitamins assayed are listed in Table 9.10. The test microorganisms and corresponding amino acids are listed in Table 9.11. Special strains, or mutants of the microorganisms, are used for these assays.

TABLE 9.10. ASSAY OF VITAMINS

Vitamin	Microorganism
Biotin	*Lactobacillus plantarum*
	Saccharomyces cerevisiae
Choline	*Neurospora crassa* (mutant)
B_6 (pyridoxine)	*Saccharomyces uvarum*
	Neurospora sitophila (mutant)
Nicotinic acid	*Leuconostoc mesenteroides*
	Lactobacillus plantarum
B_{12}	*Lactobacillus leichmannii*
	Escherichia coli (mutant)
	Euglena gracilis
	Ochromonas malhamensis
Thiamin	*Lactobacillus viridescens*
Pantothenic acid	*Saccharomyces uvarum*
	Pediococcus acidilactici
	Lactobacillus plantarum
Folic acid	*Streptococcus faecalis*
	Lactobacillus casei
Riboflavin	*Lactobacillus casei*
	Lactobacillus helveticus
Inositol	*Neurospora crassa* (mutant)
	Saccharomyces uvarum

Besides vitamins and amino acids, microorganisms can be used to assay materials for antibiotics, some types of pesticides, and bacteriolytic enzymes such as lysozyme (Bitton, Koopman, and Wang 1984; Kingdon 1985; Mueller, Reed, and Barkate 1979; Rajkowski, Peeler, and Messer 1986).

TABLE 9.11. ASSAY OF AMINO ACIDS

Amino Acid	Microorganism
Arginine	*Streptococcus faecalis*
Aspartic acid	*Leuconostoc mesenteroides*
Cystine	*Leuconostoc mesenteroides*
Glutamic acid	*Lactobacillus plantarum*
Glycine	*Leuconostoc mesenteroides*
Histidine	*Leuconostoc mesenteroides*
Isoleucine	*Lactobacillus plantarum*
Leucine	*Lactobacillus plantarum*
Lysine	*Leuconostoc mesenteroides*
Methionine	*Leuconostoc mesenteroides*
Phenylalanine	*Leuconostoc mesenteroides*
Proline	*Leuconostoc mesenteroides*
Serine	*Leuconostoc mesenteroides*
Threonine	*Streptococcus faecalis*
Tryptophan	*Lactobacillus plantarum*
Tyrosine	*Leuconostoc mesenteroides*
Valine	*Lactobacillus plantarum*

Microorganisms or their enzymes have been adapted to analyze foods for various chemicals in general (Schwimmer 1981) and specifically, such as acetaldehyde in beer (Delcour et al. 1982), cyanide in cassava (Rao and Hahn 1984), cholesterol in egg yolk (Shen, Chen, and Sheppard 1982), starch in meat products (Skrede 1983), and ethanol in beer (Waites and Bamforth 1984). In many cases the cells or enzymes are part of a biochemical electrode system (Hopkins 1985; Schwimmer 1981). These have been used to determine glucose (Ikeda et al. 1985; Wingard et al. 1984), monoamines in meat (Karube et al. 1980), ethanol in wine (Mason 1983), aspartame (Renneberg, Riedel, and Scheller 1985), and hypoxanthine and inosine in fish (Watanabe et al. 1984). Enzymes are also used in the enzyme-linked immunosorbent assay (ELISA) as described in Chapter 6.

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10

Control of Microorganisms

The control of microorganisms is one of the major concerns of food microbiologists. This control is needed to retard or prevent spoilage and to reduce or eliminate health hazards associated with foods. The control of contaminants also aids in obtaining better results when specific microorganisms or enzymes are used in food processing.

Four basic systems are used to aid in the control of microorganisms in foods. These are (1) prevent contamination (asepsis); (2) remove contaminants; (3) inhibit growth; and (4) destroy contaminants.

In most food products, two or more of these systems are used to control the microbial level. Preventing contamination is practiced for all foods, but since contamination will still occur, other safeguards are needed.

Besides controlling the microorganisms, foods must be protected from reactions that are catalyzed by inherent enzymes, from chemical degradation such as fat oxidation, loss of nutrients, and from the destruction by pests, such as insects and rodents. There are various chemicals and procedures that will control microorganisms and pests but cannot be used for foods. This is because the food must be safe for consumption, be of acceptable organoleptic quality, and have good nutritional value.

Although the control of microorganisms in food is usually relegated to the food processor, everyone involved in the production, processing, handling, warehousing, retailing, preparation, and serving should be involved in the control process as well as in maintaining a safe and nutritious food supply.

The need for this overall effort is evident from the fact that only a few outbreaks of foodborne illness are caused by problems at the processing level. Most of the outbreaks are caused by mishandling and contamination of foods at foodservice establishments or in the home.

In some cases, we assume that a food will be handled and used in a particular manner; but this assumption may not always be correct. For example, we assume that people will cook meat before eating it. However, a series of outbreaks of salmonellosis revealed that some people thought that eating raw hamburger would make them healthy, strong, and vigorous. Because of such misuse of foods by some people, it is necessary to take precautions beyond the normal processing requirements. A

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food microbiologist must be aware of all aspects of a food, from production to consumption.

A primary function of food processing is the preservation of foods. This preservation ranges from short periods of a few days or weeks to long terms of a year or more. Preservation by the removal, inhibition, or destruction of microorganisms is easier and the results are more satisfactory when the original microbial index of the food is low. Keeping the contamination low by sanitary means is very important.

CONTROL BY ASEPSIS

If all of the microorganisms in a food were either useful or inert, we would have no concern about them. Unfortunately, the spoilage and health hazard types are quite prevalent. Generally, the higher the microbial load, the greater the possibility that undesirable types are included. The exceptions are foods in which specific microorganisms are grown to produce desirable products and a high level of the useful microorganism is needed.

It is much easier to inhibit or destroy low numbers of microorganisms than high numbers. A food with a low microbial load will generally have a longer shelf life than a food with a high microbial level. The shelf life is the time after packaging during which a food maintains its best quality if it is stored under proper conditions of humidity and temperature.

Because microorganisms are ubiquitous, it is impossible to keep them from contaminating food. However, we can reduce contamination by controlling the potential sources of microorganisms.

The control of contamination is referred to as sanitation. Sanitation may be defined as a modification of the environment in such a way that maximum health, comfort, safety, and well-being are ensured for people. The sanitation involved in the food industry is only a small part of overall sanitation.

The control of the microbial quality of food must begin with the production and harvesting of food. Then it must carry over to the processor and on to the ultimate consumer.

Production and Harvesting

The raw materials used by the food processor affect the quality of the finished product. To have adequate supplies of high-quality raw materials with acceptable microbial loads, the food processor must work closely with the producer or become the producer. In some cases, a food processor who is also the producer seems to be the best method. As a result, we

now have companies that own or control food-producing and -processing systems. By owning a system, the company can integrate all of the processes to end up with a uniform, high-quality product.

With many foods, the processor must rely on the producer to deliver an acceptable raw material. Some ingredients are obtained from foreign countries where the level of sanitation is different from that in the United States. Hence, processors sometimes have to work with raw materials that are produced in areas beyond their control. Sometimes a visit to the foreign supply areas will help in obtaining more acceptable products.

When the raw materials are obtained from local areas, the processor can usually visit the sources of supply, make evaluations, and suggest procedures that will improve the quality. Sometimes the processor obtains help from local, state, and federal regulatory agencies that inspect the producing areas.

MILK. Many years ago, the occurrence of epidemics of various diseases caused by the consumption of tainted milk caused health authorities to decide that milk production should be regulated. To help control the spread of tuberculosis, cows were tested and reactors eliminated. From this beginning, many regulations and codes of operations have been devised to control the production and handling of milk at the farm. No other commodity is produced with as much regulation as grade A milk. As a result, tuberculosis and brucellosis in milk are controlled. At present, the main infectious problem at the farm is mastitis.

Besides the health of the dairy cow, standards have been developed for the housing and milking areas so that they can be maintained in a satisfactory manner. There are requirements for the equipment used to obtain, handle, and store the milk, and for the equipment to be properly cleaned, sanitized, and maintained (Guterbock, Blackmer, and Duffy 1984; Stone, Myhr, and Davie 1983). Grade A milk must be cooled to 10°C or less within two hours after milking to maintain the quality. The requirements are more stringent for the production of grade A milk than for the manufacturing of grade milk. This latter milk is used in various dairy products other than liquid milk.

Insect and rodent control are needed at the farm level as well as at the processing level. Only approved insecticides can be used in dairy barns or around other food products. Removing potential breeding places helps control insects and rodents.

ANIMAL PRODUCTION. Healthy, disease-free animals should be maintained in a relatively clean, disease-free environment. Modern husbandry practices have tended to increase rather than decrease infections in domestic animals. The practice of crowding animals into feedlots,

broiler houses, laying houses, and holding pens increases the potential for spreading diseases, including salmonellosis. Sanitation of all animal quarters on the farm may help, but will not prevent, infection.

With meat-type chickens (broilers), flocks that are *Salmonella*-free tend to result in carcasses that are *Salmonella*-free. When salmonellae are isolated from birds prior to slaughter, the carcasses tend to contain these organisms.

The transportation of animals to market or from one farm to another causes stress, and animals held in dirty pens until slaughter shed salmonellae at a higher rate than on the farm. This indicates that stress induces shedding of organisms already present in the intestinal tract, or it makes the animals very susceptible to *Salmonella* infection, or both. One solution may be to hold animals at the farm and then move them into clean, uncrowded conveyances to the slaughterhouse for immediate slaughter.

One source of contamination of carcasses during slaughter is of fecal origin. Withdrawal of feed and water 8 to 10 hr before slaughter reduces the fecal content of chickens and might reduce the potential contamination of the carcasses.

SHELL EGGS. Regardless of whether laying hens are housed on the floor or in cages, it is desirable to keep the egg shells clean. The incidence of spoilage is much greater in dirty eggs than in clean eggs, whether the shell is washed or not. When eggs are washed, the temperature of the wash water should be 20°C higher than the temperature of the egg and an approved germicide should be added to the cleaning solution. The addition of a germicide tends to reduce spoilage of the washed eggs during storage. Only odorless, nontoxic cleaner-sanitizers should be used for washing and sanitizing egg shells.

FRUITS AND VEGETABLES. The production of quality plant products starts with quality plant varieties and seeds. The disease resistance and the amount and quality of food produced are important considerations for the selection of plant varieties.

The World Health Organization (WHO 1969) recommended several hygienic practices for fruits and vegetables. The environmental sanitation during growing included sanitary disposal of human and animal wastes, sanitary quality of irrigation water, and control of animals, pests, and diseases. The sanitary harvesting aspects include the equipment and containers, which should be cleaned and maintained and not be a source of microorganisms. Unfit produce should be segregated and disposed of in a satisfactory manner. The product should be protected from contamination by animals, insects, birds, chemicals, or microorganisms.

The transportation of produce should be done in conveyances that

can be cleaned and not be a source of contamination. All handling should be done with care and, if needed, refrigeration or ice of sanitary quality should be used.

The contamination of plant products by soils is important to the food processor, since soils contain bacterial spores, including those of *Clostridium botulinum.* Bacterial spores are resistant to various sanitizing agents, as well as food-preservation treatments. Hence, it would be a definite advantage to the processor if spores could be kept out of the raw materials.

SEAFOODS. Seafoods are contaminated by the environment in which they grow, as well as during harvesting and transportation to the processing plant.

The main problem of contamination during growth involves the bivalve mollusks, such as oysters, clams, and mussels. They grow in estuaries or protected coastal waters. These are also the most convenient places for the disposal of sewage and other wastes of coastal communities. The mollusks obtain nourishment by pumping water through a complex system of gills that filter and concentrate food, such as bacteria and marine organisms. With the dumping of sewage, the bacteria may include potential pathogens. With oysters and clams, the entire animal, including the intestinal tract and its contents, is consumed, frequently raw or after superficial heating. If the bivalves have consumed enteric pathogens, they can serve as vehicles for organisms that cause foodborne illness.

Due to a series of illnesses culminating in a major typhoid fever epidemic in 1924 from the consumption of contaminated oysters, a voluntary cooperative program for the certification of interstate shellfish shippers was established in 1925 by the U.S. Public Health Service. The program is now known as the National Shellfish Sanitation Program (NSSP). The coastal states adopt laws, make sanitary and microbial surveys of shellfish-growing areas, delineate and patrol restricted areas, prevent illegal harvesting, inspect shellfish-processing plants, and conduct other activities needed to ensure that shellfish are grown, harvested, and processed in a sanitary manner.

The FDA administers the NSSP at the federal level and is responsible for evaluation of state and foreign programs, standards development, research, and training.

The survey of the growing areas has three parts: (1) the area is sufficiently removed from major sources of contamination so that shellfish are not subject to fecal contamination; (2) the area is free from pollution by potentially harmful industrial wastes; and (3) the median coliform level does not exceed 70/100 ml of water and not more than 10 percent of the samples exceed 230/100 ml for a five-tube MPN, or 333/100 ml for

a three-tube MPN. In 1974, the FDA proposed that the coliform standard be changed to fecal coliforms as follows: The median fecal coliform MPN value shall not exceed 14/100 ml of sample, and not more than 10 percent of the samples shall exceed 43 for a five-tube, or 49 for a three-tube, MPN test (Hunt 1977).

Even with safe coliform levels, occasional outbreaks of foodborne illness have occurred after the consumption of contaminated shellfish reportedly harvested from approved areas. *Vibrio cholerae* was detected in estuarine-type waters even when coliforms were not detected (Guthrie and Scovill 1984; Hood and Ness 1982), and there was no strong linear correlation between *V. cholerae* and *E. coli* or fecal coliforms (Hood et al. 1983). Also, there does not seem to be a relationship between safe water and the presence of enteric viruses (Gerba et al. 1980; Vaughn and Landry 1984). Although there are problems, Larkin and Hunt (1982) stated that the shellfish control programs are working reasonably well and that there is no guarantee that raw shellfish will be free of disease-producing organisms or toxic materials. However, Madden, Buller, and McDowell (1986) suggested using *Clostridium perfringens* as the indicator organism for contaminated shellfish.

In one study, researchers found that fish, as caught, have low bacterial loads (Huss et al. 1974). The hygienic standards on board the ship determined the level of contamination of the fish. To reduce the growth rate of the bacteria, proper refrigeration, icing, or freezing must be used. The fish should be processed as soon as possible after catching. Some ships have processing systems on board. In these cases, seawater is used to wash and process the fish. If not treated with chlorine, this water may serve as a source of microorganisms. The sanitary conditions of the working areas and equipment, and the health and hygiene of workers on the ship should be the same as in processing plants on land.

Processing

If care is taken to produce and deliver a high-quality product to the processing plant, it is the duty of the processor to maintain the quality at a high level.

The U.S. Food, Drug and Cosmetic Act of 1968 states: "A food shall be deemed to be adulterated if it contains any poisonous or deleterious substance which may render it injurious to health; or if it consists in whole or in part of any filthy, putrid, or decomposed substance; or it has been prepared, packed, or held under insanitary conditions whereby it *may* have become contaminated with filth or rendered injurious to health." With these stipulations, the presence of any microorganisms or chemical that may cause illness or that would indicate that the food may

contain harmful agents makes the food adulterated and subject to seizure. The presence of filth, such as insects or rodent hairs or droppings, is evidence of an adulterated product. However, there are tolerances for these agents (see Chapter 13). Even if the food does not actually contain filth or potentially harmful agents, if conditions exist so that these substances may be present, the food can be considered adulterated. Although this act applies only to foods in interstate commerce, many states have copied the federal regulations.

The buildings, environment, and processing facilities, as well as raw materials, personnel, equipment (including cleaning and sanitizing), and refuse disposal facilities, are important to the overall plant operations.

The level of sanitation maintained in the plant has a direct bearing on the quality of the final product and should be a basic part of the quality-control program of the processor.

Quite often, a sanitation program is considered to be a cost that is passed on to the consumer, making the price of the food product less competitive. More realistically, unsanitary conditions result in increased costs, which increase the price of the food to the consumer. Additional costs, such as those caused by loss of food due to a short shelf life and spoilage of perishable products, medical bills of consumers ingesting pathogenic organisms or poisons, legal fees, and settlement of court causes due to illness or death of former consumers, are passed on to present consumers. Regulatory agents may seize a product that violates the regulations. As a result, the least that can happen to a food processor is to have inventory tied up. The seizure may result in destruction of the product, corporation or personal fines, prison terms, and the closing of the processing plant. Hence, although sanitation may be a cost, the lack of sanitation can create a greater cost. Sanitation is a necessity in order for a food processor to stay in business and has resulted in advantages to both the processor and the consumer.

Proper sanitation reduces or eliminates potential spoilage hazards and creates a longer shelf life of perishable products. This reduces losses due to spoilage, increases the efficiency of plant opeations, results in easier maintenance of equipment, develops better employee relationships, workmanship and safety, increases consumer acceptance and public relations, improves the quality and safety of foods for the consumer, and with better consumer acceptability, there is a potential increase in sales. This results in faster turnover and less spoilage of perishable foods. As a result of these benefits, the actual cost of the product may decrease.

The "may have" section of the U.S. Food, Drug, and Cosmetic Act of 1968 (Section 402 (a) (4)) describes adulterated food. To help interpret this description and to establish criteria to aid industry in complying with this section of the act, the FDA has published regulations called current

good manufacturing practice (CGMP) for various foods. These regulations cover essentially all operations of a food-processing plant, including its outside appearance, equipment, processes, and personnel. The USDA regulates the processing of meat, poultry, eggs, and egg products.

Each food-processing operation has certain points at which failure to prevent contamination can be detected by laboratory tests with maximum assurance and efficiency. These are known as critical control points. Bauman (1974) defined hazard analysis as the identification of sensitive ingredients, critical process points, and relevant human factors, as they affect product safety. Together these form the concept called hazard analysis critical control points (HACCP).

Foods have been placed into five categories in terms of health hazards, which are based on three hazard characteristics (NAS 1969). These are described in Chapter 6 in regard to controlling *Salmonella* in foods. These hazard characteristics are used as the basis for the HACCP inspections.

According to Kauffman (1974), the HACCP inspection approach is used to determine the points in the process that are critical to the safety of the product, and these can then be used by the processor to identify critical points. HACCP inspections, guided by the CGMPs, will tell the FDA and industry what needs to be done to assure safe, high-quality food production and distribution.

The critical control points are determined for each type of food process and for each food-processing plant. The critical control points have been discussed for frozen foods (Peterson and Gunnerson 1974), canned foods (Ito 1974; Warne, Capaul, and Moffitt 1985), and a foodservice system (Cichy, Nicholas, and Zabik 1982).

With this concept of inspection, not only is visible evidence of filth (insects, rodents) or unclean equipment and facilities important, but also the microbial analysis for potential hazard or spoilage types is used at critical control points to determine the problems involved in the process. This concept increases the importance of the microbiologist.

Although the CGMP and HACCP concepts provide a basis for satisfactory operation for food processors, they do not state how to meet the requirements in all cases, or why some of the CGMPs are needed. Therefore, it seems worthwhile to discuss some of the these factors.

PLANT AND GROUNDS. The plant should be designed so that materials can be received, stored, processed, warehoused, and shipped in an efficient and satisfactory manner.

Some operations, such as washing dirt off vegetables or slaughtering animals, are not considered to be clean activities and must be separated

from areas where processed food is exposed. The segregation of facilities should be such that no cross-contamination can occur between raw products and processed products.

AIR. Fresh air is needed in food-processing areas. Since air is a potential source of microorganisms, a system or systems are needed for microbial control. The sources of microorganisms and the means of dispersal throughout the plant must be considered in designing a method for control.

The ventilating system is the main place to control microorganisms in the air. Filters have been developed to remove microorganisms, as well as other materials from air. Viruses were effectively removed from air by filters with an efficiency rating of 93 percent or higher (Roelants, Boon, and Lhoest 1968). There are filters available with ratings of 99.9 percent and an ultra filter of 99.999 percent. The filter should be rated by the dioctyl-phthalate filter test using 0.3 μm particles to challenge the filter. However, in some cases, organisms such as *Bacillus* spores (Robertson and Frieben 1984) or *Pseudomonas* cells (Leahy and Gabler 1984) have been used to test filter systems.

By pumping in more filtered air than is removed by the outlets, a positive air pressure is established in the room. With positive air pressure, when a door is opened for personnel or product to move in or out of the room, unfiltered air will not come in, since air is moving only out of the room. The clean, fresh air should enter the cleanest areas of the processing plant, move to the less clean areas, and leave the plant from the areas that are "dirty." This procedure reduces the potential airborne contamination of processed product from these less clean areas.

To eliminate turbulence factors, laminar airflow (unidirectional) has been developed. Laminar airflow devices are effective in reducing and controlling airborne contamination. Due to the relatively higher cost, they have not been readily adopted for use in the food industry. They should find usage in cabinets or special rooms where contamination of products must be tightly controlled, such as in aseptic packaging.

Besides mechanical filters, the air may be treated by chemical germicides, UV radiation, electronic air cleaners, thermal sterilization, centrifuging, or "scrubbing." When meticulous air cleaning is essential, electrostatic precipitation might be employed. The larger the particles, the easier it is to remove them. Thus, molds and yeasts can be removed by electrostatic precipitation more easily than bacteria or viruses. Due to the high cost of installation, this system has been used seldom, if at all, in the food-processing industry. Electrostatic precipitation is used to remove dust and other pollutants from effluent air.

WATER. The water may be from a municipal supply, in which case it should be potable, that is, free from potential pathogens. However, municipal water is not tested for potential spoilage organisms, and reservoirs and pipelines at the plant can serve as a place for multiplication. Some pseudomonads grow in distilled water. Hence, it may be necessary to treat the water by chlorination at the processing plant.

The chemicals in the water influence the type of cleaning agents that are used in the processing plant. When water is an ingredient of the food, chemicals causing odors and flavors may affect the characteristics of the food product. In these cases, the water should be monitored and given the necessary treatments that are indicated by the analysis and by the food that is being processed. The treatment of water ranges from filtration, flocculation, softening, demineralization, reverse osmosis, distillation, to various combined processes to obtain pure water. A water softener is not a means of removing organisms and, in some cases, the resins in softeners may act as a source of inoculum of the soft water.

EQUIPMENT AND UTENSILS. The equipment should be designed so that food does not become lodged on or in it to form pockets of microbial growth for continual contamination of foods moving over or through it. The sanitary design of equipment results in easier cleaning and sanitizing, thereby saving labor and materials.

The equipment surfaces in contact with food must be nontoxic, nonabsorbent, nonporous, and noncorrosive. Although equipment is made of several types of material, stainless steel is preferred for surfaces that are in contact with food.

Stainless steel is not just one metal, but is a group of alloys of iron with chromium, nickel, carbon, molybdenum, titanium, silicon, phosphorus, manganese, and sulfur.

Although stainless steel is corrosion resistant, it is not corrosion proof. Stainless steel is protected from corrosion by its self-repairing surface film of chromium oxide. When this film is broken down by cleaning, it will reform when exposed to the air. If abrasive materials are used in cleaning, the metal surface will be scratched, which allows corrosion to occur. If harsh chemicals are used in cleaning, they can cause pitting of the metal. Scratches and pitting make it difficult to clean, sanitize, and maintain the equipment in a satisfactory manner. Scanning electron microscopy revealed that there were flaws such as scratches, cracks, and pits in samples of stainless steel prior to any stress or wear (Stone and Zottola 1985a).

Certain materials should not be used in equipment that contacts food. Due to its porous nature and difficulty in maintenance, wood should not be used. Copper and its alloys accelerate rancidity of fats, discolor food

products, and contaminate foods with copper salts. Cadmium is toxic. Lead should not be used, except in small amounts when soldering is needed. Aluminum is attacked by acids and alkalis and it has a tendency to rub off and cause black marks. Iron can cause off-flavor in products, such as milk, and it is impossible to clean because of rusting and pitting. Zinc and antimony are not acceptable metals for surfaces in contact with food.

As an alternative to stainless steel, glass (usually borosilicate glass pipelines) or glass fiber–reinforced polyester resin are used in some food-processing plants. A unique substitute for metal is the use of liquid jets for cutting. This system is based on the principle of forcing a liquid (water, glycerine, vegetable oil, alcohol) through a tiny nozzle at extremely high pressure. These conditions accelerate the liquid to about 1,000 m/sec. This results in a more sanitary cutting method than using knives, and it is easier to maintain in a sanitary manner.

There are some areas or conditions in which flexibility is needed, and plastics or rubber products are used. Plastics may not be acceptable if they contain free phenol, formaldehyde, or any other substance that can transfer to or alter the food. Since additives used in the manufacture of plastics can be transferred to foods, the suitability of each type of plastic must be determined for each particular food.

Ease of assembly and disassembly is important if the equipment is to be cleaned by hand methods. When possible, equipment should be designed for clean-in-place (CIP) or automatic methods. These systems may cost more at the beginning but usually pay for themselves by requiring less labor and more effective cleaning and sanitizing.

The design and installation of equipment for proper drainage are important. This is needed in CIP systems, so that soil removed by cleaning solutions, as well as the cleaning solutions, sanitizing solutions, and rinse waters can be removed from the equipment.

Many years ago, 3-A Sanitary Standards for dairy equipment were developed (Atherton 1986). Since then, equipment standards have been written for other food products. These standards can be found in publications such as the *Journal of Food Protection* and *Dairy and Food Sanitation.*

Processing equipment does not have a normal microbial flora. It becomes contaminated with food residues that can support high levels of microorganisms. As the buildup of the microbial load progresses, it becomes a source for contaminating food that contacts it. Hence, it is necessary that the equipment be cleaned and sanitized at appropriate intervals to keep the microbial load at a reasonable level.

Cleaning. Equipment that is less than 100 percent clean is still dirty. It is difficult, if not impossible, to sanitize equipment that is not completely clean. Therefore, cleaning is a very important part of the operation.

Before cleaning, the food must be removed from the line and protected from the possible contamination by the cleaning and sanitizing agents, as well as the water and soil removed from the equipment.

In some processing plants, all of the equipment is cleaned and sanitized manually. In other plants, some or all of the cleaning and sanitizing may be semiautomated or fully automated. In any of these systems, the human factor is the most important ingredient, since even with a fully mechanized system, it is necessary to set up the operation and supply the system with the cleaning and sanitizing agents.

The first part of any cleaning operation is to remove waste material and to rinse loose soil from the surfaces with clean water. For fatty materials, hot water may aid in this initial rinse, but cool water is desirable for protein soils (Anderson et al. 1985; Dunsmore et al. 1981; Koopal 1985; Middlemiss et al. 1985).

Even if most of the soil can be removed by hand and by water rinsing, the equipment is still not clean, since a residue remains that is difficult to remove. An input of energy is needed to remove this residual soil. The energy forms are thermal (hot water, steam), chemical (cleaning agents), and mechanical (high-pressure spray, manual scrubbing). Not only is soil attached to the equipment surfaces by physical forces, but microorganisms are also attached and are difficult to remove (Powell and Slater 1982; Stanley 1983; Stone and Zottola 1985b).

The cleaning process consists of bringing the cleaning solution into intimate contact with the soil on the equipment surface, removing the soil from the equipment, dispersing the soil into the cleaning solution, and preventing redisposition of the soil, as well as scale due to hard water, onto the cleaned surface. Various cleaning agents are available that perform these activities. Cleaning agents are designed to reduce the amount of work needed to remove the adhering soil.

Cleaning Agents. A cleaner may consist of a single compound or be a complex mixture of various chemicals.

Only approved cleaners and sanitizers should be used in food-processing plants. Various government agencies compile lists of compounds considered to be acceptable and safe for washing food-processing equipment, as well as the floors and walls of plants.

A cleaning agent should have the following characteristics:

1. Quickly and completely soluble
2. Noncorrosive to metal surfaces
3. Nontoxic
4. Easily rinsed
5. Economical

6. Stable during storage
7. Not harsh on the hands if used for manual cleaning
8. Possess germidical action, if required.

The types of cleaners and their properties are listed in Table 10.1. Substances that lower the surface tension of a liquid, or the interfacial tension between two liquids, are known as surface-active agents, or surfactants. Chelating agents are chemicals that can form a soluble ringlike complex with a metallic cation. In this form, the metallic ions do not form insoluble films on the equipment, but are removed from the equipment by rinsing. The chief organic chelating agents, or sequestrants, are sodium salts of EDTA and related compounds, the hydroxy carboxylic compounds (citric, gluconic, tartaric, and hydroxyacetic acids) and the amino-hydroxy compounds (such as tetraethanolamine). The important inorganic sequestrants are the sodium polyphosphates (sodium pyrophosphate, sodium tripolyphosphate, sodium hexametaphosphate). They have a softening effect on hard water. The chelating agents can increase the effectiveness of sanitizing agents.

Conventional cleaners are used and rinsed with hot water. For low-temperature cleaning of poultry and meat-processing areas, proteolytic enzymes are added to surfactants. The surfactants possess a penetrating and emulsifying action on fats, while the enzymes soften and break up protein soils.

SANITIZING. If the cleaning procedure removed all of the organic material, the equipment would not need to be sanitized. The sterilization of all equipment would be ideal, but it would be neither practical nor economical. Thus, we are satisified if the equipment is sanitized.

In some cases steam, hot water, or hot air is used for sanitizing. In most processing plants, these agents are not practical, so chemical sanitizers are used. Sanitizing does not mean that all living bacterial cells are destroyed or removed. A sanitized surface does mean, however, that all pathogenic or disease-producing bacteria, as well as a large percentage of nonpathogenic ones, are destroyed.

Some sanitizing agents are added to detergents (detergent-sanitizers) so that the bacteria are killed during cleaning. This single operation of cleaning and sanitizing saves time and labor. When chlorinated cleaners are used, the chlorine not only kills bacteria, but also aids in the removal of soil and reduces redeposition.

The effectiveness of a sanitizer depends upon the concentration and the time of exposure. The pH modifies the sanitizing action by affecting both the bacteria and the chemical. An increase in temperature usually increases the rate of sanitizing. The effectiveness of the sanitizer depends

TABLE 10.1. RELATIVE CLEANING VALUES OF VARIOUS DETERGENT INGREDIENTS

Ingredients	Comparative Ability											
	Emulsification	Saponification	Wetting	Dispersion	Suspension	Peptizing	Water Softening	Mineral Deposit Control	Rinsability	Suds Formation	Noncorrosive	Nonirritating
Basic alkalies												
Caustic soda	C	A	C	C	C	C	C	D	D	C	D	D
Sodium metasilicate	B	B	C	B	C	C	C	C	B	C	B	D
Soda ash	C	B	C	C	C	C	C	D	C	C	C	D
Trisodium phosphate	B	B	C	B	B	B	A[1]	D	B	C	C+	C−
Complex phosphate												
Sodium tetraphosphate	A	C	C	A	A	A	B[2]	B	A	C	AA	A
Sodium tripolyphosphate	A	C	C	A	A	A	A[2]	B	A	C	AA	B
Sodium hexametaphosphate	A	C	C	A	A	A	B[2]	B	A	C	AA	A
Tetrasodium pyrophosphate	B	B	C	B	B	B	A[2]	B	A	C	AA	B
Organic compounds												
Chelating agents	C	C	C	C	C	A	AA[3]	A	A	C	AA	A
Wetting agents	AA	C	AA	A	B	B	C	C	AA	AAA	A	A
Organic acids	C	C	C	C	C	B	A[3]	AA	B	C	A	A
Mineral acids	C	C	C	C	C	C	A[3]	AA	C	C	D	D

SOURCE: Klenzade Dairy Sanitation Handbook.
Key: A = high value; B = medium value; C = low value; D = negative value; [1] = via precipitation; [2] = via sequestration; [3] = also stable to heat.

upon the types and numbers of microorganisms, as well as on their history. The use of a high concentration of a sanitizing agent as a substitute for thorough cleaning is inefficient and expensive, and, in some cases, may cause corrosion of the metal. Chemicals that leave bacteriostatic or bactericidal residues on surfaces may help with sanitation programs in food-processing plants.

Thousands of chemicals have germicidal powers, but only those compounds that are approved by a regulatory agent can be used as sanitizers. The four most popular types of sanitizers are the chlorine compounds, iodine or iodophors, quaternary ammonium compounds, and acid-anionic surfactants.

Chlorine. The most widely used sanitizers are the chlorine-containing compounds. Six factors determine the effectiveness of chlorine as a sanitizing agent. They are (1) type and concentration of chlorine compound used; (2) pH; (3) temperature; (4) contact period; (5) types of organisms; and (6) presence of organic material.

The types of chlorine compounds include the sodium and calcium hypochlorites, the salts of isocyanuric acid, the hydantoin derivatives and chlorinated trisodium phosphates, as well as gaseous chlorine. In powered chlorinated cleaners the source of chlorine is usually chlorinated trisodium phosphate, potassium dichlorocyanurate, dichlorodimethylhydantoin, or chloramine T. The first two give the most active chlorine in the cleaning solution. Besides these compounds, mixtures of chlorine and bromine, as well as chlorine dioxide, have been used as sanitizing agents. The most widely used agents in the food-processing area are the hypochlorites.

The advantages and disadvantages of the use of hypochlorites as sanitizing agents are listed in Table 10.2. The hypochlorites are effective at relatively high dilutions and are active against a wide range of bacteria and bacterial spores, as well as molds, yeasts, bacteriophages, and some viruses. The hypochlorites are considered to be more effective against Gram-negative than against Gram-positive bacteria. Viruses are more resistant than bacteria to the action of chlorine.

In moderately strong concentrations, the hypochlorites can cause irritation of the skin and may cause an objectionable odor. In some cases, unacceptable flavors may develop in foods exposed to hypochlorites. There is a reduction of efficiency of hypochlorites by organic compounds and oxidants. Due to their high corrosion potential, they should not be used at high temperatures or be allowed to contact the surfaces of equipment for excessively long periods.

Chlorine compounds are used in various aspects of food processing. They are added to water used for washing and conveying raw food products, for cleaning and sanitizing food-handling equipment, and for cool-

TABLE 10.2. ADVANTAGES AND DISADVANTAGES OF HYPOCHLORITES AS SANITIZERS

Advantages	Disadvantages
Relatively inexpensive	Unstable during storage
Quick acting	Inactivated by organic compounds
Not affected by hard water salts	Corrosive if misused
Harmless residue, does not form a film	Irritating to skin
Effective at high dilution	Odor may be undesirable
Active against a wide variety of microorganisms, including spores and phage	Precipitate in iron waters
Relatively nontoxic at use dilutions	Effectiveness decreases with increasing pH of solution
Nonstaining	May remove carbon from rubber parts of equipment
Colorless	
Easy to prepare and apply	
Concentration easily determined	
Can be used for water treatment	

ing heat-sterilized cans of food. Citrus packers in Florida dip or rinse fresh fruit in 200 ppm for 2 min prior to shipment to prevent the spread of citrus canker, which includes the presence of dark-brown and yellow lesions on the fruit.

In-plant chlorination is accomplished by injecting gaseous chlorine or liquid sodium hypochlorite into the water supply of a processing plant by an automatic proportioner. Sufficient chlorine is added to satisfy the demand of substances present in the water (the breakpoint), as shown in Figure 10.1, and to give a free available residual chlorine of from 2 to 20 mg per liter. The free available chlorine is in the form of hypochlorous acid (HOCl) and hypochlorous ions (OCl^-). The combined chlorine residual is that chlorine reacted with amino nitrogen (chloramines). The total available chlorine includes the combined and free chlorine.

In-plant chlorination provides a continuous bactericidal action of chlorine on food-preparation equipment during its operation. The use of chlorinated water sprays at selected points reduces or prevents accumulation of microbial slimes and off-odors in food-processing plants. The total microbial load of the finished product tends to be lower in food plants with in-plant chlorination. Chlorination cannot substitute for good sanitary practices and good plant operations.

Sodium hypochlorite reacts with water to form hypochlorous acid and sodium hydroxide.

$$NaOCl + H_2O \rightleftarrows HOCl + NaOH$$

Thus, the solution is alkaline, and the hypochlorous acid ionizes to form hydrogen ions and hypochlorous ions:

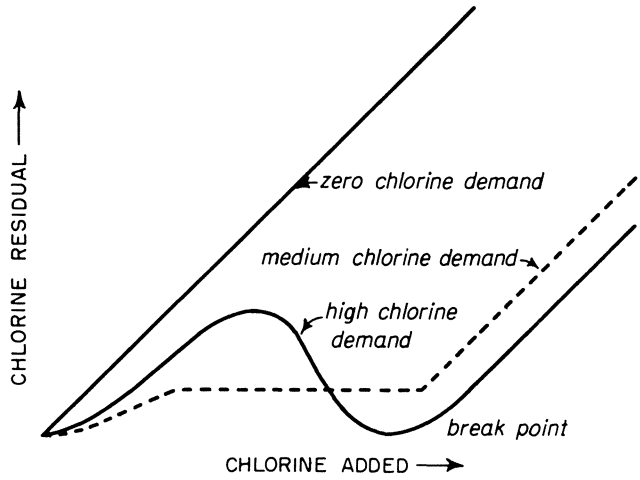


Figure 10.1. Chlorine demand characteristics of water.
Courtesy of Heid and Joslyn (1967).

$$HOCl \rightleftarrows H^+ + OCl^-$$

The hypochlorous acid can dissociate to form hydrochloric acid and nascent oxygen:

$$HOCl \rightarrow H^+ + Cl^- + O$$

The HOCl, OCl^- and O have germicidal properties. In an acid solution, the hypochlorous acid tends to remain as HOCl, which is more bactericidal than the hypochlorous ion. Hence, the bactericidal efficiency of chlorine solutions is higher in acid than in alkaline solutions (Fig. 10.2) and is increased with temperature increases.

Apparently there is more than one mechanism by which chlorine compounds can affect living cells. Chlorine reacts with cell membrane proteins, forming *N*-chloro compounds. This impairs the transportation of nutrients (carbohydrates, amino acids) into the cells. Also, damage to the cell membrane can result in the leakage of cellular components from the cell. The HOCl can penetrate the cell and oxidize sulfhydryl groups on key enzymes, interfering with cellular metabolism and resulting in cellular death. It may react with DNA of living cells, and oxidize purine and pyrimidine moieties (Wlodkowski and Rosenkrantz 1975).

The effect of chlorine on spores of *Bacillus* and *Clostridium* was investigated by Wyatt and Waites (1975). They found that chlorine disrupts the spore coat by combining with and removing protein. This inactivated the

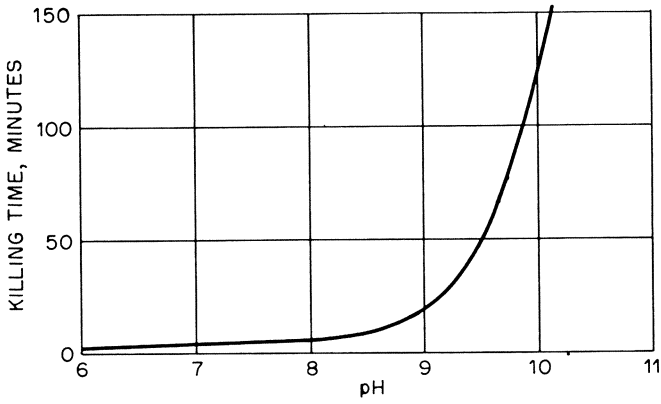


Figure 10.2. Effect of pH on germicidal efficiency of hypochlorite solution.
Courtesy of Heid and Joslyn (1967).

germination mechanism, but if spores were able to germinate, most were damaged to the extent that no outgrowth occurred. Degradation of *B. subtilis* spore cortex by chlorine appeared to correlate with loss of colony formation and may be the primary lethal effect (Gorman et al. 1984). The heat resistance of the spores was reduced by a chlorine treatment. Treatment of *B. cereus* spores with a solution of 0.25 percent NaOCl at 20°C for 19.5 min resulted in a 99 percent loss in viability (Kulikovsky, Pankratz, and Sadoff 1975). They found that the spores lost calcium ions, dipicolinic acid, ribonucleic acid, and deoxyribonucleic acid. The primary effect was the degradation of the outer spore coats, which led to a disruption of the normal permeability barriers. In general, *Bacillus* spores are more resistant than *Clostridium* spores to the action of chlorine (Foegeding 1983). However, *Bacillus* spores were more sensitive to chlorine dioxide than were *Clostridium* spores (Foegeding, Hemstapat, and Giesbrecht 1986). Spores are about ten times as resistant to chlorine as are vegetative cells.

Chlorine inactivation was due to degradation of RNA of polioviruses (O'Brien and Newman 1979). The rate of inactivation of viruses is affected by pH, temperature, chlorine concentration, viral aggregation, and the type of virus (Engelbrecht et al. 1980; Harakeh and Butler 1984; Keswick et al. 1985; Vaughn, Chen, and Thomas 1986).

For sanitizing equipment, regulatory agencies recommend a chlorine concentration of 100 to 200 mg per liter, with a holding time of 2 min. A 200-mg-per-liter hypochlorite solution circulated for 2 to 5 min is usually satisfactory for sanitizing pipes cleaned in place. Enclosed vats or tanks can be sanitized by fogging with a chlorinated (200 mg per liter)

solution. With open vessels or vats, the sanitizer can be pressure-sprayed onto the surfaces.

The use of chlorine has received a great deal of bad publicity because of its reactions with organic compounds and the resultant mutagenic and carcinogenic potentials. Due to these aspects, it has been suggested that, in some cases, the use of chlorine may create a health hazard. The use of chlorine dioxide, however, does not seem to produce these toxic compounds (Wei, Cook, and Kirk 1985).

Iodine. Iodine is well recognized for its germicidal properties against a wide variety of microorganisms. However, its corrosiveness, toxicity, instability, low solubility in water, and other unacceptable characteristics have limited the usefulness of iodine in food plants.

An iodophor is a combination of iodine and a solubilizing agent, usually a surface-active agent (surfactant), which, when added to water, releases free iodine at a low rate. The surfactant can be nonionic, cationic, or anionic, but certain nonionic types are preferred. The advantages and disadvantages of using iodophors as sanitizers are listed in Table 10.3.

As with hypochlorites, the iodophors show greater bactericidal activity at an acidic pH. Hence, phosphoric acid buffers are added to the sanitizer to maintain the pH at 4.0 to 5.0.

In the iodophor combination, the germicidal properties of iodine are maintained or enhanced, but the characteristic odor of iodine is elimi-

TABLE 10.3. ADVANTAGES AND DISADVANTAGES OF IODOPHORS AS SANITIZERS

Advantages	Disadvantages
Good stability	Effectiveness decreases with increase in pH, such as carryover of alkaline detergent solutions
Long shelf life	Less effective than chlorine against bacterial spores and bacteriophages
Surface activity	May cause off-flavors in dairy products
Generally active against all microoorganisms except bacterial spores and bacteriophages	Should not be used at temperatures above 49°C
Destroy yeast cells at a greater rate than do hypochlorites	May cause discoloration
Not affected by hard water salts	More expensive than chlorine
Relatively nontoxic	Causes staining of some material such as plastic hoses
Not corrosive	
Do not penetrate the skin	
Good penetrating and spreading properties	
Acid nature prevents mineral film formation	
Built-in color indicator	
Concentration easily determined	
Less sensitive than hypochlorites to organic matter	
Easily dispensed and controlled	

nated. At the recommended dilution, the iodophor solution has a rich amber color and, as it is used, the color fades. Hence, the iodophors have a built-in indicator of effectiveness.

The iodophors have good stability with a long shelf life. A high degree of surface activity gives them good penetrating and spreading power and aids in quick draining. Residues or films of minerals are prevented, since they are solubilized by the acid nature of the iodophors.

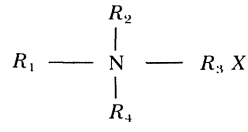
Iodophors cannot be used at temperatures above 49°C due to sublimation of the iodine, which results in loss of effectiveness.

Although the iodophors have good germicidal properties, they are less effective than hypochlorites against bacterial spores and bacteriophages.

There is less reaction of iodophors than hypochlorites with organic matter. However, the germicidal power is reduced by starch.

The widespread use of iodophors to sanitize equipment on dairy farms in Australia resulted in higher-than-normal levels of iodine in milk (Wheeler, Fleet, and Ashley 1982). This, in turn, caused thyroid problems in some consumers of dairy products.

Quaternary Ammonium Compounds. The quaternary ammonium salts (quats) have the general formula



The R stands for organic groups ranging from methyl (CH_3) to long-chain aliphatic groups (C_8H_{17} to $C_{18}H_{37}$) and phenyl groups. For bactericidal properties, at least one R is a plain or substituted long-chain aliphatic group. The X is an anion, such as a halide (Cl, Br), sulfate, or acetate.

The quats may be bacteriostatic in low concentration and bactericidal in high concentration. Their lethal effect has been attributed to various activities, including reactions with cell membranes, denaturation of essential cell proteins, or enzyme inactivation. The effects on the cell membrane may be due to membrane lysis, membrane enzyme inactivation, or interference with the transport system. Quats may act on the cells by increasing the permeability of the cell membrane, allowing water to enter until the cell bursts.

The quaternary ammonium compounds have certain advantages and disadvantages when used as sanitizing agents. These are listed in Table 10.4.

The quats are not as effective as the hypochlorites in destroying bacte-

TABLE 10.4. ADVANTAGES AND DISADVANTAGES OF QUATERNARY AMMONIUM COMPOUNDS AS SANITIZERS

Advantages	Disadvantages
Stable	Incompatible with anionic agents in detergents
Long shelf life	Expensive
Stable to temperature changes	Low activity in hard water
Effective in alkaline conditions	Less effective in activity against spores and bacteriophage, as well as coliforms and psychrotrophs
Noncorrosive	Need to rinse residual film from equipment
Odorless	Problem with foam during mechanical application
Less affected by organic matter	
Residual bacteriostatic effect	
Nonirritating to the skin	
Easily dispensed and controlled	
Control off-odors	
Nontoxic	
Active against many microorganisms, especially thermoduric types	
Good penetration qualities	
Combined with nonionics for detergent sanitizing agents	

rial spores of Gram-negative bacteria, including coliforms and psychrotrophs. They are definitely inferior in acting against bacteriophages. However, the addition of certain sequestering agents enhances the virucidal activity. In general, the quats are very effective against Gram-positive bacteria.

Generally, residues of quats are not allowed on equipment when food is being processed. Therefore, it is necessary to rinse the equipment, preferably with chlorinated water, prior to processing food.

Acid-Anionic Surfactants. These are combinations of an organic or inorganic acid (usually phosphoric) with a surfactant (usually an alkylaryl sulfonate). The low pH as well as the activity of the surfactant provide a bacteriostatic or bactericidal effect. The advantages and disadvantages of this class of sanitizers are listed in Table 10.5.

RAW MATERIALS. To control the raw material, it may be necessary for the processor to become the producer. For ingredients not directly controlled, specifications are needed, if for no other reason than to prevent the acceptance of rejects from another company. The specifications must be written so that they are flexible, but rigid enough so that they are useful. Some conditions for microbiological specifications of raw materials were described by Davies (1968) and Mossel (1969). The allowable number of the significant types of microorganisms, as well as the processing treatment that these ingredients are given, should be considered. The presence of microorganisms that may be a public health hazard should

TABLE 10.5. ADVANTAGES AND DISADVANTAGES OF ACID-ANIONIC SURFACTANTS AS SANITIZERS

Advantages	Disadvantages
Stable	Effective only at acid pH (optimum range 1.9–2.2; above 3.0, action decreases rapidly)
Long shelf life	Slow activity against sporeformers, not effective in destruction of most spores.
Active against a wide spectrum of microorganisms including thermodurics, controls phage and most yeast strains.	Foam problem in mechanical and clean-in-place applications
Residual antibacterial film	
Noncorrosive and nonstaining to stainless steel	
Relatively low toxicity	
Effective in hard water or with organics	
Action enhanced at high temperatures	

be of concern to the processor, especially if the processing treatments do not destroy these potential pathogens. Spoilage organisms cannot be overlooked. The presence of a significant number of resistant bacterial spores in ingredients such as sugar, flour, or spices may cause spoilage of canned foods.

Before worrying about microbial analysis, the raw materials should be given a preliminary examination with normal human senses for evidence of filth or decomposition. The presence of rodent pellets or hairs, insect infestation, field rot, or decay of fruit and vegetables, a diseased condition of animals prior to slaughter, parasitic infestation of fish, or decomposition of the material are causes for rejection. Since the quality generally does not improve with processing, only clean and sound materials should be brought into the processing plant.

After the raw material is inspected and passed, it should be processed as soon as possible. When delays are unavoidable, facilities for storage should be maintained to protect the raw material from contamination and infestation as well as from deterioration. Some fresh raw materials can be modified by storage to obtain a more uniform product. An example is the storage of potatoes for making potato chips. At low temperatures, reducing sugars will accumulate and result in a dark potato chip. By conditioning these potatoes for a short time at high temperatures, the reducing sugars are returned to a low level.

Many foods have an outside peel, skin, or shell to protect the interior portion of the food from microorganisms. However, the organisms on the outside surface serve as a source of contamination when the protective covering is removed during processing. Hence, the raw material should be washed, when needed, to remove field dirt, other material, and many of the microorganisms from the outside surface.

The washing of the raw food material should be done in an area separated from the rest of the processing facilities. The water should be free of microorganisms that may cause foodborne illness or spoilage.

OPERATIONS. It is important that the equipment is maintained, kept in good repair, and is properly cleaned and sanitized. When fruits and vegetables are cut or peeled, or an animal is killed and the hide removed, the natural defenses are broken down and the food is subject to contamination and the beginning of a chain of events that lead to quality loss and finally to destruction of the food.

Bacteria multiply rapidly. Therefore, the speed of handling becomes important. The operations should be timed so there is no backup of product, since long delays in processing can cause microbial problems. Bacteria develop more rapidly as the temperature rises; hence, the temperature should be kept as low as possible in keeping with processing requirements.

When organisms attain a high level in a food, even if they are inhibited or killed, their enzymes may remain active and cause changes in texture, flavor, and color.

Equipment such as trays or vats must not be used interchangeably for raw and cooked products unless this equipment is completely cleaned and sanitized before being used for cooked products.

Equipment is designed to reduce the handling of foods by workers. This equipment causes less contamination than the people handling it if it is properly cleaned and sanitized.

To reduce the temperature of poultry carcasses, continuous chillers are used. With proper sanitation during processing, washing of the carcaasses before chilling, maintaining low water temperatures, chlorination of the chill water with an acceptable input of fresh water and ice, with resultant overflow of excess water, microbial contamination of the carcasses is reduced.

PACKAGING. Food containers and packages are made from paper, plastic, glass, and metal, as well as combinations of these materials. The package should identify and advertise the contained product as well as present an attractive appearace to give it sales appeal. The package usually gives the product a standardized shape. The main function of packaging is to separate the food from the surrounding environment so that the food is protected from recontamination by microorganisms and pests and from physical and chemical damage from atmospheric oxygen, light, storage odors, and moisture.

Packaging materials must meet certain standards. The package should

be easily filled and emptied. It must be strong and durable, since damage to the package will result in contamination. The material used for packaging must not contaminate the food with foreign odors, flavors, or toxic substances. Further, the food must not corrode or weaken the package. The package must be able to withstand the processing treatment of the food (heating, freezing, cold storage), and the package must be obtainable at a reasonable cost.

If not properly handled and protected, the package may be a source of microorganisms. The food may acquire chemicals from the packaging materials, such as plastics or waxes, either alone or coated onto paper. Makinde, Gilbert, and Lachance (1976) listed chemicals such as monomers (ethylene, vinyl, propylene, styrene) and catalysts used in preparing plastics or various stabilizers, pigments, and antistatic, antifogging, bactericidal, and antifungal agents incorporated into plastics. Polycyclic aromatic hydrocarbons or carcinogens such as 3,4-benzopyrene may be associated with waxes. Polyvinyl chloride plastics are made from vinyl chloride, a human carcinogen.

With proper packaging, the greatest number of people can be supplied with a clean, wholesome supply of food. Extremes of food packaging are now available. The poisoning of packaged drug products with cyanide and the threatened contamination of foods have been responsible for an increase in tamper-evident packaging. These types of packages include film overwraps, blister-strip packs, bubble packs, shrink seals, sealed tubes, breakable caps, bottle seals, pouches, sealed cartons, aerosol sprays, and tape seals. This extra effort increases the cost of the product and in some cases makes it difficult for the consumer to open the package for use.

At the other extreme is no packaging. There is a growing trend for retail stores to sell bulk foods that are in a box, barrel, or case, or simply sitting on a shelf. The consumer selects the type and amount of foods and puts them into a container such as a plastic bag for purchase. It is questionable whether these foods are handled in a sanitary manner. Although these bulk foods are often presumed to be cheaper because of reduced labor and packaging costs, they may not actually be priced any lower per unit than packaged foods. At the same time, they offer a potential target for contamination by toxic substances. However, as long as bulk foods are purchased, the retailer will continue to allot more space for their sale, until a government agency decides that regulations are needed.

Different packaging materials vary in their physical and chemical properties. The permeability to moisture, oxygen, gases, light, and even microorganisms varies with different materials. With the various properties of packaging materials, the characteristics of a package can be engi-

neered to fit almost any food product or processing and storage requirement.

PERSONNEL. People are an important cause of microbial contamination of foods. The CGMPs stress (1) disease control; (2) cleanliness; (3) education and training; and (4) supervision. The concern of the plant personnel for sanitation and personal hygiene must begin with plant management. Management must set a good example, furnishing the proper environment and establishing within the personnel a desire to comply with the requirements necessary to maintain sanitation and personal hygiene.

All personnel should be involved in keeping the processing plant neat and orderly. However, of special concern are those people who handle ingredients, foods, or food-contact surfaces. In the handling of cream-filled pastries, cooked meat and poultry products, milk products, and salads, the hygiene of workers is a critical control point (Bryan 1974).

Disease Control. Even with the development of equipment for various purposes, there is still direct handling or close contact of food by humans. Although equipment is a major source of microorganisms, these are usually spoilage types normally associated with foods. Humans are a potential source of pathogens. In this respect, humans are more important than equipment as a source of food contaminants.

Anyone who is sick, has a communicable disease, is a carrier of such a disease, or has boils, sores, or infected wounds must not be allowed to work in any area of a food plant where there is a possibility of contaminating the food or a food-contact surface.

According to California health authorities, education in food handling and personal hygiene, along with inspection of food establishments for sanitation, will prevent more foodborne disease than any program involving the routine screening of food handlers for communicable diseases. In general, the value of routine medical examinations of food handlers is questioned for the following reasons: (1) It has not been established that most outbreaks of foodborne illness are caused by food handlers; (2) medical examinations and laboratory tests are very expensive; and (3) because of the rapid turnover of food handlers, it is difficult to assure that all employees are examined.

Cleanliness. Employees should maintain a clean appearance. The CGMPs suggest the wearing of clean outer garments. Even though the employee may dress in clean clothing at home, it is not sterile. Furthermore, by the time of arrival at the plant, the clothing has been subjected to various sources of contamination, which can include both food-spoilage and disease-producing microorganisms. Hence, in some cases, it is required

that employees wear special protective outer clothing that is not worn outside the processing plant. In the general guidelines for handling ready-to-eat products, employees who work on raw and ready-to-eat products must change any garment that has contacted a raw product. Due to human shedding of microorganisms, special clean-room clothing may be needed in selected areas. This clothing is essentially that worn in operating rooms, including coveralls or smocks with hand, foot, and head coverings. In many food plants, the employees wear company-furnished uniforms. Uniforms of different colors should be provided for different departments. In this way, there will be less chance for cross-contamination of food by personnel.

The hands are used for many things. Therefore, it is essential that before working with food, the hands are washed and, if necessary, sanitized. Simply rinsing the hands is not sufficient. Washing with soap and water does not remove all of the microorganisms; however, a thorough washing with soap and water will remove most of the transient organisms. Antimicrobial soaps will aid in removal of the resident flora. Sanitizing solutions for hand dips may contain chlorine, iodine, or any other approved bactericidal agent.

Not only are hand washing and sanitizing needed before beginning work, but also after each break, after leaving and returning to work, or when the hands may be contaminated from handling contaminated equipment or picking up debris from the floor.

The hand-washing facilities should be provided with foot-operated controls to obtain warm water. Tap handles have been found to be highly contaminated. It does little good to wash one's hands and then promptly contaminate them with enteric-type organisms from the tap handles.

Probably not surprising, McBride (1984) found that bar soap had significantly higher levels of organisms than did liquid soaps. Stiles and Sheena (1985) found that, of the products tested, only 4 percent chlorhexidine gluconate liquid detergent and iodophor containing 0.75 percent available iodine were significantly better than nongermicidal soap for reduction of transient bacteria. They also reported that a barrier cream on the hands reduced the shedding of organisms from the fingertips (Sheena and Stiles 1983).

In some cases, gloves are worn to control potential contamination. These should be disposable rubber gloves and should be maintained in a sound, clean, and sanitary condition.

Hair on the head, face, and body is a source of microorganisms, as well as loose hair that contaminates food. Hence, hair coverings, such as hair nets, beard nets, headbands, caps, or other effective hair restraints must be worn by employees.

Personnel should not have extraneous materials in the food-process-

ing area. Smoking, chewing of gum or tobacco, spitting, eating, or drinking cannot be tolerated in areas that will affect the food. Also, the food must be protected from contaminants such as perspiration, cosmetics, and medicants.

With the regulations of various government agencies relative to such things as health and safety, no one except authorized plant employees or government inspectors should be allowed in the food-handling or processing areas.

People tend to have bad habits, forget what they are supposed to be doing, and get careless. People forget to wash their hands, or merely rinse them. Food is left at room temperature rather than being put into the refrigerator or freezer. Some people believe that there is nothing wrong with salvaging unfit food. It is relatively easy to observe examples of poor sanitation as well as questionable practices when eating in a restaurant or other foodservice unit.

In a processing plant, the cleanup crew must do a complete and thorough job of cleaning and sanitizing the plant and equipment. If they are not careful, they can contaminate the equipment more than they clean it. The food-processing crew must follow rules for acceptable operations.

Government inspectors should be clean and set an example for the plant personnel. It is illogical for an unkempt person to try to explain sanitation and personal hygiene to other people.

Training. Many food-processing jobs are seasonal and often offer relatively low salaries. Hence, the food industry tends to obtain young people, part-time workers, and various others who cannot find work elsewhere. Some of them are poorly trained people in the lower socioeconomic group. With this combination of jobs and types of people, the turnover rate is rather high and sanitary training is difficult.

The employees must be properly trained and retrained in the correct use of equipment, sanitation, and personal hygiene. They must be informed of the perishable character of the food product, sources of contamination, and their role in the production of a top-quality product.

Since a majority of outbreaks of foodborne illness are due to contamination or mishandling of food at foodservice establishments and in the home, it seems logical that the general population needs training in the care and handling of food products and in personal hygiene. A course concerning these aspects given in high school might be beneficial.

Supervision. The supervision of the plant sanitation and the sanitary training of personnel should be under the direction of a well-trained sanitarian.

The sanitarian and other plant supervisors must be able to recognize unsatisfactory employee habits or health problems. If employees cannot

or will not follow the necessary rules, the supervisor must remove them from critical areas, since the supervisor is ultimately responsible for compliance with the regulations that are applicable to the product being processed.

Microbial Evaluation

A visual inspection of raw material, equipment, packages, and finished product can detect gross filth, but microbial analyses are needed to determine microbial contamination.

Samples for analysis should be obtained from the equipment surfaces, hands, raw material, the material during processing (on-line samples), finished processed product, and other things such as air, water, or packaging materials that might be involved in product contamination. Routine tests will show day-to-day variation and help reveal weaknesses in raw materials, ingredients, equipment sanitation, or personal hygiene.

Not only the total number of organisms, but also pathogenic types, such as *Salmonella* or *Staphylococcus* or indicators such as coliforms, *Escherichia coli* or enterococci, should be determined. In some processing plants the sporeforming bacteria (*Bacillus, Clostridium,* and *Desulfotomaculum*), yeasts, molds, psychrophiles (*Pseudomonas*), or other spoilage types might be evaluated. In the citrus industry, a colorimetric test for the presence of diacetyl and acetylmethylcarbinol is effective in quality control. These are metabolic products of the degradation of citric acid by species of *Lactobacillus* and *Leuconostoc.*

The total aerobic plate count of samples obtained from various stages of production may be used as a general index of plant sanitation. However, processing treatments such as heating will decrease the count, and the addition of certain ingredients may increase the number of microorganisms. If sharp increases in numbers occur in a particular area, this is evidence that something may be amiss and needs investigation and corrective action.

A significant increase in the day-to-day microbial level of equipment surfaces indicates faulty cleaning or sanitation. Close observation of the cleaning and sanitizing may reveal the problem so it can be resolved.

Microbial surveys of various foods reveal that products handled with good sanitary conditions usually have significantly lower counts than products produced by plants operating with poor sanitary conditions. There are times when poor sanitary practices are not revealed by microbial numbers of the final product, but with sufficient sampling over several sampling periods and analysis of on-line samples, a correlation can be obtained between plant sanitation and bacterial loads. When analyzed, the coliforms, *E. coli,* and coagulase-positive staphylococci counts

generally reveal the sanitation of the plant and personal hygiene of the workers.

Foodservice Establishments

Of the reported outbreaks of foodborne disease in which the place of mishandling is known, most are due to foodservice establishments. If health authorities are truly interested in reducing the number of foodborne outbreaks, then foodservice establishments need more regulation and observation than has been evident. It is unfortunate that people must be forced to do an acceptable job of producing food. Too many owners of foodservice businesses have no background in sanitation and the proper handling of foods.

The same general requirement for housekeeping, equipment, cleaning, sanitizing, and personnel problems and regulations that apply to food-processing plants also are required for foodservice establishments.

In many food-processing plants, one or only a few foods are handled and processed, whereas in a restaurant all types of foods in various stages of processing are handled and utilized. Some raw foods are served with only a little preparation, such as in salads. Other foods, such as meats, are cooked. The common organisms (salmonellae and staphylococci) that cause foodborne illness are destroyed by cooking meat to an internal temperature of 74°C (Bryan and McKinley 1974). Hamburgers or other meats may not receive sufficient heat treatment during the cooking process to destroy pathogens. Cooked meats for sandwich use may be left at room temperature rather than being refrigerated. The same knives or other equipment used to handle raw meat may be used for cooked meat. Hence, there may be cross-contamination between raw and cooked products.

Foods that are prepared just prior to serving normally do not cause health problems. It is food that is prepared a day or two before serving and then not properly refrigerated, or food that, after preparation, is held for serving during a two- or three-hour lunch or dinner period that can cause problems of foodborne illness.

To prevent growth of microorganisms, warming tables should keep hot foods at or above 60°C, and cold tables should keep cold foods at or below 7°C. If foods are to be held for any period over two or three hours, they should be refrigerated.

Various accessories (salt, pepper, catsup, sugar, steak sauce, syrup) may be found on the table or served to several customers at different tables. The customers can be a source of microbial contamination, including potential pathogens, of these supplies.

Certainly things have improved since the days when only a superficial

rinse was given to many utensils and glasses. However, further improvements are needed in the foodservice industry to eliminate contamination and foodborne illness.

Due to the rapid turnover of workers in the foodservice industry, for a national training program to work, it seemed advisable to train managers (Baker 1982), who could then train their personnel. There are fewer managers than workers, and managers tend to have less rapid turnover than the workers. A few health departments have made manager training mandatory (Wright and Feun 1986). However, the reports on voluntary manager training programs has indicated little, if any, benefit of manager training (Cook and Casey 1979; Wright and Feun 1986). Perhaps foodservice people are more interested in profit than in food service, or perhaps the training program needs to be revised so that it is more realistic.

The HACCP concept for food processing also can be used in foodservice operations (Bryan 1981). The examination of raw materials, processes, practices, personnel, products, equipment, and premises is used to determine things such as potentially hazardous foods, pathogenic foodborne organisms, and unacceptable employer practices, time-temperature combinations, procedures, and environmental conditions. Various foodservice operations have been studied with this concept in mind, including Mexican and Chinese foods, caterers and hospitals (Bryan and Bartleson 1985; Bryan and Lyon 1984; Bryan, Bartleson, and Christopher 1981; Bryan, Harvey, and Misup 1981; Bryan et al. 1982a, 1982b, 1982c; Snyder 1986).

Salad bars, smorgasbords, buffets, and Sunday brunches are the bulk food sections of the foodservice industry. Problems of salad bars, such as over-reuse of food, people's coughing and sneezing on the food, or sticking their fingers into the food for sampling were described by Dahl (1986). Fortunately, most vegetables and fruits found in salad bars are not involved in foodborne illnesses (Chapter 6). However, that is not true for many of the other self-serve foods. Surprisingly, the number of outbreaks of foodborne illness involving these foods found in self-serve systems has been relatively small (see Chapter 6). However, lesser known foodborne illnesses may be associated with consumption of these foods.

Retail Stores

The modern food supermarket resembles a group of food-processing operations as well as a foodservice unit mixed in with normal retail functions. Some stores have a meatcutting and -packaging area, a produce-packing area, a candy-making area, a bakery, a delicatessen, an orange juice squeezer, and a lunchroom with a salad bar.

With all of these activities going on, there are many chances for mishandling of food, with subsequent problems of foodborne illness. As discussed previously, the bulk food section is subject to contamination not only by store personnel, but also by the general public. Many other foods are not packaged in tamper-evident systems.

Wyatt (1979) pointed out the importance of sanitation in retail stores. However, the sanitary condition of the meatcutting area was not as important as the original bacterial load on the carcass and temperature of holding in affecting the microbial contamination and shelf life of steaks or ground meat (Greer and Jeremiah 1980; Greer, Jeremiah, and Weiss 1983; Wyatt and Guy 1980). Obviously, if the original count is 100,000 per gram, the addition of another 10,000 bacteria will not have as great an effect as the multiplication of the original bacteria. However, sanitation should be practiced to aid in the control of pathogenic organisms.

To provide industry and governments (state and local) with a uniform system for the operation and regulation of retail food stores, the Association of Food and Drug Officials (AFDO) initiated and jointly developed with the FDA a sanitation ordinance (FDA 1982). It includes such things as general provisions, food (supplies, storage, preparation, display), personnel (health, cleanliness, clothing), equipment and utensils, cleaning and sanitizing, sanitary facilities and controls (water, sewage, plumbing), physical facilities, and compliance procedures.

Sanitation at Home

The practice of sanitation and personal hygiene in the home will go far toward preventing problems in other food-handling establishments. Many people seem to be unaware of their responsibility for proper home storage, hygienic food preparation, and sanitary practices in the kitchen. Efforts to improve food protection in the home must overcome family customs and personal habits.

For a foodborne illness to occur, a potentially pathogenic organism must be in an acceptable food environment for growth or production of toxin. A sufficient amount of this contaminated food must be eaten by a susceptible person for illness to occur. Thus, not all unsanitary acts will result in illness. The person who prepares the food becomes complacent and careless. Besides carelessness, other factors that may be involved in foodborne outbreaks are apathy, ignorance, and poor judgment.

Food protection in the home encompasses essentially the same systems as in the food-processing plant. It begins with the selection of food and continues with transportation, storage, preparation, serving, and handling of leftovers and wastes.

At the store, we can make a judgment about certain foods by their

appearance. Damaged or opened packages should not be selected. If produce has a shriveled appearance, discoloration, evidence of mold, or other microbial growth, it should be avoided. In the frozen food counter, do not select packages that are piled above the freezer. Canned foods that show bulging or the leaking of contents should be pointed out to the manager. Although, in general, the food in dented cans may be satisfactory, this type of can does show evidence of rough handling. This could affect the can seams and result in spoilage or a health hazard.

Perishable food should be protected while being transported home. Allowing food to remain in a hot trunk of a car parked in a sunny parking lot for several hours will cause problems. Many people own cooling chests that give some protection to perishable foods during the period of transportation.

The home should have adequate space for dry storage as well as refrigerated storage. These areas should be kept neat and clean. The foods that are brought home must be promptly stored in the proper area, especially those that require refrigeration or freezing.

The food preparer should practice good personal hygiene. Thoroughly washing the hands is essential, and rewashing is needed when food-preparation activities are interrupted for any reason. It is expected that animals should be kept out of processing plants, but some people allow their house pets to run around the kitchen and think little about this. Rodent and insect control are as important at home as in the processing plant.

The kitchen should be kept clean. This includes the floors, walls, cupboard, stove, refrigerator, and other accessories. Pots and pans should be cleaned thoroughly. Have you ever seen anyone sanitize the utensils at home? This failure may account for some of the problems that occur in the home. Although wiping cloths in processing plants and foodservice establishments is considered to be unsatisfactory, it is unusual to find a home in which dish towels are not used to wipe at least some of the utensils or dishes.

Fresh meats, poultry, and fish should be cooked sufficiently to destroy pathogenic types of microorganisms. Eating raw animal products is not recommended. Cross-contamination between raw meats and prepared foods should be avoided.

With most meals, more food is prepared than is eaten. Hence, there are leftovers. If these are to be used again, they should be placed promptly into the correct storage conditions, and perishable foods should be refrigerated. It is not considered acceptable to prepare food ahead of time and allow it to remain at room temperature before serving. Sooner or later, this practice will result in illness.

Various sites in several homes were surveyed for microorganisms

(Finch, Prince, and Hawksworth 1978; Scott, Bloomfield, and Barlow 1982). Surprisingly, except for washcloths, tea cloths, and sink and basin drains, most sites in the home had relatively low levels of contamination.

CONTROL OF MICROORGANISMS BY REMOVAL

One means of controlling microbial levels in food is the removal of microorganisms. The systems used commercially for this procedure include washing, centrifugation, and filtration.

Washing

Washing of fruits and many vegetables is used in the preparation of these foods for canning, freezing, or fresh consumption. Washing removes soil, microorganisms, and some residual pesticides (Munsch, Simard, and Girard 1982; Wright et al. 1979).

During processing, broiler carcasses go through a spray washer. This washing process removes many of the microorganisms from the carcass surface (Mulder and Bolder 1981; Notermans, Terbijhe, and Van Schothorst 1980).

Significant reductions in bacterial numbers on beef and lamb carcasses by using high-pressure water sprays have been reported. However, spray washing was considered to be impractical (Sheridan 1982), and there was no increase in storage life of the sprayed carcasses (Kelly, Lynch, and McLoughlin 1982). Kriaa, Arthaud, and Fournaud (1985) found that the microbial contamination of carcasses was not greatly affected by spray washing. One problem seems to be that bacteria become attached to meat surfaces and are then difficult to remove. The addition of acid (acetic or lactic) or sodium hypochlorite to the spray water yielded some increased benefits (Anderson et al. 1979; Skelley et al. 1985; Smulders and Woolthuis 1985; Woolthuis and Smulders 1985).

The addition of KCl (0.1 m) to a water rinse resulted in a greater removal of microorganisms from meat surfaces (Appl and Marshall 1984). Pedraja (1973) recommended that fish or seafoods be passed under a spray of chlorinated (10 mg/L) water before and after processing to control bacteria.

Washing may not be beneficial in all cases. Fellers and Pflug (1967) reported that washing of pickling cucumbers "reduced the storage life to half that of unwashed fruit." The polishing of fruit with saran brushes can cause invisible injuries to the peel. This can result in increased decay during storage.

If the washing of shell eggs is done improperly, there is increased spoilage during storage.

Centrifugation

This operation involves the separation of a solid from a liquid or a liquid from a liquid by means of centrifugal force. This process has been used in the preparation of sugar, clarification of fruit juice, and the separation of cream from milk.

By increasing the centrifugal force of the milk separator, it is possible to separate bacteria from the milk. In this centrifuge, the bacterial cells, as well as other sediment, are thrown to the vertical wall of the bowl. The outer edge of the bowl has two small holes (0.3 mm) through which about 1.5 percent of the milk containing from 90 to 97 percent of the bacteria is removed. By using two centrifuges in series, 99 percent or more of the bacteria in the original milk is removed.

Although originally described for milk, other applications have been developed, including the separation of microorganisms from substrate in the production of single-cell protein, the purification of molasses for use in the production of baker's yeast (*Saccharomyces cerevisiae*), and the separation of distiller's yeast from the substrate prior to distillation of the alcohol.

When bacteria are removed by centrifugation, there is no damage to the flavor or other quality factors such as occurs with heat. Also, the cells are removed, not just destroyed.

Spores are more dense than bacterial cells and are readily removed from liquid foods. The removal of over 99 percent of the heat-resistant spores and vegetative cells makes it easier to destroy the remaining spores by heat treatment.

There may be problems with some applications. Centrifugation also may remove some protein, calcium, and phosphorus from milk. To aid in separation, some foods may need to be heated. To centrifuge viscous molasses, it is diluted and heated.

The use of centrifugation is limited to liquid foods and then only when the density of the bacteria is greater than that of the food components. Since not all of the microorganisms are removed, these liquid foods cannot be considered sterile. However, by reducing the microbial load, it is easier to achieve sterilization by heat treating the food.

Filtration

Filtration is the separation of solids from a liquid or gas by subjecting the mixture to a porous material, the filter. The size of the filter pores determines the particles that pass through and those that are retained.

Bacterial membrane filters (MF) have pores of 0.45 μm. However, if all but the smallest viruses are to be retained, a pore size of 0.1 to 0.2 μm is needed. Although one manufacturer indicated that the pore size varied from 0.43 to 0.47 μm, according to Olson (1983) the range for pore diameter of this size MF is 0.20 to 1.0 μm. This variation is perhaps why a 0.20-μm size is recommended for sterilization.

There are some advantages in using membrane filters. The inert behavior, high initial flow rate, particle retention due to the "sieve" effect, no shedding of fibers, negligible loss of solution, and no effect on the concentration, pH, color, or flavor of the filtrate are good features of a membrane filter (Grubert 1974).

Before the membrane or other filter is used to remove bacteria from substances, it must be sterilized. This usually is accomplished by means of steam heat, ethylene oxide, or gamma radiation. After filtration, all pipes, filling apparatus, and packages must be sterile or the product will be recontaminated.

Filtration is used in the sterilization of media or media components that are adversely affected by heat treatment. It is used in the microbial analysis of liquids, such as potable water with low microbial counts (membrane filtration method).

Millipore (1970) listed several uses for membrane filters. These include (1) the filtration of beer or wine for cold pasteurization or for analysis of yeasts and bacteria; (2) filtering air to determine microbial contamination or to remove microorganisms and other particles from air used in contact with food products, in fermentation vats, and in packaging food products; (3) in the analysis of additives, pipelines, packaging equipment, and foods, such as soft drinks, syrups, and liquid sugar for microbial contamination; (4) remove yeasts and bacteria for the biological sterilization of champagne; and (5) remove microorganisms or particulate material from water, such as that used in blending of alcoholic beverages, to achieve clarity.

Cold pasteurization or sterilization by filtration can be used for clear fruit juices, vinegar, soft drinks, syrups, and vegetable oils. Since filtration to remove bacteria does not destroy or remove enzymes, if these cause undesirable reactions in foods, heating must be used to inactivate them. With ultrafiltration, enzymes can be removed. Systems such as reverse osmosis or ultrafiltration can be used to concentrate foods prior to drying.

Filtration has some advantages, as well as disadvantages, as a means to control microorganisms in foods. Among the advantages are that less energy is needed to filter than to heat liquid foods. Not only are the microorganisms removed, but particulate matter is also removed, and clarity is thus improved. Problems include the prefiltering or clogging of the filters, the residual enzymes, the necessity for sterilizing the filter system

prior to use, and the necessity for observing aseptic handling and packaging.

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11

Control of Microorganisms by Retarding Growth

By altering the factors (discussed in Chapter 4) to retard or inhibit the growth of microorganisms, the storage life of foods can be extended. These factors included temperature, water activity, acidity, oxidation-reduction potential, chemical inhibitors, and microbial interactions. Besides these factors, microorganisms can be affected by electrical current (Shimada and Shimahara 1981) as well as homologous phage (Greer 1986).

In some cases there is a rather fine line between inhibition and death. Some chemicals are bacteriostatic at low concentrations and bactericidal at higher levels. Low temperatures will reduce or inhibit microbial growth, while high temperatures will kill the cells.

The retardation of growth means that the food product is not sterile and, if mishandled or abused, the microorganisms may grow and cause spoilage or be a health hazard.

LOW-TEMPERATURE STORAGE

Temperature is one of the most important environmental factors influencing the growth and activity of microorganisms. Both microbial and biochemical activity can be reduced by lowering the temperature of a food and holding it in a refrigerator or freezer. The lower the temperature, the lower the rate of biochemical reactions or microbial activity. Therefore, it might be assumed that freezing and freezer storage are best for all foods. However, freezing damages food tissues, and, for some types of foods, this damage is unacceptable. For some foods, low-temperature refrigeration can cause injury. Thus, the temperature at which a food can be stored depends upon the type of food as well as economic factors. Removing heat and maintaining a controlled cold environment are expensive. Although many types of foods or ingredients might benefit from low-temperature storage, we are concerned mainly with perishable foods.

Refrigeration

The minimum growth temperature for most mesophilic organisms is about 10°C. Since refrigeration generally refers to temperatures below 10°C, these mesophiles do not grow and are not a problem with refriger-

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ated food. Some mesophilic microorganisms are psychrotrophic and can grow on refrigerated foods. However, the optimum growth temperature of these organisms is 25° to 30°C, and reduced growth occurs at temperatures of 10°C or less.

The main organisms of concern on refrigerated foods are the psychrophiles. These organisms can grow at temperatures below 0°C, and some reportedly have an optimum growth temperature as low as 10°C.

Although psychrotrophic and psychrophilic microorganisms can grow at low temperatures, they grow more slowly as the temperature is reduced. The lower growth rate is very evident as the minimum growth temperature of an organism is approached. Also, as the refrigerator temperature is reduced from 10°C, fewer strains can grow and cause spoilage. Hence, food will spoil about four times as fast at 10°C and twice as fast at 5°C as at 0°C.

Some microorganisms causing foodborne illness are psychrotrophic, but most will not grow or produce toxins below 4.4°C. None of these organisms grow below 1.7°C. For safety, perishable foods should be held below these temperatures. Under no circumstances should perishable foods be left at room temperature longer than necessary, as dictated by good handling procedures.

Quite often, other systems for preserving food are used in conjunction with refrigeration. Since psychrophiles are primarily aerobic, vacuum or carbon dioxide packaging with the exclusion of oxygen will delay microbial spoilage. Salting, curing, smoking, or other chemical systems, as well as mild heat treatments, may be used to inhibit or reduce the microorganisms on refrigerated food.

COLD SHOCK. This is a form of direct chilling injury to the cells, resulting from temperature reduction without freezing of the substrate. Damage is evident immediately upon reaching the low temperature. Cold shock is dependent upon the rate of cooling and the magnitude of temperature reduction. Cells in the early exponential growth phase are the most sensitive to cold shock.

Cold shock results in a loss of one or more membrane permeability barriers. Hence, there is a leakage of amino acids and nucleotides from the cells. According to Sato and Takahashi (1970), the DNA of cold-shocked cells contains more single-strand breaks or nicks than either unshocked or recovered cells.

Injury to certain cells can occur simply by holding them at temperatures approaching 0°C (Patterson and Jackson 1979a, 1979b).

RAPID CHILLING. It is necessary to remove the field heat from fruits and vegetables, and the animal heat from milk, eggs, or meat, to prevent

spoilage of these foods. There are many ways in which the temperature of a food can be lowered rapidly. Liquid foods can be put through a plate or other type of heat exchanger. Other foods are rapidly cooled by forced air cooling, spray chilling, hydrocooling, icing, or in continuous chillers in slush ice. Fluidized beds are used for cooling vegetables after blanching and prior to freezing.

Food should be refrigerated in small containers, so it can be cooled as thoroughly as possible within a short time. Bulk refrigeration can be a contributing factor in foodborne illness.

When food is placed in storerooms, it should not be stacked tightly, since allowance must be made for air circulation. This will aid in keeping a uniform temperature and humidity in the warehouse, storeroom, refrigerator, or freezer.

LATENT-ZONE CHILLING. Refrigeration usually is considered to be storage at temperatures from 0° to 5°C. The freezing point of most foods is below 0°C (Table 11.1). Hence, storage below 0°C but above the freezing point has several advantages (Smith 1976). At these colder temperatures there is a longer shelf life than obtained with normal refrigeration. The energy requirement is lower and there is less product alteration or damage than when the food is frozen.

TYPES OF FOOD. Due to the different characteristics of foods, the cooling and refrigeration requirements vary. The potential refrigerated shelf life is influenced by the chemicals in a food.

Some foods, such as bread, should be stored at room temperature or be frozen instead of refrigerated. Crackers and certain cookies tend to lose their crispness in the refrigerator, due to moisture. The crystallization of sugar in marshmallows or marshmallow cremes occurs in refrigerated storage. When foods are refrigerated, they should be packaged in a proper manner to prevent moisture or aroma exchanges between the various foods.

Meat. It is desirable to cool and hold meat below 3°C and as close to −2°C as possible to prevent growth of potential pathogens and to retard growth of spoilage organisms and maintain the meat in a fresh condition (Table 11.2). Below −2°C, meat will freeze.

From a microbiological viewpoint, it is desirable to remove the body heat from carcasses as soon as possible after slaughter. The body temperature varies from 38° to 40.5°C, with an average of 38.9°C, depending on the animal. The rate at which a carcass is cooled depends not only on the chilling room temperature, but also on the size, heat capacity, and amount of fat covering on the carcass, and the velocity at which the air is circulated in the room.

TABLE 11.1. APPROXIMATE FREEZING POINTS OF SELECTED FOODS

Food	Temperature (°C)	Food	Temperature (°C)
Beef	−1.6 to −2.2	Oranges	−2.2
Pork	−1.6 to −2.2	Peaches	−1.4
Poultry	−2.8	Pears	−1.9
Fish	−0.6 to −3.3	Pineapples	−1.2 to −1.6
Shellfish	−2.2 to −2.7	Asparagus	−1.2
Shell eggs	−2.2 to −2.8	Beans	−1.1 to −1.3
Egg yolk	−0.6 to −0.7	Beets	−2.8
Egg white	−0.4 to −0.5	Carrots	−1.3
Milk	−0.5 to −0.6	Celery	−1.1
Cheeses	−2.2	Corn, sweet	−1.7
Brick	−8.7	Cucumbers	−0.8
Cottage	−12.7	Lettuce	−0.4
Ice cream	−2.8	Melons	−1.5 to −1.7
Apples	−2.0	Peas, green	−1.1
Apricots	−2.2 to −2.3	Peppers, green	−1.1
Avocados	−2.7	Potatoes	−1.7
Bananas	−1.0 to −3.3	Pumpkins	−1.1
Berries	−0.9 to −1.7	Rhubarb	−2.0
Cherries	−2.2 to −4.2	Spinach	−0.9
Grapes	−2.5 to −3.8	Sweet potatoes	−1.9
Lemons	−2.2	Tomatoes	−0.9
Limes	−1.5	Turnips	−0.8
Nuts	−4.5 to −10.5	Fruit pies	−3.0 to −4.0
Olives	−1.9	Meat pies	−1.4 to −1.9

TABLE 11.2. STORAGE CONDITIONS OF SELECTED ANIMAL PRODUCTS

Product	Storage Temp (°C)	Conditions Humidity (%)	Approximate Storage Life
Meat			
Beef			
Fresh	−2.0 to 1.11	88–92	1–6 weeks
Pork			
Fresh	−2.0 to 1.11	85–90	3–12 days
Bacon	1.11 to 4.44	85	2–4 weeks
Sausage			
Smoked	4.44 to 7.22	85–90	
Lamb			
Fresh	−2.0 to 1.11	85–90	5–12 days
Poultry			
Fresh	−2.0 to 0		1 week
Eggs			
Shell	−1.7 to 0.6	85–90	8–9 months
Dried	1.67	Low	6–12 months
Cheese	1.67	65–70	
Brick	−1.11 to 1.11	50	
Cottage	0 to 1.11	45	
Fish			
Fresh	0.56 to 4.44	90–95	5–20 days
Smoked	4.44 to 10.0	50–60	6–8 months

SOURCE: Adapted from Farrall (1976).

Light carcasses (lamb, pork, veal, and small beef) will cool in one-third to one-half the time required for heavy beef carcasses. Increasing the air velocity will reduce the cooling time but may increase shrinkage from moisture loss. High humidity will help reduce shrinkage but may stimulate microbial growth on the carcass surface. Although for microbiological reasons the carcasses should be chilled as rapidly as possible, doing so may result in less tender meat (James, Gigiel, and Hudson 1983).

Beef carcasses are generally aged to increase the tenderness of the meat. Smith, Arango, and Carpenter (1971) suggested that chilling the carcasses for the first 16 to 20 hr at 16°C appears to be the most practical system to use to develop tenderness. The use of ultraviolet light has made possible higher room temperatures to obtain a more rapid tenderization of the meat. The UV light, as well as the ozone produced by these lamps, helps retard microbial growth. Processes to circumvent the aging room include electrical stimulation of beef muscles and the hot boning of carcasses. Raccach and Henrickson (1978) reported that electrical stimulation of beef carcasses extended the shelf life of resultant ground beef by three days over meat from control carcasses. The disassembly or breaking of carcasses before chilling (hot boning) has economic advantages. Rapid chilling of this cut-up meat can be accomplished with solid carbon dioxide (Gigiel, Swain, and James 1985). Reportedly, hot-boned cuts of meat had higher microbial counts than did meat from chilled carcasses (Kennedy, Oblinger, and West 1982).

The vacuum packaging of beef and lamb tends to minimize shrinkage and discoloration as well as retard microbial growth. The low O_2 and relatively high CO_2 developed in the packages inhibit pseudomonads, the usual spoilage organisms of refrigerated meats. Low gas permeable bags containing lamb carcasses were flushed with N_2 or filled with CO_2 (Grau, Eustace, and Bill 1985). The addition of CO_2 significantly reduced the growth of Gram-negative oxidative (pseudomonads) and fermentative (enterobacteria), *Brochothrix thermosphacta,* and lactic acid bacteria.

Lower-grade carcasses are boned and the meat is used in sausage and other processed products. The USDA requires that all fabrication areas (beef breaking, pork cutting, or boning rooms) be maintained at 10°C or lower to retard bacterial growth. Smith (1985) stated that maintaining the boning rooms at 10°C or less does not appear to be necessary, providing the meat is processed with time limits which he determined for various temperatures, for example, 8.4 hr at 15°C.

Fresh meat is packaged for retail sale and put into a refrigerated display case. The salable life for this meat is about three days. In the home, refrigeration (2.2° to 4.4°C) of meat should be limited for maximum quality. Ground meats, liver, and poultry should not be stored more than

two days, roasts and steaks for five days, and ham or bacon for seven days (Table 11.3). For longer storage, the meat should be frozen.

Cured Meats. The psychrophilic flora is inhibited by the curing salts so these products are more stable than fresh meats during refrigerated storage. They often receive a pasteurization treatment that also increases the stability. Vacuum packaging aids in preventing discoloration, and the anaerobic environment inhibits aerobic bacteria and molds (see Tables 11.2 and 11.3).

Poultry. The storage life of poultry carcasses is limited by surface microbial activity that is influenced by the initial bacterial load and holding temperature.

After processing, it is important to remove body heat rapidly. This may be accomplished by continuous immersion chilling, spray chilling,

TABLE 11.3. HOME STORAGE TIMES FOR MEAT AND POULTRY (TO MAINTAIN QUALITY)

Product	Refrigerator at 1.7° to 4.4°C (days)	Freezer at −17.8°C (months)
Meat, fresh		
Roasts (beef and lamb)	3 to 5	8 to 12
Roasts (pork and veal)	3 to 5	4 to 8
Steaks (beef)	3 to 5	8 to 12
Chops (lamb and pork)	3 to 5	3 to 4
Ground and stew meats	1 to 2	2 to 3
Variety meats	1 to 2	3 to 4
Sausage (pork)	1 to 2	1 to 2
Meat, processed		
Bacon	7	1
Frankfurters	7	½
Ham (whole)	7	1 to 2
Ham (half)	3 to 5	1 to 2
Ham (slices)	3	1 to 2
Luncheon meats	3 to 5	Freezing not recommended
Sausage (smoked)	7	
Sausage (dry and semi-dry)	14 to 21	
Meat, cooked		
Cooked meats and meat dishes	1 to 2	2 to 3
Gravy and meat broth	1 to 2	2 to 3
Poultry, fresh		
Chicken and turkey	1 to 2	12
Duck and goose	1 to 2	6
Giblets	1 to 2	3
Poultry, cooked		
Pieces (covered with broth)	1 to 2	6
Pieces (not covered)	1 to 2	1
Cooked poultry dishes	1 to 2	6
Fried chicken	1 to 2	4

SOURCE: USDA (1970).

or air chilling. Immersion chilling is the usual practice in the United States. It is more efficient, effective, and economical than other systems.

For retail sale, the carcasses are packed on a fiberboard base and overwrapped or are packed in a plastic bag. They may be whole or cut up. At −2.0°C, the approximate storage life is one week (Table 11.2) and home storage at 1.7° to 4.4°C is only one to two days (Table 11.3). Poultry carcasses are more subject to spoilage than beef, pork, or lamb carcasses. The surface-to-volume ratio is greater for poultry carcasses, which may account for some of this difference. The method of slaughter and processing may result in a greater contamination by psychrophilic types of spoilage bacteria.

Shell Eggs. As soon as an egg is laid, its quality begins to decline. With our present system of producing, distributing, and marketing shell eggs, the main concern is with maintaining quality rather than with microbial spoilage. Low temperature, shell treatments, thermostabilization, carton overwraps, controlled humidity, and various other innovations have been suggested or used to help maintain egg quality.

When shell eggs are stored at −1.1°C, the rate of moisture loss is decreased, the amount of carbon dioxide lost is reduced, the rate of chemical change is slowed, and microbial growth is retarded. Soft-cooked eggs could be stored at 10°C for thirty days or at 25° to 35°C for two to three days (Imai 1981).

Egg Products. These are obtained by carefully breaking the egg shell and removing the liquid. Most of the egg products are obtained by using egg-breaking machines. The highly perishable nature of egg yolk and whole eggs makes it necessary that this liquid be cooled to 4.4°C or less within 1½ hr after mixing, and straining (to remove egg shell particles). Egg whites can be held at a higher temperature than yolk or whole egg due to the microbial inhibitors and the less available nutrients in egg white.

Liquid eggs can be held at 0° to 5°C with relatively little change in the microbial load up to 24 hr. However, at a temperature of 10°C or more, significant increases occur in the number of microorganisms.

Milk and Milk Products. When obtained, milk is at the body temperature of the cow. This raw milk should be cooled and held near 0°C. At 3.3°C or less, the coliform count tends to decrease, while in three days at 7.2°C, the psychrotrophs can show a tenfold increase. Gram-negative bacteria are the organisms in raw milk held in cold storage. These include *Pseudomonas, Acinetobacter, Aeromonas, Alcaligenes,* and members of the family Enterobacteriaceae. To obtain the maximum shelf life, pasteurized milk should be held at 5°C or less.

Buttermilk, yogurt, cottage cheese, and other soft cheeses are chilled

promptly after manufacture to near 0°C. When held at 0°C, they have a shelf life of a week or more. Unfortunately, due to the practice of stacking of cartons in the display case, those packages at the top may reach 5°C or even 10°C. This shortens the useful life of the product.

The refrigerator in the home may not be as cold as desired, and the food is removed from the refrigerator during preparation of the meal. Allowing these products to remain at room temperature for one hour each day can reduce the storage life.

Some dairy products such as hard cheeses can have a refrigerated storage life of several months or more. Mold growth can occur on the surface of hard cheese. This growth is not desirable due to the formation of off-flavors in the immediate area of mold growth. Also, the potential production of mycotoxins by the molds must be considered.

Fishery Products. These products are very perishable. Fishery products must be cooled promptly and held in this condition until either consumed or used in further processed items. The methods for cooling include ice, chilled seawater, refrigerated seawater, cold-air refrigeration, and cryogenic gas or liquids.

There is a marked extension in the storage life of fish by small reductions in the temperature in the range of 0°C. A reduction in temperature from 3° to 0°C doubles the storage life, and a reduction from 10° to 0°C increases the storage life by a factor of 5 or 6. Fish spoils rapidly at temperatures of 15° or 20°C.

Atlantic mackerel were held satisfactorily in ice for nine days, after which bacteria increased rapidly (Jhaveri, Leu, and Constantinides 1982). Molin and Stenström (1984) found that the shelf life of herring fillets was one day at 15°C or seven days at 0°C. They found that the shelf life increased to three days when it was packed in CO_2 at 15°C or thirty-three days at 0°C. Molin, Stenström, and Ternström (1983) reported that, at 2°C, CO_2 storage prolonged the shelf life of herring fillets by a factor of 3.5. The storage life of trout and herring at 0°C can be extended from one to two weeks by vacuum packaging (Hansen 1972). This packaging also retards the oxidative rancidity of fats and oils in the fish.

According to Cobb, Vanderzant, and Hyder (1974), shrimp boats remain at sea for four to twenty days. To maintain the quality, the shrimps are packed in ice. As with other fishery products, icing can cause a certain amount of leaching of nutrients from the shrimp. With ice storage, the shrimp in the lower layers tend to have a higher bacterial load because the melting ice carry bacteria from the upper layers. The weight of the shrimp and ice will tend to crush the shrimp at the bottom of the pile. This crushing, plus the bacteria washing down from the upper layers, increases the rate of deterioration.

Fruits and Vegetables. Substantial losses occur in the shipping, distribution, and retailing of fresh fruits and vegetables. These losses are due to biochemical, enzymatic, and bacteriological activities that cause overripeness and decay. Prolonging the storage life of fresh produce is advantageous, not only to the food industry, but also to the consumer, because it is the consumer who pays for the losses in the form of higher prices.

To help control these losses, it is essential that the field heat be removed from the fresh produce as soon as possible after harvesting. Then the produce should be maintained in a cool condition.

Two systems for removing field heat are hydrocooling and vacuum cooling. Hydrocooling is applicable to both fruits and vegetables. In this system, cold water sprayed over the fresh product reduces the temperature of the produce at a very high rate. The warmed water is recirculated through ice or a refrigeration system to cool it before respraying. In some cases, hydrocooled products have developed more decay than untreated produce. The addition of chlorine helps control the microorganisms in the water as well as those on the surface of the produce.

Vacuum cooling is adaptable to products that have a large surface-to-volume ratio. This system works in a manner similar to human perspiring. The evaporation of water from the surface produces a cooling effect. To prevent dehydration of the fresh produce, water is sprayed onto it. Then the vacuum treatment evaporates this sprayed-on water and cools the produce. The amount of cooling is roughly proportional to the water loss. Products in which vacuum cooling is beneficial include lettuce, spinach, and cabbage.

Besides cold storage, controlling the atmospheric gases (oxygen and carbon dioxide) and the relative humidity (RH) are beneficial in maintaining quality during storage of fruits and vegetables (Tables 11.4 and 11.5). Hypobaric storage (precisely controlling the pressure, temperature, and humidity of storage) has been discussed for various fruits and vegetables (Jamieson 1980).

As is generally true of all foods, most fruits and vegetables keep their fresh quality longer if stored at slightly above their temperature of freezing. The freezing temperature of fruits varies from about $-3.9°$ to $-1.0°C$ and vegetables, from about $-3.7°$ to $-0.5°C$ (Table 11.1).

Some fruits and vegetables are injured by low-temperature storage (Table 11.6). When injured produce is returned to room temperature, the respiration rate rises, then declines rapidly, due to the death of cells in the tissues.

Tropical and subtropical fruits are especially subject to low temperature injury. Even some fruits that are grown in temperate climates can be affected.

Oranges are less susceptible to chilling injury than are other citrus

TABLE 11.4. RECOMMENDED STORAGE TEMPERATURES, RELATIVE HUMIDITIES, AND APPROXIMATE STORAGE LIFE OF FRUITS AND NUTS

Fruit or Nut	Storage Temperature (°C)	Relative Humidity (%)	Approximate Storage Life
Apples	−1.1 to −0.6	85 to 90	—
Apricots	−0.6 to 0	85 to 90	1 to 2 weeks
Avocados	2.8 to 8.9	85 to 90	—
Bananas	11.7 to 15.6	85 to 90	1 to 3 weeks
Berries			
Blackberries	−0.6 to 0	85 to 90	7 to 10 days
Dewberries	−0.6 to 0	85 to 90	7 to 10 days
Gooseberries	−0.6 to 0	85 to 90	3 to 4 weeks
Loganberries	−0.6 to 0	85 to 90	7 to 10 days
Raspberries			
Black	−0.6 to 0	85 to 90	7 to 10 days
Red	−0.6 to 0	85 to 90	7 to 10 days
Strawberries	−0.6 to 0	85 to 90	7 to 10 days
Cherries	−0.6 to 0	85 to 90	10 to 14 days
Coconuts	0 to 1.7	80 to 85	1 to 2 weeks
Cranberries	2.2 to 4.4	85 to 90	1 to 3 months
Figs	−0.6 to 0	85 to 90	10 days
Grapefruit	0 to 10	85 to 90	—
Grapes			
Vinifera	−1.1 to −0.6	88 to 92	3 to 6 months
American	−0.6 to 0	80 to 85	3 to 8 weeks
Lemons	12.8 to 14.4	85 to 90	1 to 4 months
Limes	7.2 to 8.9	85 to 90	6 to 8 weeks
Mangoes	10.0	85 to 90	15 to 20 days
Nuts	0 to 2.2	65 to 70	8 to 12 months
Olives	7.2 to 10	85 to 90	4 to 6 weeks
Oranges	−1.1 to 1.1	85 to 90	8 to 10 weeks
Papayas	7.2	85 to 90	15 to 20 days
Peaches	−0.6 to 0	85 to 90	2 to 4 weeks
Pears	−1.7 to −0.6	88 to 92	2 to 7 months
Persimmons	−1.1	85 to 90	2 months
Pineapples			
Mature green	10 to 15.6	85 to 90	3 to 4 weeks
Ripe	4.4 to 7.2	85 to 90	2 to 4 weeks
Plums, prune	−0.6 to 0	85 to 90	3 to 8 weeks
Pomegranates	−0.6 to 0	85 to 90	2 to 4 months
Quinces	−0.6 to 0	85 to 90	2 to 3 months

SOURCE: Adapted from Desrosier and Desrosier (1977).

fruits. With the banning of ethylene dibromide fumigation, cold exposure (0°C for ten days or 2.2°C for sixteen days) is the only quarantine treatment for the Mediterranean fruit fly. This treatment causes peel pitting in grapefruit. However, preconditioning or delayed cooling (holding the fruit at 17°C for six days or 16°C for seven days prior to the cold treatment) reduced the incidence of chilling injury (Chalutz, Waks, and Schiffman-Nadel 1985; Hatton and Cubbedge 1981). Film wrapping or

TABLE 11.5. RECOMMENDED STORAGE TEMPERATURES, RELATIVE HUMIDITIES, AND APPROXIMATE STORAGE LIFE OF VEGETABLES

Vegetable	Storage Temperature (°C)	Relative Humidity (%)	Approximate Storage Life
Artichokes, Jerusalem	0	90 to 95	2 to 5 months
Asparagus	0	90 to 95	3 to 4 weeks
Beans			
Green	7.2	85 to 90	8 to 10 days
Lima—unshelled	0	90 to 95	2 to 4 weeks
shelled	0	90 to 95	15 days
Beets			
Topped	0	90 to 95	1 to 3 months
Not topped	0	90 to 95	10 to 14 days
Broccoli, Italian	0	90 to 95	7 to 10 days
Brussels sprouts	0	90 to 95	3 to 4 weeks
Cabbage	0	90 to 95	3 to 4 months
Carrots			
Topped	0	90 to 95	4 to 5 months
Not topped	0	90 to 95	10 to 14 days
Cauliflower	0	90 to 95	2 to 3 weeks
Celery	−0.6 to 0	90 to 95	2 to 4 months
Corn, green	−0.6 to 0	90 to 95	4 to 8 days
Cucumbers	7.2 to 10	90 to 95	10 to 14 days
Eggplants	7.2 to 10	90 to 95	10 days
Endive	0	90 to 95	2 to 3 weeks
Garlic	0	70 to 75	6 to 8 months
Horseradish	0	90 to 95	10 to 12 months
Leeks, green	0	85 to 90	1 to 3 months
Lettuce	0	90 to 95	2 to 3 weeks
Melons (ripe)			
Watermelon	2.2 to 40	80 to 85	1 to 2 weeks
Muskmelon	4.4 to 10	80 to 85	10 to 14 days
Honeydew	4.4 to 10	80 to 85	2 to 4 weeks
Casaba, Persian	4.4 to 10	80 to 85	4 to 6 weeks
Mushrooms	0	85 to 90	5 days
Okra	10	85 to 95	2 weeks
Onions	0	70 to 75	6 to 8 months
Parsnips	0	90 to 95	2 to 4 months
Peas, green	0	85 to 90	1 to 2 weeks
Peppers, green	7.2	85 to 90	8 to 10 days
Potatoes	3.3 to 4.4	85 to 90	6 to 9 months
Pumpkins	10 to 12.8	70 to 75	2 to 6 months
Rhubarb	0	90 to 95	2 to 3 weeks
Rutabagas	0	90 to 95	2 to 4 months
Spinach	0	90 to 95	10 to 14 days
Squashes			
Summer	4.4 to 10	85 to 95	2 to 3 weeks
Winter	10 to 12.8	70 to 75	4 to 6 months
Sweet potatoes	12.8 to 15.6	80 to 85	4 to 6 months
Tomatoes			
Ripe	4.4 to 10	85 to 90	7 to 10 days
Mature green	12.8 to 21.1	80 to 85	3 to 5 weeks
Turnips—topped	0	90 to 95	4 to 5 months

SOURCE: Adapted from Desrosier and Desrosier (1977).

TABLE 11.6. COMMODITIES SUSCEPTIBLE TO COLD INJURY WHEN STORED AT ONLY MODERATELY LOW TEMPERATURES

Fruit or Vegetable	Approximate Lowest Safe Temperature (°C)	Character of Injury When Stored Between 0°C and Safe Temperature
Apples		
Certain varieties	1.1 to 2.2	Internal browning, soggy breakdown
Avocados	7.2	Internal browning
Bananas		
Green or ripe	13.3	Dull color when ripened
Beans—snap	7.2 to 10	Pitting increasing on removal, russeting on removal
Cranberries	1.1	Low-temperature breakdown
Cucumbers	7.2	Pitting, water-soaked spots, decay
Eggplants	7.2	Pitting or bronzing, increasing on removal
Grapefruit	7.2	Scald, pitting, watery breakdown, internal browning
Lemons	12.8 to 14.4	Internal discoloration, pitting
Limes	7.2	Pitting
Mangoes	10	Internal discoloration
Melons		
Cantaloupe	7.2	Pitting, surface decay
Honeydew	4.4 to 10	Pitting, surface decay
Casaba	4.4 to 10	Pitting, surface decay
Crenshaw and Persian	4.4 to 10	Pitting, surface decay
Watermelons	2.2	Pitting, objectionable flavor
Okra	4.4	Discoloration, water-soaked areas, pitting decay
Olives, fresh	7.2	Internal browning
Oranges, California	1.7 to 2.8	Rind disorders
Papayas	7.2	Breakdown
Peppers, sweet	7.2	Pitting, discoloration near calyx
Pineapples		
Mature—green	7.2	Dull green when ripened
Potatoes		
Chippewa and Sebago	4.4	Mahogany browning
Squash, winter	10 to 12.8	Decay
Sweet potatoes	12.8	Decay, pitting, internal discoloration
Tomatoes		
Mature—green	12.8	Poor color when ripe; tendency to decay rapidly
Ripe	10	Breakdown

SOURCE: Adapted from Desrosier and Desrosier (1977).

holding at 21°C for three days reduced the chilling injury of lemons when stored at 1°C for twenty-one days (McDonald 1986). However, film wrapping of mature green bell peppers did not reduce the incidence of chilling injury (Miller and Risse 1986). The treatment that might be used to alleviate chilling injury depends upon the fruit or vegetable. Systems for controlling chilling injury have been discussed for avocados (Chaplin,

Wills, and Graham 1983), pineapples (Paull and Rohrbach 1985), and muskmelons (Lipton and Aharoni 1979).

When cucumbers are stored at 7°C or lower for ten days or more, they develop pits, dark-colored watery patches, and tissue collapse. These damaged tissues can readily become infected with microorganisms that cause rotting. A temperature of 10°C and 95 percent RH are the best conditions for storing and transporting cucumbers (Etchells et al. 1973).

Potatoes can heal injured areas of the skin if held at normal room temperatures. This wound healing is very slow at 10°C. Sweet potatoes need a curing treatment at about 29.5°C and a humidity of 85 percent to heal cuts and abrasions. After this treatment, 12.5°C is recommended, since chilling injury occurs at lower temperatures. The injury causes darkening of the tissues and makes the potatoes susceptible to rotting.

Some vegetables deteriorate rapidly if not placed promptly into low-temperature storage. Quick cooling and storage at 0.5° to 1.5°C are needed for corn, peas, and green lima beans. Leafy vegetables, beets, and carrots need quick cooling and storage at 0.5° to 4.5°C (the temperature depending on the vegetable) to maintain their quality.

Canned Food. Various biochemical reactions can occur in canned foods during storage. These changes include loss in organoleptic quality, alterations in chemical constituents, and losses in nutrients. These reactions are generally reduced when the canned food is held in cold storage (10°C or less), than at ambient-temperature storage. The extent of any benefit depends upon the type of food.

Freezing

Foods are frozen to extend their storage life over that obtained in a refrigerator. The food is preserved by using temperatures low enough to stop or greatly reduce the deterioration caused by microorganisms, enzymes, or chemicals such as oxygen. It is generally recognized that freezing offers one of the best means of maintaining the fresh color, flavor, and appearance of many foods.

GENERAL ASPECTS. Pure water at atmospheric pressure freezes at 0°C. As the temperature is lowered below 0°C, the vapor pressures of water and ice are reduced so that the water activity below −10°C is lower than that necessary for the growth of most bacteria (Table 11.7).

The fluids in plant or animal tissues are aqueous solutions. The freezing points of these solutions may range from 0° to −10°C (Table 11.1). When food is subjected to a temperature below freezing and the extracellular water begins to freeze, the solutes in this water tend to migrate to the remaining liquid. This increases the concentration of solutes and low-

TABLE 11.7. VAPOR PRESSURES AND WATER ACTIVITY OF WATER AND ICE

Temperature (°C)	Vapor Pressure		Water Activity (VP_{ice}/VP_{water})
	Liquid Water (mm Hg)	Ice (mm Hg)	
0	4.579	4.579	1.00
−5	3.163	3.013	0.95
−10	2.149	1.950	0.91
−15	1.436	1.241	0.86
−20	0.943	0.776	0.82
−25	0.607	0.476	0.78
−30	0.383	0.286	0.75
−40	0.142	0.097	0.68
−50	0.048	0.030	0.62

ers the freezing point of this liquid. The extent of solute concentration is influenced by the product characteristics and the rate of freezing. Some solid material is occluded within or among the ice crystals, the amount depending upon the freezing rate.

As the solute concentration of the extracellular liquid increases, an osmotic differential occurs and water tends to leave the plant, animal, or microbial cells. Slow freezing causes a greater degree of solute concentration and dehydration of the cells than does rapid freezing.

When a liquid is cooled slowly, solidification commences at a relatively small number of places, and the resultant crystals grow to large sizes. Rapid cooling, on the other hand, results in the initiation of a large number of smaller crystals. The rate of freezing, therefore, may alter the physical structure of the tissue. Rapid freezing of food is desired.

The rupture of the cellular structure may result in a poor texture of the thawed food. The exudation of juices, resulting in "drip," may occur in frozen and thawed plant and animal tissue. It contains dissolved protein, other nitrogenous substances, vitamins, and minerals.

Although most of the water in a food is frozen at −20°C, there is usually a certain amount of bound water that does not freeze. This unfrozen water is considered to be unavailable for biological activity, so it should not affect microbial deterioration of the frozen food.

SYSTEMS FOR FREEZING. Heat can be removed from foods by convection, conduction, evaporation, or radiation. Commercially, freezing systems use convection, conduction, or combinations of these. The product may be frozen before packaging, after packaging, or even after casing. Packaged foods freeze more slowly than unpackaged foods, given the same conditions of freezing, due to the insulating effect of the packaging material.

Systems developed for freezing food include the following:

Plate freezer (single, double, or pressure)
Still air (sharp freezer)
Air blast (room, tunnel)
Slush freezer
Fluidized bed
Liquid immersion (brine, dichlorodifluoromethane)
Liquefied gases (CO_2, N_2)
Freezant spray

The system selected for use depends on the type of food, speed of freezing, and cost factors. Although relatively slow, tunnel air-blast freezing can be used for most foods.

Fluidized bed freezing is a means of rapidly freezing small particulate material. It is the best system for the production of individually quick-frozen (IQF) foods. Mattick, Rawnick, and Ellis (1982) described a fluidized system in which the food travels through a tunnel on a stainless steel wire-mesh belt. Blasts of cold air (−34°C) from underneath the belt agitate, fluidize, and freeze each individual food particle. Symons (1982) described a fluidized bed with no belt. The holes in the trough carrying the food were angled so that the air coming through moves the food upward and forward. According to Symons, there is no moving part in the fluidized bed; only the air moves. Advantages are that the food is frozen quite rapidly and individually, so an entire carton of food does not need to be thawed when only a small portion is needed for use.

Liquid is a better conductor of heat than is air, so more rapid freezing is accomplished by liquid immersion freezing than by air-blast freezing. Since the liquid refrigerant is in direct contact with the food, it must meet the FDA criteria for food additives. Only a few liquid refrigerants are acceptable. Solutions of NaCl or $CaCl_2$ have been used for immersion fluids. Another fluid used for immersion freezing is dichlordifluoromethane (Freon 12), at −29.8°C. This freezant is under scrutiny by the Environmental Protection Agency because it is suspected to react with and damage the ozone layer in the atmosphere, which is needed to protect the earth from the sun's UV radiation. Although the system is enclosed so the fluorocarbon can be recycled, some losses may occur.

For very rapid freezing (cryogenic freezing), liquefied gases (CO_2 or N_2) are used, either as immersion liquids or sprays. There are certain advantages for using liquid nitrogen freezing. A nitrogen spray system is shown in Figure 11.1. Since the food is frozen in an inert atmosphere of nitrogen, surface oxidation is prevented. The rapid freezing and low temperature (−196°C) prevent microbial growth. There are problems associated with this cryogenic freezing. The sudden reduction in the temperature of food by direct contact with liquid nitrogen can cause shatter-

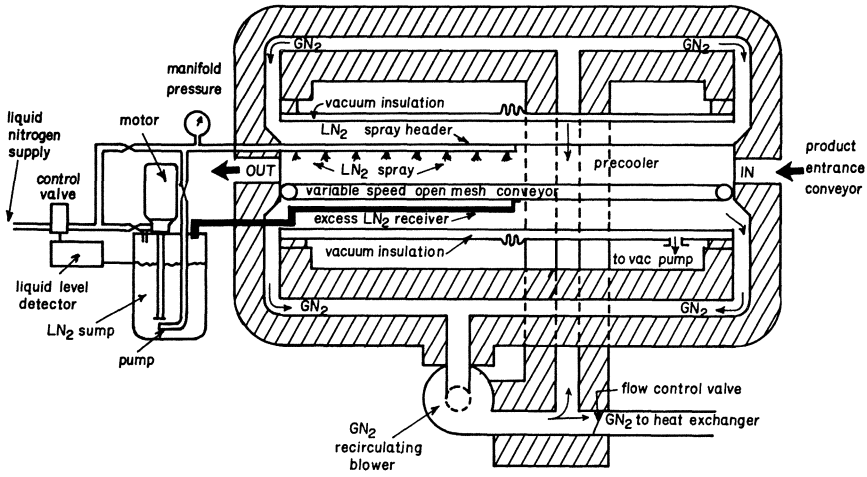


Figure 11.1. Liquid nitrogen flash freezer.
Courtesy of Farrall (1976).

ing of the food. Other materials, such as plastics, rubber, or glass, become brittle and can crack or shatter due to the thermal shock caused by liquid nitrogen.

STABILITY OF FROZEN FOODS. Some aspects that affect the storage life of frozen foods are the treatments prior to freezing, the type of packaging, the temperature, fluctuations in the temperature, and conditions for thawing of the frozen food. The frozen storage life of selected foods is listed in Table 11.8.

The lower the temperature of storage, the longer the storage life of the product. The deterioration of foods can occur at high freezer storage temperatures. Many enzymes are not inactivated by these temperatures. To help control enzymatic activity, some foods are given a heat treatment (blanching) to destroy the enzymes prior to freezing.

Good packaging materials prevent the diffusion of oxygen into the package to retard rancidity or oxidation. Also, the package will prevent evaporation of moisture from the frozen food. The addition of ascorbate or various antioxidants to certain frozen foods enhances the storage life by retarding oxidative changes.

The freezing rate affects the storage life of frozen foods. Slow freezing conditions may allow a food to remain at temperatures above 0°C for several hours. The quality lost due to slow freezing may be equivalent to that lost during several months of storage at −18°C.

TABLE 11.8. APPROXIMATE STORAGE LIFE (MONTHS) FOR QUALITY MAINTENANCE OF SELECTED FOODS

	Temperature °C		
Product	−17.8	−12.2	−6.7
Orange juice (blanched)	27	10	4
Peaches	12	<2	0.2
Strawberries	12	2.4	0.3
Cauliflower	12	2.4	0.3
Green beans	11–12	3	1
Green peas	11–12	3	1
Spinach	6–7	<3	0.75
Raw chicken	27	15.5	<8
Fried chicken	<3	<1	<0.6
Turkey pies or dinners	<30	9.5	2.25
Beef (raw)	13–14	5	<2
Pork (raw)	10	<4	<1.5
Lean fish (raw)	3	<2.25	<1.5
Fat fish (raw)	2	1.5	0.8

If a malfunction occurs during storage, transportation, distribution, or retailing, the frozen food can be subjected to defrosting temperatures. In many cases, it is necessary to defrost the frozen food prior to use. Thus, defrosting is of concern whether it is accidental or deliberate.

If a food is only partially defrosted, there is usually no harmful effect, at least from a public health viewpoint. This is also true concerning a product that is thawed but still cold. These foods usually can be refrozen with no potential problems. If a food becomes thawed and warmed to room temperature, it may or may not be safe to refreeze, depending upon the type of food and the time that it has been at room temperature.

For usage, fast thawing of frozen foods helps retard bacterial growth, as well as loss of quality attributes. The rate of thawing usually is lower than the rate of freezing. This is because of the lower temperature differential during thawing than during freezing, and because the rate of heat transfer is lower through the defrosted surface than through the frozen surface during freezing.

Some foods can be cooked from the frozen state without intermediate thawing. With other foods, thawing in the refrigerator usually is suggested because the temperature of the defrosted food will remain low enough to prevent the growth of potential pathogens. Thawing in warm water will hasten the process, but if the warm water is in direct contact with the food, nutrients from the food will be dissolved into the water and lost. With this method of thawing a packaged frozen turkey, the outer part of the turkey will be warm enough to support the growth of spoilage and potentially pathogenic organisms while the inside is still frozen. A

microwave oven is useful for tempering frozen foods used in further processed items.

ADVANTAGES AND DISADVANTAGES OF FREEZING. The advantages of freezing as a method of preserving food are that it (1) neither adds nor removes any components; (2) imparts no new flavor nor alters the natural flavor; and (3) does not diminish the digestibility or cause a significant loss of nutrient value.

Some disadvantages of freezing are as follows: (1) microorganisms may be reduced in number but not destroyed; (2) spores are very resistant and toxins are not destroyed; and (3) frozen foods not properly wrapped dehydrate very rapidly, which causes marked deterioration in the flavor and general appearance of the food.

SPECIFIC FOODS. Besides the general aspects of freezing foods, certain types of foods have some special problems, conditions, or characteristics.

Red Meats. The storage life of frozen meat depends upon the initial quality, condition, and pH of the meat, the interval between slaughter and freezing, the method of preparation, the temperature of holding before freezing, package method and materials, method and temperature of freezing, and the temperature and other conditions of frozen storage. The cryogenic freeezing of meat was discussed by Sebranek (1982).

The quality of red meats can be maintained if the meat is packaged in a wrapper impermeable to oxygen and moisture, rapidly frozen, and then stored below $-18°C$. Beef and lamb can be stored for one year or more if properly protected from dehydration (Table 11.8). Pork, especially smoked or cured, has poorer storage life. However, it can be held for six months or more if oxygen is excluded from the product and dehydration is prevented. The fatty components of meats are susceptible to changes in odor and flavor, with pork fat being more susceptible than beef or lamb.

Fresh meats are seldom frozen for retail sale to consumers. Surveys indicate that consumers prefer to buy fresh meat. However, most of this fresh meat is then frozen in home freezers. Freezing in commercial freezers would produce a more satisfactory product.

Meats preserved by freezing are stored at temperatures low enough to prevent microbial growth. Freezing with subsequent thawing will kill some microbial cells. Microorganisms tend to die slowly during storage, but those that survive will grow in the thawed meat at a rate similar to that exhibited by the same strain in raw meat (at the same temperature) that had never been frozen. The rate of microbial growth in thawed meat is dependent upon the temperature of the meat surface.

A condition called freezer burn may occur on the surface of frozen meat. It appears as patches of light-colored tissue and gives the meat a bleached, unattractive appearance. The cause is similar to a freeze-drying system in which ice crystals are sublimated. The control of freezer burn is aided by packaging in a moisture-impermeable film.

Poultry. Poultry meat that is frozen should be fresh, well finished, and properly packaged. Although the quality is not improved, freezing should maintain the quality as closely as possible to the value at the time of freezing. Freezing can affect the appearance, the exudation of fluid (drip) on thawing, and a progressive loss of flavor, tenderness, and juiciness.

Freezer burn can occur on frozen poultry in a manner similar to that in red meat. With poor packaging, this dehydration can occur in one month or less of storage.

The long bones in young frozen poultry (under fifteen weeks of age) usually are discolored. The freezing and thawing processes hemolyze the red blood cells, releasing hemoglobin. The hemoglobin is able to pass out of the inner porous bone. This darkening is a normal condition found in frozen poultry and in no way affects the food value or flavor of the meat. It does affect the attitude of the consumer concerning frozen poultry, however. Bone darkening was nearly absent when poultry was frozen in liquid nitrogen at $-196°C$ (Cunningham 1974).

There was little change in the quality of broilers through five thawings and refreezings (Baker et al. 1976). The microbial load increased slightly after the five thawings and refreezings. The bacterial load on the chickens before freezing was 3.8 (10^7) CFU per chicken, and after five thawings the loads were 2.1 (10^8) and 3.0 (10^8) CFU per bird for fast and slow freezings, respectively. Thawed poultry does not spoil more rapidly than poultry that has not been frozen.

Chickens are generally retailed in the refrigerated fresh form, but turkeys are usually packaged in an evacuated, tight-fitting plastic film in the frozen condition. Turkey freezes at a faster rate in evacuated packages than in unevacuated packages. Microwave-defrosted frozen chicken breasts have been found to be more tender than those thawed in a home refrigerator (Younathan, Farr, and Laird 1984).

Seafoods. The storage life of frozen fish is significantly longer than that of iced or refrigerated fish. It is desirable to lower the temperature rapidly. The frozen fish should be stored at $-20°C$ and, for some fish, even $-40°C$ (Gates et al. 1985; Jiang, Ho, and Lee 1985).

The main deteriorative alterations in frozen seafoods are flavor changes such as rancidity, toughening or loss of tenderness, formation of drip, development of freezer burn, and gaping. Some of these changes

are due to protein denaturation and breakdown of fat. Changes of fish during storage have been the subject of studies (Kelleher et al. 1982; Ronsivalli and Baker 1981).

The storage life for seafoods is generally shorter than that for either red meat or poultry (Table 11.8). The storage life of frozen salmon or whitefish was improved by vacuum packaging in low oxygen permeable film (Josephson, Lindsay, and Stuiber 1985; Yu, Sinnhuber, and Crawford 1973).

Egg Products. When egg yolk is frozen and thawed, it becomes a viscous, rubbery mass. This irreversible increase in apparent viscosity is known as gelation. Certain additives, such as sugar (sucrose), fructose, galactose, or salt, will reduce gelation of the frozen and thawed yolk. Proteolytic enzymes added to the liquid will partially inhibit gelation. If the egg yolk is frozen rapidly, such as with liquid nitrogen, there will be less gelation than when it is frozen by slower freezing systems. Rapid thawing also reduces gelation. Rapid freezing, combined with rapid thawing, is more effective in preventing gelation of frozen and thawed egg yolk than either one alone.

When sugar or salt is added to yolk, the freezing point is lowered. The temperature of the liquid must be reduced to $-12°C$ within 60 hr after pasteurization. The eggs freeze from the outside of the carton toward the middle, so it is a race between ice formation and microbial growth. If the bacteria win, the product becomes sour. With pasteurization to destroy *Salmonella,* many of the spoilage bacteria are destroyed. Hence, there should be less spoilage today than in the prepasteurization era.

Dairy Products. These are frozen as a means of preservation, or in the manufacture of ice cream or ice milk.

The freezing point of milk varies within relatively narrow limits. The average freezing point is considered to be $-0.55°C$. The determination of the freezing point provides a method of detecting added water. If water is added to milk, the freezing point is closer to $0°C$. There is a linear relationship between the hydrolysis of lactose to monosaccharides and depression of the freezing point of milk (Jeon and Bassette 1982).

The deterioration in physical stability is one of the principal defects encountered in frozen milk. During frozen storage, the proteins become unstable and when the milk is thawed, the proteins settle out. Besides the appearance, the milk has an undesirable chalky texture.

Fruits and Vegetables. Most of these products can be frozen and stored at $-17.8°C$ for several months. The quality losses include color, texture and flavor. The factors that influence the quality of the frozen product include the initial quality, the type and variety of the fruit or vegetable,

processing operations, freezing temperature, packaging, and storage temperature (Ahvenainen and Mälkki 1985; Kramer 1979).

Some varieties of certain fruits and vegetables withstand freezing and frozen storage much better than other varieties. Freezing does not improve quality, so only high-quality products should be frozen.

Most vegetables must be heat treated (blanched) before freezing to inactivate the enzymes, or these catalysts may cause flavor or color deterioration during storage.

Baked Goods. Bakery products are especially adaptable to frozen storage. Although products such as crackers and cookies generally are not frozen commercially, cookies can be frozen at home to extend the storage life. The stability of frozen dough was affected more by final freezing temperature than freezing rate. Freezing without fermentation produced better-quality bread than freezing after fermentation. Damage to yeast cells is a major problem in frozen doughs (Hsu, Hoseney, and Seib 1979; Wolt and D'Appolonia 1984a, 1984b).

The freezing point of most bakery products is between $-7°$ and $-2°$C. The richer doughs freeze at the lower temperatures. Some adjuncts, such as jams or frostings, require even lower temperatures for freezing.

Miscellaneous Foods. The freezer cases in the retail grocery stores contain frozen dinners, pot pies, pizzas, and various mixtures of many types of foods. Although there has been much concern about the possible health hazards of certain precooked foods, they have had a good record for noninvolvement in foodborne disease outbreaks.

MICROBIOLOGICAL ASPECTS. Although freezing is not a method for sterilizing food, many of the associated microbial cells are killed or damaged by freezing, frozen storage, and thawing. Various treatments before freezing can affect the microorganisms associated with food.

Calcott and MacLeod (1974) listed reasons for the cryoinjury of cells. These are (1) thermal shock; (2) effect of concentration of extracellular solutes; (3) toxic action of concentrated intracellular solutes; (4) dehydration; (5) internal ice formation; and (6) attainment of a minimum cell volume.

Freezing and thawing can damage the cell wall, membrane, ribosome, and DNA. Cell wall damage results in loss of resistance of Gram-negative cells to various agents such as bile salts, sodium lauryl sulfate, and lysozyme (Ray, Johnson, and Wanismail 1984; Takano et al. 1979). Cell membrane damage allows leakage of various compounds (proteins, phospholipids, lipopolysaccharides) from the cell (Calcott et al. 1984; Souzu 1980). The wall and membrane damage can be repaired (Calcott, Steenbergen,

and Petty 1979). Freezing of yeast induced a decrease of respiratory activity (Mori, Suzuki, and Nei 1986). A loss of ribonucleic acid (RNA) from the cell is associated with ribosome damage. The DNA may have both single- and double-strand breaks (Grecz and El-Zawahry 1984; Grecz et al. 1980). According to El-Zawahry and Grecz (1981) freezing caused substantially greater cell injury to *Y. enterocolitica* than did radiation.

We are concerned with the lethal effects of freezing foods because we would like to destroy as many cells as possible without destroying the organoleptic or physical quality of the food. On the other hand, freezing is used to preserve cultures for research as well as those organisms used in food fermentations, as a source of enzymes, or as a source of single-cell protein.

There are many conflicting reports in the literature concerning the effects of freezing on microbial cells. Part of the problem may be due to the use of different types of microbial cells in various suspending media which are cooled at various or unknown rates. The response of microorganisms is dependent upon many factors. Those that have been considered include (1) the type, species, and strain of the microorganism; (2) the age, population density, and nutritional status of the cells; (3) the cooling rate; (4) the minimum temperature to which the cells are cooled; (5) composition, pH, and type of suspending medium, including the presence of protective agents; (6) the temperature and time of storage; (7) the rate of warming or thawing; and (8) the methods used to determine cell viability.

Bacterial spores are the most resistant microbial entity to freeze damage. Gram-positive cells are more resistant to freezing and thawing than are Gram-negative cells. Survival of microorganisms is greater in a supercooled than in a frozen environment.

The cooling rate during freezing can affect the survival of microorganisms. At low cooling rates, the extracellular water begins to freeze and form ice crystals. The solutes are concentrated into the remaining water. This causes the water to emigrate from the cells due to the osmotic pressure gradient between the intracellular and extracellular solutions or the vapor pressure difference between the supercooled intracellular water and the extracellular ice. The slower the process of cooling and the greater the permeability of the cells, the greater will be the loss of water from the cells. The movement of water out of the cells concentrates the solutes of the cells, lowers the intracellular vapor pressure, and dehydrates the cells. The cell volume also decreases and solutes precipitate.

The maximum survival of *Escherichia coli* is at a cooling rate of about 6°C/min, and the minimum at about 100°C/min (Calcott and MacLeod 1974). Calcott, Lee, and MacLeod (1976) determined the survival of *Streptococcus faecalis, Salmonella typhimurium, Klebsiella aerogenes, Pseudomonas aer-*

uginosa, and *Azotobacter chroococcum* at various cooling rates. The cooling rate for optimum survival varied from 7°C/min for *A. chroococcum* to 11°C/min for *P. aeruginosa*. The minimum survival for these five organisms was found to be at about 100°C/min. At higher cooling rates, the survival increased.

Meyer, Sinclair, and Nagy (1975) studied the effect of cooling rate on a mesophilic yeast and a psychrophilic yeast. The optimum cooling rate for survival of both yeasts was between 4.5° and 6.5°C/min. The level of survival varied considerably even at the optimum cooling rate. The survival of the psychrophilic yeast varied from 5 percent when cooled from the exponential phase, to 71 percent when in the stationary phase. The mesophilic *Candida utilis* survival for exponential phase cells was 47 percent, and for those in the stationary phase it was 71 percent.

Four Gram-negative bacteria studied by Calcott, Lee, and MacLeod (1976) showed less survival when frozen in saline than in water. However, the Gram-positive *S. faecalis* did not show this reduction in survival. They believed the difference might be due to the respiratory systems of the bacteria. The Gram-negative bacteria respire aerobically with cytochrome systems integrated into the cytoplasmic membranes. The *S. faecalis* is anaerobic and has no cytochrome system.

At very high cooling rates, small crystals of ice are formed. Due to their high surface energies, the small ice crystals tend to enlarge during warming. Mazur (1970) stated that a forty-fold decrease in survival of yeast cells occurred when the time for warming from −70° to 0°C was increased from 0.06 sec to 3.6 sec. The damage apparently can be very rapid and is thought to be due to the growth of the intracellular ice crystals, rather than to the initial ice formation.

The enlarging intracellular ice crystals exert sufficient force to rupture plasma membranes. Cells killed by intracellular freezing suffer membrane damage and become leaky.

High thawing rates result in greater survival of fast-frozen cells than do low thawing rates. Enhanced recovery of slowly frozen cells by slow thawing is believed to be due to the ability of cells to reconstitute more readily during slow thawing. Rapid thawing of the dehydrated cells causes a rapid entrance of water, which may cause excessive internal pressure and resultant rupture of the membranes.

Cryoprotective Agents. There are many substances that protect the microbial cell during freezing and thawing. Some, such as glycerol and dimethylsulfoxide, penetrate the cells, while others do not. Besides glycerol and dimethylsulfoxide, protective agents include egg white, carbohydrates, peptides, serum albumin, malic acid, milk, glutamic acid, yeast extract, diethylene glycol, Tween 80, dextran, sodium glutamate, glucose, polyeth-

ylene glycol, erythritol, polyvinyl-pyrrolidone, ionophores, acetamide, and urea. Also, mixtures of cryoprotectants have been used.

Glycerol reduces damage to the cell wall and membrane, while Tween 80 prevents only membrane damage (Calcott and MacLeod 1975). The mechanism of action of glycerol is not fully understood. The presence of glycerol decreases the mole fraction of solutes in the suspending fluid at freezing temperatures and prevents the high concentration of solutes that might affect the cell wall and membranes.

Microorganisms in Frozen Foods. Many factors influence microorganisms in foods that are frozen. During freezing, the solutes and some microorganisms become concentrated in the unfrozen portion. Until the temperature is reduced below the minimum temperature for growth, presumably some microorganisms can multiply. However, the solutes can become concentrated to the level that microbial growth is inhibited. Microbial growth can occur in frozen foods stored at $-7°C$ and possibly even at $-12°C$.

The greatest reduction in the microbial load occurs during or shortly after the freezing of foods. During frozen storage, the numbers are gradually reduced further.

Although food that is being thawed is susceptible to microbial growth, there is generally a long lag phase before growth commences if the temperature is kept near 0°C.

Although salmonellae, staphylococci, and other potential pathogens can survive freezing and frozen storage, the saprophytic flora tends to inhibit their growth. During freezing and thawing of food, the temperature favors the growth of the psychrophilic organisms. Hence, in nearly all cases, if a frozen product is mishandled, spoilage is apparent before the food becomes a health hazard.

There have been suggestions that freezing foods disrupts the cellular structure of flesh foods and that bacteria can grow more readily on thawed meat that was frozen than on unfrozen meat. Sulzbacher (1952) found no indication that frozen meat was more perishable after thawing than was fresh meat. Elliot and Straka (1964) reported that frozen and thawed chicken decomposes at about the same rate as unfrozen chicken. Thawed meats, like fresh meats, are perishable and should not be held very long before cooking.

In some studies of the microorganisms associated with frozen foods, the total number of aerobic bacteria was determined by incubation at 37°C. At this temperature, mesophilic types are enumerated, but the psychrophiles are the important potential spoilage organisms in frozen foods. Incubation should be conducted at 20°C or less to obtain an estimate of psychrophilic types.

The microbial characteristics of frozen seafoods are similar to those of red meat and poultry. During freezing of seafoods, from 60 to 90 percent of the bacterial population is inactivated. During frozen storage, there is a gradual decline in the number of cells. Gram-positive cells survive frozen storage better than do Gram-negative cells. As with red meats and poultry, defrosted fish spoils at about the same rate as unfrozen fish.

Poliovirus inoculated into oysters showed a gradual decline in plaque-forming units (PFU) during frozen storage at −36°C (DiGirolamo, Liston, and Matches 1970). About 10 percent of the original PFU survived twelve weeks of storage.

Vibrio parahaemolyticus is a potential pathogen associated with fishery products. It is generally stated that this organism does not survive freezing and frozen storage. *V. parahaemolyticus* cells were inoculated into oysters, fillets of sole, and crabmeat (Johnson and Liston 1973). Although there was a sharp reduction in vibrios during freezing, these organisms persisted in these seafoods stored at either −15° or −30°C, with better survival at −30°C. They were found in oysters after 130 days, the end of the experiment.

The microbiology of frozen egg products is similar to that of other foods. During freezing, there may be microbial growth and, in poorly frozen product, spoilage may occur. With proper freezing, there is usually a reduction in the microbial load. During frozen storage at −15° to −18°C, the microbial load may show a slight decline. There is a 50 to 85 percent reduction in bacteria within sixty days of frozen storage.

The microbial load on raw vegetables may reach 10^6/g. Washing and blanching (95°C for 1 to 3 min) will reduce the number of microorganisms. In commercial practice, the vegetables are recontaminated by equipment, so that the microbial load at the time of freezing may be higher than that of the raw vegetables.

When slow cooling rates are used for freezing vegetables such as corn or peas microbial growth may occur before the product is frozen. However, the inactivation of cells during the freezing process helps compensate for any increase in numbers. Overall, there may be a slight decrease or increase in the microbial load. Fast cooling rates and freezing result in essentially no change in the microbial count of vegetables. The 30°C aerobic plate count of frozen vegetables was reported as 45,000/g of cauliflower, 8,500/g of corn, and 6,800/g of peas (Barnard et al. 1982). The highest aerobic plate count of various frozen potato products was 3,600/g (Duran et al 1982).

There is a greater reduction in the bacterial load if the product is stored at −12°C rather than at either −18° or −23°C. However, quality may suffer at the warmer temperature. A temperature of −4°C is not sufficiently low to inhibit microbial growth in vegetables such as peas.

Some of the dominant organisms in frozen vegetables are species of *Micrococcus, Flavobacterium, Streptococcus, Pseudomonas, Lactobacillus,* and *Enterobacter.*

When frozen fruits are held at −4°C or higher, yeasts and molds have a tendency to grow. Growth usually does not occur if the fruit is packed in an airtight container. At these temperatures, the respiration of the fruit replaces the oxygen with carbon dioxide. This, along with the low pH, inhibits the growth of most types of microorganisms.

DRYING

Drying is thought to be the oldest method used to preserve food. Early in time, man observed that naturally dried seeds could be stored from one season to the next. The sun drying of meat, fish, and fruit has been practiced for many centuries. These dried foods tended to have their own characteristic flavor, aroma, and texture. The drying process caused irreversible changes; for example, grapes became raisins, and the rehydrated food was different from its fresh counterpart.

In order to compete with fresh foods or with other preservation methods, irreversible reactions of dried foods should be kept to a minimum. Most commercially dried foods are produced with controlled conditions of temperature, humidity, and airflow. Since removal of moisture by heat is relatively expensive and uses much energy, systems such as screening, pressing, centrifuging, settling, filtration, reverse osmosis, osmotic solutions, or freeze concentration may be used to eliminate water prior to final drying of the food.

Reasons for Drying

The obvious reason for drying food is to prevent microbial growth. However, there are other reasons, including (1) to preserve the product from physical or chemical changes induced or supported by excess moisture; (2) to reduce the packaging, storage, and transportation costs; (3) to prepare a material for a process when only dry materials can be used; (4) to remove moisture added in previous operations; (5) to bring the product to a moisture level at which it is normally graded, bought, and sold; (6) to recover waste products; and (7) to provide convenience and a saving of time to users.

Dried foods weigh less and have reduced bulk. A thirty-dozen case of shell eggs weighs about 26 kg, while the dried product weighs about 5.1 kg. The dried eggs occupy about 16 percent of the space of the fresh product. Thus, costs of transportation and storage are lower.

If properly packaged, most dried foods can be stored at ambient con-

ditions, with no special freezing or refrigeration requirements. One advantage is that the package of dried food can be opened and part of it removed for use, and the remainder can be expected to have an acceptable storage life. Due to their concentrated character, dried foods can be used as ingredients to create new types of foods.

The reduced weight and bulk, the good storage life, and the versatility of dried food are important, especially during wartime. In the past, dried foods were exploited during wartime and then more or less forgotten. However, after World War II, although the amount of drying decreased, the technology that had been developed was used to produce consumer goods, such as instant dried milk and dried potato products. Also several mixes, such as various dried soups and cake mixes, have been developed.

Pretreatments

Only food of good quality should be used for drying. Generally, quality is not improved by drying.

The usual and necessary washing, cleaning, breaking, peeling, slicing, and other operations are carried out prior to drying. The operations needed depend upon the type of food to be dried.

Some foods require a heat treatment prior to drying. Quite often, foods such as meats are cooked before dehydration. Cooking meat can reduce the moisture content by 20 percent or more. Cooking before drying adds a useful convenience factor to dried foods. Also, cooking reduces the number of viable microorganisms.

Potatoes are cooked prior to drying, so the dried product is ready for use merely by adding water and mixing. Instant rice is prepared by a cooking process and redrying of the product.

Blanching is an important pretreatment for dehydrated vegetables. This process inactivates enzymes, expels air from tissues, yields a more tender product on reconstitution, minimizes loss of palatability, delays the development of odors, reduces color loss, helps retain carotene and ascorbic acid during dried storage, and lengthens the storage life.

The peel, as well as the waxy coating on fruits, inhibits the removal of water from fruits. Dipping fruits and berries into a weak sodium carbonate or lye solution produces fine cracks in the skin, removes the outside waxy covering, and facilitates drying.

Fruits are given a sulfite treatment to prevent browning during drying and storage. Antioxidants may be added to foods to help prevent oxidative changes in the dried product.

For eggs and egg white, the reducing sugars are removed by fermentation or treatment with glucose oxidase. This is to prevent the Maillard reaction in the dried products.

Other pretreatments may consist of concentration by reverse osmosis,

ultrafiltration, or other methods. Milk may be concentrated to 20 to 40 percent solids by evaporation in vacuum pans before spray drying. The osmotic concentration of potatoes in a 45 percent sucrose/15 percent salt solution before solar drying resulted in a two- to fivefold increase in the throughput of product (Islam and Flink 1982). This system can be used for various foods.

Various ingredients may be blended before drying to impart desired functional traits to the dried product.

Drying Aspects

Water is associated with foods as (1) water containing solids in solution (dissolved salts or other solutes); (2) water containing solids in suspension; (3) water occupying the pores or interstices of a solid; (4) water adhering to the surface of a solid; (5) chemically combined water (water of hydration); and (6) hydroscopic water that is naturally absorbed by a substance from an atmosphere containing water vapor. Generally, water is considered to be bound or free (Chung and Chang 1982; Rizvi and Benado 1984).

The removal of water during drying requires the addition of heat for evaporation. The water migrates within the material to the surface. Here it is evaporated into the atmosphere surrounding the product and then removed from the vicinity of the food. The rate of removal of moisture from the surface is dependent upon the temperature, humidity, and movement of the surrounding air. The size, shape, and other characteristics of the food particle will influence moisture migration from the center to the surface.

The usual drying curve shows a constant rate period followed by a falling rate period (Fig. 11.2). According to Szentgyörgyi and Örvös (1985), four periods are involved: a period of constant drying rate, a first falling rate period, a second falling rate period, and a final period of drying. During the constant rate period, moisture travels to the surface of the food quite rapidly so that the food is kept moist. When the rate of migration of water from the interior cannot maintain a sufficient flow to the surface to sustain the initial rate of evaporation, the constant rate period is terminated. The drying rate declines, becoming very small as the water content of the food approaches equilibrium.

During the drying process, some portions of the food will dry more quickly than others. The wet areas will be at lower temperatures than the dry portion, due to evaporative cooling.

In dehydrated foods, the sugars, acids, and inorganic salts are concentrated, which also lowers the a_w of the foods and increases the preservative action.

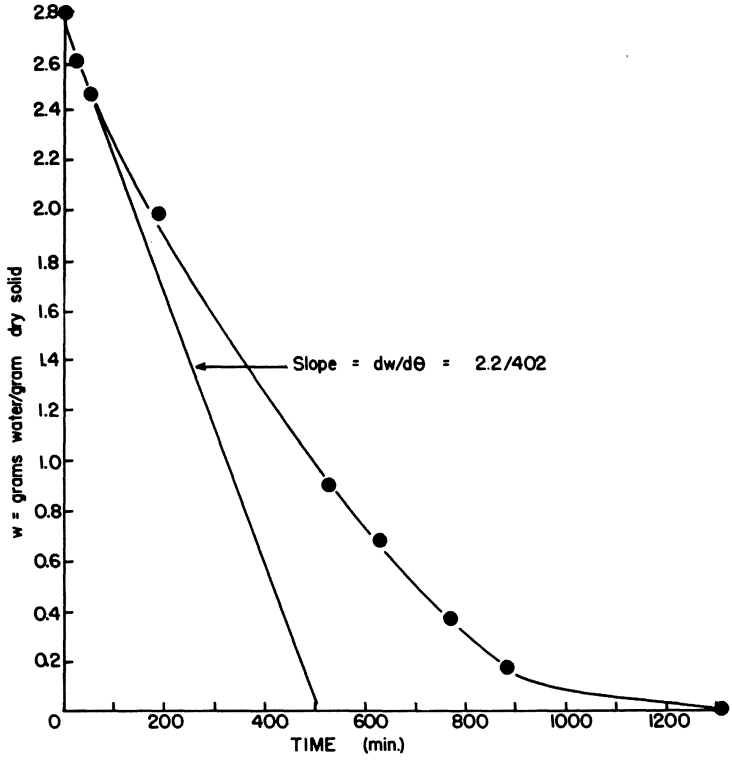


Figure 11.2. Drying curve showing slope for constant rate period. *Courtesy of Charm (1978).*

Chemical Stability

Dried foods may have adverse quality characteristics when they are compared to fresh foods. The alterations in quality may be due to enzymatic activities, protein denaturation, oxidation of lipids, or other reactions during drying, storage, or reconstitution. The three factors that generally determine the type and extent of deterioration of dried food are residual moisture, oxygen content, and storage conditions (time and temperature) (McWeeny 1980). Quite often, a number of preservation techniques must be used to obtain the desired dried food. For example, a food may be heated to inactivate enzymes, dried to a low moisture, treated with a chemical antioxidant, and packaged in an inert gas such as nitrogen in a sealed container.

There is an optimum moisture content for the best storage stability of each food. Complete dehydration may result in destruction of the

food. The monomolecular layer of moisture, referred to as bound water, may be regarded as a protective film that protects the particles of food from attack by oxygen.

Dried foods must be packaged in moisture-impermeable materials or stored in an atmosphere with a relative humidity similar to that of the food. In mixtures of various dehydrated foods, moisture is transferred from items of high moisture vapor to those of lower moisture vapor until equilibrium is attained. Quite often, dried mixes include smaller packages of ingredients within the main package in order to maintain the components at the desired moisture level (Hong, Bakshi, and Labuza 1986; Iglesias, Viollaz, and Chirife 1979).

Drying preserves food as long as the material is kept dry. Reconstituted dry products should be used as soon as possible since, after reconstitution, these foods are perishable, being subject to microbial spoilage.

Types of Drying Systems

There are several systems that can be used to remove water from foods. Some types of drying systems are listed in Tables 11.9 and 11.10, and several are depicted in Figures 11.3 through 11.7. The choice of the method depends upon the qualities that must be maintained, the sensitivity of the food to heat damage, the rehydration characteristics, and the cost of the process. In some cases, only one of the systems can be used to obtain a commercially acceptable or marketable product.

The four principal methods of drying are (1) hot air drying, for foods such as vegetables; (2) spray drying for liquids and semiliquids; (3) vacuum drying for juices; and (4) freeze drying. There are various types of

TABLE 11.9. SOME SYSTEMS FOR THE DRYING OF FOODS

Types of Drier	Usual Food Types
Natural (solar)	Grapes (raisins), meats, fish
Bin	Grain, peanuts
Kiln	Apples, some vegetables
Cabinet or compartment	Fruits and vegetables
Tray or pan	Egg white
Tunnel	Fruits and vegetables
Continuous belt (atmospheric or vacuum)	Vegetables
Fluidized bed	Potato granules, coffee whiteners, spices
Roller or drum (atmospheric or vacuum)	Milk, fruit and vegetable purees
Spray	Milk, eggs, pureed foods
Foam-mat	Fruit and vegetable juices
Freeze vacuum	Chicken, shrimp, meat, coffee
Microwaves	Pasta products, finish drying of foods

TABLE 11.10. COMPARISON OF SOME DRYING SYSTEMS

Type	Advantages	Disadvantages
Continuous belt (conveyor-convection)	Relatively low cost; high production rate; continuous process	Slow drying; large space needed; slow product rehydration
Fluidized belt	Gentle to product; low-energy operation	Difficult cleaning of equipment
Spray	For liquid products or high moisture content; can be used for heat-sensitive foods, short drying time, and retention of flavor, color, and nutritive values	May require product concentration prior to drying and product must be in solution, paste, or slurry form.
Freeze vacuum (freeze dry)	High retention of color, flavor, and nutritive values; rapid rehydration of product	Time consuming and costly; batch operation
Microwave	Efficient energy usage; very short drying time; high production rates; space savings	Delicate products may puff; less economical for high-moisture (over 50%) products; may be difficult to work with microwaves

driers that use these principal methods. For some products, combinations of these methods are used.

A combination of explosion puffing with hot air drying increases the efficiency of dehydration as well as rehydration of the dried product (Sullivan and Craig 1984). Also, the addition of low-frequency sound waves allows the use of lower temperatures during drying (Swientek 1986).

NATURAL. Natural air drying is simply the exposure of the product to the sun and the wind.

Fish and meats are dried by natural methods in some areas of the world. Since natural drying of fish is a slow process, if the temperature is not satisfactory, microbial growth can occur and cause spoilage before the moisture content is reduced to a level that will inhibit growth. It is difficult to preserve fatty fish by this method.

Raisins can be produced by sun drying of grapes. Besides the simple placement of grapes in trays in the sun, solar driers have been developed for various agricultural products so that the food is protected from rain (Eissen, Muhlbauer, and Kutzbach 1985; Wagner, Coleman, and Berry 1984).

To prevent spoilage of coconut during sun drying, a thin coat of acetic acid was applied to the kernel surface (Nathan et al. 1979).

CONTROLLED. Controlled air drying is used to reduce the moisture content of various crops to a level acceptable for storage. Warm or hot air is blown through bins of the product. High temperatures affect the enzymes, germination, and nutritive value of grains. Barley may become unacceptable for malting when used in beer making if it is subjected to very high air temperatures.

Air-suspension drying is used as a finishing stage of some spray-drying operations to obtain very low moisture levels of food. This is similar in principle to the fluidized bed dryer.

The vibrating conveyor dryer is used to redry powdered products that are moistened in the instantizing process.

Vacuum Drying. In this process, the food is placed in a chamber that can be evacuated. Water is evaporated at a lower temperature than at atmospheric pressure. At lower temperatures, less damage to the quality of the food occurs.

Tunnel Drying. This system uses drying tunnels through which carts or trucks containing trays of product to be dried are moved (Fig. 11.3). Hot air is blown parallel or countercurrent to the movement of the food.

Fluidized Bed Drying. Fluidized beds are used for freezing, blanching, transporting, and other procedures, as well as for drying (Fig. 11.4).

Fluidized bed dryers hold granular material in a turbulent suspension by aeration. Hot air supports and dries the food particles. Due to the intimate mixing, the process has a high thermal efficiency. It is particularly suited for the final drying of partially dried particles.

To overcome problems of the fluidized bed, centrifugal fluidized beds have been developed (Hanni, Farkas, and Brown 1976). In this system, the food particles are subjected to a centrifugal force greater than gravitational force. This increases the apparent density of the particles and, by varying the centrifugal force, achieves smooth fluidization of the food at any air velocity. With the increased air velocity, heat transfer is improved, which helps prevent scorching or surface heat damage of the food.

Drum Drying. In this system, the liquid or liquefied food is run as a thin layer or film onto a revolving heated drum. The moisture is removed almost instantaneously. The dried product is scraped off in sheets or broken into flakes as the roller is turned slowly (Fig. 11.5).

The drum drier is used in the production of dried foods, such as sweet potato flakes, tomato flakes, instantized breakfast cereals, starch, and soup mixes (Manlan et al. 1985).

Spray Drying. In this process, a liquid is sprayed with heated air into a drying chamber. The atomized substance gives up its water very rapidly.

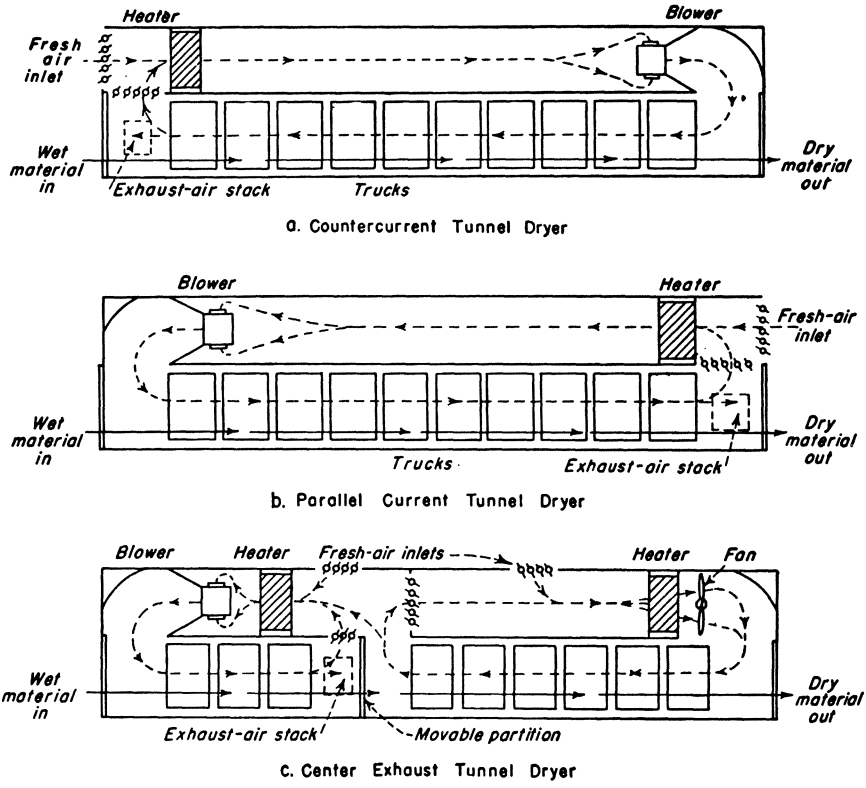


Figure 11.3. Three types of tunnel driers.
Courtesy of Charm (1978).

The dry powder falls to the floor or is separated by a bag or cyclone collector, and the air stream goes out of the drying chamber carrying the water vapor (Fig. 11.6).

Since the drying occurs almost instantaneously (5 to 30 sec), spray drying is useful for heat-sensitive as well as other liquids. The very rapid evaporation of water from the atomized particles produces a cooling effect that helps counterbalance the heating effect of the hot air.

Spray drying can be used to dry any liquid that can be atomized. The main foods are milk and eggs, but coffee, tea, fruit juices, cake mixes, flavorings, garlic, extracts, whiteners, and other foods have been spray dried.

Foam Spray Drying. This process is applicable to liquids and slurries. The injection of an inert gas into the feed line of a conventional spray drier produces a foam that dries rapidly.

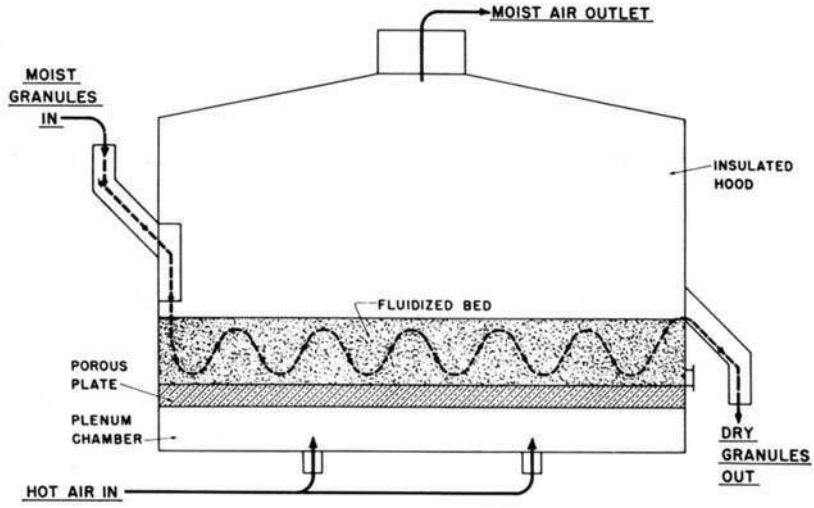


Figure 11.4. Fluidized bed drier.
Courtesy of USDA.

Foam spray drying is said to increase the efficiency and capacity of a spray drier. The product has enhanced flowability, is bulkier, and reconstitutes readily, even in cold water. Milk and eggs have been dried by this system.

Foam-Mat Drying. In this process, the liquid food is concentrated, converted to a stable foam, spread in a thin layer on a supporting surface, cratered by blowing air or an inert gas through the foam, and dried with warm air.

The process has been used for drying a wide range of foods, including fruit juices, egg white, whole egg, whole milk, fruit purées, potatoes, and tomato juice. Most fruit juices should be concentrated to 40 to 45 percent solids prior to foaming.

Freeze Drying. In this process, frozen food is dried under a high vacuum. This is also called sublimation drying, lyophilization, or vacuum contact plate drying (Fig. 11.7).

Sublimation is the conversion of ice to water vapor without passing through the liquid phase (Fig. 11.8). At normal atmospheric pressure, sublimation is a very slow process, but it can be speeded up by reducing the pressure. As ice is sublimed, energy is used. Therefore, it is necessary to supply heat to the food to compensate for this latent heat of sublimation. The heating must be carefully controlled so that the food remains

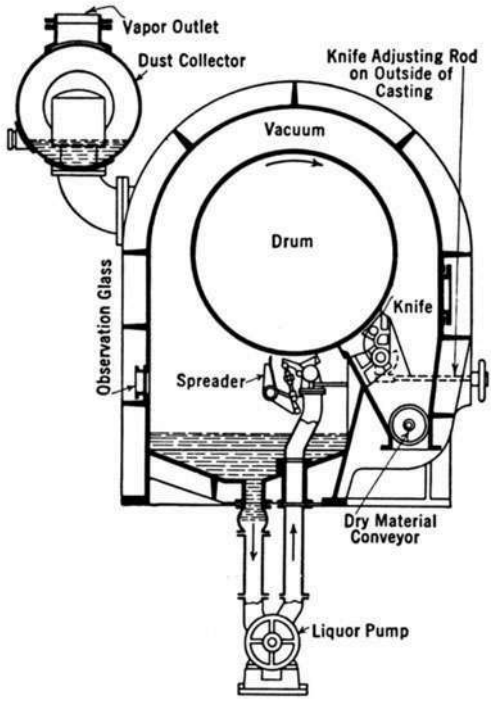


Figure 11.5. Vacuum roller drier.
Courtesy of Farrall (1976).

frozen. Sunderland (1982) discussed the use of microwaves as a source of heat for sublimation of the ice. This system is reported to be more rapid and also less expensive than conventional freeze drying.

Due to the low temperatures involved in freeze drying, the method is especially suited to the dehydration of heat-sensitive foods. Freeze drying occurs with a minimum of discoloration and loss of flavor and nutrients, such as vitamins. Foods retain more of their volatile constituents when freeze dried than when dried by other systems.

Microwave Drying. In this process, microwaves are used to heat the product being dried. With microwave heating, the material being treated is exposed to a high-frequency field. This causes polar molecules of the material to oscillate rapidly. It is this molecular motion that produces heat within the material rather than in the air surrounding the material. The internal heating permits drying from the inside out and avoids extreme surface hardness (case hardening).

Liquid water selectively absorbs the radiation. Hence, as the water is removed from the food, there are fewer water molecules to absorb the

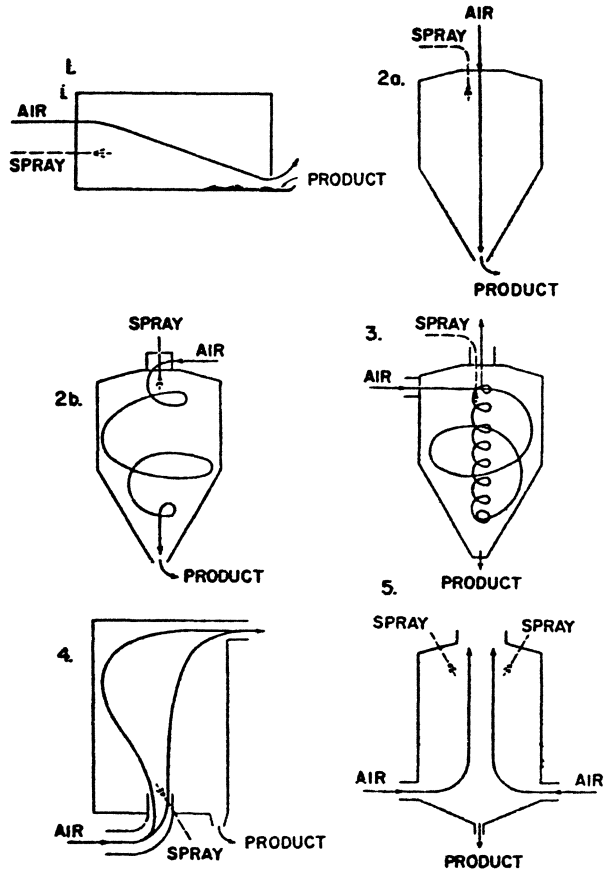


Figure 11.6. Types of spray driers.
Courtesy of Van Arsdel and Copley (1973).

microwave energy, and the process becomes self-regulating. In conventional drying with hot air, the entire food mass, even when dry, absorbs the heat energy.

Heat loss to the surrounding air is minimal, since the heat generated is inside the food product. Maurer, Tremblay, and Chadwick (1971) indicated that the cost of microwave drying was only three-fourths of the cost of conventional drying. Another study (anon. 1974) reported that the energy used was only half of that in conventional drying. Due to the potential for case hardening, conventional air drying of noodles, macaroni, or other pasta products requires from 5 to 10 hr. With microwave heating and drying, this time is reduced to between 15 and 36 min (Maurer, Tremblay, and Chadwick 1971).

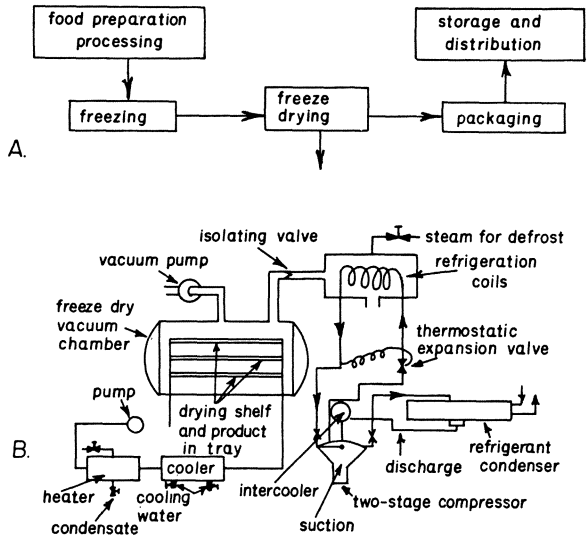


Figure 11.7. Freeze-drying relationships: (*A*) the freeze-drying cycle; (*B*) process and cycle plant flow; (*C*) arrangement of vacuum freeze-drying chamber and condenser.
Courtesy of Farrall (1976).

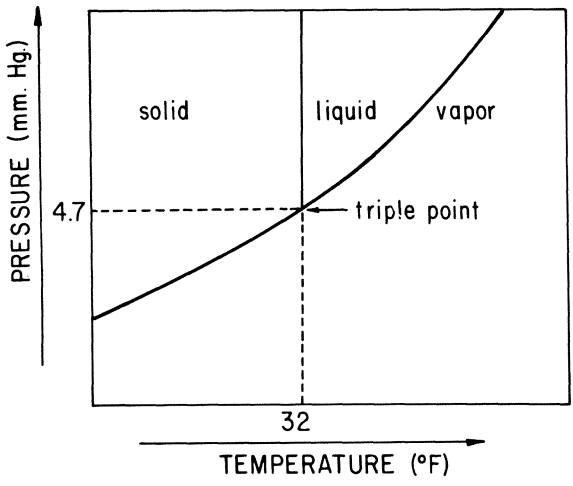


Figure 11.8. Triple point of water. Considerations in freeze drying.
Courtesy of Desrosier and Desrosier (1977).

There may be some adverse reactions that occur during and after the drying of food. Thijssen (1979) considered these to be material losses, physical and chemical changes, and microbial growth and toxin production.

Food Products

Dried milk products and potato products account for most of the commercially dried foods produced in the United States. Milk represents the main dehydrated food in countries in which dairies are prominent. There are many other types of dehydrated foods, including both animal and plant products. Condiments, such as spices, garlic, and onions, along with modified starches, coffee, and beans are prominent dried foods.

Even when dried meat, poultry, and fish products are microbiologically stable, they are subject to oxidative and browning reactions. The removal of all fat and the packaging of the dried product in an inert gas or vacuum helps retard oxidative rancidity. Browning can be controlled in some dried foods by removing the reducing sugars before drying.

The browning of dried fruit may be accompanied by the impairment of flavor, texture, and nutrient value. The rate of browning is dependent on the moisture content, availability of oxygen, storage temperature, and concentration of added SO_2. If the fruit is treated with SO_2, enzymatic browning is inhibited and nonenzymatic browning is retarded. The dehydration and storage of food was reviewed by Cunningham (1982). The drying and storage of dried milk (Ford, Hurrell, and Finot 1983; Hurrell, Finot, and Ford 1983), apples (Bolin, Steele, and Davis 1985; Magee, Hassaballah, and Murphy 1983) and blueberries (Yang and Atallah 1985) have been discussed more recently.

Microbial Aspects

In considering the microbial aspects of dried foods, we should examine the growth or death of cells before drying, during drying, during storage, during rehydration, and in the reconstituted product.

BEFORE DRYING. Foods that are to be dried should be considered perishable foods. They should be handled so that the microbial load is kept as low as possible before drying. The finding of salmonellae in dried milk was related to the isolation of these organisms in the milk plant (Jarl and Arnold 1982). Obviously, the sanitary condition of the processing environment will influence the presence not only of salmonellae, but also of other organisms, in dried foods. The contamination of dried fruit

with mold spores is difficult to prevent because of the prevalence of molds in air around fruit-processing plants.

Heat treatments, such as the blanching of fruits and vegetables, the cooking of meat, poultry, or fish, or the pasteurization of liquid eggs or milk reduces the total microbial count. Freezing in the freeze-drying system and atomization during spray drying may lower the number of viable microorganisms in foods.

DRYING. Drying removes the moisture from a food and lowers its water activity (a_w). If no death occurred during drying, there would be a higher microbial load per gram of dried product than in the original food, due to the loss of water. Usually, the dried product has a lower microbial level than the original food due to the death of microorganisms. Of those organisms that survive drying, some have been stressed and have sublethal lesions.

The lethal action during drying is influenced by the organism (species, strain, physiological age, state of the cells, cell concentration), the condition of drying (freeze, spray, roller, air, moisture, temperature, time, the rate and extent of water loss, residual water content), and the type of food or suspending medium (protective factors, pH, inhibitors).

If drying proceeds at a slow rate and at acceptable temperatures, the microorganisms may multiply during the process. In general, foods being dried should be either at a high or low temperature, so that bacterial growth cannot occur until the water content is reduced to a level that will inhibit microbial action. This is not always possible because some products become overheated and lose their physical qualities if temperatures are too high.

Although the viable microbial load is decreased during drying, the dried product is not sterile, regardless of the system used for drying. In fact, drying is used to preserve cultures of microorganisms.

Freeze drying is the most gentle of the drying methods, with respect to both foods and the microbial population. A high microbial count in the frozen product will be reflected in the dried food. During freezing, drying, storage, and rehydration, there can be a shift in the microbial population. Gram-negative bacteria such as *Pseudomonas, Escherichia,* and *Vibrio* do not survive freeze drying as well as do Gram-positive cocci and the sporeforming bacteria.

Freeze drying reduced the total viable count of 10^4/g in acceptable shrimp to 10^3/g (Moorhouse and Salwin 1970). Shrimp of questionable quality or spoiled shrimp with total counts of 10^6/g had viable counts of 10^3 to 10^4/g after freeze drying. Although no specific microorganisms were determined, it can be assumed that the spoiled shrimp contained pseudomonads.

Freeze drying includes a number of stresses that may cause injury to microorganisms. Cells of *Lactobacillus acidophilus* that survived freeze drying revealed damage to the cell membrane and cell wall (Brennan et al. 1986). The formation of mutants by freeze drying indicates that cellular DNA is altered or damaged. Asada, Takano, and Shibasaki (1979) suggested that the removal of unfreezable water from the cells during drying might induce DNA strand breaks.

During spray drying, aerosols of the liquid food are dried in hot air. Microorganisms are affected by the aerosolization, heating, and drying. When cells are aerosolized there may be changes in the outer layer of the cell envelope, cytoplasmic constituents may leak out, and they may lack control of ion transport. The exact nature of the lesions resulting in permeability changes is unclear.

The lower the temperature of spray drying liquid egg, the better the physical qualities of the powder, but the greater is the survival of bacteria, including the salmonellae. Over 99 percent of the salmonellae in liquid egg are killed during spray drying if the inlet air temperature is 121°C and the outlet temperature is about 60°C. There is a difference in the survival rate of various strains or serotypes of *Salmonella.* Rapid cooling of the powder is essential to maintain the quality of the powder. However, quick cooling also allows greater survival of the microorganisms.

The microorganisms associated with various dehydrated foods were determined by Powers, Ay, and El-Bisi (1971). Of 129 samples, 93 percent had less than 10,000/g (total aerobes), 98 percent had less than one coliform in 10 g, and 99 percent revealed no fecal coliforms. Of 104 samples, all of the 5-g subsamples were negative for coagulase positive *S. aureus,* and 98 percent of the 10-g subsamples yielded no *Salmonella.* Only 102 samples were tested for fecal streptococci, and 93 percent of these had less than 100/g.

Graikoski (1973) reviewed the microbiology of dried fish. The dried product could be produced with less than 100 microorganisms per gram. However, the analysis of commercial dried fish revealed aerobic bacteria at 10^4 to 10^6/g (Ito and Abu 1985), or 10^6 to 10^7/g (Valdimarsson and Gudbjörnsdóttir 1984). The majority of surviving bacteria were Gram-positive heat-resistant micrococci, with Gram-negative cocci comprising the remainder.

The plate count of dried vegetables is usually 10^3/g or less. The flora varies with the vegetables. The organisms include molds, actinomyces, micrococci, staphylococci, bacilli, enterococci, and coliforms. Clostridia may be found in some samples.

STORAGE. The removal of moisture lowers the a_w of a food. The a_w is related to the spoilage rate and type of spoilage that may occur in a food

during storage. Lowering of the a_w can result in the selective growth of microorganisms. Thus, it is common to see mold growth on dried prod ucts stored in humid conditions.

At 0.70 a_w and below, most microbial growth is inhibited. However, some xerophilic fungi can grow at a_w of 0.65 and osmophilic yeasts at an a_w of 0.60. To recommend safe a_w limits for foods, consideration must be given to the nature of the food, its history, processing, packaging, and conditions of storage and handling.

Foods are usually dried below the moisture levels that will inhibit microbial growth in order to retard chemical deterioration, which occurs more readily at 0.65 to 0.70 a_w than at lower a_w levels.

The unequal distribution of moisture can occur in various types of dried foods. Thus, there may be areas or pockets that contain more moisture than is indicated by the analysis of the entire sample. Molds, or even bacteria, might grow in these pockets of moisture.

Factors affecting survival include the species and strain of the organism, morphology (spores), age and mass of the culture, treatments before storage, moisture content of dried material, composition of the food or substrate, and the conditions of storage (temperature, time, and atmosphere).

Although organisms causing spoilage tend to lose viability during storage, components of the cells, such as enzymes, remain active and contribute to undesirable changes in food. Hence, keeping the initial microbial count low is important in the maintenance of quality during storage of dried food.

Species of *Salmonella* that survive drying of eggs die during storage of the dried product, even at low temperatures. The storage of dried albumen at elevated temperatures (50° to 60°C) will inactivate any salmonellae in a matter of a few days to a week. The time needed for this inactivation is influenced by number of survivors, the temperature and the moisture content, or the a_w of the dried powder. Inactivation is greater at 10 percent moisture than at 2 percent moisture.

The initial death rates of salmonellae in dried fish meal were not influenced appreciably by a_w from 0.54 to 0.71 (Doesburg, Lamprecht, and Elliott 1970). Lowering the a_w to 0.34 reduced the death rates, especially in meal stored at 15° or 20°C.

During storage of dried milk, the viable bacteria decrease in number. *Streptococcus durans* dies quite rapidly in spray-dried milk powder, while *S. thermophilus* remains viable for longer periods. Micrococci survive better than streptococci and, as the storage time is lengthened, aerobic sporeformers (bacilli) tend to dominate the viable population. Storage of dried milk for three months will inactivate more than 70 percent of the bacteria.

Survival is highest for organisms stored in vacuum and lowest for those stored in air or oxygen (Cox and Heckly 1973). Following dehydration, oxygen can kill *Serratia marcescens* (Cox, Gagen, and Baxter 1974). The site for the toxic action of oxygen lies in the interspace between the cytoplasmic membrane and the cell wall. The presence of reducing substances helps prevent the lethal action of oxygen.

Because microorganisms die during the storage of dried foods, it is difficult to correctly assess the microbial quality of these products. This is especially true if there is an unknown interval between drying and examination for viable microorganisms. A total microscopic count may be useful in determining the microbial character of dried foods.

REHYDRATION. Normally, no bacterial growth occurs in dried food. Reconstitution or rehydration of such food results in a perishable product. This rehydrated food should be used immediately or refrigerated as any other perishable food.

The growth of microorganisms in rehydrated food is influenced by the initial number and types that are present, as well as by the temperature and time for rehydration.

The microbial population of dried animal products becomes essentially mesophilic types. When rehydrated and held at 4°C, there is a shift in types so the psychrotrophs dominate. If the rehydrated food is held at 20° to 30°C, mesophilic types—including potential pathogens—will grow and become prominent.

With the total viable population reduced in dried food, any surviving potential pathogen will have less competition and fewer microbial interactions in the rehydrated food. If not properly handled before use, the food may become a vehicle for foodborne illness.

Due to osmotic forces and plasmolysis, rehydration can cause the apparent death of some cells. Plasmolysis is considered to be the result of the retraction of the plasma membrane away from the cell wall due to osmotic forces.

Sublethal injury prevents some cells from growing as readily as others in the rehydrated food. Since rehydration is necessary for microbial analysis, the effect of rehydration on these injured cells or other cells is important.

The microbial analysis of dried food should be interpreted with certain facts in mind. If inhibitory media are used, injured cells will not grow and be detected. Bacteria play no significant part in the microbiological deterioration of dried foods under normal conditions. This is because there is not sufficient water for their growth. The bacteria that dominate the microbial flora are important only when the food is rehydrated and stored at temperatures acceptable for their growth.

INTERMEDIATE-MOISTURE FOODS

An intermediate-moisture (IM) food is one that is preserved by lowering the water activity by drying or adding solutes, often in combination with chemical preservatives, anaerobic packaging, or both. IM foods have been defined as having moisture levels ranging from 15 to 50 percent and a_w values of 0.6 to 0.9. Regardless of the exact moisture or a_w value, IM foods can be eaten without rehydration, but they are microbiologically stable without refrigeration.

IM foods have certain advantages:

1. They are relatively low in moisture. Hence, they can be considered concentrated in regard to bulk, weight, and caloric content.
2. Their plastic characteristic allows them to be molded into uniformly shaped cohesive blocks to facilitate packaging and storage.
3. They can be consumed with no preparation by the user.
4. They are similar to normal foods in texture.

Foods with a_w 0.60 or less are essentially free from microbial growth. If solutes are used to obtain the low a_w, the food may have unacceptable flavors as a result of the high concentration of solutes needed.

Foods with a_w over 0.85 readily support the growth of microorganisms. Between a_w 0.60 and 0.85, most of the microbial growth is due to molds, with some yeasts growing in this range. By adding antimycotic agents to the food, the microbial growth can be controlled. Since molds are primarily aerobic, reducing the availability of oxygen will limit their growth. Thus, there can be good storage stability at room temperature.

The types of IM foods include high-moisture dried fruits (figs, dates, raisins, prunes, apricots, apples), fig newtons, marshmallows, soft candies, jams, jellies, molasses, honey, syrups, fruit cake, brownies, fruit-filled toaster products, diet bars, pepperoni and other dry sausages, pemmican, jerky, and the semimoist pet foods.

Most of these foods are well known. With the creation of semimoist pet foods, there has been an effort to develop more human foods that have IM characteristics. The effect of blanching and sulfite treatment on IM bananas was discussed by Levi, Ramirez-Martinez, and Padua (1980). Substances that have been added to lower the a_w to obtain IM foods include glycerol, propylene glycol, sodium chloride, sucrose, dextrose, corn syrup, and sorbitol, as well as lactose, sodium lactate, lactic acid, and other hydroxy acids. Various IM foods were reviewed by Erickson (1982).

There are many ways in which an IM food can be developed. So far, systems that have been used are the following:

1. Mix ingredients so that an acceptable moisture level and a_w are obtained. Usually antimycotic agents and solutes to keep the a_w low are included in the formula.
2. Dry a food below a_w 0.85 and, if needed, add antimycotic agents for preservation.
3. Mix ingredients, such as sugar added to fruit, and then evaporate water to a desired consistency and sugar concentration, as in the manufacture of jam.
4. Dry a food and then rehydrate it in a solution containing solutes and antimycotic agents so that the final food has a normal moisture content, but a lowered a_w and preservatives.
5. Soak small pieces of food in a low a_w solution of various chemicals and antimycotic agents. Equilibration will produce a food with lowered a_w and preservatives.
6. Add water-binding agents to maintain a higher water content but a lower a_w.
7. Use mild drying conditions to lower water content in conjunction with the water-binding capacity of natural plant colloids (Gee, Farkas, and Rahman 1977).

By incorporating an effective antimycotic agent, heating to destroy vegetative microorganisms, and adjusting the a_w to 0.85, pet foods sealed in plastic pouches have an excellent record for stability under market conditions.

Equilibration of foods by soaking in solutions of low a_w was discussed by Brockman (1970) for tuna, pork, beef, carrots, celery, pineapple, and macaroni. Water activities of 0.81 to 0.86 were obtained for these foods.

The usual antimycotic agent is potassium sorbate. However, other substances also have antimycotic activities. Propylene glycol acts as a plasticizing humectant for texture of the food as well as contributing to the soluble solids of the aqueous phase. It is also an antimycotic agent. Together, potassium sorbate and propylene glycol act as an antimycotic system to protect IM foods from mold and yeast deterioration. Calcium propionate also may be used to aid in antimycotic activity. The addition of sorbic acid to the surface with a zein coating, and lowering the surface pH were used to control microbial activity of an IM cheese analog (Torres and Karel 1985).

With IM foods, small temperature changes may cause moisture condensation inside the package. Microbial growth can occur in these condensates which have high a_w and usually sufficient nutrients.

One problem with the bacteriological picture of IM foods is the possibility of growth of *S. aureus*. This organism can begin to grow at a_w 0.83 to 0.84, and to produce enterotoxin at 0.86 (Tatini 1973; Troller 1973).

The growth and toxin production depend upon other growth factors being at acceptable levels.

A reduction in the bacterial load has been reported for IM foods stored at 38°C (Brockmann 1970; Kaplow 1970). *S. aureus* at a level of 10^4/g was inoculated into both chicken à la king and ham sauce at a_w 0.85. The number of *S. aureus* was reduced to less than 2,000/g in one month and to about 0.4/g after four months. Inoculation with *E. coli, C. perfringens,* or *Salmonella* gave comparable results.

Labuza, Cassil, and Sinskey (1972) challenged IM foods with various microorganisms. The IM foods were prepared by direct mixing or by freeze drying, and then absorption of moisture to the desired level. The microorganisms showed better growth in the direct mix than in the freeze-dried food system, even though both foods were at the same a_w. *S. aureus* grew at 25°C in the direct-mix food at a_w 0.84. In the freeze-dried system at 0.84, death occurred, but survivors were evident for one month. Pseudomonads showed growth at a_w 0.84 in direct-mixed food, but no growth in food adjusted after freeze drying, even at a_w 0.90. These data indicate that microbial growth is affected by more than the a_w of a food. Presumably, the moisture content must be considered.

CHEMICALS

The chemical preservation of food has a long history. The preservative action of acids, salt, and sugar was recognized by primitive peoples. As discussed in Chapter 4, there are chemical inhibitors naturally present in some foods.

Public Concern

Some consumer groups have argued against the presence of chemicals in foods. There will always be chemicals in foods, simply because foods are composed of chemicals. Some chemicals naturally present in foods are toxic to humans (Chapter 1).

Food Additives

There are many definitions for food additives. According to the U.S. Food, Drug, and Cosmetic Act, a food additive is a substance the intended use of which results or may be reasonably expected to result directly, or indirectly, in its becoming a component or otherwise affecting the characteristics of any food. In 1958, Congress amended the act to restrict food additives to substances specifically approved after that date by the

FDA. Many substances had been used for years without FDA approval, so Congress exempted substances that, through long use, were "generally recognized as safe" (GRAS) by experts in the field. Since then, those chemicals that have been used outside the United States and whose experience has established safety can be included as GRAS. In 1969, President Nixon called for an evaluation of GRAS substances, and in 1979, the FDA began a comprehensive program to determine the safety of these chemicals through information already published.

Food additives have been defined by the FAO/WHO as substances that are added intentionally to food, generally in small quantities, to improve the appearance, flavor, texture, or other storage properties. This definition does not include substances that are not deliberately added but nonetheless find their way into food.

Additives may be intentional or incidental. Intentional additives are added on purpose to perform specific functions. Incidental additives have no function in the finished food but become part of it through some phase of production, processing, storage, or packaging (Davis 1979; Koros and Hopfenberg 1979; Larsson et al. 1983; Miller 1985; Mock and Cress 1986; Pariza 1982). Normally, these residues are present in foods only in trace amounts.

Chemicals Added to Food

There are several major classes of chemicals that are added to foods. These have been listed variously, but usually the function in food is considered. Examples are acidulants, antioxidants, colors, enzymes, flavors or flavor enhancers, functional protein additives, leavening agents, nutritive agents or supplements, preservatives, stabilizers or thickeners, surfactants, and miscellaneous (including anticaking agents, dough conditioners, and sequestrants). According to an FDA press conference in 1973, the FDA uses thirty-one categories for food ingredients or additives. Of the total food additives, chemical preservatives comprise only 3 percent or less on either a weight or a dollar basis.

According to Middlekauff (1974), sugar is the main substance added to foods, being used at an annual rate of about 46 kg per person. Of the other major substances per person, we use salt (6.8 kg per year), corn syrup (3.8 kg per year), and dextrose (1.9 kg per year). All of the other substances total about 4.1 kg per year for each person. Of this amount, the next thirty-two additives, including nutrient supplements (vitamins and amino acids), account for 3.5 kg per year. Although there are thousands of additives, most of them account for a very small portion of the 910 kg of food that a person living in the United States consumes each year.

A more recent estimate is that we consume an average of 681 kg of food per year (Emerson 1981). Of this amount, food additives account for about 500 g. According to Rehwoldt and Van der Have (1986) the food additive used most often is salt, although its use has decreased in recent years. The researchers did not include sugar, corn syrup, or dextrose in their list of the ten most heavily used food additives.

Reasons for Using Additives

There are certain reasons for adding and for not adding chemicals to foods. The reasons for adding chemicals are to

1. Maintain or improve the nutritional quality of food.
2. Reduce waste by enhancing the keeping quality.
3. Make foods more attractive to the consumer.
4. Provide essential aids in food processing.
5. Minimize public health hazards.

With our present technology of food preservation, the use of chemicals is essential. Without food additives, the cost of food would increase. Besides the increased cost, most foods would be less wholesome and lack their present keeping quality.

Reasons to Not Allow Additives

Chemicals cannot be added to foods if they

1. Disguise faulty processing or handling methods.
2. Deceive the consumer.
3. Reduce the nutritive value.
4. Replace good manufacturing practices that could accomplish the same effect.
5. Are unsafe for human consumption.

Strong flavors cannot be added to foods to disguise spoiled or partially spoiled food.

Safety of Chemicals

The safety of chemicals in foods is very important, regardless of whether the chemical is added or is naturally present in the food. The absolute proof of safety of a chemical for human use is difficult to demonstrate. It is much easier to show that a chemical may be unsafe than to

prove absolute safety. For this reason, consumer action groups have a definite advantage over the food industry in the battle against food additives.

To petition for the use of a chemical additive for foods, one must submit certain information to the regulatory agency. This information includes the chemical identity of the substance, the purity, a quantitative method of assay for the amounts that would be expected in a food, its reactivity and stability, solubility, manufacturing process, its intended purpose, and, if it accomplishes this purpose, an environmental impact analysis concerning its manufacture, use, and consumption, suggested tolerances for use, toxicology, biochemistry, and allergic responses (Clausi et al. 1982; Lechowich 1981; NAS 1980).

According to the safety decision tree developed by NAS (1980), the first step in toxicity testing is for acute toxicity, then genetic toxicology as well as metabolism and pharmacokinetics, followed by subchronic toxicity and reproduction, and finally, chronic toxicity studies. Some tests, such as the Ames mutagenicity test, may use microorganisms or cell cultures, while others use one or more species of animals, such as rats, mice, dogs, or hamsters. Even if these tests show a chemical to be positive for toxicity, it does not necessarily mean that the chemical is toxic to humans. The level of feeding to animals is generally considerably higher than would be used in human food. In some cases, one test might indicate that the chemical is carcinogenic in animals, but if the experiment were repeated, a negative result might be obtained. Thus, the safety of a chemical is difficult to guarantee. It is obvious that some chemicals should never be used in foods, while others may be used safely at low levels. Almost anything at a very high level has a toxic effect. There have been instances of people dying from kidney failure after drinking too much water. If a chemical passes the acute and subchronic tests, the chronic effects are determined next.

The chronic feeding tests require a minimum of two years but may go on forever. Two species, such as the rat and dog, are used. A reproduction study through three generations may be needed. As in the subacute studies, complete work-up is done of biochemical and histopathological effects. The metabolic fate of the chemical should be determined, such as the absorption, distribution, concentration in organs or tissues, and metabolic products that result. The mutagenic, carcinogenic, and teratogenic properties of the chemical are determined. The possible interactions with other chemicals in foods or drugs, as well as dietary restrictions, must be known.

It should be evident that to acquire all of this information requires considerable time and money. Even if the tests indicate that the chemical is safe, there is no assurance that it will receive favorable consideration.

When a chemical is deemed to be safe, a tolerance is established. The tolerance or limitation does not exceed the smallest amount needed for the intended purpose, even though a higher amount would be safe. After determining the amount of the chemical that produces any undesirable effect on the test animals, only 1/100th of that amount is normally the maximum allowed for humans. It is likely that humans might react differently than the test animals to the chemical.

Since there are certain chemicals that do have a real value when added to food, and since humans cannot be used as test animals, it is necessary to develop acceptable tests, extrapolate the data to human beings, and determine the risks as well as the benefits that could result from the use of the chemical (Anon. 1978; Casciano 1982; Devoret 1979; Flamm and Winbush 1984; FSC 1978; Gibbs and Kahan 1986; Hattan and Rulis 1986; Schramm 1984; Smith 1984; Taylor 1982; Wodicka 1980). Because regulations for food additives of various countries differ, there is much interest in having some type of international regulations to help world trade of food products (Feldberg 1983; Gangolli 1983; Kroes and Feron 1984; Wonnacott and Jukes 1986).

Chemical Preservatives

A chemical preservative can be defined as a substance that is capable of inhibiting, retarding, or arresting the decomposition of food. This definition includes microbial inhibitors, as well as antioxidants, acidulants, and sequestrants. In this text we are concerned mainly with controlling the deterioration of foods by microorganisms. Besides the requirements described for all chemical additives, there are requirements for food preservatives. Chemical preservatives should

1. Provide an economical means of preservation.
2. Be used only when other preservation methods are inadequate or not available.
3. Extend the storage life of food.
4. Not lower the quality (color, flavor, odor) of the food.
5. Be readily soluble.
6. Exhibit antimicrobial properties over the pH range of the particular food.
7. Be safe at levels that are needed.
8. Be readily identified by chemical analysis.
9. Not retard the action of digestive enzymes.
10. Not decompose or react to form compounds of greater toxicity.
11. Be easily controlled and uniformly distributed in the food.

12. Have a wide antimicrobial spectrum that includes the spoilage types of organisms associated with the food to be preserved.

Such factors as how the food is to be processed, packaged, stored, and distributed, composition of the food, other preservation processes, the types and number of organisms to be controlled, shelf life, and cost should be considered when selecting the antimicrobial agent or agents to be used.

Chemical preservatives should be used only when other methods to control microorganisms are lacking, damaging to the product, or very expensive. It has been suggested that, except in a limited number of cases, the use of chemical preservatives would be unnecessary if strict attention were paid to sanitation and hygiene during processing and adequate refrigeration were used during the various stages after processing. Chemical preservatives do add a margin of safety from possible abuses by people involved in any of the postprocessing stages. This is applicable especially to the actions of the consumer.

In considering the possible hazards associated with the use of chemical preservatives, it is important to consider the foods in which they are used. The possible adverse effects to humans are likely to be greater if such preservatives are used in foods such as bread, meat, milk, or fresh fruits and vegetables, which may be consumed in substantial amounts daily, than in foods that are eaten only occasionally in small amounts.

Ideally, the chemical will inhibit or kill the important microorganisms and then break down to harmless, nontoxic substances. The chemical should not decompose so fast that it is ineffective. Slow inactivation of microorganisms may lead to unsuccessful preservation. Nonsterilizing amounts of antimicrobial agents are selective and influence the type of microbial flora that can survive in a food system. The degree of inhibition varies with the preservative, and the amount of inhibition is determined by the concentration of the chemical.

ACTIVITY OF PRESERVATIVES. The factors that affect the microbial activity of chemical preservatives include such things as the type of chemical and its concentration, the type of organism and its physiological state, the numbers of organisms, the composition of the food and its pH, and the temperature of storage.

Bacterial spores are the most resistant type of microorganism. Fungal spores are more resistant than vegetative cells to the action of preservatives. In many cases, molds are more susceptible than yeasts to inhibitors. There is a variation in resistance to chemicals between species and between strains of the same species. The higher the microbial load, the

greater the amount of chemical preservative needed to accomplish inhibition or death of the cells.

Actively growing microbial cultures are susceptible to killing agents. As the culture ages and becomes inactive, the cells tend to become more resistant to antimicrobial conditions. Sporicidal agents can be considered as chemical sterilizers, since they have the potential to destroy all microbiological forms of life. The apparent lethal or inhibitory action on microbial spores may be due to either inhibition of germination or of outgrowth. Only a few agents are capable of killing all types of microorganisms, including spores.

Many preservatives have an increased activity in acid foods. Organic material of the food may react with the preservative, rendering it less toxic or inert to microorganisms. Liquid foods allow better contact between the inhibitor and the microorganisms than do solid foods. In cases in which the microorganisms are inside food particles, they are protected from the effects of the chemical preservatives.

In general, increasing the temperature increases the effect of preservatives on microorganisms. If, however, a low temperature is increased toward the optimum of the organism, the stimulatory effect on growth may outweigh the increased action of the preservative. If the temperature is above the optimum for growth, the increased preservative effect is more pronounced.

Time is an important factor in the use of chemical agents. Chemicals may react with the organisms quite rapidly, or they might require several hours to achieve the desired action on the microorganisms. Hence, the longer the contact time, the more effective will be the action of the chemical preservative.

Chemical preservatives may inhibit the growth of microorganisms (bacteriostat, fungistat), or kill them (bactericide, fungicide, sporicide, or virucide). Whether a chemical is bacteriostatic or bactericidal may be relative. In very dilute amounts some chemicals may act as food sources for the microorganisms. Increasing the amount may be inhibitory, and still higher levels may kill some or all of the microbial cells. In general, the more concentrated the chemical agent, the more effective will be its action. Usually, very high levels are not desired due to the potential adverse effect on food quality and toxicity to humans consuming the food.

We determine the viability of a microbial cell by its ability to multiply. The apparently lethal action on cells may be reversible, so that there is only a bacteriostatic effect. In some cases, the number of killed cells is balanced by the number of cells that are able to multiply, so that, although there is a lethal effect, the action appears to be only bacteriostatic.

The bacteriostatic agents most frequently used are those that lower the a_w below the level needed for bacterial growth. Sugar and salt are two of these important compounds. Besides retarding growth by affecting the chemical or physical nature of foods, preservatives affect microorganisms by several mechanisms, including the denaturation of protein, inhibition of enzymes, alteration or destruction of DNA, the cell wall, or the cytoplasmic membrane, suppression of cell wall synthesis, or competition with essential metabolites.

Interference with Genetic System. In this case, the chemical enters the cell. Some chemicals can combine with, or attach to, the ribosome and inhibit protein synthesis. If genes are affected by the chemical, enzyme synthesis, which the genes control, will be inhibited.

If the logarithm of surviving bacteria is plotted against time, at least a portion of the curve is a straight line. This indicates that the lethal action is a result of the reaction of a single molecule of the cell. The alteration or destruction of the DNA fits the monomolecular reaction requirement. Hence, the effect of inhibitors on the genetic mechanism was thought to be the primary action on the cell. The fact that the curve is not exactly a straight line function indicates that other actions occur.

Cell Walls or Membranes. The chemical does not need to enter the cell to inhibit growth. A reaction on the cell wall or membranes can alter the permeability of the cell. This can impair the passage of nutrients into the cell, as well as allow leakage of cellular constituents from the cell.

Damage to the cell wall alone does not kill the microbial cell. Due to the increased permeability, the chemical is able to enter the cell. Once inside, the chemical may lyse the cell membrane or coagulate the cytoplasm of the cell. Damage to the cell membrane can occur by preservatives reacting with chemically active sites or by dissolving lipid constituents.

Since cell walls are complex polymers, some chemicals can interfere with cell wall production, by affecting synthesis of simple components or by inhibiting the polymerization of the components. Or, the cell wall, when developed, may not be able to satisfy the requirements of the cell.

Inhibition of Enzymes. Since enzymes are proteins, they can be denatured by heavy metals, alcohols, phenols, or surface-active substances. Generally, these chemicals are too toxic for food preservatives.

Wetting agents alter the colloidal properties of enzymes and interfere with enzyme action. High concentrations of salts will curtail the biological activity of enzymes. Sufficient alteration of the pH, either up or down, will inhibit the action of enzymes and prevent the multiplication of microorganisms. Oxidizing agents can inactivate certain enzyme systems that contain active sulfhydryl (-SH) groups. These -SH groups, when oxi-

dized, form disulfide (S-S) bonds that cause the enzymes to become inactive. Reducing agents split the S-S bonds and reproduce the -SH form. The halogens, hydrogen peroxide, and ozone, can oxidize the -SH group. Other mild oxidizing substances also may inhibit microbial cells in this manner.

On the other hand, certain active systems have S-S bonds. These enzymes can be inactivated by reducing agents.

The hydroxyl (OH) group is essential for the activity of certain enzymes. This group can react with chemicals and cause inactivation of the enzyme. Other groups on enzymes that can react with chemical inhibitors include the amine, amide, and carboxyl groups, as well as metals in respiratory enzyme systems.

Another action on enzyme systems is due to antienzymes, such as the trypsin inhibitor of soybeans. These antienzymes are found in a number of natural products, especially in cereals and legumes.

Some chemicals selectively inhibit certain enzyme systems. Although both microorganisms and humans can perform a given enzymatic reaction, the actual enzymes involved, and even the associated coenzymes, can be quite different in many features. Thus, it is possible to selectively inhibit specific enzyme reactions of microorganisms without affecting those of humans.

Binding of Essential Nutrients. Since microorganisms differ in their nutrient requirements, the binding of essential nutrients affects different organisms in different ways. If an organism has few requirements, it will be less affected than an organism that needs many preformed nutrients.

Although hundreds of antimetabolites (analogues of vitamins, amino acids, purines, and pyrimidines) have been synthesized, only a few have been useful to inhibit microbial growth. The sulfonamides that prevent the conversion of para-aminobenzoic acid to folic acid are the classic example of these antimetabolites.

ACTION ON HUMANS. The toxicity of chemicals to humans may be either due to the same mechanism that causes the effect on microorganisms, or the person can develop a hypersensitivity to the compound. This results in an allergic reaction. Inevitably some people will develop a hypersensitivity to any type of substance that is introduced into the body.

When a foreign chemical enters the body, it has three possible fates:

(1) The chemical may undergo spontaneous reactions to other products without the intervention of enzymes. (2) The chemical may be metabolized and transformed into other compounds by means of enzymes of the body. This is the fate of most compounds. The enzymes that metabo-

lize foreign chemicals are located in the liver, intestines, kidneys and lungs. Bacteria in the gut play a role in this metabolism. (3) The inhibitor may be excreted with no alteration. These compounds are usually highly polar types. Many other foreign compounds are metabolized to highly polar types, which are then eliminated from the body.

ACTIVITY OF COMBINATION TREATMENTS. In certain circumstances, combinations of two or more chemical preservatives are more effective than indicated by testing each one individually. For the treatment of diseases, combinations of antibiotics may be used, since, if there is resistance to one antibiotic, there may not be resistance to the second or third antibiotic.

Combinations of chemicals and other treatments may increase the potential for control of microbial spoilage of food. Using low water activity to control bacteria and antimycotic agents to control fungi is discussed for IM foods.

TYPES OF CHEMICAL PRESERVATIVES. Some chemicals that help prevent bacterial and fungal spoilage have been used for centuries. Sodium chloride was used by our primitive forebears to prevent spoilage of meat and fish. Sugar has been used for many years in the making of jams and jellies.

Mold inhibitors, such as the sodium or calcium salts of propionic acid or sodium diacetate, are helpful in controlling mold in bakery products. Mold control in bread was a serious problem for the baker, both in the bakery and after the bread left the bakery. Chemical agents give a margin of safety in cases such as this.

Besides chemicals that are added directly to food, some are applied to the wrapping material to control contamination on the surface of the food. Antimycotic agents that have been used for food wraps include sodium benzoate, sodium and calcium propionates, methyl and propyl parahydroxybenzoates, sorbic acid, and diphenyl.

Some chemicals are added to water that is used to wash food. In these cases, some chemical residual may remain on the food. Fumigants, such as SO_2, used for grapes and in the drying of fruit, leave a residual material on the food.

Some chemicals that are used as preservatives, such as salt and sugar, are not classed as preservatives from a legal viewpoint. Chemicals that are antioxidants and may or may not have antimicrobial properties, are listed as chemical preservatives. There are other chemicals that are added to foods for various reasons that have some antimicrobial properties.

ACIDS. Acids have many functions in foods. An acid condition is unfavorable for the growth of many microorganisms. The preservative effect

of acids may be due to the pH, the undissociated acid molecule, or the anion. With short-chain fatty acids, a decrease in pH enhances the activity to a greater degree than is the case for acids with longer chains. At low pH levels, the undissociated molecules of the weak, short-chain organic acids enter the cell where they dissociate and affect the pH of the cell. This can interfere with the mechanism that transports chemicals across membranes. Also, these acids can interfere with intracellular enzymes. The ionized acid does not enter the cell as well as does the undissociated form.

Branched-chain fatty acids are less active than the corresponding straight-chain acids. Unsaturated acids have been reported to have the same activity or to be more active than the corresponding saturated acids. The substitution of a hydroxyl (OH) group for a hydrogen has been reported to both increase and decrease the activity of the fatty acids. Researchers reported that the OH group was particularly active, but a -SH group caused loss of activity (Kabara et al. 1972). They found that amine derivatives of fatty acids showed activity against both Gram-positive and Gram-negative microorganisms.

It has been suggested that short-chain fatty acids can inhibit or kill both Gram-positive and Gram-negative bacteria. The long-chain fatty acids affect primarily Gram-positive bacteria since they cannot penetrate the lipopolysaccharide layer of Gram-negative bacteria.

The growth inhibition of bacteria by these lipophilic acids has been attributed to the effect on membrane transfer systems of amino and keto acids (Freese, Shev, and Galliers 1973). The long-chain lipophilic acids are more readily partitioned into the cell membrane. This attachment is reversible.

It is well established that an increase in acidity or in hydrogen ion activity inhibits the growth or even destroys certain microorganisms, reduces the germination and outgrowth of spores, and reduces the heat resistance. The thermal death point of yeasts and molds is less affected by low pH than is that of bacteria.

The pH and the type of acid are important in the inhibitory or lethal action of these chemicals. Acetic acid has greater inhibitory and lethal effects on the basis of pH than either hydrochloric or lactic acids.

The exact order of effectiveness depends upon various factors such as the type of microorganism, whether inhibition or death is determined, the pH, temperature, and other environmental conditions of the substrate, as well as the concentration of acid that is used. Comparisons have been made on molarity, normality, percentage, as well as at specified pH levels. These various systems influence the relative amounts of the acids that are present to inhibit the microorganisms. By selecting the correct microorganism, conditions of study, and the basis for comparison, any acid may be or may not be as effective as any other acid as an antimicro-

bial agent. Comparisons should be made on the basis of molarity of undissociated acids, and not on the percentage of the acids added to the system.

Respiratory viruses are generally acid labile. Acids may be quite important in preventing the transmission of viruses by foods. Acids with only one carboxyl group are less active against viruses than those with more than one carboxyl group (Poli et al. 1979).

Acetic Acid (CH_3COOH). This acid is soluble in water and has a pK value of 4.76. The reasons for using acetic acid as a food preservative are its low cost, availability, and low toxicity. Acetic acid as well as its salts—sodium acetate, calcium acetate, sodium diacetate, and calcium diacetate—are GRAS. They can be added to food in accordance with good manufacturing practice. They are effective at pH 4.5 or less. Acetic acid is effective against a wide spectrum of bacteria, being bactericidal to some, such as coliforms and salmonellae. It is less effective against molds than against yeast.

Acetic acid is the principal organic component of vinegar. Vinegar is used in many foods, including mayonnaise, salad dressing, prepared mustard, pickles, marinades, catsup, and pickled beets.

Although acetic acid is an effective inhibitor, it cannot be used on some foods because of its pungent odor. Moreover, it can cause discoloration and textural changes when applied to poultry or meat (Bell, Marshall, and Anderson 1986). However, acetic acid is permitted as an acidifier in processed meat and poultry.

The addition of 0.1 percent acetic acid to poultry scald water reduced the D_{52} values of two salmonellae and *C. jejuni* (Okrend, Johnston, and Moran 1986). At the 1 percent level, acetic acid caused instantaneous death of these organisms. At 0.1 percent acetic acid inhibits most organisms causing foodborne illness, and at 0.3 percent it inhibits molds that produce toxins.

Dehydroacetic acid inhibits bacteria, yeasts, and molds at concentrations of 0.005 to 0.4 percent. It appears to be a better fungistatic agent than a bacteriostatic agent. It can be used as a fungistatic agent in cheese wrappers. Autoclaving of this acid causes only a slight decrease in biological activity.

Dipping certain fruits and vegetables in a solution of 0.1 to 0.2 percent dehydroacetic acid extends the storage life of the produce. At 0.1 percent it inhibits undesirable secondary fermentations in alcoholic beverages, such as beer and wine. In bread and baked pastries, a level of 0.2 percent inhibits microbial growth.

The sodium salt of dehydroacetic acid at pH 5.0 is twice as active as sodium benzoate against *Saccharomyces cerevisiae* and twenty-five times more effective against *Penicillium glaucum* or *Aspergillus niger.*

Sodium diacetate is a loose, equimolecular compound of acetic acid and sodium acetate. It tends to enhance the antimicrobial activity of acetic acid. The action of sodium diacetate is not as pH dependent as is that of acetic acid. This compound is listed as GRAS and can be used at 0.4 parts per 100 parts of flour for use in bread. It is effective in the bread-making industry as an inhibitor of mold and of preventing ropiness of bread. This inhibitor does not interfere with the yeast fermentation. It also may be applied as a dust to the outer surface of bacon or ham or other cuts of red meat or poultry. It inhibits storage molds that commonly occur in foods and feeds (Glabe and Maryanski 1981).

Ascorbic Acid. This is vitamin C. It is present in many foods, especially citrus fruits, and is safe to use as a food additive.

According to one study, ascorbic acid stimulated diacetyl but not acid production by lactic acid bacteria in milk (Richter et al. 1979). Ascorbic acid inactivated enteroviruses (Salo and Cliver 1978), inhibited growth and toxin production of *C. botulinum* type B (Notermans, Dufrenne, and Keijbets 1985), enhanced the antibotulinal effect of nitrite in canned cured meat (Tompkin, Christiansen, and Shaparis 1978a), and showed some inhibitory properties toward *C. jejuni* (Fletcher et al. 1983). The standard plate count of ground beef increased significantly during five days in a display case, even when treated with 0.1 percent ascorbic acid (Shivas et al. 1984). It is evident that ascorbic acid has some inhibitory properties against certain types of microorganisms.

The antibacterial action may be due to breakdown products of oxidized ascorbic acid, or ascorbic acid may cause inhibition in a manner similar to other organic acids.

Benzoic Acid. Besides benzoic acid, sodium benzoate, potassium benzoate, and the parahydroxy esters of benzoic acid (parabens) are used as chemical preservatives in food. Due to its better solubility, sodium benzoate is preferred to benzoic acid, but to reduce the sodium level in foods, the potassium salt is used.

The FAO/WHO unconditional acceptable daily intake of benzoic acid for humans is 5 mg/kg body weight. Cats and rats are particularly susceptible to high levels of benzoic acid. This chemical is excreted by many mammalian species either as hippuric acid or as benzoyl glucuronide. Benzoic acid is naturally present in many plant products, especially cranberries. The acid and the sodium and potassium salts are GRAS with maximum usage in food set by the FDA at a level of 0.1 percent. Since potassium is heavier than sodium, for the same benzoate content and antimicrobial effect, a higher level of potassium benzoate is needed than of sodium benzoate or benzoic acid.

The antimicrobial activity of benzoates is dependent upon the pH of

the substrate, since it is the molecular form that is effective in preservation. Using the formula

$$pH = pK + \log \frac{(\text{ionized})}{(\text{molecular})}$$

and a pK of 4.19, the pH at which various amounts of undissociated acid are present can be calculated, and is listed in Table 11.11. For benzoic acid, the inhibition of microorganisms was due to both the dissociated and undissociated acid (Eklund 1985). However, the optimum antimicrobial activity is between pH 2.5 to 4.0. At low pH values, the dissociated acid is present in larger amounts than at higher pH levels.

Benzoates have been used in fruit juice, jellies, jams, soft drinks, purees, condiments, fruit syrups, margarine, puddings, pickles, and prepared sauces. Benzoic acid suppresses the formation of trimethylamine, a chemical indicator of spoilage of fish. Besides having legal limitations, the use of benzoates may be limited by flavor considerations.

Primarily, the benzoates are used to inhibit yeasts and molds, since most bacteria are inhibited by the low pH levels at which these chemicals are effective. The benzoates cannot be used to correct poor sanitation because they are not effective in highly contaminated foods. Although considered to be fungistatic, Romano and Suzzi (1985) reported the death of *S. cerevisiae* cells at a concentration of 0.05 percent benzoic acid. The yeast spores were not affected. The addition of sodium benzoate to the substrate reduced the heat resistance of yeasts (Beuchat 1981).

Combinations of benzoate with other preservatives may be beneficial. Sulfur dioxide helps prevent discoloration and flavors attributed to ben-

TABLE 11.11 THE pH VALUES NEEDED FOR VARIOUS LEVELS OF UNDISSOCIATED ORGANIC ACIDS

Undissociated Acid (%)	Acids		
	Benzoic	Propionic	Sorbic
99	2.19	2.87	2.75
95	2.91	3.59	3.47
90	3.24	3.92	3.80
80	3.59	4.27	4.15
70	3.82	4.50	4.38
60	4.01	4.69	4.57
50(pK)	4.19	4.87	4.75
40	4.37	5.05	4.93
30	4.56	5.24	5.12
20	4.79	5.47	5.35
10	5.14	5.82	5.70
1	6.19	6.87	6.75
0.5	6.49	7.17	7.05

zoate. Combinations of benzoate and sorbate are believed to be more effective than either chemical used alone.

The esters of parahydroxybenzoic acid have been suggested for preservation of foods. It is only the esters and their salts that are of value as preservatives. The free parahydroxybenzoic acid is of little use in practical concentrations. The antimicrobial activity of the parahydroxybenzoates increases with the length of the alkyl chain of the alcohol moiety, up to the amyl ester. Some mixtures of the esters show greater activity than would be expected by their individual action. These esters are called parabens. The methyl and propyl esters are GRAS for use up to 0.1 percent and the *n*-heptyl ester can be used at 12 ppm in malted beverages and at 20 ppm in noncarbonated fruit-based beverages.

The parabens have an advantage over the benzoates in that they are more effective at higher pH levels, with an active range of pH 3 to 8. They are extremely stable in food systems and can be autoclaved or retorted or frozen without losing their effectiveness.

The parabens inhibit yeasts and molds but are of limited value against Gram-negative bacteria. The inhibition of yeast precludes their use in yeast-leavened baked products.

At levels of 0.03 to 0.06 percent, a 3:1 ratio of methyl to propyl ester is used in cakes, pastry, toppings, or fillings. At 0.03 to 0.05 percent, a 2:1 ratio of methyl to propyl ester is used in soft drinks, fruit extracts, and flavorings. A 0.07 percent level of a 2:1 ratio of methyl to propyl ester is used in synthetic jam, jelly, or preserves. Antimicrobials are not permitted in standardized sugar-sweetened jam, jelly, or preserves.

Parker (1969) suggested the parabens inhibit germination rather than the outgrowth of bacterial spores. The methyl and propyl parabens allow a limited degree of germination of *B. subtilis* spores followed by some outgrowth of vegetative cells.

Citric Acid. Citric acid ($COOH\text{-}CH_2\text{-}C(OH)\text{-}(COOH)\text{-}CH_2\text{-}COOH$) is a common constituent of citrus fruits. It is an acidulant with unique flavoring characteristics and is water soluble.

Although citric acid has been tested as an antimicrobial agent, it is considered to be not as effective as other acids or preservatives. However, at 0.3 percent, citric acid significantly lowered the level of inoculated salmonellae remaining on poultry carcasses (Thomson et al. 1967). The microbial load of fish flesh was lowered to a greater extent by citric acid than by potassium sorbate (Debevere and Voets 1972). Citric acid (0.1 or 0.3 percent) added to fish flesh inhibits the formation of total volatile nitrogen and trimethylamine. The storage of hard-cooked, peeled eggs in a 0.75 percent citric acid solution (pH 5.0) reduced the microbial pop-

ulation and inhibited the growth of inoculated organisms (*S. typhimurium, E. coli, Y. enterocolitica,* or *S. aureus*) (Fisher et al. 1985).

Lactic Acid. This acid (CH_3-CHOH-COOH) is a natural constituent of some foods and is formed by lactic acid bacteria in several fermented foods.

Depending upon the substrate and microorganisms, lactic acid has been reported as having very good, just average, or rather poor antimicrobial properties.

Using lactic acid solutions as a wash for livers or as a spray for carcasses resulted in reduced microbial load on these meats (Smulders and Woolthuis 1985; Woolthuis and Smulders 1985; Woolthuis et al. 1984). Gill and Newton (1982) observed that the inhibitory effect of lactic acid on Gram-negative psychrotrophs on meat appeared to be due to the decrease of pH and not to the undissociated acid. According to Van Netten, Van der Zee, and Mossel (1984), the lethality of cells due to lactic acid has been overestimated because many cells exposed to the acid can repair injury due to the acid in a rich infusion agar containing catalase.

Malic Acid. This is hydroxysuccinic acid (COOH-CHOH-CH_2-COOH), a natural component of apples. It is permitted by the FDA as an optional acidifying ingredient in mayonnaise and other salad dressings. In these products, it has a microstatic action against yeasts and certain bacteria.

Peracetic Acid. This is known also as peroxyacetic acid (CH_3-COOOH) or acetyl hydroperoxide. It is a strong oxidizing agent and will explode if heated to 110°C.

It is freely soluble in water, and the aqueous solutions are quite stable. It reacts slowly with water to form acetic acid and hydrogen peroxide but shows greater bactericidal action than either of these. A 1 percent solution is active against vegetative bacteria, bacterial spores, fungi, and viruses.

The main problem with the use of this chemical is the pungent odor of its vapors and the off-flavor associated with acetic acid that occurs due to residues.

Propionic Acid. This acid (CH_3-CH_2-COOH) is soluble in water and has a pK of 4.87. It is produced by propionibacters during the aging of Swiss cheese. Propionic acid is a constituent of human body fluids as the free acid or its salts. The propionates are GRAS to be used in accordance with good manufacturing practice. In bread and rolls, propionates are limited to 0.32 percent of the weight of the flour; in whole wheat products, the limit is 0.38 percent; and the limit in cheese products is 0.3 percent.

The bacteriostatic and fungistatic activity is greater in acid than in neutral or alkaline substrates, indicating the antimicrobial action is due

mainly to the undissociated acid. Eklund (1985) reported that some inhibition is due to the dissociated acid.

The propionates are effective against molds and have a limited antibacterial activity and relatively no effect on yeasts. Since propionates do not affect yeast fermentation, but do inhibit molds as well as *Bacillus* species causing ropiness, they are ideal for yeast-leavened bakery products.

Propionates are used to control mold in various types of cheese, malt extracts, chocolate, fruits, vegetables, bread, cakes, pies, jellies, and preserves.

Sorbic Acid. This unsaturated carboxylic acid (CH_3-CH=CH-CH=CH-COOH) is water soluble (0.25 percent at 30°C). The sodium and potassium salts are more soluble than the acid, and the potassium salt is usually selected for use. Sorbic acid is more soluble in fat or oil than in water. The sorbates are more effective against fungi than bacteria and at pH values of 6.5 or less. However, Eklund (1983) reported that bacterial inhibition is due to both the undissociated and dissociated acid, and the effect of dissociated acid becomes appreciable with increasing pH levels. Since sorbic acid sublimes when heated, it should be added to food after heat processing. Other precautions for its use have been listed by King (1981).

The sorbates are listed as GRAS. These chemicals may be added to foods in accordance with good manufacturing practice. They are used in foods such as baked goods, toppings (fruit, icing), beverages (fruit juice, carbonated), cheese products, dried fruit, jams, jellies, preserves, margarine, dry sausage, cucumbers, sauerkraut, olives, smoked and salted fish, syrups, wine, and cake (batter). Usually, it is used at levels ranging from 0.01 to 0.1 percent, although higher levels are used in some foods, such as cheese products and fats.

Sorbic acid may be applied to foods in a variety of ways, including direct incorporation into the food, dipping the food in a solution or slurry of the acid, spraying a solution onto the food, application by means of impregnated packaging materials, or addition as a solution in ethanol, propylene glycol, or vegetable oil.

The antimicrobial effect of sorbates and their use in foods has been reviewed (Dziezak 1986; Liewen and Marth 1985; Sofos et al. 1986). The effectiveness of sorbates is influenced by the type of food, pH, water activity, types and numbers of microorganisms, type of atmosphere, and temperature.

To inhibit mold on food such as bread, a 3 percent solution of potassium sorbate is sprayed over the surface of the freshly baked product. The heat from the product evaporates the water leaving the sorbate as an inhibitor. With propionates inside and sorbates outside, the usual

mold-free shelf life can be extended significantly. Spraying a 5 or 10 percent potassium sorbate solution onto cured hams protected them from mold growth for sixty days (Baldock et al. 1979).

Yousef and Marth (1981) reported that potassium sorbate (200–300 ppm) delayed growth of *A. parasiticus* but did not inhibit biosynthesis of aflatoxin. Similar results were reported for higher levels of potassium sorbate (0.05 to 0.15 percent: 500 – 1,500 ppm) on the growth of *P. roqueforti* and the production of patulin (Bullerman 1984).

The addition of potassium sorbate at 500 to 1,000 ppm to the heating menstrum significantly reduced the *D* value of yeasts (Beuchat 1981).

The dipping of beef steak into a 10 percent potassium sorbate solution extended the shelf life for two days (Greer 1982).

The use of potassium sorbate increased the shelf life of poultry (Cunningham 1981; Robach 1979), and seafoods (Bremner and Statham 1983a; Shaw, Bligh, and Woyewoda 1983). Potassium sorbate inhibited the growth of *S. aureus* (Elliott, Tomlins, and Gray 1982; Pierson, Smoot, and Stern 1979).

In an effort to reduce the nitrite level of cured meats, various chemicals were tested for their ability to inhibit growth and toxin production of *C. botulinum.* The sorbates were found to be an effective additive for this purpose (see sodium nitrite).

Treatment of salad vegetables by immersion in potassium sorbate (0.2 percent) solution is not effective and, in some cases, is detrimental in maintaining these foods during refrigerated storage (Priepke, Wei, and Nelson 1976).

The dipping of dates in a 2 percent solution of potassium sorbate does not prevent yeast growth during storage (Bolin et al. 1972). This is due to the pH of dates being near 5.9, so that the sorbates are less effective than in other fruits. The addition of methyl bromide in conjunction with the sorbate dip is effective in preserving dates.

The growth of yeasts in apple juice is retarded by addition of 0.025 to 0.035 percent sorbic acid. However, this level does not control certain species of *Acetobacter.* For protection of apple juice, levels of 0.05 to 0.10 percent potassium sorbate in conjunction with refrigeration have been recommended.

Sorbic acid was added to wine to inhibit yeasts (De Rosa et al. 1983). However, in sparkling wine the researchers found an odor described as pineapple-celery, due to the formation of ethyl sorbate. Hence, they did not recommend the use of sorbates in sparkling wines.

Some microorganisms are able to grow in the presence of large amounts of sorbic acid or its salts. Some organisms, especially molds, can metabolize this additive. Hence, sorbates are not a substitute for good plant sanitation.

Succinic Acid. This acid ($HOOC\text{-}CH_2\text{-}CH_2\text{-}COOH$) was as effective as acetic acid in its ability to inhibit microorganisms on poultry carcasses (Mountney and O'Malley 1965). Although treatment with a 3 or 5 percent succinic acid solution at 60°C effectively reduced the microbial load of poultry, the appearance was impaired (Cox et al. 1974).

Succinic acid occurs naturally in foods such as asparagus, beets, broccoli, rhubarb, sauerkraut, and cheese.

Acetaldehyde. This chemical ($CH_3\text{-}CHO$) has a boiling point of 21°C and is naturally present in some fruit tissues. The vapors are irritating to mucous membranes. The chemical has a general narcotic action. Large amounts may cause respiratory paralysis and result in death.

Acetaldehyde vapor has bactericidal and fungicidal properties. It is effective in the control of various microorganisms causing postharvest decay of fruits and vegetables, as well as yeasts associated with spoilage of fruit juice (Barkai-Golan and Aharoni 1976). The extent of fungicidal properties depends upon the concentration of the vapor and the length of exposure.

Ethyl Alcohol. This is one of the most widely used substances to control microorganisms. It acts by denaturing protein.

Alcohol is bactericidal but not sporicidal. It has poor penetrating power and is readily inactivated by organic matter. It is not considered to be a food preservative, although it can increase the antimicrobial effectiveness of other substances such as sorbic acid in wine. Shapero, Nelson, and Labuza (1978) found ethanol effective against *S. aureus* and suggested that it might be useful in foods with lowered a_w.

Fungicides. Certain fungicides are used for the postharvest treatment of fruits and vegetables to control spoilage. Also, they may be used in foods with low a_w to control mold growth. Fungicides were discussed by Hall (1979). The fungicides include benzimidazole derivatives (benomyl, thiabenzadole), and polyene macrolides (nystatin, natamycin [pimaricin]).

Benomyl is the name given to methyl-1-(butylcarbamoyl)-2-benzimidazolecarbamate. Benomyl dips reduce the rotting of apples by *Gloeosporium* species. Applied to tomatoes as a dipping treatment, benomyl controlled *Botrytis, Penicillium,* and *Cladosporium* (Domenico, Rahman, and Westcott 1972). Benomyl did not control *Alternaria,* which seemed to grow better with benomyl treatment. This was believed to be due to reduced competition from the inhibited fungi.

Thiabenzadole has been widely adopted by apple growers to reduce rotting by *Gloeosporium* species (Edney 1973).

The FDA has permitted the use of natamycin on the surface of cuts and slices of cheese when the standards of identity allow antimycotic

agents. Application can be made by dip, spray, or incorporation into edible or plastic coatings.

Antioxidants. Antioxidants such as butylated hydroxyanisole (BHA) and butylated hydroxytoluene (BHT) also have certain antimicrobial properties. Both of these chemicals are classified as GRAS at levels currently used. In fact, there is evidence that antioxidants prevent the formation of carcinogens in foods such as fried hamburger. They retard the development of oxidative rancidity and extend the shelf life of foods containing fat. Hence, they are chemical preservatives.

One team of researchers reported that BHT is toxic to rodents at doses of 0.25 or 0.50 percent of the diet (Vorhees et al. 1981). These levels are about 100 times higher than would be used in any food.

In their reviews, Kabara (1981) and Davidson and Branden (1981) reported the antimicrobial effects of BHA on bacteria (*P. fluorescens, E. coli, V. parahaemolyticus, S. typhimurium, C. perfringens, C. botulinum, S. aureus*) and molds (*A. flavus, A. parasiticus, Penicillium, Geotrichum*) and BHT on bacteria (*S. senftenberg, S. typhimurium, Mycoplasma synoviae*) and viruses, as well as tertiary butylhydroquinone (TBHQ) on *S. aureus. Pediococcus pentoaceus* was inhibited by TBHQ with the minimum inhibitory concentration (MIC) at 15 ppm (Raccach and Henningsen 1981). According to Post and Davidson (1986) two of the most resistant species to BHA were *S. aureus* and *P. fragi,* while two of the most susceptible they tested were *C. perfringens* and *P. fluorescens.*

Protease secretion of *Aeromonas hydrophilia* was inhibited by BHA (Venugopal, Pansare, and Lewis 1984). When *S. aureus* was treated with BHA, there was a leakage of nucleotides (Degré and Sylvestre 1983) and inhibition of respiratory activity (Degré, Ishaque, and Sylvestre 1983).

Chlorine. The use of chlorine in the sanitation of processing plants and equipment is discussed in Chapter 10.

There have been many hypotheses regarding the site of action of hypochlorite on the bacterial cell. It is generally agreed that the action of chlorine involves the penetration of the cell. There are suggestions that once inside, chlorine reacts with cellular protoplasm, which causes inactivation. There are other suggestions that the chlorine reacts with key enzyme systems and prevents normal respiration. These reactions may involve oxidation or denaturation of the protein. The -SH groups of essential enzymes are possible sites of action. The oxidation theory has been questioned, since oxygen from other sources does not kill bacteria as readily as does chlorine.

There have been suggestions that cell membranes are altered or disrupted by chlorine. This allows diffusion of cell contents out of the cell or toxic materials into the cell.

Sodium hypochlorite preferentially inhibits polymerase deficient bacteria. Polymerase I is active in repair of damaged DNA. Rosenkranz (1973) suggested that the hypochlorite enters the cell and reacts with cellular DNA. According to Roller, Olivieri, and Kawata (1980), the primary site of action is not dehydrogenase enzymes, the protein-synthesizing complex, or DNA.

Using chlorine (1,000 ppm) as a dip for fresh fish extended the storage life of the fish (Kosak and Toledo 1981). When added to chiller water for poultry, the microbial load was reduced in the water and on the poultry carcasses (Lillard 1979; Thiessen, Usborne, and Orr 1984).

Diphenyl. This is also biphenyl or phenylbenzene ($C_{12}H_{10}$). It is colorless with a pleasant but peculiar odor and is insoluble in water.

In experimental animals, the diphenyl is excreted mainly as 4-hydroxy diphenyl. High levels cause an intoxication, with increased respiration rate, loss of appetite and body weight, muscular weakness, paralysis, and convulsion. Respiration difficulties can terminate in coma or death. The LD_{50} for rats is 2.2 g/kg.

Diphenyl is effective against molds but is not very active against yeasts or bacteria. It is used on the inside of shipping containers of citrus fruits, especially oranges. The diphenyl gradually volatilizes and inhibits the growth of molds on the fruit surface.

Although diphenyl may migrate into the peel, very little is found in the juice (usually less than 1 μg/ml). The amount allowed in the whole fruit (including the peel) is 110 μg/g (0.011%).

Ethylenediaminetetraacetic Acid (EDTA). This chemical is a chelating agent. The chemical formula is $(HOOCCH_2)_2\text{-}NCH_2CH_2N\text{-}(CH_2COOH)_2$. The principal function of EDTA is to chelate trace metals that have an adverse effect on the quality of foods. EDTA is colorless, odorless, and tasteless. EDTA is not GRAS, but it has been approved as a food additive by the FDA. The toxicity of EDTA was reviewed by Kabara (1981). There is only a low order of toxicity in animals, and no effects were noted in humans. Apparently EDTA is not carcinogenic.

The main effect of EDTA on microorganisms is as a synergist for other antibacterial compounds. By chelating the trace metals, EDTA deprives the microorganisms of these essential cellular components as well as affecting the metal-containing enzymes.

Although EDTA had only a slight effect on the total bacterial count of fresh fish (Baltic herring and rainbow trout), it had a favorable effect on the keeping quality (Kuusi and Loytomaki 1972). The volatile basic nitrogen, trimethylamine and hypoxanthine, increased at a slower rate in EDTA-treated fish than in the untreated controls.

A combination of lysozyme and EDTA inhibited microbial growth on

shrimp (Chander and Lewis 1980) and *S. typhimurium* on broiler parts (Samuelson, Rupnow, and Froning 1985). At 500 ppm EDTA, *B. cereus* was inhibited, but the addition of iron, zinc, and calcium reversed the inhibition (Bulgarelli and Shelef 1985). Houben and Krol (1985) stated that EDTA was promising for use in pasteurized ham to control *Streptococcus faecium* spoilage.

Hydrogen Peroxide (H_2O_2). Hydrogen peroxide is available as stabilized solutions at various concentrations. Aqueous solutions of H_2O_2 gradually deteriorate to water and oxygen. Excess H_2O_2 can be removed from food by adding catalase to produce water and oxygen. The addition of H_2O_2 to certain foods is a simple and efficient method to inactivate certain undesirable microorganisms.

Hydrogen peroxide has been approved by the FDA for use as an antimicrobial agent in raw milk to be used to make certain types of cheese, and in whey processing. Also, it can be used for sterilizing food-contact surfaces made with polycarbonate resins. The milk can be treated with no more than 0.05 percent H_2O_2. When added to liquid egg white, H_2O_2 is an aid in pasteurization to destroy salmonellae. Milk or liquid egg white treated with H_2O_2 can be pasteurized at a lower temperature than that normally used for this purpose.

The effectiveness of H_2O_2 is influenced by the concentration, the microbial population, the temperature, pH, inorganic ions, the time of contact, and other treatments given the microorganisms, such as heating, ultraviolet light, or other preservatives.

The actual effect of H_2O_2 on cells is not completely known, other than its oxidative action. According to Rosenkranz (1973), H_2O_2 is a mutagen and an inducer of chromosomal aberrations. It is capable of reacting with DNA and its bases. The results of Hoffmann and Meneghini (1979) favored the hypothesis that H_2O_2 reacts in the cell, producing a radical that causes the formation of single-strand breaks of the DNA. Reportedly H_2O_2 affects certain enzyme systems, and it can affect the ultrastructure of certain spores. According to Baldry (1983), H_2O_2 is more effective as a sporicide than as a bactericide. The sporicidal effects of H_2O_2 were discussed by Stevenson and Shafer (1983) and Wang and Toledo (1986).

Lysozyme. This is an enzyme that is naturally present in various biological systems. Egg white lysozyme was suggested as a preservative for saké (Yajima, Hidaka, and Matsuoka 1968). The minimum inhibitory concentration against lactobacilli in saké varied from 1.25 μg/ml for *L. fermentum* to 100 μg/ml for *L. acidophilus.* In sake, lysozyme was very stable. Lysozyme has its optimal activity near pH 7.0.

Systems have been suggested that allow lysozyme to attack the cell walls of Gram-negative bacteria. The use of EDTA in combination with

lysozyme was suggested as a means of controlling salmonellae (Anon. 1976). Miller (1969) suggested that treating Gram-negative bacteria with a mixture of H_2O_2 and ascorbic acid renders the organisms sensitive to lysozyme.

The relatively high cost of lysozyme limits the use of this enzyme as an antimicrobial agent for food preservation.

Methyl Bromide. This chemical is used for fumigation of dried fruits to prevent insect infestation. Methyl bromide has weak bactericidal properties. It is active only on fungi and vegetative bacterial cells.

At a level at which methyl bromide protected dates against microorganisms, the dates developed an off-odor (Bolin et al. 1972). A combination of sorbate and methyl bromide gave good protection of dates with no detectable odor or flavor changes. Methyl bromide is allowed in dates if the residual inorganic bromide is below 100 μg/g.

The fumigation of poultry feed with methyl bromide eliminated salmonellae from the feed. There were no adverse effects on the chicks fed this treated feed (Tucker, Brown, and Goodship 1974). Methyl bromide can cause mental and neurological disturbances in humans and concentrations above 500 mg/kg can be fatal.

Propylene Glycol. This chemical (CH_3-CHOH-CH_2-OH) has many functions in the food industry. It has some antimicrobial properties and has been used as a food preservative. The chemical is relatively nontoxic.

Propylene glycol exhibits activity toward Gram-positive and Gram-negative bacteria. Acott, Sloan, and Labuza (1976) found that the only approved chemical effective in controlling aspergilli and staphylococci in an IM food was propylene glycol. Usually a mixture of 2 percent propylene glycol and 0.3 percent potassium sorbate is used as the antimycotic system in IM meat products.

Sodium Chloride (NaCl). Treating food with salt was one of the earliest methods of food preservation. Salt preservation has been important in the development of the fishing industry. In some areas of the world, salted fish remains an important food product. Besides its preservative function, salt is added to foods to enhance flavor, to aid in the development of color in processed meats and sauerkraut, to aid in the texture of processed meats, yeast-raised baked goods, and certain cheeses, to solubilize myofibrillar proteins that act as binders in cured meats, and to help in the regulation of the rate of fermentation in pickles, sauerkraut, sausage, cheese, and yeast-leavened dough.

In some individuals, the ingestion of sodium has been associated with hypertension. As a result, there is an effort to reduce the consumption of salt by various means. One possibility is to substitute KCl or $MgCl_2$ for

TABLE 11.12. SALT CONCENTRATION IN THE WATER PHASE (OR BRINE) IN FOODS WITH VARIOUS WATER CONTENT

Salt Added (%)	Brine Concentration with Water Content of Food (%)				
	40	50	60	70	80
2.5	5.88	4.76	4.00	3.45	3.03
3.0	6.98	5.66	4.76	4.11	3.61
3.5	8.05	6.54	5.51	4.76	4.19
4.0	9.09	7.40	6.25	5.41	4.76
5.0	11.11	9.09	7.69	6.67	5.88
6.0	13.04	10.71	9.09	7.89	6.98
7.0	14.89	12.28	10.45	9.09	8.04
8.0	16.67	13.79	11.76	10.26	9.09
9.0	18.37	15.25	13.04	11.39	10.11

NaCl in cured meats (Barbut, Tanaka, and Maurer 1986; Keeton 1984; Pelroy et al. 1985; Sofos 1983; Wagner and Busta 1985; Whiting, Benedict, and Cangialosi 1986). In some cases, it appears that sodium can be reduced or partially replaced with other chemicals or metallic ions, but further research is needed before widespread changes are made in food formulations. As discussed by Sofos (1983), there are many factors that influence the effect of salt on microorganisms in a food. The level of salt presently used does not completely inhibit sporeforming organisms, but other things must be considered (refrigeration, pH, heat treatment, and other chemicals or treatments).

Microorganisms vary in their tolerance to salt. The least tolerant are organisms such as *Spirillum,* which can tolerate only about 1 percent NaCl. Most normal organisms have a slightly greater salt tolerance. Generally these are inhibited by 3 to 10 percent salt. The mesophilic Gram-negative rods and psychrophiles are inhibited by 4 to 10 percent salt. The lactic-acid–producing bacteria vary in tolerance from 4 to 15 percent salt. Sporeforming bacteria can tolerate from 5 to 16 percent salt. The organisms in these upper ranges are called halotolerant. Although they grow well in low levels of salt, they can tolerate these high levels of salt and even display some growth. These organisms include the micrococci, staphylococci, and sporeforming bacteria.

The halophilic (salt-loving) bacteria need relatively high concentrations of salt for growth. Some may grow at levels approaching saturation (26.4 percent). Most of the halophiles contain red, pink, or other pigments and are classified in the genera *Halobacterium* or *Halococcus.*

Yeast species vary greatly in their tolerance to NaCl. The two most important factors are thought to be a mechanism to prevent entrance of NaCl to retain a low concentration of salt within the cells and the ability of enzyme systems to react in the presence of high levels of salt. A number of yeast types are able to propagate in pickle brine with 19 to 20

percent NaCl. They are more resistant to salt at pH 3.0 to 5.0 than at higher or lower values.

Salt has been reported to both increase lethality due to heat and to have a protective effect for the cells. Generally, low levels (1 to 2 percent) of salt may protect, while higher levels (6 to 10 percent) may increase lethality of heat.

Although salt at a level of 10 percent in glucose-yeast-salt medium inhibited *A. parasiticus,* it did not prevent the formation of aflatoxin production over a ten-day period at either 13° or 28°C (El-Gazzar, Rusul, and Marth 1986).

There are three main types of microbial spoilage of salted fish. These are sliming, pink (red), and dun. Sliming is characterized by a semigreasy, sticky, glistening layer that is yellow-gray or beige and results in a pungent odor. The dominant organisms involved are Gram-negative rods, although Gram-positive and Gram-negative cocci also are present. These organisms grow optimally with 6 to 8 percent salt.

The pink or red defect is due to growth of red halophilic bacteria, which are species of *Halobacterium* or *Halococcus.* Besides the red color, an off-odor is evident due to the production of indole.

The defect called dun consists of peppered spots ranging in color from chocolate to fawn. It is caused by a halophilic mold, *Sporendonema epizoum,* which has optimum growth at 10 to 15 percent salt.

Sodium Phosphates. Sodium and phosphate combine to form a number of compounds such as sodium hexametaphosphate, sodium tripolyphosphate, and tetrasodium pyrophosphate. These are referred to as polyphosphates. The polyphosphates are added to foods for various reasons. The antimicrobial effects are mainly due to the chelation of metallic ions.

Soaking chicken carcasses in solutions of 8 percent polyphosphates retards the production of off-odor, slime, and discoloration during refrigerated storage. Lipolytic types of bacteria on the poultry seem to be inhibited to a greater extent than proteolytic types.

Dipping of cherries in 10 percent polyphosphate solution inhibited fungal growth up to thirty days at 1.1°C (Post et al. 1968). Untreated fruit showed fungal growth at fourteen days. Sodium tetraphosphate appeared to be the most effective antimycotic of the polyphosphates that they tested.

The type of phosphate and pH influenced the inhibition of *S. aureus* (Jen and Shelef 1986). Inhibition was reduced or eliminated by the addition of magnesium, calcium, or ferrous salts. Sodium acid pyrophosphate improved the antimicrobial properties of ground meat with reduced salt content (Madril and Sofos 1986). Various other antimicrobial activities

of the polyphosphates were reviewed by Sofos (1986), Tompkin (1983), and Wagner (1986).

Sucrose. Although there are many sugars, the common name *sugar* denotes sucrose. Low levels of sucrose are utilized by many microorganisms as a source of energy, but high levels of this chemical can have inhibitory properties. The preservative effect of sucrose is due to the development of osmotic forces and a lowered water activity.

Sugar is used in the preservation of several foods, including sweetened condensed milk, jams, and jellies. Sweetened condensed milk is made by adding up to 40 percent sucrose to fresh milk. The milk is heated in evaporating pans to remove water and then canned as a product with 70 to 75 percent solids. Yeasts and molds are killed by the heat treatment during evaporation, the can keeps out further contamination, and the high sugar content prevents the growth of surviving microorganisms.

The main preservative factor in normal jams is the combination of sucrose concentration and low pH. In an acid medium, sucrose is inverted to glucose and fructose. This increases the number of solute molecules, which assist in lowering the a_w of the product. The low a_w, along with a pH of about 3.0, prevents the growth of most microorganisms.

The control of microbial growth is helped by filling these products while still hot, by heating the surface by steam injection, spraying the surface with a chemical fungicide, or using a cover paper impregnated with a fungicide.

Sulfites. The GRAS list of FDA has included sulfiting agents. The sulfites include sulfur dioxide, sodium sulfite, and the sodium and potassium salts of bisulfite and metabisulfite.

Besides their antimicrobial effects, the sulfites are added to foods to control enzymatic and nonenzymatic browning, for bleaching, antioxidant properties, and as dough conditioners. Hence, they have been added to many foods, but in the United States they have not been allowed as additives of fresh red meats or of foods known to contribute more than 10 percent of the RDA of thiamin.

There have been reports of more than 1,000 adverse reactions, including some deaths, presumably caused by the ingestion of sulfites. These chemicals are not harmful to most people, but some asthmatic people, particularly those who are on steroid drugs, may be hypersensitive to sulfites and experience anaphylactic shock or acute asthma attacks after ingestion of these chemicals. Most of the problems with the chemical have been a result of the consumption of sulfites used on fruits and vegetables in salad bars. As a result, in 1986, the FDA issued regulations banning the use of sulfites on fresh fruits and vegetables, including those in

salad bars. Furthermore, since January 1987, the presence of sulfiting agents in any food at levels of 10 ppm or over must be declared on the label, whether the chemical was added directly or indirectly. Besides legal limits of sulfites, there are technical limits, because, at high levels, they may produce off-odors or off-flavors in certain foods.

There are other chemicals that can be used to help maintain the appearance and retard browning of fruits and vegetables in salad bars.

The antimicrobial action is pH dependent. Sulfur dioxide combines with water to form sulfurous acid ($SO_2 + H_2O \rightarrow H_2SO_3$). This acid has two dissociation constants. These are near pH 2 and pH 7. At pH 7, the HSO_3^- and $SO_3^=$ are in about equal proportions. At pH 5, most of the compound is in the form of HSO_3^-. At lower pH values, protonation of the bisulfite ion results in molecular sulfur dioxide. The lower the pH, the higher is the proportion of SO_2. The free, molecular SO_2 is about sixty times more inhibitory than is the bound form. The molecular SO_2 form has the greatest and the $SO_3^=$ the least antimicrobial effect. The undissociated H_2SO_3 is the only effective molecular species against yeast (King et al. 1981).

Sulfites inactivate certain enzyme systems, such as cytochrome oxidase. The cytoplasmic membrane of bacterial cells is attacked by SO_2, which alters the permeability.

Sulfites combine with acetaldehyde, the normal hydrogen acceptor required for glycolysis, and inhibit fermentation. There is evidence that bisulfite reacts with pyrimidine bases, so the antimicrobial activity may involve the RNA and DNA of microbial cells.

The sulfites show antimicrobial activity against both fungi and bacteria. Gram-negative bacteria are more susceptible than Gram-positive types. Some lactic acid bacteria are fairly resistant to the action of SO_2, even at high concentrations. Yeasts are more resistant to SO_2 than either bacteria or molds.

Antibiotics. The therapeutic value of antibiotics in medicine stimulated research to determine their effectiveness as food preservatives. This revealed that broad-spectrum antibiotics, such as chlortetracycline (CTC) or oxytetracycline (OTC), had greater potential for use in foods than antibiotics such as penicillin. In November, 1955, a tolerance of 7 μg/g for CTC in or on uncooked poultry was established. The following year, the same tolerance was established for OTC. The petition requesting the establishment of the tolerance stated that the cooking of poultry destroyed residues of CTC when used at the accepted level. In 1959, the authorization to use CTC or OTC was extended to raw, whole, or gutted fish, shucked scallops, and unpeeled shrimp. A maximum tolerance of 5 μg/g was established for these products.

The original tolerances were established due to the antibiotics having a useful function in the food and because their use in food was seen as safe. However, due to the hazard of the development of resistant strains of pathogens, the increasing resistance of these organisms, the potential for human hypersensitivity to the antibiotics, the finding of residual antibiotics even after cooking (although these levels were very low and probably insignificant), the lack of significant benefits when used in commercial operations, and the costs and difficulties involved in monitoring all of these aspects, the tolerance of CTC, OTC, or similar antibiotics on or in food was rescinded, and they are no longer permitted in foods in the United States. Antibiotics were never approved for use in fresh meat.

Although antibiotics are not permitted as direct additives, this does not mean that there are no antibiotics in our food, nor does it mean that all antibiotics will be banned in the future.

Several antifungal antibiotics have been suggested to control mold or yeast damage on crops in the field or on fruits and vegetables immediately postharvest. Since the antibiotics are on the surface, they can be washed off quite readily, so there should be little, if any, residual in the food.

As long as microorganisms are associated with food, there is a possibility that antibiotics are present, since, after all, microorganisms produce antibiotics.

One antibiotic that occurs naturally in food products such as cheese, and has attracted much attention, is nisin. This antibiotic is a polypeptide produced by certain strains of *Streptococcus lactis*. It has inhibitory properties against Gram-positive microorganisms.

Some advantages concerning the use of nisin in foods are that it is not toxic to humans, it has no use in medicine, and it does not form cross-resistant mutants. Its medical value is of no significance because it is practically insoluble in blood at physiological pH. It is digested by intestinal enzymes and bacteria, and the large size of the molecule precludes absorption if intramuscular injection is used. It is unlikely that nisin has any effect on the intestinal flora. Nisin has no effect on Gram-negative bacteria and does not inhibit all of the Gram-positive bacteria. Some bacteria produce the enzyme nisinase, which inactivates nisin.

The primary application of nisin is in heat-processed foods. This is due to the action of nisin on spores. Heat fixes nisin on spores and thereby reduces their heat resistance. This makes it possible to heat sterilize foods with reduced heat treatments.

Nisin does not influence the heat resistance of some spores, but its continued presence in the recovery medium reduces the count of survivors. In sufficient concentration, nisin inhibits either the germination of spores or the outgrowth of the cells. According to the review by Hurst

(1972), nisin prevents the outgrowth rather than the germination of spores.

Clostridium botulinum is one of the more nisin-resistant species. Somers and Taylor (1981) found that nisin was less effective in preventing the outgrowth of *C. botulinum* spores in cooked meat medium than in TPYG broth. They suggested that nisin binds to meat particles. Rayman, Malik, and Hurst (1983) reported that up to 550 ppm nisin in combination with 60 ppm nitrite failed to prevent outgrowth of *C. botulinum* spores when added to pork slurries at pH 5.8. Hence, with canned foods, a sufficient process (time and temperature) is needed to assure that *C. botulinum* spores are controlled. The use of nisin would need to be evaluated for each type of heat-processed food.

Nisin-sensitive cells of lactobacilli were killed within 1 min of contact with the antibiotic, apparently because of the loss of ATP (Ogden and Waites 1986). Ogden and Tubb (1985) found that most Gram-negative bacteria and none of the brewing yeasts were resistant to nisin. They suggested the use of nisin to prevent spoilage of worts or beers by lactic acid bacteria.

Nisin is not allowed as a food additive in the United States; however, some countries allow its use in one or more foods.

Since antibiotics can become associated with foods in many ways, some countries, and the Food and Agricultural Organization of the World Health Organization of the United Nations, have set tolerances for some of these agents. For the FAO, these include ampicillin (0.06 μg/g), penicillin (0.06 μg/g), oxytetracycline (0.25 μg/g), and neomycin (0.50 μg/g).

Sodium Nitrite ($NaNO_2$). The history of meat curing by nitrites and nitrates was discussed by Binkerd and Kolari (1975). Besides these chemicals, various ingredients, including NaCl, are used to produce cured meats. In recent years, many chemicals have been tested as adjuncts to or substitutes for nitrites in these foods.

An additive should have a useful function in the food and be safe for the consumer. In cured meats, nitrites are involved with the development of the typical color and flavor as well as having antimicrobial activity, especially against *C. botulinum.* However, the safety of nitrites has been questioned, mainly due to the formation of carcinogenic nitrosamines.

Nitrite is added to the curing mixture to develop the typical color of cured meat. In acid conditions, nitrite forms nitrous acid, which is reduced to nitric oxide. By a series of reactions, the nitric oxide combines with the muscle pigment myoglobin to form nitric oxide myoglobin. Upon heating, this pigment forms the more stable red pigment nitrosyl hemochrome, associated with cooked ham.

According to Kemp, Fox, and Moody (1974), good-quality hams can be produced without either nitrate or nitrite. Hams cured with 182 μg/g nitrite had a more typical ham color than those cured with 91 μg/g nitrite (Brown, Hedrick, and Bailey 1974). Eakes and Blumer (1975) obtained hams with acceptable color when 90 μg/g nitrite was used in the cure. Higher levels of nitrite produced higher concentrations of the cured pigment, but these higher values were not considered to be necessary for color acceptability.

Hence, relatively low levels of nitrite are sufficient for color development. Nitrite levels of 10 μg/g and 50 μg/g of meat have been suggested as sufficient for complete color development of cured meat. Reducing agents such as ascorbic acid accelerate color formation and increase color uniformity and stability.

A cured meat flavor in ham was developed by as little as 50 μg/g sodium nitrite (MacDonald et al. 1980). This level of nitrite was as effective as 500 μg/g in retarding the development of off-odors or off-flavors during aerobic storage at 4°C for seven days. The role of nitrite in cured meat flavor has been reviewed in detail (Gray et al. 1981).

The antimicrobial action of nitrite is due primarily to the presence of undissociated nitrous acid (HNO_2). The antimicrobial activity is pH dependent. Above pH 7.5, nitrite may aid bacterial growth. Between pH 6.0 and 7.0, only a small portion of undissociated HNO_2 is present, so there is only a slight antibacterial effect. Between pH 4.5 and 5.5, the presence of nitrous acid shows a definite antimicrobial effect. Below pH 4.5, the nitrous acid appears to decompose to nitric acid, nitric oxide, and water. Neither nitric oxide nor nitric acid shows the same activity against microorganisms as nitrous acid. In the presence of oxygen, the nitric oxide is oxidized to nitrogen dioxide. This combines with water to form nitric acid and nitrous acid. In canned or vacuum-packaged meat, as well as in the interior of meat, there would not be sufficient oxygen for these reactions to occur. Below pH 4.5, nitrite is less effective, but pH effects help control bacterial growth.

The main benefit of nitrite is the inhibition of *C. botulinum* in canned cured meat. If sufficient heat is used to destroy *C. botulinum* spores, the quality of ham may be affected. The nitrite concentration in vacuum-packed hams delays the onset of toxin formation and spoilage. Despite the fact that both aerobic and anaerobic spores can be grown from canned cured meats, according to an FDA consumer memo, there have been no known outbreaks of botulism caused by processed food containing nitrites.

The inhibition of *C. botulinum* involves the interaction of the concentration of salt and sodium nitrite, the pH, the amount of heating, the number of spores that are present, and the temperature of storage. Ap-

parently, heat injury is not necessary for $NaNO_2$ to inhibit toxigenesis of *C. botulinum* (Pivnick et al. 1967). Curing salts do not alter the heat resistance of spores, but they tend to inhibit the subsequent outgrowth of surviving spores.

Tompkin, Christiansen, and Shaparis (1978b) reported that iron affects the inhibition of *C. botulinum* in canned cured meat. Their hypothesis is that nitric oxide reacts with iron, such as in ferredoxin, in the germinated spore and prevents outgrowth of the cells. When the available iron was increased, the antibotulinal effect was reduced (Tompkin, Christiansen, and Shaparis 1979). The addition of ferric chloride or myoglobin decreased the effect of nitrite on *C. botulinum* (Vahabzadeh et al. 1983). For both studies, the addition of EDTA, which chelates iron, restored the antibotulinal effect of nitrite. Reddy, Lancaster, and Cornforth (1983) also suggested that the inhibition of *C. botulinum* by nitrite is due to the inactivation of iron-sulfur enzymes (especially ferredoxin) by the binding of nitric oxide.

Another team of researchers stored cured bacon inoculated with *C. botulinum* at 7° and 27°C (Christiansen et al. 1974). No toxin was found in samples stored at 7°C for eighty-four days. However, samples formulated with 120 μg/g nitrite or less became toxic when held at 27°C. At low levels of inoculation (210 spores before processing and 52 spores/g after processing), levels of 170 μg/g nitrite or more prevented toxin formation. At higher levels of inoculation (4,300 spores/g after processing), toxin was detected even with 340 μg/g nitrite added to the meat. Increased levels of nitrite decreased the rate of toxin production. The level of nitrite needed to prevent toxin production depended upon the number of *C. botulinum* spores.

The importance of a low pH on nitrite inhibition of toxin production has been established (Christiansen et al. 1975). At normal pH levels (pH 5.8–5.9) in a summer sausage, nitrite levels of 150 μg/g did not prevent toxin production. At lower pH (4.2–4.7), no toxin was evident with nitrite at 50 μg/g.

Sorbic acid or potassium sorbate, at 0.2 percent or higher, alone or in combination with nitrite, has been effective in retarding spore germination, outgrowth, and toxin production (Huhtanen et al. 1983; Ivey et al. 1978; Roberts, Gibson, and Robinson 1982; Sofos et al. 1980; Wagner et al. 1982). With the addition of sorbate, the level of nitrite can be reduced and still be effective in inhibiting *C. botulinum.* There have been several reviews on the effect of nitrites on *C. botulinum* in cured meats (Benedict 1980; Holley 1981; Sofos, Busta, and Allen 1979).

Many organisms, such as some strains of lactobacilli, pediococci, enterococci, and various Gram-negative species, including salmonellae, are relatively resistant to nitrite. Under anaerobic conditions, *L. plan-*

tarum was sensitive to nitrite (Collins-Thompson and Thomson 1986). Nitrite at 150 ppm did not inhibit *Y. enterocolitica* at 5°C and *B. cereus* or salmonellae at 10 to 12°C (Nielsen and Zeuthen 1984). Nitrite did inhibit active transport, oxygen uptake, and oxidative phosphorylation of *P. aeruginosa* (Rowe et al. 1979).

The concentration of nitrite decreases during storage of cured meat and may disappear after several months. Sodium nitrite is considered to be heat labile, and some of it is destroyed during the heat processing used in cooked or canned cured meats. Growth of organisms in stored cured meat may be due to a tolerance for nitrite, or it can be due to the loss of nitrite to such a low level that there is essentially no antibacterial activity.

The loss of nitrite is important not only because there is a reduction in the amount of inhibitor, but also because the type of compounds that are formed may be desirable or undesirable. In a meat model system, about 74 percent of the nitrogen from added nitrite was found as nitrite, nitrate, nitrosothiol, denatured nitrosomyoglobin, and gaseous nitrogen (Miwa et al. 1978). The other 26 percent was found in unidentified compounds. Other researchers suggested that about 30 percent of the added nitrite is bound to protein (Woolford et al. 1976). Woolford and Cassens (1977) reported that between 10 and 15 percent of the nitrite was in the adipose tissue of bacon.

Besides the consumption of cured meats, there are other dietary sources of nitrate and nitrite. According to Rooma (1971), many vegetables contain higher levels of nitrate than do cured meats. High levels of nitrate and nitrite in vegetables were reported by Eerola, Varo, and Koivistoinen (1974), Lin and Yen (1980), and Hall, Stall, and Hicks (1978). Well water and rainwater runoff from fertilized cropland may contain high levels of nitrate (Nicolson 1979; Super et al. 1981). The major source of nitrate is vegetables, while nitrite is primarily from cured meats, cereals, baked goods, milk, fruit juice, and water (McCormick 1982).

According to Emerick (1974), nitrite is ten times as toxic as nitrate. Bartholomew and Hill (1984) stated that ingested nitrate is absorbed from the upper digestive tract and is secreted in sweat and saliva. However, they also stated that 65 to 70 percent of nitrate is excreted in the urine within 24 hr.

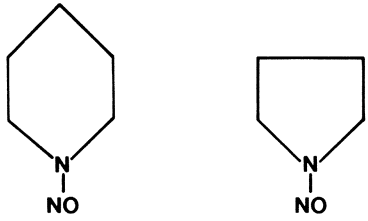
Some nitrite reacts with hemoglobin in the blood, oxidizing it to methemoglobin (CDC 1975). In this form, it cannot carry oxygen to the tissues. The normal reducing systems of the blood can handle some increase in methemoglobin, but when these systems become overtaxed, cyanosis occurs, followed by methemoglobinemia, which can be fatal. Numerous cases of infant methemoglobinemia have been related to excessive nitrite in the diet.

In recent years, the main concern about sodium nitrite has been the reaction of nitrite with secondary and tertiary amines to form nitrosamines. Some, but not all, nitrosamines are mutagenic, teratogenic, and carcinogenic. The mutagenic and carcinogenic aspects of nitrosamines were reviewed by Olajos and Coulston (1978). Various carcinogenic and noncarcinogenic nitrosamines have been listed (Nagao et al. 1977).

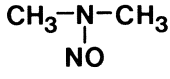
The widespread use of nitrites, coupled with the natural occurrence of amines in foods, has resulted in the possibility that these carcinogens are present in our food supply or are produced in the digestive tract.

There are many nitrosamines, but the common ones are *N*-nitropyrrolidine, *N*-nitrosodimethylamine, and *N*-nitrosopiperidine (Fig. 11.9). Nitrosamines have been detected in rubber baby bottle nipples and pacifiers (Havery and Fazio 1983; Österdahl 1983), vegetable oils and margarine (Hedler, Schurr, and Marquardt 1979; Sen and Seaman 1981a), malt and beer (Goff and Fine 1979; Havery, Hotchkiss, and Fazio 1981; Sen, Tessier, and Seaman 1983), seafoods (Fong and Chan 1976; Knøchel and Huss 1984), dairy products (Havery, Hotchkiss, and Fazio 1982; Lakritz and Pensabene 1984; Libbey, Scanlan, and Barbour 1980; Sen and Seaman 1981b; Weston 1983), various Chinese foods (Fong and Chan 1977; Gough et al. 1977), and animal feed (Ragelis et al. 1983). Generally, the nitrosamines are at very low levels or in trace amounts. Also, research has shown the sources for some of these compounds so that contamination can be reduced further (Havery and Fazio 1985; McWeeny 1983; Sen and Baddoo 1986).

When nitrosamines are found in cured meats, they are usually at levels below 10 ng/g (Fiddler et al. 1975; Panalaks et al. 1974). However, some products had levels of nitrosodimethylamine from 90 to 105 ng/g.



Nitrosopiperidine **Nitrosopyrrolidine**



Dimethylnitrosamine

Figure 11.9. Nitrosamines.

Due to the reduction of the amount of nitrite in cured meats, as well as other procedures, most cured meats now have either nondetectable levels of volatile nitrosamines or only trace amounts. However five of nine samples of cured meats contained the moderately volatile nitrosothiazolidine (Sen, Seaman, and Baddoo 1985).

The analysis of spice-cure mixtures yielded nitrosamines at levels of 2 μg/g (Havery et al. 1976) to 25 μg/g (Sen et al. 1974). Due to findings such as these, the FDA and USDA ordered meat packers to quit mixing sodium nitrite with spices in advance of placing the curing mixture into meat products. Federal regulations prohibit the marketing of spice-curing mixtures unless the nitrites and spices are separated by chemical buffer or by packaging.

Although various foods and cured meats have been found to contain low levels of nitrosamines, bacon has been the greatest concern in this regard due to the presence of relatively high levels of nitrosodimethylamine and nitrosopyrrolidine, both of which are carcinogenic. The formation of nitrosamines in cooked bacon is influenced by preprocessing procedures, pH, salt concentration, nitrite concentration, the presence of other additives, ratio of lean to fat, slice thickness, smoking, and method and temperature of cooking (Bharucha, Cross, and Rubin 1979; Lee, Gray, and Pearson 1983; Pensabene and Fiddler 1983, 1985; Pensabene et al. 1980; Reddy et al. 1982; Robach et al. 1980; Sen, Seaman, and Miles 1979; Skrypec et al. 1985; Spinelli-Gugger et al. 1981; Theiler et al. 1981).

One team of researchers reported different levels of nitrosamine formation in bacon cooked by various methods (Pensabene et al. 1974). Baking of bacon resulted in the detection of 12 to 35 ng/g, while microwave-cooked bacon contained from 0 to 3 ng/g. Broiling, pan frying, and the baconer system resulted in levels of nitrosamine between 7 and 20 ng/g.

There has been concern that the mere presence of nitrites and amines in food can result in formation of nitrosamines by reactions in the food, by microorganisms, or in the acidic environment of the stomach. It has been suggested that these precursors should be at a low level in the diet. Lijinsky (1984) found an increase in tumors in rats fed nitrosatable amines, and he suggested reducing the intake of nitrite-containing cured meat. However, Wagner and Tannenbaum (1985) reported that humans on a low-nitrate diet excreted significant levels of nitrosoproline in their urine. When they compared the endogenous production of nitrite, they found that the amount consumed in cured meats was not significant. Another study did not find an increase in the number of tumors in mice fed nitrite with pyrrolidine (Pearson et al. 1982).

To reduce or eliminate nitrosamines, there has been an effort to lower the nitrite level of cured meats without sacrificing the organoleptic characteristics or the antibotulinal effects. Widdus and Busta (1982) de-

scribed six antibotulinal alternatives (α-tocopherol and ascorbate, sorbate, acidulation by lactic acid bacteria, sodium hypophosphite, fumarate esters, and irradiation).

Bacon with acceptable quality can be made with no added nitrite (Tichivangana, Morrissey, and Buckley 1984; Williams and Greene 1979). Researchers found that sodium ascorbate, ascorbic acid, or sodium erythorbate (isoascorbate) markedly inhibit nitrosamine formation in a model system as well as in frankfurters (Fiddler et al. 1973). Both ascorbic acid (vitamin C) and alpha-tocopherol (vitamin E) act as reducing agents or antioxidants and are effective in blocking the formation of nitrosamines. Vitamin C is water soluble and acts in the lean tissue, while vitamin E is fat soluble and acts in the fatty tissue of bacon. Alpha-tocopherol is an effective inhibitor of nitrosamine formation in bacon (Bernthal et al. 1986; Gray et al. 1982). The presence of up to 1,000 ppm of tocopherol did not retard the antibotulinal action of nitrite (Tanaka 1982). Up to a level of 500 ppm, alpha-tocopherol has been approved for use in cured bacon to inhibit nitrosamine formation.

Besides inhibiting nitrosamine formation, according to Raevuori (1975), erythorbate increases the antimicrobial effect of nitrite. Tompkin, Christiansen, and Shaparis (1978c) reported that 50 μg/g nitrite with erythorbate was as effective as 156 μg/g of $NaNO_2$ alone to inhibit *C. botulinum.*

Potassium sorbate has been proposed as an adjunct with nitrite to control the growth and toxin production of *C. botulinum.* Reports that 0.26 percent potassium sorbate with either 40 or 80 ppm nitrite inhibits *C. botulinum* as well as does 120 ppm nitrite were reviewed by Robach and Sofos (1982). According to this review, different types of bacon produced with any of these chemicals were equivalent in organoleptic quality. Although sorbates are listed as GRAS by the FDA, they have not been approved as additives in bacon by the USDA.

Wierbicki and Heiligman (1973) suggested using lower levels of nitrite in conjunction with radiation pasteurization (radappertization). It was determined that the radiation level would need to be less than 0.8 megarad or the treatment should be done at $-51°C$ or less to reduce the adverse effects of the treatment (Terrell et al. 1981, 1982). The irradiation treatment of foods for microbial destruction has not been approved in the United States.

Treating an inoculated meat model with 3,000 ppm of sodium hypophosphite closely resembled the antibotulinal effect of 150 ppm of nitrite (Wood et al. 1986). At 1,250 ppm, monomethyl fumarate scored better than did other treatments tested.

Studies have reported that a system using a lactic acid producer (*Pediococcus acidilactici*) with 40 or 80 ppm nitrite and 0.7 percent sucrose added

to bacon (Wisconsin Process) resulted in a comparable and safer product than bacon produced with 120 ppm nitrite (Tanaka et al. 1985a, 1985b).

Lowering the a_w of bacon to 0.92 or less will inhibit *C. botulinum* growth and toxin production and allow lowered nitrite levels in the curing mixture (Konstance and Panzer 1985).

Compounds such as sodium bisulfite, tannic acid, hydroquinone, ammonium sulfamate, methionine, cysteine, glutathione, and propyl gallate can block the nitrosation reaction (Gray and Dugan 1975). Treatment of bacon with 1,000 μg/g of propyl gallate, piperazine, ascorbyl palmitate, or sodium ascorbate just prior to frying reduces the formation of nitrosamines during heat treatment (Sen et al. 1976).

The legal limits of nitrite have been lowered in recent years. Originally, the limit was 200 μg/g residual nitrite. With this limit, higher levels of nitrite could be added, as long as the residual was 200 μg/g or less. After research showed that nitrosamines were formed in cooked bacon and may be present in other cured meats, an expert panel was established to advise the Secretary of Agriculture.

With the panel's recommendation, the USDA proposed various levels of nitrite, depending upon the type of cure and the specific product. Since bacon is the cured meat of most concern as a source of nitrosamines, the acceptable nitrite levels are most stringent for this food. In 1986, the USDA specified alternative procedures for bacon. Processors may now use any one of the following: (1) 120 ppm of sodium nitrite (or 148 ppm of potassium nitrite) and 550 ppm of sodium ascorbate or sodium erythorbate (isoascorbate); (2) 100 ppm of sodium nitrite (or the equivalent amount of potassium nitrite) and 550 ppm of sodium ascorbate or sodium erythorbate, with an approved partial quality control program to precisely control the level of nitrite and other factors; or (3) 40 to 80 ppm of sodium nitrite (or the equivalent amount of potassium nitrite) and 550 ppm of sodium ascorbate or sodium erythorbate plus added sugar and starter cultures (lactic-acid–producing bacteria), with an approved partial quality-control program to precisely control the level of added nitrite, added starter cultures, and other factors.

In 1978, the USDA established a 10-ppb violative level for volatile nitrosamines in fried, cure-pumped bacon. Hence, any process used must result in a level of volatile nitrosamines of less than 10 ppb.

Smoking. In ancient times it was noticed that meat exposed to smoke from a fire had improved keeping qualities. Since the beginning of refrigeration, the preservation of meat by smoking has not been as important. However, smoking is used to develop distinctive and desirable flavors in certain foods, such as meat, fish, poultry, and cheese. The chemical composition of wood smoke depends upon factors such as the type of

wood, the amount of air, the temperature, and the time. Hickory or other hardwoods are preferred. Soft woods cause unpleasant flavors in the food.

The preservative effect is a result of the combination of drying and the deposition of the chemicals resulting from the thermal decomposition of wood. The amount of smoke deposited on a food is a complex function of the composition and concentration of the smoke, the temperature and humidity, and the nature of the food surface. To enhance the deposition of particles of wood smoke onto the food surface, electrostatic charges can be used.

Several hundred chemicals have been identified in the smoke condensate. These chemicals include acids, alcohols, aldehydes, ketones, phenols, carbonyl compounds, waxes, resins, and tars. The acids, phenols, and carbonyls are recognized as the components that contribute most significantly to the odor and flavor of smoke. Volatile phenolic compounds in the vapor phase of the smoke are primarily responsible for the smoke flavor.

Several of the chemicals found in smoke or on smoked foods normally would not be allowed as food additives. Hence, smoking is one means of using these chemicals in foods.

A number of investigators have detected 3,4-benzopyrene and 1,2,5,6-dibenzanthracene in wood smoke. These polycyclic hydrocarbons are carcinogenic to laboratory animals. Although it is thought that smoked foods are unlikely to be a health hazard, according to Moodie (1970), there is a relatively high incidence of carcinoma of the alimentary tract in populations in which smoked foods are consumed in large quantities. The presence of a nitrosamine, nitrothiazolidine, found in bacon is related to the smoking process. The exact mechanism for the formation of this compound has not been determined.

In a survey, researchers detected polycyclic aromatic hydrocarbons in many smoked foods at low levels (0.5 to 7.0 ng/g) (Malanoski et al. 1968). Fretheim (1976) reported benzopyrene at levels of 0.15 ng/g, as well as several other polycyclic hydrocarbons in smoked sausages. The carcinogenic polycyclic hydrocarbons are derived from the lignin portion of wood. Their production is minimized when the temperature of combustion is maintained below 350°C.

Using liquid smoke is one means of developing a smoked flavor in meat. In liquid smoke production, the smoke is fractionally distilled and the selected fraction diluted with water. This results in a product reportedly essentially free of the water-insoluble 3,4-benzopyrene. However, White, Howard, and Barnes (1971) found concentrations of benzopyrene ranging from 25 to 3,800 ng/g in condensates that settled out of liquid smoke flavors during storage.

Formaldehyde, acids, and phenols are antimicrobial agents. They are regarded as being responsible for much of the antibacterial action in smoked meats. As would be expected, bacterial spores are quite resistant to smoking. Many vegetative bacteria are destroyed by smoking for two or three hours, but most of the spores survive smoking for seven or more hours. The destruction of bacteria is correlated to some extent with smoke density, duration of smoking, temperature of the meat, fish, or poultry, and the humidity. Spoilage of smoked meat is often due to contamination that occurs after processing, with subsequent exposure to conditions that permit microbial growth.

Since smoking does not control spores, including those of *C. botulinum,* the good manufacturing practices for smoked fish (GSA 1977) require heating at 82.2°C for 30 min.

Liquid smoke at 0.08 percent retards but does not stop growth of bacteria. At a level of 0.1 percent liquid smoke has a preservative effect and gives a mild smoke flavor. Together, liquid smoke with NaCl effectively inhibited outgrowth and toxin production of *C. botulinum* in whitefish (Eklund et al. 1982).

Other Chemicals. There are many other antimicrobial agents. Some of these may be acceptable, while others may not be acceptable as food additives. These chemicals include calcium (Conway and Sams 1985; McGuire and Kelman 1984), diacetyl (Jay, Rivers, and Boisvert 1983), dimethyl dicarbonate (Porter and Ough 1982), fumaric acid, methyl and ethyl fumarates (Huhtanen and Guy 1984; Islam, Lirio, and Delvalle 1984), glutaraldehyde (Collins 1986; Gorman, Scott, and Russell 1980; Gorman et al. 1983; Mast and MacNeil 1978), hydroxycinnamic acid (hydroxycinnamates) (Baranowski and Nagel 1984), monolaurin (Beuchat 1980; Kabara 1984), poly(hexamethylenebiguanide hydrochloride) (Thomson et al. 1981), and xylitol (Mäkinen and Söderling 1981). Some of these inhibit only certain types of microorganisms. Although calcium does not inhibit organisms, it tends to make the treated produce more firm and less susceptible to attack by spoilage types.

Combinations. In meat curing, a combination of NaCl and $NaNO_2$ as well as ascorbate or erythorbate is used. There are other cases in which a synergistic antimicrobial effect is noted when two or more chemicals are added together. Doty (1968) reported that a mixture of 10 percent propionic acid, 10 percent lactic acid, and 5 percent citric acids in 75 percent water killed salmonellae. When this mixture was added to meat meal at the rate of 5 percent, all salmonellae were destroyed.

Webster, Fowler, and Cooke (1985) studied a combination of sodium citrate and sodium benzoate at various a_w and pH levels. A synergistic

effect with lactic and acetic acids was noted by Rubin (1978). Sodium salts of these acids plus sodium propionate were synergistic in their effect on yeast (Moon 1983). Potassium sorbate has been used in combination not only with nitrite in meat curing, but also with antioxidants, acids, salt, and EDTA (Davidson, Brekke, and Branen 1981; Kondaiah, Zeuthen, and Jul 1985; Lahellec, Fung, and Cunningham 1981; LaRocco and Martin 1981; Restaino, Lenovich, and Bills 1982; Robach and Stateler 1980).

CONTROLLED ATMOSPHERES

The gaseous environment surrounding a food can influence the types of microorganisms associated with the food. The alteration of the air may result in a controlled atmosphere (CA), modified atmosphere (MA), CO_2-enriched atmosphere, vacuum packaging (VP), or a hypobaric environment.

If a chemical preservative is added to a food, its presence is listed with the ingredients. However, since gases such as CO_2 are naturally present and leave the food when the package is opened, they do not need to be included as an ingredient.

Fruits and vegetables respire during storage, and this respiration contributes to the loss of quality. As the temperature of storage is decreased toward 0°C, the rate of respiration is reduced, and storage life is increased. During respiration, O_2 is used and CO_2 is given off. The regulation of the concentration of O_2 and CO_2 for storage of fresh food is called CA storage.

Developing the desired atmosphere by natural respiration is satisfactory for fruits such as apples. However, this is not satisfactory for perishable vegetables. Deterioration occurs before the optimum atmosphere is obtained by respiration. Rather than depending upon the natural conversion of O_2 to CO_2 through respiration, the gases can be altered by adding CO_2 and removing O_2 until the desired concentrations are obtained. This is useful for produce as well as for nonrespiring foods such as meat, poultry, and fish. The use of controlled or modified atmospheres to store fresh produce was reviewed by Brecht (1980) and Kader (1986).

Different types of fruits and vegetables, and even the varieties of the same type, have different optimum concentrations of O_2 and CO_2 for storage. When the O_2 concentration is reduced low enough to appreciably reduce mold growth, many fruits are affected adversely.

High CO_2 and low O_2 concentrations become a safety hazard for workers. Therefore, controlled atmospheres are not practical in warehouses when produce is constantly being moved in and taken out.

The low O_2 level in CA affects the growth of some molds, and the CO_2 level limits the growth of some microorganisms. In addition, microorganisms are less able to attack ripening fruit than to attack overripened fruit. Hence, the microbial spoilage is retarded by the slowing of the ripening process. The direct inhibition of microorganisms due to gas concentrations seems to be secondary.

Carbon Dioxide

When the concentration of CO_2 is increased slightly above normal, microbial growth is usually stimulated. At high concentrations, CO_2 has been recognized as a means to control the growth of microorganisms in carbonated beverages, beef, poultry, fruits, vegetables, and shell eggs.

Edney (1973) reviewed the storage of apples in CA. Refrigeration and CO_2 will retard the rate of apple rotting. However, if the CO_2 concentration is increased beyond the optimum, increased rotting may result. There were fewer rotten apples at 8 percent CO_2 than at either 4 or 12 percent CO_2. The lowering of the O_2 level from the 21 percent in air to 10 percent reduced the rotting of apples, and a further reduction in rots was noted as the O_2 concentration was reduced from 10 to 3 percent.

The storage of blueberries in CO_2-enriched atmospheres (10 to 20 percent) significantly reduced decay (Ceponis and Cappellini 1985). Storage of grapefruit in low O_2 resulted in increased alcohol in the juice and an undesirable stale flavor (Barmore, Purvis, and Fellers 1983). An atmosphere of 2 percent O_2 and 5 percent CO_2 retarded the decay of immature apricots but was of little benefit when ripe apricots were stored (Brecht et al. 1982). They also reported that different results were obtained for different varieties of peaches when stored in a 2 percent O_2, 5 percent CO_2 environment. Hence, the benefits of CA storage depend not only on the type of fruit or vegetable, but also on the variety. Besides the composition of gases, the humidity, temperature, and permeability of the package are important considerations (Geeson et al. 1985; Wilson, Jay, and Hill 1985).

The packaging of foods subjected to mold spoilage, such as bread, in high CO_2 or N_2 atmospheres with no O_2, may be beneficial in extending the shelf life.

Wilson, Huang, and Jay (1975) stored high-moisture corn (19.6 or 29.4 percent moisture) inoculated with *Aspergillus flavus* in various mixtures of O_2 and CO_2. They found that reduced O_2, increased CO_2, or both delayed deterioration, but growth was not completely stopped for *A. flavus*. In their modified atmospheres, the level of aflatoxin in the corn was always less than 20 μg/kg. They suggested that, in naturally infested corn, the

A. flavus might do better than these inoculated cultures in the modified atmospheres.

Lillehoj, Milburn, and Ciegler (1972) studied the effect of CO_2 on germination and growth of *Penicillium martensii* on corn. At 30°C, the germination fell from 36 percent in air to 2 percent in 60 percent CO_2, and no germination was found with the combination of 10°C and 60 percent CO_2. With either 40 percent or 60 percent CO_2, and storage at 5° or 10°C, there was essentially no penicillic acid formed by this mold. Penicillic acid is a carcinogenic mycotoxin.

The shelf life of meat, poultry, and fish products is extended by storage in a CO_2-enriched atmosphere. The greatest antimicrobial effect is obtained at a level of 100 percent CO_2. However, this concentration affects the color of red meat, but not poultry or fish. Even at 10 percent O_2, beefsteaks did not maintain their bright red color during storage (Bartkowski, Dryden, and Marchello 1982). According to a review by Finne (1982), CO_2 at 15 to 20 percent partially inhibits bacterial growth and maintains the bright red color of fresh meat. Luiten, Marchello, and Dryden (1982) suggested a combination of 60 percent CO_2/40 percent O_2 to increase the shelf life and maintain the color.

Broilers stored at 2°C in 65 percent CO_2 had an increased shelf life of five days over broilers in ice, but this was only one day more than the shelf life of broilers stored in 20 percent CO_2 (Bailey et al. 1979). However, Statham (1984) stated that microbial growth in a CO_2-enriched environment is inversely related to the CO_2 tension in the atmosphere.

For fishery products, rather high levels of CO_2 have been used. Stenström (1985) suggested a 50 percent CO_2/50 percent O_2 mixture for retail packages of cod fillets. In a CA with 100 percent CO_2, the bacterial counts were lower, but weight losses were higher, than for spotted shrimp stored in air (Matches and Layrisse 1985). Storage in CO_2 did not prevent toxin production in fish fillets inoculated with *C. botulinum* (Post et al. 1985). In some cases, toxin was present in the fillets before they were rejected by sensory evaluation.

In muscle foods, high CO_2 atmospheres inhibit the normal spoilage types (*Pseudomonas, Acinetobacter, Moraxella*) but allow the growth of lactic acid bacteria, which results in a sour and acid condition of these foods.

King and Nagel (1967) found a linear correlation between CO_2 concentration and the amount of inhibition on the rate of growth of *Pseudomonas aeruginosa.* The inhibition by CO_2 was observed only when other environmental factors (O_2 tension, pH, nutrients, and temperature) were not optimum. Compared to growth in air, the relative inhibition of 100 percent CO_2 for those organisms tested was greatest for *B. cereus, Brochothrix thermosphacta,* and *Aeromonas hydrophilia* (Molin 1983). Further inhibi-

tion by 100 percent CO_2 was lowest for *E. coli, S. faecalis,* and *Lactobacillus.* The effect of CO_2 on microbial growth and food quality was reviewed by Daniels, Krishnamurthi, and Rizvi (1985).

Vacuum Packaging

This type of package is evacuated before sealing, either by inserting a vacuum probe into the neck of the package, or by placing the package into a chamber and evacuating it. With flexible films, this results in a package that is drawn tightly around the contents. The development of packaging materials impermeable to oxygen and other gases has made vacuum packaging possible. The integrity of the package must be maintained, or the vacuum will be lost. It should be evident that a complete vacuum is not attained, but any residual oxygen is soon utilized by the microorganisms on the food and replaced with CO_2.

Vacuum packaging has been the method for packaging table-ready processed meats. These meats have a longer shelf life than when packaged with air. There is a tendency for a greenish discoloration of vacuum-packaged cured meats. The use of nitrogen packaging retarded the greening of meat loaves and frankfurters (Lee et al. 1984b; Simard et al. 1983a). Vacuum-packaged fresh beef tends to become purple-red rather than bright red. According to Lynch, Kastner, and Kropf (1986), consumers will accept the purple-red color of vacuum-packaged meat.

As with packaging in CO_2 atmospheres, the microbial flora of meat is altered by vacuum packaging. At the beginning, pseudomonads and *Brochothrix* are dominant, but during refrigerated storage, lactobacilli become dominant (Lee et al. 1984a; Nielsen 1983; Simard et al. 1983b). Molds do not grow on vacuum-packaged meat (Carr and Marchello 1986).

Vacuum packaging extends the refrigerated storage life of fresh fish (Clingman and Hooper 1986; Scott, Fletcher, and Summers 1984), as well as poultry (Anderson et al. 1985; Jones et al. 1982; Patterson, Gillespie, and Hough 1984). Notermans, Dufrenne, and Keijbets (1981) reported that potatoes inoculated with *C. botulinum* spores, then cooked and vacuum packed, became toxic unless stored at temperatures below 4°C.

Hypobaric

Hypobaric storage refers to a system with a controlled environment involving low pressure, low temperature, high humidity, and ventilation. As compared to conventional coolers, this system can extend the shelf

life of beef and lamb carcasses, ham, pork loins, and broilers (Mermelstein 1979, Restaino and Hill 1981).

Ozone

Ozone is formed by electric discharge, such as an electric arc through air or by UV radiation at wavelengths of 200 to 210 nm. This production occurs especially at higher altitudes.

Ozone contains three atoms of oxygen. The third atom is loosely bound to the other two. This third atom unites easily with cellular material, resulting in inhibition or death.

Since ozone is a powerful oxidizing agent, it has been postulated that part of the cellular damage is due to the oxidation of -SH bonds. It is likely that the ozone reacts with the double bonds of cell wall lipids, since ozone causes leakage of cellular components and lysis of some microbial cells. The leakage or lysis of the cells depends on the extent of the reaction with cell wall lipids.

Ozone inactivates viruses by breaking the protein capsid into subunits (Kim, Gentile, and Sproul 1980) or by damaging the viral RNA (Roy et al. 1981).

The lethal threshold concentration of ozone for washed cells of *Bacillus cereus* was 0.12 mg per liter and for *E. coli* and *B. megaterium,* it was 0.19 mg per liter. For spores of the bacilli, the lethal threshold concentration was 2.29 mg per liter (Broadwater, Hoehn, and King 1973). Unwashed cells were not killed during five minutes exposure up to 0.71 mg per liter of ozone. Apparently, organic material associated with unwashed cells affected the activity of the ozone.

A part of the microbial effect of UV lamps in meat-storage rooms is due to the production of ozone and the resultant inhibition or death of bacteria. An ozone level of 0.1 mg per liter has little effect on microorganisms in air, but on moist meat surfaces, it exerts a bacteriostatic effect. The continual exposure of the organism to sublethal doses of ozone may eventually cause the death of the cell, particularly at low temperatures. Ozone was effective in destroying microorganisms in water used for chilling broiler carcasses and those on the carcasses as well (Sheldon and Brown 1986). Generally, ozone is not recommended for use on fresh produce.

The ozone level that is toxic to small animals is about 3 to 12 mg per liter. At 0.02 to 0.04 mg per liter, most people can detect the odor of ozone, and at 0.1 mg per liter, it may cause headache, dryness, and irritation of the throat, respiratory passages, and eyes. Ozone is a powerful mutagenic and oxidizing agent.

MICROBIAL INTERACTIONS

The use of microbial fermentations as a means of preservation is well known and is discussed in Chapter 9. Besides the production of acids or alcohols, some microorganisms form bacteriocins and antibiotics that inhibit other bacteria (Chapter 4).

The inoculation of foods with certain microorganisms to control other microorganisms might be unique, although this has been used in fermentation such as in dairy products. The addition of a lactic culture to ground meat significantly retarded the growth of Gram-negative spoilage bacteria (Reddy, Chen, and Patel 1975). There was a less dramatic effect when the lactic culture was used on beefsteaks. Besides a lower count of spoilage bacteria, the pH, volatile nitrogen, and free fatty acids were lower in inoculated than in control samples of meat. The inoculation of lactic acid bacteria onto scallops did not retard spoilage (Bremner and Statham 1983b).

This type of inhibition is limited to certain types of foods. In most instances, it would probably be more beneficial to add the chemicals directly than to depend on inoculated microorganisms to produce them in the food.

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12

Control of Microorganisms by Destruction

The death of some microbial cells occurs during refrigeration, freezing, drying, or chemical treatment. However, these systems are not expected to produce a sterile food. Sterilization is considered to be a process by which all forms of life are destroyed. When subjected to a lethal process, a bacterial culture is reduced at a rate that is approximately logarithmic. This indicates a first-order reaction. The inactivation or death of cells can be determined by the formula

$$K = \frac{\log a - \log b}{t}$$

where K is the death rate, a is the initial number of organisms, and b is the number of cells remaining at time t.

Considering the microorganisms of public health significance, *C. botulinum* is the most difficult to destroy. The main objective is to eliminate this organism from processed and packaged foods.

The inactivation of microorganisms has been attributed to gases, thermal processing, ionizing radiation, and light waves.

GAS TREATMENTS

Gases can be used to sterilize materials that cannot withstand the high temperatures of heat sterilization. Many organic compounds begin to degrade at 71°C and this degradation accelerates as the temperature is increased to 121°C. Volatile food flavors can be altered by heat. Some plastic materials are softened by heat.

Gaseous sterilization offers a means for packaging heat-sensitive products in an essentially microbe-free condition. A gas can not only affect airborne and exposed surface bacteria, but it can also penetrate porous materials to attack the microbial cells. Each gaseous substance has certain advantages and disadvantages.

It is necessary to evaluate the physiological effectiveness and the toxic

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hazards, as well as the retained residuals of any particular gas. Important aspects in the use of gas sterilization are comparative efficiency of different methods of application, effect of humidity and atmospheric gases, resistance of microorganisms, synergism or potentiation of gas mixtures, the mode of action, reaction kinetics, physical and chemical sorption, effects on food quality, photodecomposition, air pollution, worker safety, permeability of materials such as packages, and the nature and physiochemical properties of the material being treated (Berck 1965).

Besides the gases used to inactivate microorganisms, fumigants such as phosphine, ethylene dibromide, and methyl bromide are employed to destroy insects in stored grain.

Ethylene Oxide

Ethylene oxide is a cyclic ether with the structural formula

Ethylene oxide is a gas at ambient temperature and pressure. It boils at 10.7°C and freezes at −111°C. It dissolves in and reacts with water to form ethylene glycol. Mixtures of 3 to 100 percent ethylene oxide in air are highly flammable and explosive. To eliminate this hazard, ethylene oxide is mixed with dichlorodifluoromethane, carbon dioxide, or formic acid methyl ester.

For an effective sterilization process, the gas concentration, time, temperature, and humidity are parameters that must be considered and controlled. Factors such as stratification effects of various gases, diffusion barriers, moisture-reducing effects, temperature gradients, and ancillary gaseous reactions can enhance or limit sterilization. The sterilizing action of ethylene oxide requires three hours or more but is effective against spores as well as vegetative bacteria, yeasts, molds, and viruses. Some advantages and disadvantages of ethylene oxide treatment are listed in Table 12.1.

The ability of ethylene oxide to penetrate certain packaging materials (paper and some flexible plastics) allows foods to be sterilized in sealed containers. The packaging material must be gas permeable, but impermeable to bacteria.

The mechanism by which ethylene oxide inactivates microorganisms is not completely known. A commonly accepted hypothesis is that the microbicidal action is directly related to alkylating activity. The enzymes and other proteins essential for cellular metabolism contain carboxyl, amino, sulfhydryl, hydroxyl, and phenolic groups. Ethylene oxide reacts with these groups by replacing an available hydrogen atom with the alkyl

TABLE 12.1. ETHYLENE OXIDE STERILIZATION

Advantages	Disadvantages
Good penetration of materials	Flammable
Attacks and kills all microorganisms	Process is slow
Little, if any, damage to materials (compared to heat)	Residues (analysis may be needed)
	Expensive
Effective at room temperature	Each process must be closely monitored
Effective with dry products	Toxic
Sterilizes packaged items	May alter nutrients and other quality factors of foods

hydroxyethyl group. The bacterial cell cannot utilize the new chemical and therefore it dies. According to Ernst (1974), the biologically critical reaction occurs with RNA and DNA within the cells. Winarno and Stumbo (1971) observed an impairment of RNA and protein synthesis. They considered this to be an indirect effect resulting from alkylation of DNA components. This alkylation reaction is thought to be important in the inactivation of spores. Spores are usually several times more resistant than vegetative cells to chemical agents. With ethylene oxide, the resistance of some spores is comparable to that of some vegetative cells, while the more resistant spores are only about five times as difficult to inactivate as the less resistant vegetative cells.

Bacillus subtilis var *niger* spores are used to determine the effectiveness of ethylene oxide sterilization. However, these are not the most resistant spores (Dadd and Daley 1980). The apparent resistance of bacterial spores is influenced markedly by environmental conditions that exist during spore production and their preparation and evaluation (Dadd, McCormick, and Daley 1983). Rupture of the spore coat did not sensitize the spores to ethylene oxide (Dadd and Daley 1982).

Ethylene oxide and propylene oxide have been used to sterilize spices, cereals, colloid gums, fruits, dried yeasts, and other items. For the sterilization of paprika, gamma irradiation was more effective than was treatment with ethylene oxide (Franco et al. 1986).

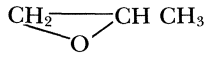
The use of ethylene oxide to sterilize food has been questioned. Ethylene oxide is chronically toxic at levels not detected by smell. The toxicity of ethylene oxide was reviewed by Glaser (1979). Human exposure to ethylene oxide can cause irritation of the skin, throat, and eyes as well as respiratory problems, headaches, lachrymation, nasal discharge, salivation, and labored breathing. Further effects may include nausea, vomiting, diarrhea, lung edema, paralysis, convulsions, and, according to White (1977), death. Rats exposed to ethylene oxide developed various neoplasms (Lynch et al. 1984). A mutagenic effect on human fibroblasts was noted (Star 1980). Embree, Lyon, and Hine (1977) reported mutagenic effects on various cells, including mammalian cells.

Besides formation of the toxic ethylene glycol when hydrated, Wesley, Rourke, and Darbishire (1965) reported that ethylene oxide reacts with naturally occurring chlorides in food to form chlorohydrins. These are toxic and nonvolatile and persist in the food during processing. The researchers reported ethylene chlorohydrin at levels of over 1 mg/g in spices sterilized with ethylene oxide.

Ethylene oxide sterilization should be used only when other sterilization techniques are not practical.

Propylene Oxide

Propylene oxide has the structural formula



It is a colorless liquid with a boiling point of 35°C. It is used like ethylene oxide since it must be diluted with inert gases for safety. It has weak penetrating ability, but it is microbicidal. It requires a higher temperature or longer treatment time to accomplish the same microbial kill as ethylene oxide.

Propylene oxide has been used in various foods to kill 90 percent or more of the microorganisms. Propylene oxide treatments of pecans reduced the surface flora by 80 to 92 percent and the internal microorganisms by 64 percent (Blanchard and Hanlin 1973). Even with high concentrations of propylene oxide, neither bacteria nor fungi could be eliminated completely.

The hydration of propylene oxide yields propylene glycol. This should be acceptable since propylene glycol is used to lower the a_w in IM foods.

However, as with ethylene oxide, propylene oxide can react with chlorides to form chlorohydrins in foods (Wesley, Rourke, and Darbishire 1965).

Propylene oxide can be used as a package fumigant on or in dried prunes or glacé fruit. It can be used on bulk quantities of cocoa, gums, processed spices, starch, and processed nutmeats (except peanuts) when these foods are to be further processed. There are limitations on temperature, time, and residues (FDA 1977).

Formaldehyde

Formaldehyde (HCHO) gas is present in wood smoke, especially in that used for smoking ham or fish. It is flammable in air at concentra-

tions of 7 to 73 percent by volume. Like ethylene oxide, it is an alkylating agent. As such, bacterial spores are only slightly more (2 to 15 times) resistant than are vegetative cells. Optimum activity occurs at about 80 to 90 percent relative humidity (RH).

Formaldehyde vapor is not very penetrating when used under the normal conditions of fumigation. Its action is confined primarily to airborne bacteria and those on exposed surfaces. Even a thin film of organic matter will protect microorganisms from formaldehyde.

HEAT

The most common method of killing microorganisms is to subject them to a heat treatment. When properly done, this is an effective means of improving the microbiological quality of foods. Sometimes, relatively mild heat treatments are used in conjunction with other processes, such as refrigeration, freezing, drying, or acidification. Since the destruction of microorganisms is incomplete, poor sanitation and mishandling of these mildly heat treated products may result in a higher total load of microorganisms than was present before the heat treatment.

The destruction of spoilage microorganisms increases the storage life of the product. However, if this food is contaminated with pathogenic types of microorganisms, the natural competing flora is not present, and the food may become a health hazard.

Compared to other means of sterilization, heat may be considered more efficient and convenient. Besides killing microorganisms, heating will inactivate enzymes (tissue or microbial) that can cause deterioration of food during storage. The one condition needed in using heat is that the material being sterilized must be resistant to heat damage.

Food microbiologists must be concerned not only with the preservation of foods, but also with the sterilization of laboratory materials. The heat treatment of instruments used for sampling and analysis is discussed in Chapter 2. Moist heat (steam or water) is much more effective in destroying microorganisms than is dry heat. The effect of heat on microorganisms is generally believed to be due to enzyme inactivation, protein denaturation, or both.

There are many processes in which foods are subjected to heat. Each of these systems may influence the microbial population. The extent of this influence depends on the type of heat treatment (heat source, temperature, time), the type of food, and the types of microorganisms that are present.

Heat may be transferred to and through foods by conduction, convection, or radiation. The type of heat transfer depends upon the source of heat and the type of food being heated.

Heating Systems

The processes in which heat is applied to foods include cooking, scalding, pasteurizing, blanching, and canning, as well as drying, distillation, evaporation, and concentration.

In most of the processes employing heat, complete destruction of all microbial forms is neither attempted nor attained. Only in thermal processing of foods with steam under pressure, such as in canning, is an attempt made to approach a sterilized product. When sufficient heat is applied to assure a sterile food, the organoleptic and nutritional properties of the food suffer. In many cases, the food would no longer be acceptable to the consumer.

COOKING. Although some fresh fruits and vegetables are eaten raw, most of our foods are given a heat treatment prior to consumption. Cooking is an example of temporary preservation.

Normal cooking may destroy most of the vegetative bacteria, except for thermophiles. Most of the spores will survive normal cooking procedures. The guidelines established by the USDA for cooking roast beef suggest a minimum end-point temperature of 63°C or an alternate processing procedure as listed in Table 12.2. These processes will destroy some pathogenic organisms but will not affect bacterial spores, the enterotoxin of *S. aureus,* or the neurotoxin of *C. botulinum.* The data of Blankenship (1978) revealed that some salmonellae may survive in beef roasts

TABLE 12.2. ALTERNATIVE PROCESSING PROCEDURES FOR COOKED BEEF AND ROAST BEEF

Minimum Internal Temperature		
°F	°C	Minutes[a]
130	54.4	121
131	55.0	97
132	55.6	77
133	56.1	62
134	56.7	47
135	57.2	37
136	57.8	32
137	58.4	24
138	58.9	19
139	59.5	15
140	60.0	12
141	60.6	10
142	61.1	8
143	61.7	6
144	62.2	5

[a]Minimum processing time after minimum temperature is reached.

heated to a temperature of 64°C. Heating beef in a water bath to an internal temperature of 60°C and holding for 12 min decreased the level of *C. perfringens* by about three log cycles (Smith, Evans, and Buck 1981). The cooking of various foods lowers or eliminates pathogenic cells but not spores (Bryan et al. 1982; Cornforth et al. 1982; Parry and Gilbert 1980; Ritter, O'Leary, and Langlois 1979).

The D values for *S. typhimurium* in ground beef were reported as 61 to 62, 3.8 to 4.2, and 0.6 to 0.7 min at 51.6°, 57.2°, and 62.7°C, respectively (Goodfellow and Brown 1978), or 7.1, 5.1, 1.2, 0.9, and 0.6 min at 50°, 55°, 60°, 65°, and 70°C, respectively (Tamminga, Beumer, and Kampelmacher 1982).

The effect of cooking oysters in destroying inoculated polioviruses was reported by DiGirolamo, Liston, and Matches (1970). Stewing (placing oysters in boiling milk and stewing for 8 min), frying (placing oysters in oil at 177°C and frying for 10 min), baking (20 min in an oven at 121.5°C), and steaming (held in flowing steam for 30 min) did not eliminate the inoculated viruses. From 7 to 10 percent of the viruses survived these treatments. Coxsackievirus B2 and poliovirus 1 were inactivated by broiling hamburger patties to 60°C and holding at room temperature for 3 min (Sullivan et al. 1975). At high inoculation levels (10^7 pfu/g), Sabin type 1 poliovirus survived in ground meat held at 80°C for 5 min (Filppi and Banwart 1974).

It is important that foods are cooked sufficiently to destroy certain pathogenic organisms, but they should not be heated to the extent that the nutritional value is reduced significantly. Also, the longer certain foods are heated at an excessively high temperature, the more likely is the formation of mutagens (Bjeldanes et al. 1982; Knize et al. 1985; Lin, Lee, and Huang 1982).

Since cooking does not eliminate all microorganisms, it is essential that the cooked food be held above 50°C or promptly cooled below 4°C to prevent the growth of potential pathogens.

MICROWAVE HEATING. Microwave ovens are prominent in home kitchens throughout the United States. Also, there are many industrial applications for microwaves. Microwave ovens usually operate at either 915 or 2,450 megacycles. At 915 megacycles, the electric current is reversed 915 million times each second.

Polar molecules, such as water, have positive and negative charges concentrated at opposite ends of the molecule. As the microwaves pass through food, these molecules attempt to become aligned with the alternating positive and negative field of the microwaves. This rapid movement back and forth creates molecular friction, which appears as heat.

Microwave heating differs from the more conventional methods of

heating. Heating occurs internally rather than from the outside. This type of heating system is extremely rapid.

Other heating sources must be at a higher temperature than the food so that the heat will move from the source to the food. Microwaves, per se, are not hot, so by using suggested cooking procedures, there is less chance for microwaves to burn the food surface. Only the polar molecules become hot directly due to the microwaves. The nonpolar molecules are heated indirectly by conduction or convection from the heated polar molecules.

There have been several reports implying that microwave heating is more bactericidal than conventional cooking methods. Conversely, some reports infer that microwave cooking could result in a health hazard due to survival of microorganisms that might cause illness. An explanation of these diverse results is that there is a lack of uniform distribution of microwaves, which results in hot or cool spots. These areas cause more or less destruction of microorganisms than would be expected by determining the average temperature of the food.

Rosen (1972) very convincingly reported that the killing of microorganisms is due only to thermal effects. Janky and Oblinger (1976) found no apparent difference in the microbial population of turkey rolls heated in either hot water or in a microwave oven. Fruin and Guthertz (1982) found no significant difference in the distribution of bacteria by microwave, conventional oven, or slow cooker heating of meatloaf.

A major concern is the survival of *Trichinella spiralis* larvae in pork heated in a microwave oven (Kotula et al. 1983; Zimmermann 1983). However, there is no problem if microwaves are used to reheat cooked pork products.

PASTEURIZATION. The heat treatment of foods below temperatures needed for sterilization may be referred to as pasteurization. Quite often, a temperature below 100°C is called pasteurization, while above 100°C, the process is sterilization. In most of the pasteurization processes, the food is heated to between 60° and 85°C for a few seconds up to an hour. Generally, the heat treatment given to a food is designed to inactivate specific types or groups of microorganisms. With pasteurization, some organisms are killed and some are attenuated (sublethal injury), while the spores may be stimulated to germinate. The lethal effect depends upon the heat resistance of the organisms. Pasteurization can be used for foods in which quality is affected adversely by a more severe heat treatment.

The pasteurization process is named for Louis Pasteur, the French chemist who found that heating wine to 50° or 60° C for a short time inactivated spoilage microorganisms without seriously affecting the qual-

ity of the wine. Since his work, the main objective of pasteurization has evolved to destroy certain pathogenic microorganisms in specific foods. When spoilage organisms are not very heat resistant, pasteurization can extend the storage life of products, especially if the treated food is refrigerated, frozen, or otherwise treated to control surviving microorganisms.

Pasteurization is used to control microorganisms in foods such as liquid egg products, dairy products, alcoholic beverages (beer, wine), crab, smoked fish, and certain high-acid products (fruit juice, pickles, sauerkraut, vinegar).

Egg Products. The USDA regulations stipulate that all liquid, frozen, and dried whole egg, yolk, and white be pasteurized or otherwise treated to destroy all viable salmonellae. Depending on the nature of the liquid egg product, various times and temperatures are used, as listed in Table 12.3. Salmonellae are less heat resistant in egg white than in whole egg or egg yolk. Higher temperatures are needed to pasteurize egg yolk products than liquid whole egg. A variety of egg products are produced, and they have different degrees of heat sensitivity. Salted eggs used in salad dressings do not have to be pasteurized. They must be properly labeled, and the salad dressing must contain not less than 1.4 percent acetic acid and have a pH of 4.1 or lower. The final product must be held for 72 hr. Salmonellae will die at room temperature under these conditions.

These pasteurization conditions ordinarily will reduce the standard plate count by about 99.9 percent and the number of salmonellae essentially to zero. The heat sensitivity of salmonellae is affected by the pH. When 60°C is used, the D value at pH 9.0 is about 0.1 min and at pH 5.5, it is about 1.0 min. Hence, pasteurization of egg white at pH 9 requires less heat treatment than at lower pH levels. The addition of 10 percent salt or sugar to egg yolk increases the heat stability of salmonellae by five to ten times. Fortunately, the stability of the egg proteins to heat also is increased by sugar or salt.

TABLE 12.3. PASTEURIZING CONDITIONS FOR EGG PRODUCTS

Product	Temperature (°C)	Average Holding Time (min)
Whole egg, plain	60	3.5
Yolk		
Plain	60	7.0
	or 62.2	3.5
Sugared or salted	62.2	7.0
	or 64.4	3.5
Egg white		
Plain, pH 9.0	56.7	3.5
Plain, pH 9.0, treated with H_2O_2	51.7	3.5
Stabilized with $Al_2(SO_4)_3$ at pH 7.0	50	3.5

Successful pasteurization is based on a critical time-temperature relationship. If the temperature is lowered by 1°C, the efficiency of pasteurization is decreased. If the temperature is allowed to increase, there is danger of coagulating the egg, forming a film on the heat-exchanger surfaces (Fig. 12.1), or damaging the functional properties of the treated egg.

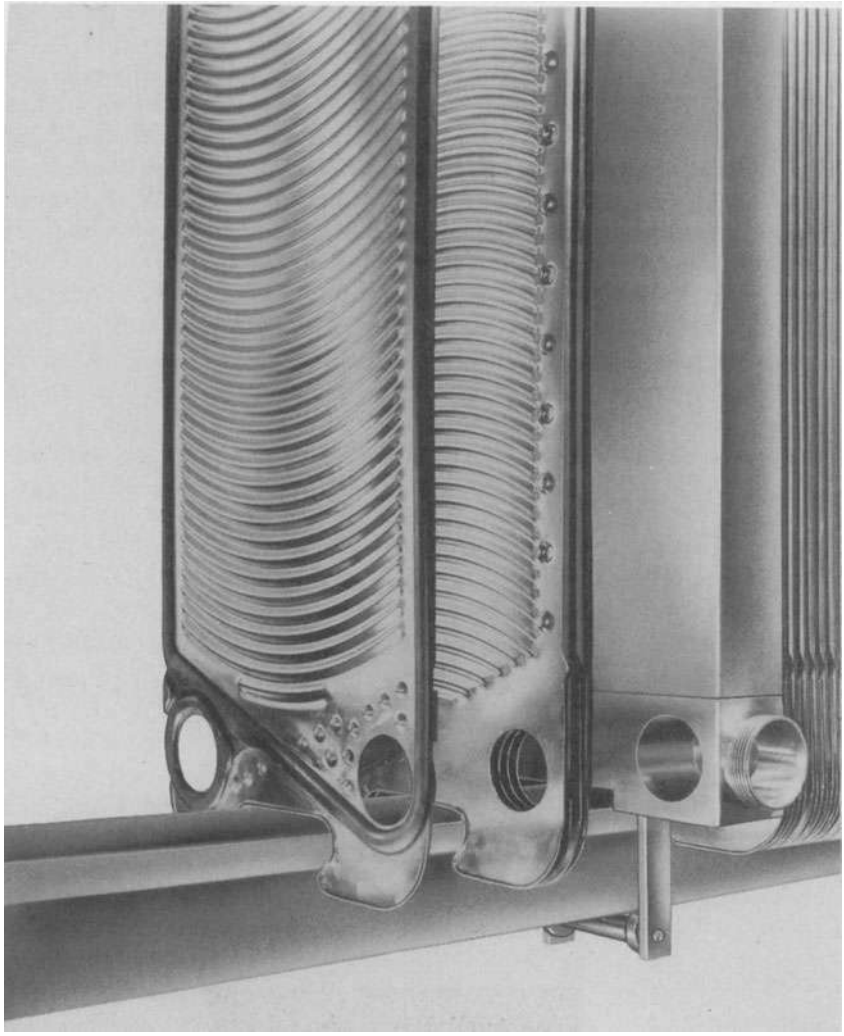


Figure 12.1. Plate heat exchanger opened to show arrangement of plates and gaskets.
Courtesy of CREPACO, Inc.

As liquids flow through the holding tubes, the flow may be laminar, turbulent or transitional. The latter has some characteristics of both laminar and turbulent flow. In any flow, the material in the center of the tube is flowing at a faster rate than that near the side of the tube. Hence, not all of the liquid egg passes through the pasteurizer at the same rate. Researchers suggested that the holding tube requirements should be based on the fastest, rather than the average, particle to traverse the tube (Scalzo et al. 1969). The pasteurization specifications (USDA 1969) require that every particle be held for at least a specified time and temperature. The normal control of plant operations uses the average holding times. These times are considered to be twice that of the fastest particle time. Hence, the minimum time for whole egg at 60°C would be 1.75 min, and the average time 3.5 min.

Liquid egg white is more heat sensitive than whole egg or yolk. If egg white is heated above 56° to 57°C, there are problems of coagulation. The addition of hydrogen peroxide to egg white reduces the heat resistance of the salmonellae so that lower pasteurization temperatures can be used.

Except for conalbumin, the proteins in egg white are sufficiently heat stable at pH 7.0. Adding an aluminum salt such as aluminum sodium sulfate stabilizes the conalbumin so that it can be heated to 60°C without coagulation (Cunningham and Lineweaver 1965).

Shafi, Cotterill, and Nichols (1970) found *Micrococcus, Alcaligenes, Pseudomonas, Bacillus,* and unidentified bacteria in pasteurized egg white. These genera, as well as species of *Staphylococcus, Streptococcus, Arizona (Salmonella)* and unidentified yeasts and molds, were reported in pasteurized whole egg (Shafi, Cotterill, and Nichols 1970). The researchers reported similar types in pasteurized egg yolk. Payne, Gooch, and Barnes (1979) found *Microbacterium,* various cocci and coccobacilli, and some *Bacillus* in pasteurized (65°C for 3 min) whole egg. The presence of egg yolk protected *S. aureus* from the effects of heat (Verrips and van Rhee 1983).

Milk. Before milk pasteurization was a requirement, diseases due to drinking contaminated milk were a common occurrence. There were epidemics of typhoid fever, scarlet fever, diphtheria, and infant diarrhea, as well as outbreaks of tuberculosis, brucellosis, and other illnesses. Today, pasteurized milk is one of our safest foods. Although the elimination of diseased animals and better sanitation and refrigeration have helped improve our milk supply, much of the credit should go to pasteurization.

Originally, the pasteurization of milk was designed to destroy *Mycobacterium tuberculosis.* Since the beginning, the process has also been used to eliminate *Coxiella burnetii,* the causative agent of Q fever. Pasteurization destroys pathogenic organisms, reduces total microbial numbers, inactivates certain enzymes, and extends the storage life of milk. Since milk

proteins are much more heat stable than egg proteins, a higher temperature can be used for pasteurization.

In the vat process, the temperature is increased to 62.7°C and maintained for 30 min. In the high temperature short time (HTST) process, the milk is heated rapidly to 71.7°C and held for 15 sec. Equivalent temperatures and times for HTST approved by the FDA (1977) are (1) 89°C for 1 sec; (2) 90°C for 0.5 sec; (3) 94°C for 0.1 sec; (4) 96°C for 0.05 sec; or (5) 100°C for 0.01 sec.

Ultrapasteurized (UHT) dairy products are heated at or above 138°C for at least 2 sec (FDA 1977). These products have an extended refrigerated shelf life. With proper heating for sterilization and aseptic packaging, this milk may be held at ambient temperature (Hansen, Swartzel, and Giesbrecht 1980; Jelen 1982; Mehta 1980; Perkin 1985; Zall, Chen, and Fields 1986). However, storage time is never indefinite, since there might not be complete sterility or destruction of enzymes, which can result in biochemical and physical changes (Burton 1984; Keogh and Pettingill 1984; Mitchell and Ewings 1985). The time-temperature relationships of various pasteurization systems are shown in Figure 12.2.

Some particularly heat-resistant micrococci and microbacteria as well as spores of *Bacillus* and *Clostridium* may survive UHT. Heating milk to 149°C is expected to kill even some of the most heat-resistant spores.

Pasteurization of milk destroys several enzyme systems. Milk is analyzed for residual alkaline phosphatase to determine the adequacy of heat treatment. One problem with using this enzyme assay is the reactivation of the enzyme following heating by the HTST system. The reactivation is small compared to the activity in raw milk.

The contamination of milk after pasteurization can affect the shelf life and even the safety of this food (Griffiths and Phillips 1984; Schröder 1984).

Animal Tissue. Crabmeat may be given heat treatments somewhat below those used for cooking in order to eliminate a sizable proportion of the microbial population.

Cockey and Tatro (1974) found that pasteurization at 85°C for 1 min was sufficient to reduce the level of inoculated *C. botulinum* type E spores from 10^8/100 g to six or fewer per 100 g. The pasteurized meat was not toxic after storage for six months at 4.4°C. The $D_{82.2°C}$ varied from 0.49 to 0.74 min for four strains of *C. botulinum* type E spores (Lynt et al. 1977). For a 12D treatment, this means the crabmeat would need to be heated to 82.2°C for 8.9 min.

The average plate count of pasteurized crabmeat is reportedly 3,000 bacteria per gram. In retail samples, Lee and Pfeifer (1975) found crabmeat to range from 2.1×10^4 to 8.2×10^7 bacteria per gram. The anaerobes plus facultative anaerobes were reported at about 10^5/g for pasteur-

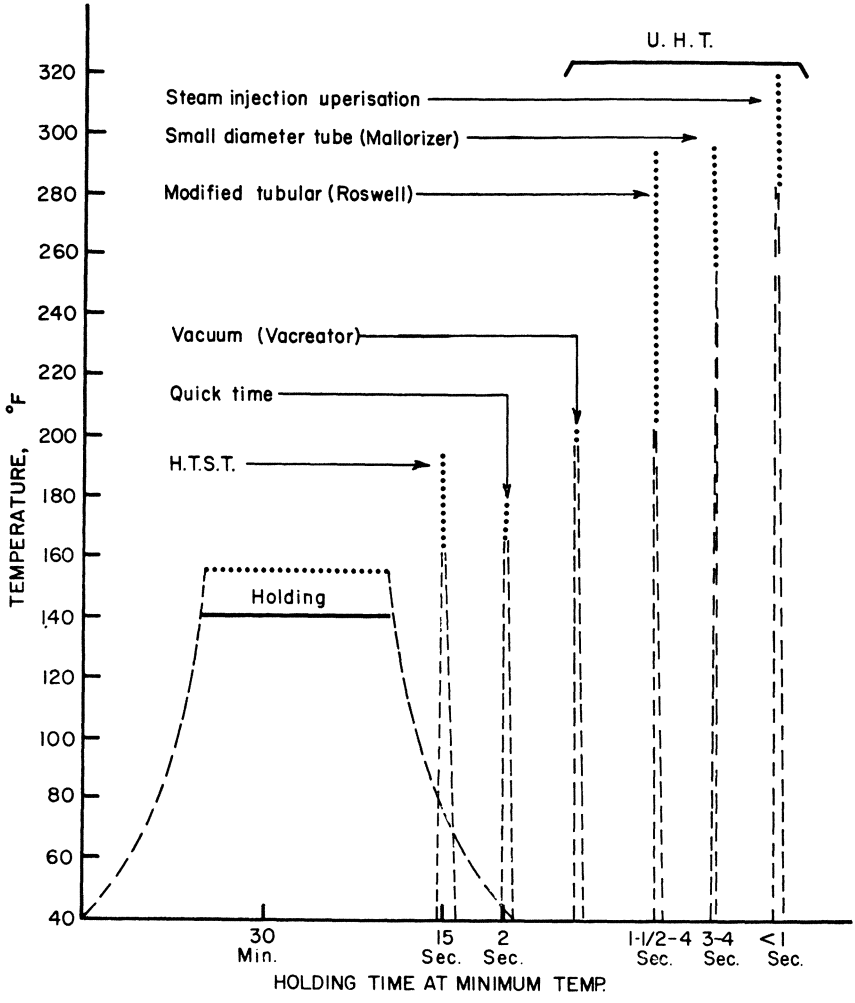


Figure 12.2. Pasteurization time and temperature.
Courtesy of Farrall (1976).

ized crabmeat (Ward, Pierson, and Van Tassell 1977). The prominent genera in canned crabmeat samples obtained from retail stores were *Moraxella, Pseudomonas, Acinetobacter, Arthrobacter,* and *Micrococcus* (Lee and Pfeifer 1975). Equivalent processes for crabmeat in cans and pouches were reported by Ward, Pierson, and Minnick (1984).

The problems associated with pasteurized fishery products became very apparent in 1963 with seventeen cases of botulism and five deaths due to consumption of smoked fish that had been given a pasteurization

treatment. *C. botulinum* spores were not killed and, due to mishandling and holding at warm temperatures, the spores germinated, multiplied, and produced the lethal toxin.

The pasteurization process for ham causes the destruction of most vegetative cells, except for heat-resistant strains of *S. faecalis, S. faecium, L. viridescens,* and micrococci. Ham appears to act as a heat protective agent for *L. viridescens* (Milbourne 1983). The surviving microflora of pasteurized chub-packed luncheon meat was dominated by species of *Lactobacillus, Brochothrix,* and *Micrococcus* (Bell 1983).

Alcoholic Beverages. Pasteurization is used to destroy spoilage organisms in beer and wine. Heating of beer in the final container is effective, but it is also costly and inefficient. Overpasteurization can adversely affect the flavor of beer (Fricker 1984).

Pasteurization just before bottling controls the growth of microorganisms in wine. The heat treatment needed often has an adverse effect on high-quality wines. As with any heat-treated product, there is no residual effect as there is with chemical additives.

Acid Foods. Microorganisms are less heat stable as the acidity of foods is increased. Hence, pasteurization is an effective means of destroying the microorganisms in acid foods.

For fruit juices, the heat treatment selected for pasteurization often is based on inactivation of enzymes rather than destruction of microorganisms. The microbial population usually consists of yeasts, molds, lactic acid, and acetic acid bacteria that are generally heat sensitive. Other organisms usually are inhibited by the acid in fruit juices. Even intermediate acidities are suitable for the growth of only a few sporeformers, such as *B. thermoacidurans (B. coagulans)* or *C. pasteurianum.*

In grape juice, yeasts are destroyed in a few minutes at 55° to 65°C. Mold spores are more resistant, with some withstanding temperatures of 74° to 80°C for up to 15 min. Commercially, grape juice is flash heated to 75° to 85°C.

Fresh pickles packed in solutions of acetic or lactic acids are called fresh-pack pickles. These pickles are given a heat treatment to aid in preservation. Underpasteurization results in spoilage, while overpasteurization can cause a softening of the pickles. Heating to a temperature of 74°C for 15 min was recommended for pasteurization of acidified fresh-pack pickles. Pickles covered with acetic or lactic acids at pH 3.7 remained firm when heated to 74°C for 30 min (Bell, Turney, and Etchells 1972).

DRY HEAT. Moist heat is more effective than dry heat for destroying microorganisms. Dry heat is believed to cause death of the cells by the destructive oxidation of cell components.

There are significant reductions in the total count, and salmonellae can be eliminated when dried egg white is stored at temperatures of 50° to 70°C. Commercially, the dried albumen is stored at 51.6° to 54.4°C for five to seven days, which not only eliminates salmonellae, but also improves the functional performance of the product.

BLANCHING. This is a heat treatment given to many foods prior to drying, freezing, or canning. Blanching inactivates enzymes, aids in cleaning, expels internal gases, removes mucilagelike substances that contribute to off-flavors, helps preserve the organoleptic quality during storage, aids in filling of containers by wilting, shrinking, or softening the food, helps assure adequate can vacuum, is essential when the specific gravity of certain products is used in grading, and destroys many vegetative microbial cells.

There are disadvantages, including the loss of nutrients, especially vitamin C, and the need for water and energy during blanching of fruits and vegetables (Poulsen 1986; Williams et al. 1986).

Blanching can reduce the total microbial load of raw vegatables by 99.9 percent. The product generally has a lower microbial count when steam blanched than when hot water blanched. A steam blancher is shown in Figure 12.3.

CANNING. This is the main system used for the long-term preservation of food. The food is packaged in hermetically sealed containers of metal, glass, thermostable plastic, or a multilayered flexible pouch (Adams, Peterson, and Otwell 1983; Berry and Kohnhorst 1983; Beverley, Strasser, and Wright 1980; Chia, Baker, and Hotchkiss 1983; Tandon and Bhowmik 1986, either before or after heat treatment. Aseptic packaging is used for those products packaged after heating (Carlson 1984; Ito and Stevenson 1984; Teixeira and Manson 1983; USDA 1984; von Bockelmann and von Bockelmann 1986).

Flowing steam is at a temperature of 100°C or less. The temperature of the steam increases as the pressure increases. The usual temperature for sterilizing media in a bacteriology laboratory is 121.1°C. This corresponds to a gauge pressure of 103.5 kPa or absolute pressure of 207 kPa. At this temperature (121.1°C), steam will kill bacterial spores in fifteen min. Saturated steam is more penetrating than hot air. Because of its dryness, superheated steam is about as effective as hot air for sterilizing.

It is important to remove all of the air from the sterilizing chamber. If 50 percent of the air remains, a pressure of 207 kPa will be only 112°C instead of 121°C.

The temperatures used for heat processing or canning vary from 100°C for high-acid foods to 121°C for low-acid foods. For HTST processes, temperatures in the range of 120° to 150°C or higher may be used.

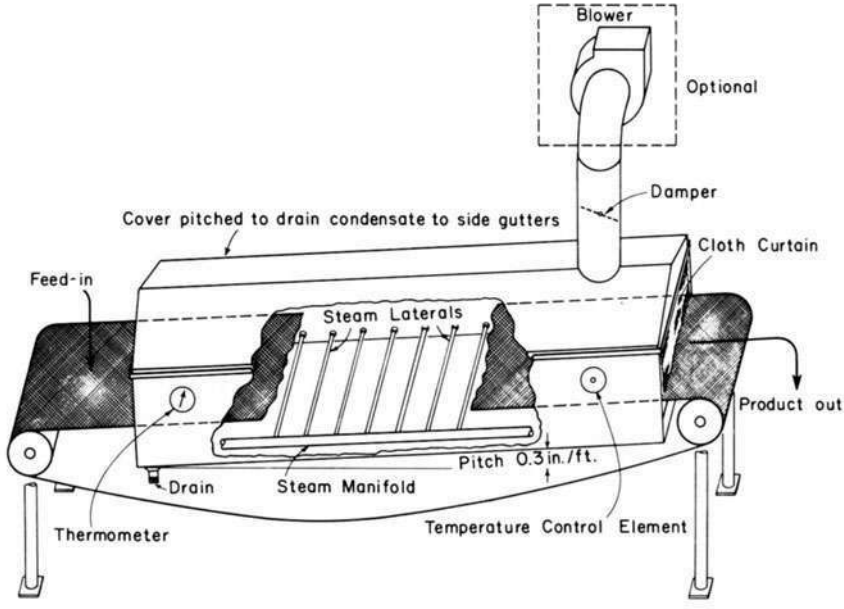


Figure 12.3. Controlled-temperature steam blancher.
Courtesy of Tressler, Van Arsdale, and Copley (1968).

A short time at a high temperature generally results in a product of better quality than equivalent processes at lower temperatures.

The use of heat for food preservation and the history of canning was reviewed by Goldblith (1971, 1972).

Fully sterilized canned products generally are less palatable and less nutritious than less severely heat treated products. Complete sterilization of food may not be attained, and generally it is not needed. The heat-processed products are considered to be commercially sterile. This means that all organisms that might be a health hazard (*C. botulinum*), as well as spores that might cause spoilage under normal nonrefrigerated storage conditions, are destroyed. Some extremely heat-resistant thermophilic spores might remain in the product. With normal storage temperatures, these will not germinate and grow. However, there may be problems in tropical areas, in vending machines that dispense hot foods, or when heat-processed food is not cooled properly before warehousing.

The time necessary to obtain commercial sterility at a specified temperature depends upon the number and type of sporeforming organisms that are present. By practicing good sanitation and keeping the heat-resistant spores at a low level, less time is needed to obtain a commercially sterile product.

When the food is processed in a jar or a can, a good vacuum inside the container is desired. A vacuum reduces the strain on the container during heat processing, holds the ends in a collapsed concave position during subsequent storage, and reduces the amount of headspace oxygen. The vacuum is obtained by exhausting with steam or by vacuum sealing. The loss of vacuum or springer formation is one of the principal types of pack failure. The evolution of gas within the container causes the can ends to become distorted beyond the normal concave position. The formation of gas may be either from hydrogen produced by corrosion on the inside of the can by the food product or by microbial action on the food product. Microbial degradation of the food is usually recognized by the odor or appearance of the product when the container is opened. If gas formation due to microorganisms occurs in canned products stored below 38°C, the food is not commercially sterile, since the spores that germinate and grow at these temperatures should have been destroyed.

It is not the intent of this text to suggest times and temperatures for processing foods. Such things as the type of retort or process used for heating, the size of the container, the composition of the product, thermal conductivity of the food (rate of heat penetration), the number and heat resistance of bacterial spores, the initial food temperature, and the temperature of the retort must be considered when developing a thermal process to obtain a commercially sterile product.

There are various types of retorts for heat processing canned foods. These include still retorts (vertical or horizontal) (Fig. 12.4), agitating retorts, hydrostatic retorts (Fig. 12.5), rotary cookers, helical pump can sterilizers, as well as aseptic canning systems and flame sterilizers.

The food may be heated directly by steam either before filling into the container or by injection of steam through the open top of the container prior to closing and sealing. When heated prior to filling, the food product is flash sterilized by direct steam injection or in a high-pressure continuous heating system at 120° to 150°C or higher for brief periods. This process is particularly valuable for food to be packed in large cans which, with still retorting, require a long process time. Heat sterilization before filling is limited to foods that can be pumped. The heated foods are usually cooled slightly and filled into sterile containers in an aseptic manner.

Certain canned foods can be heated directly with a flame. In this process, the cans are rotated rapidly to prevent burning of the food and to induce convection heating. The direct flame process reportedly yields processed products with improved quality as compared to those processed conventionally (Leonard et al. 1983; Seet et al. 1983).

Mathematical evaluations can be used to determine either the steriliz-

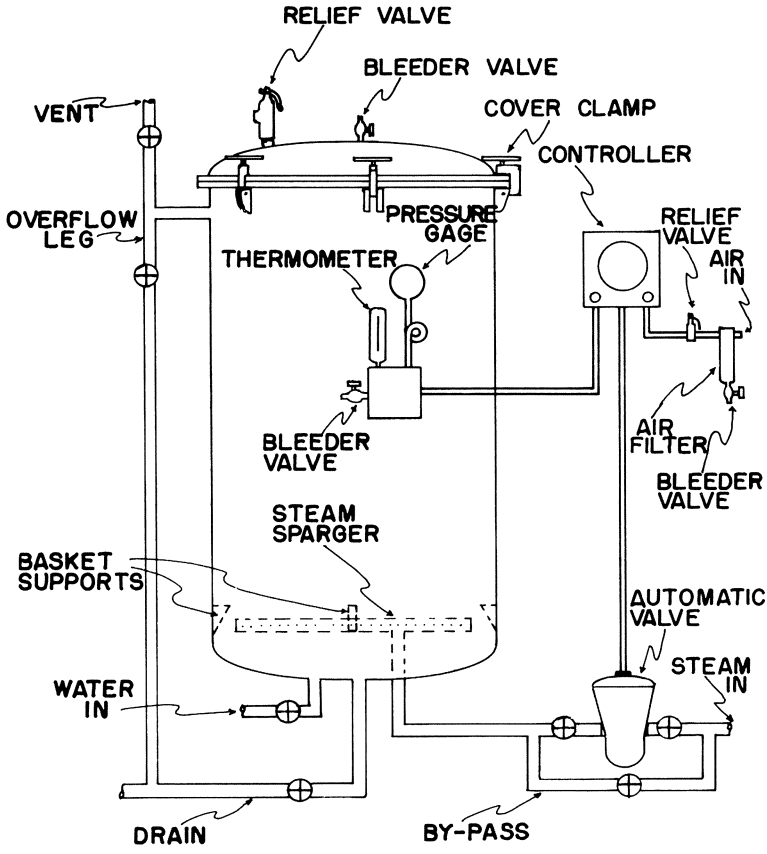


Figure 12.4. Conventional retort.
Courtesy of Nickerson and Ronsivalli (1979).

ing effect of a heat process or the necessary heat process that will produce a specific sterilizing effect. Many suggestions have been presented for estimating either of these two considerations, as well as heating and other aspects of thermal processing of foods (Dail 1985; Hayakawa 1978; Merson, Singh, and Carroad 1978; Pflug and Christensen 1980; Smith and Tung 1982; Spinak and Wiley 1982; Young and Steffe 1985).

After processing, the cans are cooled in water to a temperature of 36° to 42°C. If cans are cooled much below 36°C, they may not dry thoroughly and rusting will result. If the cans are cased at temperatures much over 42°C, thermophilic spoilage may occur.

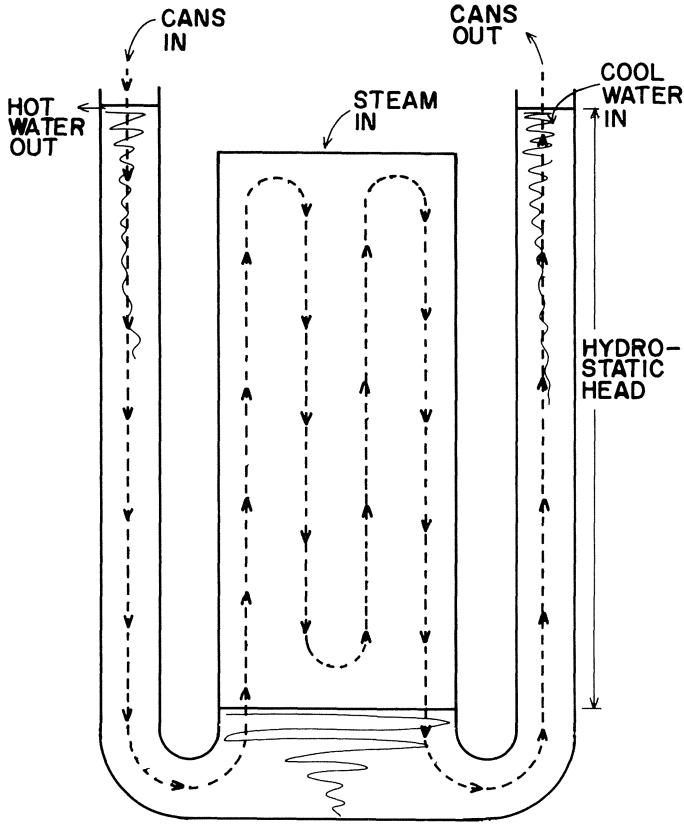


Figure 12.5. Hydrostatic retort.
Courtesy of Nickerson and Ronsivalli (1979).

Heat Penetration

Foods do not become hot or cold instantly. Therefore, a process calling for a retort temperature of 121°C for 15 min means that the product is heating during part of the process and cooling after the process. The temperature in part of the food may never reach 121°C, but it approaches this temperature during the 15 min process. The temperature attained depends upon the rate of heat penetration and other factors. The transfer of heat into foods depends upon the thermal properties of the food, the geometry of the container of the food, and the thermal processing conditions (Fig. 12.6). Liquid foods or particulate matter in

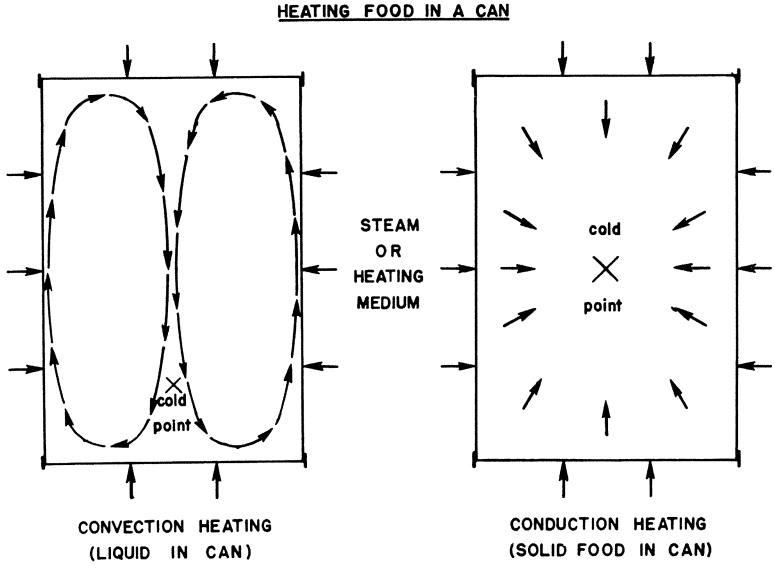


Figure 12.6. Cold point of heating for convection and conduction products.
Courtesy of Desrosier and Desrosier (1977).

fluids is heated in a much shorter time by convection than solid or semi-solid products that are heated by conduction.

Heat Resistance

The heat resistance of microorganisms in a food is a major consideration in establishing processing temperatures and times.

Several methods have been described and used to determine the heat resistance of microorganisms. These include the thermal death time tube or can methods, the rate of destruction method, capillary tube, special heat exchangers, and the miniature retort system. These various methods attempt to minimize the come-up time and cooling effects on destruction of the microorganisms.

Vegetative cells of bacteria, molds, and yeasts are destroyed by temperatures 10° to 15°C above the optimum temperatures for growth. For appropriate times at temperatures of 60° to 80°C, most vegetative cells, as well as viruses, are destroyed. Somewhat higher temperatures may be needed for thermophilic or thermoduric microorganisms.

All vegetative cells are killed in 10 min at 100°C, and many spores are

destroyed in 30 min at 100°C. However, some spores will resist heating at 100°C for several hours.

Spores are as much as 100,000-fold more heat resistant than the corresponding vegetative cells (Gould and Dring 1975a). Some mold spores are only slightly more resistant, while others are much more resistant than the mycelium cells. Reportedly, spores of *Byssochlamys fulva* can survive 5 hr at 88°C. Bacterial spores are the most resistant entities. Some of these spores, such as those of *Bacillus stearothermophilus,* may survive at 100°C for up to 20 hr with an initial population of 10^5 to 10^6 spores/g. The spores of *B. stearothermophilus* are of particular interest to the canning industry because of their high heat resistance and the flat sour spoilage of low-acid canned foods caused by this organism.

It has been suggested that the heat resistance of bacterial spores may result from dehydration of the central protoplast (Algie 1980; Gould and Dring 1975a, 1975b). This dehydration is brought about and maintained by osmotic activity, and the cortex may act as an osmoregulatory organelle. Mineralization increased the heat resistance of bacterial spores (Beaman and Gerhardt 1986; Marquis and Bender 1985), but dehydration was the dominant factor. Acids may interfere with osmoregulation and render the spores heat sensitive (Gould 1977).

Different microorganisms differ in their heat resistance, and different strains of the same species also show differences. The conditions under which the organisms are grown, the nature of the product in which the microorganisms are suspended during the test, and the methods used to determine the survivors can influence the apparent heat resistance of a microbial culture.

Effect of Heat on Microbial Cells

Although moist heat has been used for many years as a method of sterilization, the primary lethal event in thermal inactivation of microorganisms has not been fully determined. The denaturation or coagulation of proteins, breaks in deoxyribonucleic acid (DNA), lesions in ribonucleic acid (RNA), and damage to the cytoplasmic membrane have been suggested as possibilities.

The denaturation or coagulation of proteins involved in cell respiration or cell multiplication usually is suggested as a cause of death.

In the range of 50° to 60°C, the leakage of cellular components into the suspending medium indicates that there is damage to the permeability barrier of the cell. However, at higher temperatures, death precedes leakage. Scheie and Ehrenspeck (1973) suggested that heat caused denaturation of protein(s) in the cell envelope of *E. coli.* This resulted in a

weakened peptidoglycan layer, which would be sufficient to prevent multiplication with the conditions they provided. Further injury results from internal osmotic pressure rupturing the cell membrane at the weakened areas. As long as heat is applied, the rupture allows cellular material to escape. Hence, it might be assumed that destruction of cells at temperatures that cause sublethal injury may involve the cell membrane. If death precedes leakage at higher temperatures, apparently other mechanisms are involved.

Russell and Harries (1968) reported that for nonsporeforming cells such as *E. coli,* RNA degradation is closely related to heat-induced death. The heating of *E. coli* caused the release of part of the outer membrane, resulting in a partial disruption of the permeability barrier function of the membrane (Tsuchido et al. 1985). Both RNA and membrane lesions were reported in *Pseudomonas fluorescens* (Gray, Witter, and Ordal 1973). For commencement of growth, repair of both the cell membrane and RNA damage was required.

The heating of spores at 70° to 100°C results in a loss of dipicolinic acid (DPA), proteins, and other cellular constituents. This indicates damage to the spore coat. However, reportedly, the death of spores proceeds faster than the release of DPA.

The apparent death of spores may be due to the inability of the spore to germinate or, after germination, the inability for outgrowth to occur with the production of more cells. Hashimoto, Frieben, and Conti (1972) suggested that the primary cause of thermal inactivation of spores is due to physical and chemical alterations that interfere with the imbibition of water into the core during germination. The data of Flowers and Adams (1976) suggested that a spore component destined to become cell membrane or wall was the site of injury.

Other researchers concluded that thermal inactivation of phage or viral infectivity was due to protein coagulation (DiGioia et al. 1970).

Protein coagulation, if not the primary lethal event, is important in the thermal destruction of microorganisms. The denaturation of protein can occur in several areas of the cell or spore.

Factors Affecting Heat Resistance

The heat resistance of microorganisms is influenced by the type of heat treatment, the type of cell, as well as the conditions of growth and age, the suspending medium, and the method for enumerating the survivors.

TYPE OF CELL. Spores are more heat resistant than vegetative cells, usually by a factor of 10^4 to 10^5. Although there are many theories for

the high resistance of spores, the exact mechanism is still conjectured. Even with spores, the method of production can affect the apparent heat resistance. Yeast ascospores were 30- to 350-fold or 100-fold more heat resistant than the corresponding vegetative cells (Put and DeJong 1982; Splittstoesser, Leasor, and Swanson 1986).

GROWTH OR SPORULATION MEDIUM. According to a review by Gould and Dring (1974), calcium and manganese ions in sporulating media increase the heat resistance of the spores. They reported no direct evidence implicating monovalent ions in increasing the thermal properties of intact spores. No increase in heat resistance was observed with manganese ions in the sporulation medium of *Bacillus stearothermophilus* (Cook and Gilbert 1968).

The pH of the sporulation medium affects the heat resistance of resultant spores (Pang, Carroad, and Wilson 1983).

The types of peptone, carbon source, and fatty acids in the medium influence the apparent heat resistance of microorganisms. Spores are more resistant when produced in cooked meat medium than in raw meat. Doyle and Marth (1975) found that *Aspergillus* conidia produced on media low in protein and high in glucose were more heat resistant than those produced in high-protein–low-glucose media.

According to Roberts (1984), cadmium in the growth medium induced heat resistance in *E. coli*. The preincubation of arizonae at 35°C in a medium containing NaCl resulted in an increased heat resistance (Ng 1982).

TEMPERATURE OF GROWTH. Spores produced at high temperatures are generally more heat resistant than those produced at low temperatures. Hence, spores produced by thermophiles tend to be more heat resistant than spores produced by either mesophiles or psychrophiles. Although heat resistance is directly related to the growth and sporulation temperature, there are some limitations and exceptions. Phages react similarly to spores. A higher propagation temperature enhances the thermostability of viruses.

Increasing the incubation temperature increased the heat resistance of vegetative cells of bacteria and yeasts (El-Banna and Hurst 1983; Katsui et al. 1982; McAlister and Finkelstein 1980; Roy, Busta, and Thompson 1981).

AGE OF CELLS. Young, actively growing cells are more sensitive to almost any stress than are older, more mature cells. Young spores tend to be less heat resistant than older spores.

SUSPENDING MEDIUM. The chemical and physical characteristics of the suspending medium in which microorganisms are heated influence the heat resistance. It is well established that spores are significantly more resistant when heated in distilled water than when heated in various buffers.

Perhaps the most important aspect that influences heat resistance is the pH of the suspension. Besides this, the a_w, the presence of various solutes, fatty materials, and proteins affect the heat resistance of microorganisms.

Since foods vary in their chemical and physical characteristics, the heat resistance of a microorganism would be expected to vary when suspended in different foods. At 70°C, the D value for *Bacillus licheniformis* spores was almost doubled when 4 percent salt was added to the heating medium (Bell and DeLacy 1984). Compared to the addition of NaCl, some solutes increased heat resistance, while others decreased heat resistance (Smith, Benedict, and Palumbo 1982). Juven, Ben-Shalom, and Weisslowicz (1983) suggested that pectin in tomato juice protected *Lactobacillus fermentum* from the effects of heating.

Effects of pH. Generally, microorganisms have a greater heat resistance near pH 7.0 than above pH 8.0 or below pH 6.0 (Fig. 12.7).

To take advantage of the lower heat resistance at low pH values, acids are added to certain food products prior to processing. Due to the variable pH of foods such as tomatoes, acids are added to adjust the pH to acceptable levels. At pH 3.9, tomatoes can be processed at 100°C for 34 min to kill a normal spore load, but at pH 4.8, a process of 110 min at 100°C is required.

Part of the preservative effect of low pH may be due to damage to the cells or spores with sublethal treatments, and part may be due to the inability for growth of these damaged cells at low pH values.

Spores derived from *Bacillus* that grew at pH 10 but not at pH 7 had an optimum pH for heat resistance at pH 8 to 10 (Kudo and Horikoshi 1983). Increasing the pH of chicken scald water to 9.0 from 7.0 lowered the D value (52°C) of *S. typhimurium* from 34.5 min to 1.25 min (Humphrey, Lanning, and Beresford 1981).

Water Activity. The water activity of the suspending medium in which the organisms are heated influences the heat resistance. In general, the microorganisms are more susceptible to heat in foods with high a_w. In foods with very low a_w, the heat effect is similar to that of dry heat.

One of the effects of heat on microbial cells is the coagulation of protein. Coagulation occurs more readily at high a_w and is reduced as the a_w is lowered.

With moist conditions, spores of *B. stearothermophilus* are about 50,000

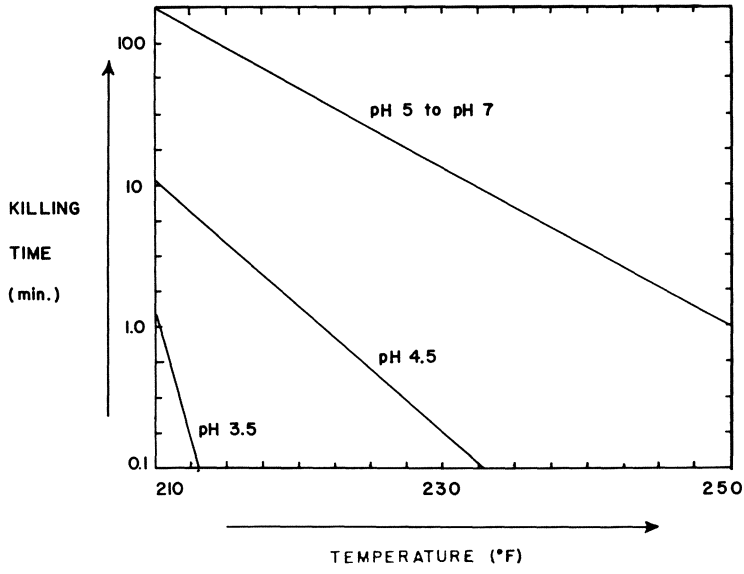


Figure 12.7. Influence of pH of heating medium on heat resistance of spores. The more acidic the substrate, the less heat-resistant the spore suspensions.
Courtesy of Desrosier and Desrosier (1977).

times more heat resistant than spores of *C. botulinum* type E. However, at a_w values less than 0.5, the ratio falls to about ten times (Murrell and Scott 1966). The heat resistance of spores of *B. stearothermophilus* in egg powder, fish protein concentrate, and wheat flour was greater at a_w 0.33 than at 0, 0.68, or 0.99 (Härnulv, Johansson, and Snygg 1977). In an egg macaroni product, Hsieh, Acott, and Labuza (1976) found that both *Salmonella anatum* and *Staphylococcus aureus* had a maximum heat resistance in the a_w range of 0.75 to 0.80. As the a_w is reduced from 0.4 to 0, the heat resistance of spores decreased (Alderton, Chen, and Ito 1980).

The water activity of the suspending medium is often adjusted by the addition of salt, carbohydrates, or other solutes. These substances can influence the heat resistance of microorganisms. Hence, the effect of these adjusted suspensions on heat resistance may be due to the solutes, to lowered a_w, or to both. When salmonellae were heated in solutions of sugars and polyols, there was no direct relationship between heat resistance and a_w (Corry 1974).

Carbohydrates. These substances can influence the heat resistance of microorganisms (Fig. 12.8). The thermostability of *C. botulinum* spores is di-

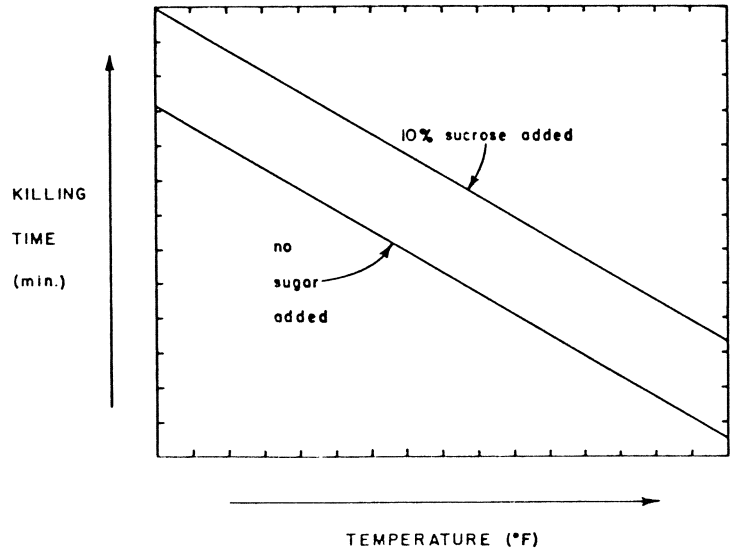


Figure 12.8. Influence of sugar on the heat resistance of bacterial spores. High concentrations of sugar protect spores.
Courtesy of Desrosier and Desrosier (1977).

rectly related to the concentration of sucrose in the suspending medium. As the concentration of sucrose increases, the heat resistance increases.

The protection afforded the cells by sucrose may be due to the reduction of the a_w, the medium surrounding the cells, as well as the extraction of water from the cell. Corry (1976b) found that the degree of protection of the solutes was correlated with plasmolysis or cell shrinkage.

Corry (1976a) determined the heat resistance of *Saccharomyces rouxii* and *Schizosaccharomyces pombe* in solutions of sugars and polyols. The resistance was maximum with sucrose, less in sorbitol, and least in solutions of glucose, fructose, or glycerol. The higher heat resistance of *S. cerevisiae* in sucrose than in either fructose or glucose was reported by Torreggiam and Toledo (1986). Sucrose increased the heat resistance of *S. cerevisiae* spores, but alcohol decreased it (Splittstoesser, Leasor, and Swanson 1986). Sucrose protects various vegetative bacteria from heat (Kwast and Verrips 1982; Smith et al. 1983).

The presence of higher carbohydrates, such as starch and pectin, reportedly gives some protection to spores and cells and increases the heat resistance.

Proteins. There is a tendency for proteins to protect microorganisms from the lethal effect of heat. Peptones, yeast extract, and albumin can provide

protection for certain organisms. When ground beef was supplemented with 30 percent textured soy protein, the heat resistance of salmonellae was increased (Craven and Blankenship 1983).

Antimicrobial Substances. Various antibacterial substances tend to increase the destruction of microorganisms by heat. An example is hydrogen peroxide used in conjunction with heat pasteurization of liquid egg white.

The addition of sulfur dioxide to fruit and fruit products tends to lower the heat resistance of the microorganisms. The effect is less apparent with yeast than with bacteria. Yeasts are relatively tolerant of sulfur dioxide. Potassium sorbate and sodium benzoate acted synergistically with heat to inactivate various molds (Beuchat 1981).

Lipids. Microorganisms suspended in fat or oil are more difficult to destroy than when they are in an aqueous medium (Fig. 12.9). Because of the poor heat conductivity of oil and the reduction in moisture, microorganisms in a fatty medium have a much greater resistance to heat than those heated in an aqueous medium without fat. The heat destruction in fats resembles dry heat sterilization.

Increasing the fat content of hamburger increased the D value of

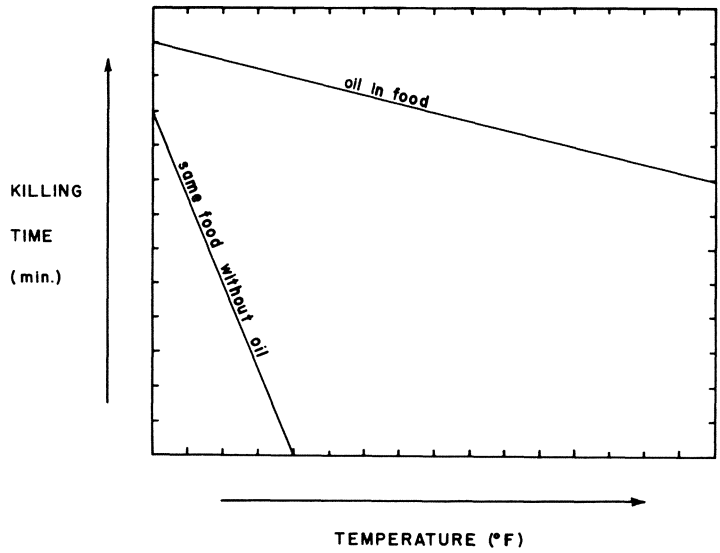


Figure 12.9.. Influence of oil on heat resistance of yeasts. (Organisms trapped in oil phase are killed by dry heat and have much heat resistance in comparison with organisms in water phase.
Courtesy of Desrosier and Desrosier (1977).

poliovirus (Filppi 1973). This protection of the virus was more evident at 80°C than at 50°C.

ENUMERATION. When spores or vegetative cells receive a sublethal heat treatment, they may be injured in some manner. The spores may remain dormant for extended periods before germination. Dormancy can be reduced or eliminated by using a suitable recovery medium. The addition of starch is thought to absorb fatty acids that inhibit outgrowth of germinated spores. Vegetative cells that are sublethally damaged require special environmental conditions for repair, growth, and multiplication.

Since the recovery medium can affect the number of survivors of a heat treatment, the apparent heat resistance of an organism may vary to a significant extent.

Spores that survive sublethal heat treatments generally are recovered better at a lower temperature than the optimum for unheated spores. The addition of lysozyme to recovery media aids the germination and colony formation of spores surviving extended heat treatments. The exposure of heated spores to EDTA sensitized the spores to lysozyme and increased the number of recovered survivors (Adams 1974). Calcium dipicolinate also can initiate germination of heat-treated spores.

Survival Curve

Since microorganisms differ in their resistance to heat and this resistance can vary according to age of cells, growth medium, and suspending medium, we need a system to mathematically express survival. Such a system makes it possible to compare the heat resistance of different species at the same temperature, or the heat resistance of one species at different temperatures, and in various suspending media. It also enables the establishment and evaluation of thermal processes for foods in relation to the types of organisms that must be controlled.

By plotting the logarithm of the number of survivors at each time of sampling, a survival curve is obtained. Generally, bacteria exhibit a logarithmic survival curve when exposed to heat or other destructive agents. If a microorganism is either alive or dead, then death is a first-order reaction. However, various shapes of survival curves have been reported. Various explanations have been given for nonlogarithmic death (Han, Zhang, and Krochta 1976; Cerf 1977).

Another aspect is the tendency for the survival curves to show a tailing effect (Fig. 12.10). As survivors approach a low number, they appear to be very heat resistant. This can result in microorganisms surviv-

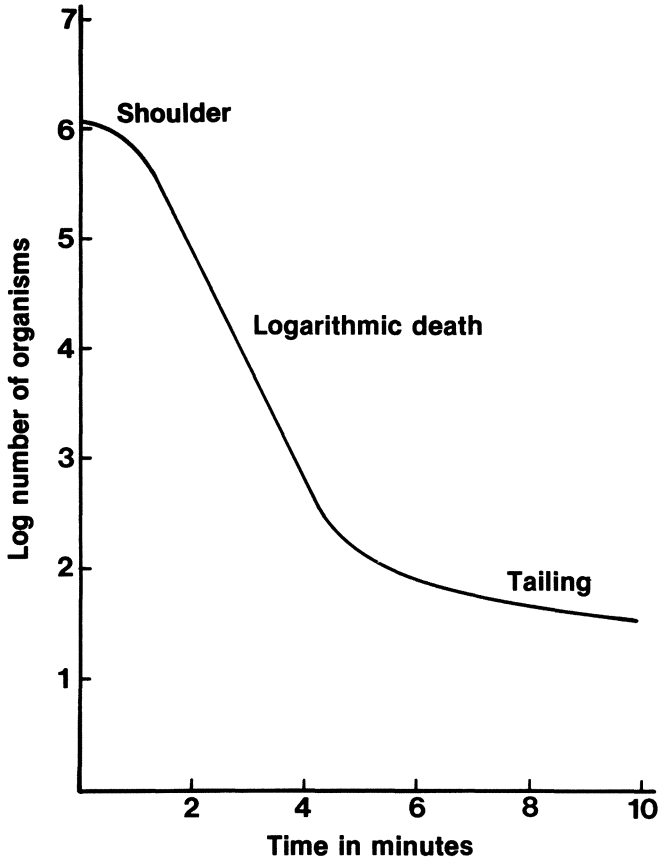


Figure 12.10. Survival curve showing a shoulder and tailing.

ing a heat process that is in excess of that calculated to destroy the entire microbial population. The exact reason for the tailing effect is not known.

Although deviations from the logarithmic order of death do occur, for the purposes of further discussion, we will assume that a straight line for the survival curve is obtained (Fig. 12.11).

The number of survivors is directly proportional to the initial number present for any given treatment. The higher the initial number of cells, the longer the time necessary to cause their complete destruction. Since the logarithm of survivors never reaches zero, theoretically complete destruction of a microbial suspension can never be achieved.

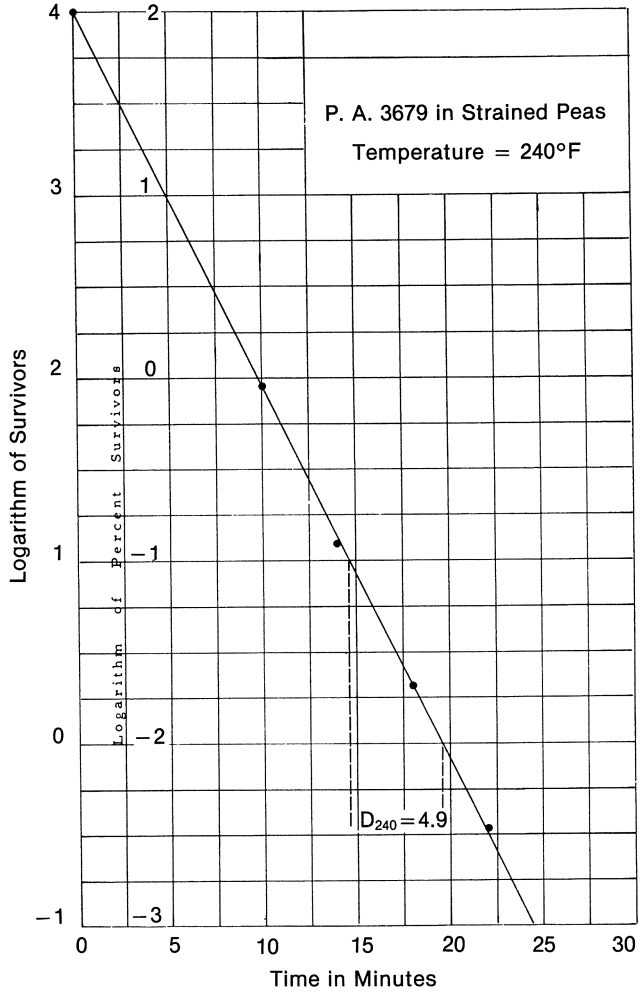


Figure 12.11. Death rate curve showing the determination of D.
Courtesy of National Canners Association Research Laboratories (1968).

Decimal Reduction Time (D)

The decimal reduction time is the time required to reduce the microbial population by 90 percent at a specified temperature. The decimal reduction time is designated by D. This is also the time required for the survival or death rate curve to traverse one logarithmic cycle (Fig. 12.11). In this case, D is 9 min. D can be determined mathematically by the formula

$$D = \frac{t}{\log a - \log b}$$

where a is the initial number of cells and b is the number remaining after time t. When the difference between a and b is one log cycle, then ($\log a - \log b$) is equal to 1, and $D = t$.

The D value is not constant, but varies with species and strains of microorganisms, temperatures, and other factors that affect the heat resistance. The temperature at which the D is determined may be designated by a subscript. For example, $D_{121°C} = 1.0$ min means that 90 percent of the spores are destroyed in 1.0 min at 121°C, and 99 percent are destroyed in 2.0 min. The D values for several microorganisms determined in various suspending media are listed in Table 12.4. As can be seen in this table, the D values vary due to type of cell, temperature, and suspending medium. The growth conditions, recovery medium, and testing different strains of the same organism can result in different D values (Doores and Westhoff 1981; Feeherry, Munsey, and Rowley 1987; Feig and Stersky 1981; Grieme and Barbano 1983; Hyun et al. 1983; Johnson, Nelson, and Busta 1982; Lovett, Bradshaw, and Peeler 1982; Lynt, Solomon, and Kautter 1984; Scott and Bernard 1982, 1985).

The destruction of *S. aureus* is accomplished more readily in whole milk than in skim milk. In skim milk at 60°C, the D value for *S. aureus* strain 161-C was 1.30 min (Walker and Harmon 1966). In whole milk, the D value for strain 161-C was 0.75 min. Firstenberg-Eden, Rosen, and Mannheim (1977) reported a D_{60} (D value at 60°C) in whole milk of 0.87 min for *S. aureus* when salt-free media were used to detect heated cells. When salt was present in the recovery medium, the apparent D_{60} was 0.62 min. Sublethal heat treatments cause *S. aureus* to lose salt tolerance. These latter workers reported a z value of 9.46°C so that if D_{60} is 0.87 min at 50.54°C, the D is 8.7, and at 69.46°C, it is only 0.087 min.

According to Bean and Roberts (1975), levels of 8 percent salt (w/v) in the heating medium protect *S. aureus* from heat. They reported the D_{60} for a meat emulsion at pH 6.5 was 4.61 to 9.62 min. When 8 to 8.5 percent salt was added to the meat emulsion, the D_{60} increased to 18.62 to 26.38 min. The addition of sodium nitrite had little effect on the heat resistance of this organism.

In thermally processed canned foods, the destruction of spores of *C. botulinum* is a major concern. To have reasonable assurance that the spores of *C. botulinum* are destroyed, a 12 D concept was established. This means that the number of spores will be reduced by 12 log cycles. Hence, with an original level of 10^6 *C. botulinum* spores, the 12 D thermal process would reduce this to 10^5 spores. It is evident that the lower the original contamination, the lower the number of spores after the process and the

TABLE 12.4. DECIMAL REDUCTION TIMES (D) FOR HEATING VARIOUS MICROORGANISMS

Organisms	Suspending Medium	D Value	
		Temp. (°C)	(min.)
Vegetative Cells			
Salmonella senftenberg 775W	Custard	60	11.3
	Chicken à la king	60	9.6
Salmonella manhattan	Custard	60	2.4
	Chicken à la king	60	0.40
Staphylococcus aureus	Custard	60	7.7–7.8
	Chicken à la king	60	5.2–5.4
Escherichia coli	Raw milk	57.3	1.3
	Ice cream mix	57.3	5.1
Spores			
Aspergillus flavus (conidia)	Phosphate buffer pH 7.0	50	16.2
		55	3.1
Clostridium botulinum Type A	Phosphate buffer pH 7.0	104.4	17.6
		110.0	4.4
		115.6	1.3
		121	0.2–0.4
	Tomato juice pH 4.2	104.4	6.0
		110.0	1.6
		115.6	0.4
Type B		110.0	0.7
Type E	Phosphate buffer pH 7.0	70.0	29.3–37.5
		80.0	0.4–3.3
Bacillus stearothermophilus	Water	115.0	17.5–18.3
	4% NaCl	115.0	11.3–12.7
	Buffer—pH 7.0	115.0	9.2–11.3
	Buffer—pH 5.0	115.0	4.2

SOURCES: Cook and Gilbert (1968, 1969); Doyle and Marth (1975); Odlaug and Pflug (1977); Read, Schwartz, and Litsky (1961).

less the chance that a can of food will contain a potentially viable spore. Even if one spore does survive the process, there is a chance that it is injured or damaged and is not able to germinate, outgrow, and produce the neurotoxin in the environment of the canned food. The spores of *C. sporogenes* and *B. stearothermophilus* are more heat resistant than those of *C. botulinum.*

The thermal inactivation of viruses has been difficult to characterize. It has been suggested that, like bacteria, viruses are inactivated by denaturation of protein. If first-order kinetics prevail, the survival curve should be a straight line. The tailing effect, or decrease in inactivation rate as the time is extended, is a common occurrence in viral survival curves. The formation of aggregates or clumps of viruses is a prominent sugges-

tion for the occurrence. Clumping tends to protect the viruses located in the middle of the clump from the heating effect. During analysis, the clumps are disintegrated to release the surviving viruses (Cliver 1971; Sullivan et al. 1971b). It is not likely that viruses can withstand the heat treatment given to thermally processed canned foods.

Thermal Death Time

The thermal death time (TDT) is the time required to achieve sterility of a suspension containing a known number of cells or spores at any predetermined temperature. At any particular temperature, the TDT depends upon the resistance of the cells, as well as the number of cells in the suspension.

Thermal death times are determined for suspensions of a particular organism at several temperatures. When the times necessary for killing are plotted on a logarithmic scale and the corresponding temperatures plotted on a linear scale, a straight line is obtained. This is the TDT curve for that particular organism.

When the D values are plotted on a logarithmic scale, with the corresponding temperatures on a linear scale, a phantom TDT curve is obtained. This should be a straight line (Fig. 12.12). These lines can be described by a D or TDT at some reference temperature (usually 60° or 121°C) and the slope (z). The symbol z is defined as the temperature necessary to bring about a tenfold change in the TDT or D value. The value of z corresponds to the number of degrees of temperature passed over by the curve in traversing one logarithmic cycle. In Figure 12.12, the z is listed for each curve. The symbol F is used to designate the time necessary to destroy a given number of microorganisms at a reference temperature, usually 121°C for spores, or 60°C for vegetative cells. To avoid confusion, the temperature can be added as a subscript to F. Hence, F_{60} is the TDT at 60°C and F_{121} is the TDT at 121°C. In Figure 12.12, °F are used, and the F_{250} for curve A_1 is 5.8 min and for curve A_2 is 7.4 min.

For spores of *C. botulinum* suspended in phosphate buffer, the value of z is about 10°C. Researchers reported that z values ranged from 5.7° to 7.6°C for five strains of *C. botulinum* type E in fish paste (Crisley et al. 1968).

For the microorganisms of most importance in canned foods, the z values are usually between 6° and 14°C. The z value varies for different strains and types of microorganisms, as well as the chemical and physical characteristics of the suspending medium. A strain of *Clostridium thermosaccharolyticum* had a z of 6.94°C in distilled water and 10°C in molasses

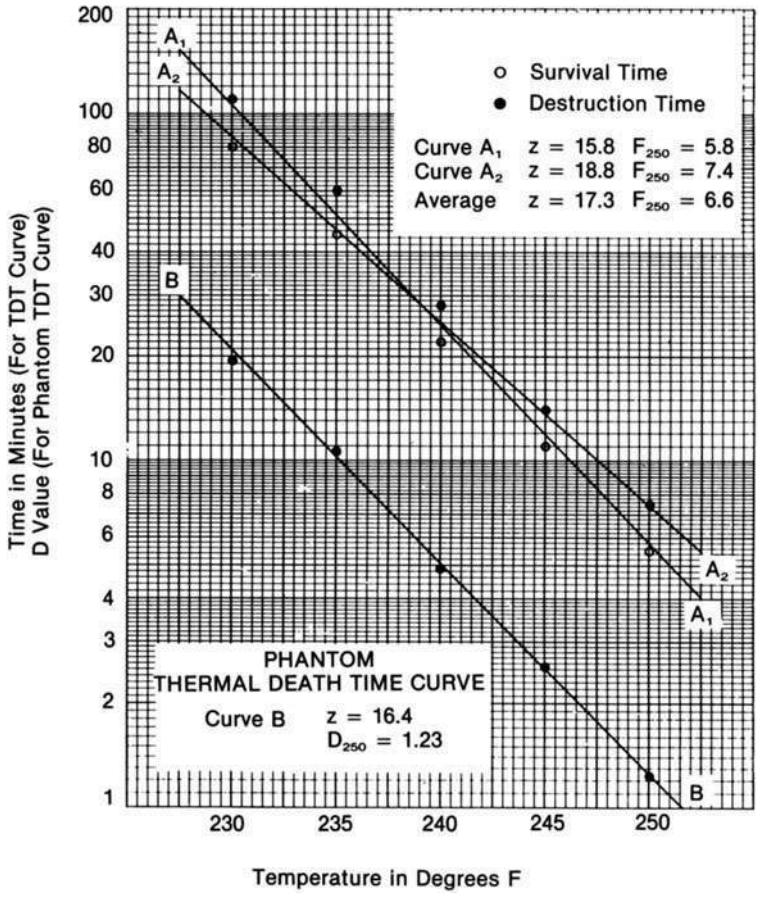


Figure 12.12. Thermal death time curve.
Courtesy of National Canners Association Research Laboratories (1968).

(Xezones et al. 1965). The z values for three strains of *Byssochlamys* varied from 3.9° to 7.8°C (Hatcher et al. 1979).

The work by Esty and Meyer in 1922 showed that the *F* was 2.78 min for spores of *C. botulinum.* Due to this, the generally accepted minimum heat treatment for a 12 D destruction of spores of *C. botulinum* in low-acid foods is considered to be 3 min at 121°C, or its equivalent. This is termed the minimum safe process or botulinum cook. Commercially canned cured meat products are given thermal processes well below this treatment. In these products, the heat treatment is supplemented by the effects of curing salts on the outgrowth of the spores, so that the treated

products are not likely to be a health hazard. Low- acid noncured meats are subjected to a treatment of $F_{121} = 6$ min.

Thermal Processes

With D, *F,* and *z* values, the rate of heat penetration, the pH of the food, the retort temperature, and other information, a thermal process can be calculated. Experimental inoculated packs are processed to determine the validity of the calculations. It is evident that the determination of acceptable, safe processing schedules for foods can become quite complex. Thermal processing schedules can be obtained from the National Canners Association, Washington, D.C. The Food and Drug Administration also has guidelines for acceptable processes. Due to the complex interactions of factors concerning the determinations of processing schedules, computers have become an important tool for the calculation of needed processes.

Contaminants and Spoilage

Heat-processed canned foods are subject to various types of spoilage. The appearance of the container will indicate probable spoilage. The internal formation of gas will cause conditions of the can termed *flipper, springer, soft swell,* or *hard swell.*

For a flipper, only one end of the container is slightly bulged, or the end of the can bulges when the container is struck on a hard surface. In either case, by applying finger pressure, the end may be pressed back to remain flat.

For cans that are swelled, both ends bulge and cannot be pressed flat. Bulged ends of soft swells yield slightly to finger pressure, but those of hard swells do not.

Even if there is no bulging of the can, the contents may be spoiled due to flat sour, sulfide, or mold spoilage.

Spoilage may occur prior to processing or it may be due to inadequate heat treatment, improper holding temperature, or leakage through defective closures, other can defects, or rough handling of the cans.

The organisms causing spoilage of underprocessed products are usually heat-resistant spores of bacteria that survive the heating process, germinate, and grow during storage of the canned food. Inadequate cooling or high storage temperatures can result in spoilage by the growth of thermophilic bacteria. The normal commercial heat treatment is not designed to destroy the spores of all obligate thermophilic organisms.

Spoilage due to leakage involves nonsporeforming bacteria, yeasts, and molds that could not have survived the heat process. They enter the

can after processing. Also, organisms involved in foodborne illness may enter cans after processing. Leakage into the container occurs when the hot cans are cooled in contaminated water. When hot, the sealant for the lids and seams is soft and not set, so organisms can enter. This entrance of organisms is aided by the vacuum created in the can during cooling.

Pflug, Davidson, and Holcomb (1981) examined 1,104 swollen cans of food obtained from supermarkets. They found that 314 of these cans had major container defects. Of the remaining 790 cans, 86 percent had typical leaker spoilage. The other defects were due to underprocessing (7 percent), nonmicrobial spoilage (6 percent), and thermophiles (1 percent). From their data, it is obvious that postprocess leakage of organisms into the cans is very important. Put, Witvoet, and Warner (1980) listed the three main factors causing spoilage from postprocess operations as the condition of the double seams, contamination of cooling waters, and can abuse during handling. The effectiveness of the addition of germicidal agents to the cooling water was reported by Ito and Seeger (1986). Their use depends upon the operation of the processing plant.

The two organisms that can cause flat sour spoilage of canned foods are *Bacillus stearothermophilus* and *Bacillus coagulans* (Ito 1981; Thompson 1981).

To prevent the postprocessing contamination of retort pouches, it is important that they be dried immediately after processing, be stored dry, and be handled carefully (Michels and Schram 1979).

Sterility Testing

Canned foods should be commercially sterile. That is, there should be no microorganisms present that have public health significance or that are capable of reproducing in the food under normal nonrefrigerated conditions of storage and distribution.

Contamination of the canned product during sampling and analysis has caused many erroneous results. One of the main sources of contamination is the investigator (Denny 1972; Evancho, Ashton, and Briskey 1973). These workers discussed the necessary precautions that must be taken for sterility testing of canned foods, as well as the media, equipment, and procedures for this test.

IONIZATION RADIATION

Shortly after Roentgen discovered X-rays in 1895, it became known that ionizing radiations kill microorganisms. However, it was not until

the early 1940s that research employed these radiations as a means of food preservation.

Ionizing radiation consists of corpuscular and electromagnetic radiations (Fig. 12.13) with sufficient energy to strip or cause ejection of electrons from atoms or molecules. Corpuscular radiation is due to subatomic particles of various types which can transfer their kinetic energy to anything they strike. The energy of these subatomic particles is based upon their velocity. Beams of electrons that have been accelerated to speeds approcahing that of light are called cathode rays. These are related to the beta particles emitted from atomic nuclei. The other subatomic particles are not usable in food preservation. Alpha particles do not have sufficient penetrating ability (Fig. 12.14). Neutrons cannot be used because they can induce radioactivity into the food.

Electromagnetic waves produced by high-voltage equipment are called X-rays. Gamma radiation is an electromagnetic wave from atomic nuclei and is essentially the same as X-rays. The electromagnetic waves have an energy that is inversely related to the wavelength of the radiation. Thus, the shorter the wavelength, the higher the energy content. The longer wavelength electromagnetic radiations (infrared, microwave, radio) are nonionizing.

The process of food irradiation uses gamma rays from radioactive isotopes, such as cobalt-60 or cesium-137, or electrons from linear accelerators. The radionuclides emit both gamma and beta rays. However, the energy level of the beta rays is relatively low for food sterilization, so gamma rays are usually the only ones considered. The electrons are limited in their ability to penetrate food materials. The depth they penetrate depends upon their energy level. Electron beams over 10 million electron volts may induce radioactivity into some of the food constituents or the container, so they are not considered to be practical for food preservation. Cobalt-60 has been the most acceptable radiation source because of its availability, price, and properties.

The energy of radiation is expressed in either rads or grays. A dose of 1 rad is obtained when 0.01 joule (j) of energy is absorbed per kilogram of material, or one rad is equal to 100 ergs of energy absorbed per gram of material. When 1.0 joule is absorbed per kilogram of material, this is 1 gray (Gy). Hence, 1 Gy is equal to 100 rads. For convenience, the kilorad (Krad) for 1,000 rads and megarad (Mrad) for 1,000,000 rads, are used. A typical sterilizing dose is 4.5 Mrad or 45 kGy, which are equal to 45 kj/kg.

For any method of food preservation to be successful, it must be proved to be technically advantageous and economically competitive (the benefits are greater than the costs), and the treated food must be wholesome and safe for human consumption. For this purpose, the biological

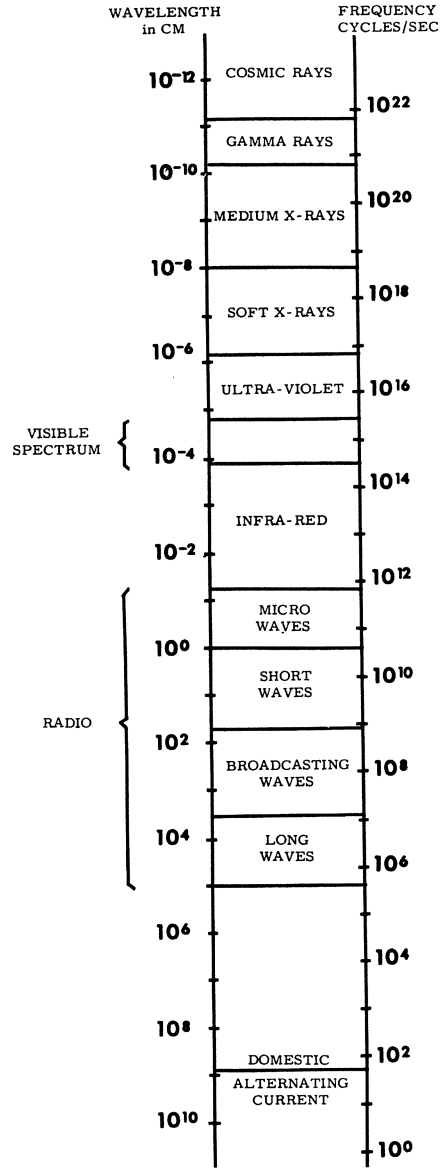


Figure 12.13. The electromagnetic spectrum.
Courtesy of Robert Taft Sanitary Engineering Center.

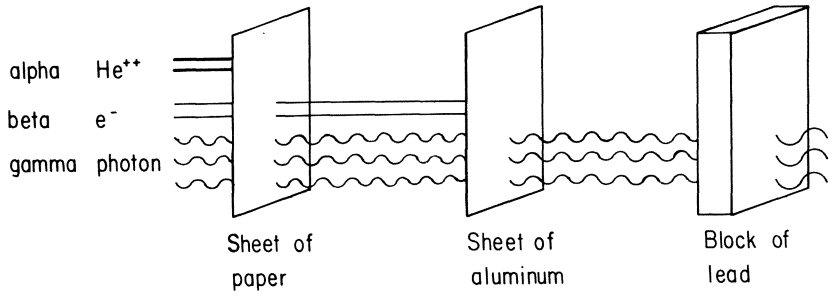


Figure 12.14. Relative penetrating power of alpha, beta, and gamma radiations. Usefulness of various radiations is dependent upon requirements. Where depth of penetration is a factor, gamma radiations and high-energy electrons must be used. *Courtesy of Desrosier and Desrosier (1977).*

effects, the stability of the treated foods in storage, the organoleptic and nutrient effects, the safety of the treated food, and the relative costs of the process are discussed.

Biological Effects

The lethal action of ionizing radiations on living cells is due to either a direct action on genetic material or to an indirect effect. With indirect action, there is an initial change in the suspending medium or some nongenetic molecule.

Ionizing radiation causes breaks in the DNA. Restitution can occur in most of the single-strand breaks, as long as the repair process is operative (Hittelman and Pollard 1982; Iwanami and Oda 1985). Within the first hour after irradiation, about 90 percent of the DNA-breaks are rejoined (Graubmann and Dikomey 1983). According to Bresler, Noskin, and Suslov (1984), most double-strand breaks occur due to enzymatic incision of primary damages.

Generally, the more complex the organism, the more sensitive it is to ionizing radiations (Table 12.5). Living cells have always been subjected to natural radiation from cosmic and terrestrial sources. The extent of this radiation depends upon the geographic location (altitude and distribution of radionuclides). When shielded from natural ionizing radiation, the cell growth rate of blue-green algae was reduced (Conter, DuPouy, and Planel 1983).

Uses in Foods

Ionizing radiation can inhibit sprouting of vegetables (potatoes, onions); delay ripening of fruits and vegetables; destroy insects, their larvae,

TABLE 12.5. APPROXIMATE LETHAL DOSES OF IONIZING RADIATION

Organism	Dose (Gy)
Human being	6–10
Insects	250–1,000
Bacteria	1,000–10,000
Fungi	1,000–10,000
Bacterial spores	10,000–50,000
Viruses	30,000–50,000

and eggs in grains, spices, fruits, and vegetables; destroy parasites in pork, beef, or fish; eliminate pathogens in meats and poultry; reduce the number of viable microorganisms in dried products such as spices and herbs; destroy organisms to extend the shelf life of various foods; or commercially sterilize various foods. Besides these processes, other actions of gamma radiation include the reduction of insecticides (Cin and Kroger 1982), the reduction of polychlorinated biphenyls (Cichy, Zabik, and Weaver 1979), the reduction or elimination of flatulence factors in green gram (Rao and Vakil 1983), the alteration of seeds resulting in plants resistant to diseases (Utkhede and Rahe 1982), and the reduction of seed germination (Wang et al. 1983). Ionizing radiation could be used to sterilize packaging materials used in aseptic processes.

The potential usage of ionizing radiation on foods gained momentum in 1980 when the Joint Expert Committee of the Food and Agricultural Organization, the International Atomic Energy Agency, and the World Health Organization concluded that the irradiation of any food up to an overall average dose of 10 kGy presents no toxicological hazard. Further, this level of treatment causes no special nutritional or microbial problems. Even then, relatively few nations have approved the use of ionizing radiation on foods.

The United States FDA approved the radiation of grain for insect control in 1963 and of potatoes for sprout inhibition in 1964. However, there have been no commercial applications of these approved uses in the United States.

In 1981, the FDA published an advance notice of proposed procedures for the regulation of irradiated foods for human consumption. Since then, the organization has approved ionizing radiation for various food uses. These include the reduction of the microbial level and control of insect infestation of dried herbs, spices, and vegetable seasonings as well as blends of these; the control of microorganisms in enzymes used in food processing; the killing of the parasite *Trichinella spiralis* in fresh pork; and the control of insects on and the delay of ripening of fresh fruits and vegetables to extend shelf life.

It is expected that there will soon be further approvals, probably for the elimination of pathogens on poultry and meat and the elimination of parasites in meat and fish.

For most approvals, there is a limit of 1 kGy, but for dried spices, 30 kGy can be used. Since the safe level for international trade has been approved at 10 kGy, the level approved by the FDA may be increased.

Mangoes were the first irradiated fruit sold in the United States (September 1986). Only about 1 percent of the spices sold are being irradiated, and even less pork is being treated.

The irradiation of food does not prevent enzymatic degradation, oxidative reactions, or recontamination by organisms. However, after a heat treatment such as blanching, to inactivate enzymes, the food can be irradiated after sealing in an oxygen-impermeable package so that recontamination does not occur. It is expected that radiation will not supplant other methods of food preservation, but rather, it will be used in conjunction with these systems.

INHIBITION OF SPROUTING. One of the major problems of storage of foods such as tubers or bulbs is sprouting.

The irradiation level needed to inhibit sprouting ranges from 50 to 150 Gy, depending on the cultivar. Limited sprouting has been observed on some types of potatoes receiving 5 to 10 Krad of irradiation.

High-level irradiation tends to increase the tendency of potatoes to rot. The rotting progressively increases as the irradiation dose increases. There is a tendency for gamma-irradiated potatoes to darken after cooking.

When performed within one or two weeks after harvest, the irradiation of onions with as little as 20 Gy inhibits sprouting. However, if stored for several weeks before treatment, irradiation even at higher levels will not prevent sprouting, but will soften the onions. The usual gamma ray treatment is with doses of 70 to 150 Gy. In one study, a level of 30 Gy prevented sprouting of garlic bulbs for 300 days (Croci and Curzio 1983), although Kwon, Byun, and Cho (1985) reported that 75 Gy is the minimal optimum dose.

Although irradiation was approved in 1964 to prevent sprouting of potatoes, the process has not been used commercially in the United States.

DELAY RIPENING. Irradiation can be used to extend the storage life of many fresh fruits by controlling the rate of maturation. To transport fruit to the market, it is necessary to pick it while it is in the green stage and sell it before it becomes overripe. Often, the ripening of the fruit is so rapid that the area of marketing is limited.

Irradiation with 250 to 500 Gy will delay ripening and extend the market life of most tropical and subtropical fruits. Temperate fruits require higher doses (1,000 Gy) to delay ripening, but this treatment may cause uneven ripening and excessive softening.

DESTRUCTION OF INSECTS. Insects can consume or ruin 25 percent or more of the grain stored in some countries. Chemicals can be used to destroy the insects, but there has been public concern that residues of fumigants and insecticides may be a health hazard. Irradiation not to exceed 1 kGy has been approved by the FDA for insect disinfestation of food. However, for herbs and spices, up to 30 kGy can be used. The adult state of insects is the most resistant and the egg phase the least resistant to radiation.

A dose of 250 Gy has been suggested as a quarantine treatment for fresh fruits and vegetables. According to Sommer and Mitchell (1986), enormous logistic problems, irradiation-induced injury, and adverse changes in quality will limit the number of foods that can be treated.

DESTRUCTION OF PARASITES. The most common parasitic diseases transferable from meat to human beings are cysticercosis of cattle and trichinosis of pigs. In 1985, the FDA approved gamma radiation at 0.3 to 1.0 kGy, to control trichinae in pork. This action, also approved by the USDA, allows the irradiation of fresh pork carcasses and cuts such as pork roasts and pork chops but does not allow the treatment of processed products. At the present time, radiation cannot be used as an alternative to curing, canning, or freezing of fresh pork for destruction of trichinae. Since *Trichinella spiralis* is killed by 0.3 kGy (Brake et al. 1985), this is the lower limit. A higher level gives a margin of safety.

Cysticerci of *Taenia solium* and *Taenia saginata* require higher doses of radiation for destruction. They were still infective after treatment with doses of 0.2 to 1.2 kGy. Various nematode larvae found in fish can infect the intestinal tract of humans. In fish with 6 percent salt, irradiation with only 10 Gy can result in a 100 percent kill of these larvae.

DESTRUCTION OF MICROORGANISMS. Ionizing radiations, especially from linear accelerators, are not necessarily uniform. Therefore, all treated microbial cells might not be exposed to equal levels of these random radiations. The death or injury of microorganisms is due to either direct or indirect effects. The direct effect is due to a direct hit with the radiation causing ionization within the microorganism while the indirect effect results from ionization and diffusion of free radicals and peroxides produced in the vicinity of the organism.

The ionizing radiation doses for destruction of microorganisms vary

with the type and concentration of cells, the type of substrate or suspending medium, oxygen tension, temperature, or the presence of protective compounds. Certain pre- and postirradiation treatments can influence the inactivation of microorganisms.

Cellular death from irradiation is related directly to lesions in the nuclear DNA. The damage caused to DNA is in the form of single- or double-strand breaks. The resistance and survival of microorganisms depend upon the capacity to repair any damage to the genetic material or other parts of the cell.

Death due to radiation is considered to be a first-order reaction, so that if the logarithm of survivors is plotted with the corresponding radiation dose on a linear scale, a straight line should be obtained. As with destruction by heat, various types of survivor curves are obtained, including shoulders at the beginning of irradiation and a tailing effect.

The D values should be relatively constant, regardless of dose. However, researchers found considerable variation with different levels of irradiation (Anellis et al. 1969). The D values can be used to reveal the relative resistances of microorganisms. Some of the reported D values are listed in Table 12.6.

RELATIVE RESISTANCES. The radiation resistance of strains and species of microorganisms can vary widely. Bacterial spores and viruses are among the more resistant of the microbial contaminants of foods. Generally, Gram-positive bacteria are more resistant than are Gram-negative types. Certain streptococci, deinococci, and *Acinetobacter-Moraxella* have very resistant vegetative cells. The D values for yeasts and molds are similar to those of most bacterial cells. Young cells are more sensitive to ionizing radiation than are old cells, with certain exceptions (Hastings, Holzapfel, and Niemand 1986; Keller and Maxcy 1984). The presence of oxygen during irradiation enhances cellular damage, apparently due to a reduction of a repair mechanism (Vinicombe, Moss, and Davies 1978).

Chemicals such as cysteamine, sodium formate, glycerol, and thiourea can protect cells from ionizing radiation (Antoku 1983; Billen 1983). Except for cured meats, higher doses of radiation are needed to destroy microorganisms in foods than in buffer solutions.

The removal of water generally increases the resistance of microorganisms to irradiation. This is thought to be due to reduced formation of H* and *OH radicals, the reduction in the mobility of free radicals that are formed, and to the protection given the cells by the dried substances.

Some of the least resistant bacteria are species of *Pseudomonas, Proteus, Escherichia,* and *Vibrio.* Ito and Iizuka (1971) isolated from rice a pseudomonad which they named *Pseudomonas radiora* because of its unusual resistance to radiation.

TABLE 12.6. REPORTED D VALUES FOR IRRADIATION OF SOME MICROORGANISMS

Microorganism	Suspending Medium	D Value (kGy)
Spores		
Clostridium botulinum		
Type A	Cured ham	2.18–2.35
	Bacon	2.14
Type B	Cured ham	1.64–1.75
	Pork loin	2.86–3.36
	Chicken	3.70
Type E	Water	0.80–1.60
Type F	Water	2.50
C. sporogenes	Water	1.60–2.20
C. perfringens	Water	1.20–3.40
Bacillus cereus	Broth	2.50
B. stearothermophilus	Buffer	2.52
	Buffer (anoxic)	4.25
Vegetative Cells		
Escherichia coli	Buffer	0.09
Salmonella typhimurium	Buffer (air)	0.20
	Buffer (anoxic)	0.60
	Dried bone meal	0.91–1.00
	Dried fish meal	1.70
	Broth	0.28–0.56
Salmonella oranienburg	Dry	0.37
Staphylococcus aureus	Buffer	0.20–0.25
	Crab meat	0.40
	Broth	0.22–0.29
Lactobacillus plantarum	Broth	0.47–0.69
Enterobacter cloacae	Dry	0.18
Pseudomonas sp.	Dry	0.38
Aeromonas hydrophilia	Fish (22°C)	0.11–0.15
	Fish (−15°C)	0.22–0.34
	Beef (2°C)	0.14–0.19
Yersinia enterocolitica	Broth	0.097–0.118
	Beef (25°C)	0.195
	Beef (−30°C)	0.388
Campylobacter jejuni	Beef (−30°C)	0.315
	Turkey (−30°C)	0.293
	Turkey (30°C)	0.162
	Broth (30°C)	0.174
Deinococcus radiodurans	Buffer	1.94–3.08
	Raw beef	3.5–6.0
Vibrio parahaemolyticus	Buffer, pH 7.0	0.05–0.14
Molds		
A. flavus (conidia)	Water	0.38
	Dry	0.50–0.55
	Cacao beans	0.66
Penicillium citrinum		
(conidia)	Wet	0.18
(mycelium)	Dry	0.55
	Cacao beans	0.88
Yeasts		
Saccharomyces cerevisiae	Buffer	0.36
Candida sp.	Buffer	0.32

TABLE 12.6 (*Continued*)

Viruses		
Coxsackievirus B-2	Eagles medium	4.40–6.90
	Water (−90°C)	5.30
	Cooked beef (16°C)	7.00
	Cooked beef (−90°C)	8.10
Adenovirus 2	Eagles medium	4.10
Adenovirus 3	Eagles medium	4.90
Echovirus 18	Eagles medium	4.40
Toxins		
Botulinum		
Type E toxin		
washed cells	Buffer	17–21
cell extract	Buffer	2.10
purified	Buffer	0.40
Type A toxin		
purified	Buffer	0.50–2.00
purified	Broth	40
purified	Cheese	60
Staphylococcal		
Enterotoxin B purified	Buffer	27
	Milk	97

SOURCES: Anellis et al. (1965, 1967, 1969); Ito, Iizuki, and Sato (1974); Kopelman, Markakis, and Schweigert (1968); Miura et al. (1970); Read and Bradshaw (1967); Roberts (1968); Sullivan et al. (1971a, 1973); Hastings, Holzapfel, and Niemand (1986); Tsuji (1983); Restaino et al. (1984); El-Zawahry and Rowley (1979); Lambert and Maxcy (1984); Thomas, Ouwerkerk, and McKercher (1982); Palumbo et al. (1986).

Escherichia coli is quite sensitive to ionizing radiation, although there are resistant strains. Due to the rapid demise of *E. coli,* this organism is a poor indicator of sanitation for irradiated food. Nearly all of the organisms involved in foodborne illness have a higher resistance to irradiation.

Serotypes of *Salmonella* vary in their resistance to irradiation. The relative resistance of serotypes shows that the mechanisms are different for the destructive action by heat and radiation. *S. senftenberg* 775W is the most heat-resistant strain, while the radiation resistance of this organism is more similar to other strains and serotypes of salmonellae.

A radiation-resistant coccus was isolated from irradiated meat (Anderson et al. 1956). Although originally named *Micrococcus radiodurans,* this organism, like some other radiation-resistant cocci, is distinct from other micrococci. Therefore, the name *Deinococcus radiodurans* was suggested by Brooks and Murray (1981). Reportedly, the high radiation resistance of these organisms is due to a very efficient system to repair DNA lesions.

Moraxella and *Acinetobacter* are more resistant than pseudomonads to radiation. Ito, Sato, and Iizuka (1976) reported the D values for *Moraxella* species to vary from 44 to 54 Krad and *Acinetobacter* to be 12 Krad. These

values are much lower than the 477 to 1,000 Krad for *Moraxella* and 405 to 814 Krad for *Acinetobacter* reported by Welch and Maxcy (1975).

C. botulinum spores are generally more resistant than other clostridial spores. This is very important if ionizing radiations are used for sterilization of food. Spoilage types of clostridia could be eliminated from the food, while *C. botulinum* spores could remain viable and produce toxins. Hence, the consumer might not be forewarned of possible danger by gas formation or putrid odors normally developed in contaminated food by spoilage types.

The required minimum radiation dose to achieve commercial sterility of food is based on the resistance of *C. botulinum* spores. In general, type A spores are more resistant than type B spores. Type A and B spores are more resistant to radiation than are types C, D, E, or F spores.

The temperature at which irradiation occurs influences the resistance of *C. botulinum* spores. From 65°C down to −196°C, the spore resistance increases with decreasing temperature (Grecz et al. 1971).

The presence of curing salts (sodium nitrite, sodium chloride, and sodium nitrate) reduces the irradiation dose needed to inactivate spores of *C. botulinum.* This is evident by the lower D values for *C. botulinum* in ham or bacon than in pork loin or chicken.

Viruses tend to be more resistant than other organisms to gamma irradiation. Viral inactivation demonstrates a one-hit or first-order reaction curve, when the suspending medium contains free radical scavengers or is in a frozen state. Freezing increased the resistance in water and cooked beef, but this increase was not observed in raw beef (Sullivan et al. 1973). Fortunately, viruses are not found in most foods. If present, they tend to be at low levels.

One team of researchers suggested that the irradiation inactivation of Newcastle disease virus at low temperature (2.2C) was due to nucleic acid degradation, while at higher temperatures (60°C), it was due to protein denaturation (DiGioia et al. 1970). Survival is enhanced in the presence of impurities. These impurities apparently bind free radicals and peroxides giving protection to the viruses.

Radappertization. Radiation sterilization or radappertization is the destruction of all or practically all of the organisms. To accomplish commercial sterility, relatively high levels of irradiation are required. Of all the microorganisms of concern that are associated with food, the spores of *C. botulinum* are the most resistant to irradiation. The spores of *C. botulinum* are more resistant to ionizing radiation than spores of organisms that have a high heat resistance and cause spoilage.

For ionizing radiation as with heat treatment, a 12D concept is used

for producing commercially sterile foods that are reasonably safe from *C. botulinum* spores or toxin. If the D used is 3.75 kGy, for commercial sterility, a 12D treatment is 45 kGy. According to Josephson (1969), the minimum radiation dose (MRD) needed for commercial sterilization varies from 23 kGy for bacon irradiated at ambient temperature to 57 kGy for beef irradiated at −80°C. The 12D irradiation doses at −30°C listed by Rowley and Brynjolfsson (1980) varied from 25.5 kGy for pork sausage to 43.7 kGy for pork, with beef at 41.2 kGy.

Five spices with bacterial levels of 10^2 to 10^5/g were sterilized with 7.5 to 10 kGy (Sharma et al. 1984). However, for commercial sterility of several spices, the treatment varied from 6 kGy for poultry seasoning to 13 kGy for black pepper and thyme (Grecz, Al-Harithy, and Jaw 1986).

Radiation doses for radappertization can cause losses in the organoleptic quality of food. The higher the dose, the greater is the potential for development of undesirable flavors, odors, and colors, as well as losses of nutrients.

Of the common meats, beef is the most sensitive and pork the least sensitive to the production of off-flavor. Although these changes in foods due to irradiation have been of concern, often they are less than the alterations due to heat processing.

To minimize alterations of foods due to irradiation, various combination processes have been suggested. These processes include addition of chemicals that are free radical acceptors, use of anaerobic conditions, elimination of moisture, or irradiation at freezing temperatures. Each of these systems reduces the objectionable activity of the free radicals.

The amount of nitrite needed in cured ham can be reduced when radappertization is used to control *C. botulinum* (Wierbicki and Heiligman 1973). For this treatment, the ham is frozen and treated at −30°C with 3.7 Mrad. The nitrite was reduced to 25 μg/g, but to prevent fading of the cured color, 100 μg/g of nitrate was added.

Even though nearly all of the bacteria are killed by radappertization, enzymes are not destroyed as readily. Hence, enzyme activity continues, especially when the food is held at ambient temperature.

Radappertization of meat involves a preirradiation treatment with heat (65° to 80°C) to inactivate proteolytic enzymes, sealing in a package under a partial vacuum, freezing to −30°C or lower, and exposing the packaged food to ionizing radiation until the desired dose is absorbed.

Freezing prior to irradiation reduces the rate at which free radicals diffuse through the tissue. This minimizes the side reactions of irradiation and reduces the alteration of odor and flavor. Generally, the lower the temperature of irradiation, the more the quality factors are protected. However, as the temperature is lowered, the microorganisms, in-

cluding *C. botulinum* spores, have increased resistance to irradiation, so that higher doses are needed to sterilize frozen meats than meats at ambient temperatures.

Radappertization appears to be beneficial for sterilizing and storage of foods when refrigeration and frozen storage are not available. There are disadvantages and problems associated with this treatment. These include the need for heating and freezing during the process, the alterations in organoleptic and nutrient quality, and the cost of the operation. Thus, it will be difficult for radappertization to compete with the present preservation procedures used in the United States. However, radiation-sterilized foods may be produced for hospital patients with suppressed immune systems. Also, radiation-sterilized foods have been used to feed astronauts during space flights.

Radurization. The use of low doses of radiation to destroy a sufficient number of microorganisms and enhance the storage life of foods is called radiation pasteurization or radurization.

The doses of radiation used for radurization (1 to 10 kGy) will destroy from 90 to 99 percent of the microorganisms. Some resistant vegetative cells and spores will survive. Hence, refrigeration or some other combination treatment is needed to retard growth and possible toxin production of the survivors.

The dose used for radurization is a compromise. High doses prolong storage life, but deleterious effects and costs increase as the dose increases. Quite often, the dose used is the maximum amount of irradiation that produces no detectable change in the product. The other possibility is to determine the storage life needed to market a product and fit the dose of irradiation to this requirement.

The psychrophilic pseudomonads are the main spoilage organisms of protein foods (meat, poultry, fish). These bacteria are very sensitive to irradiation, being essentially eliminated by doses used for radurization. Enterobacteriaceae, pseudomonads, and enterococci are eliminated from vacuum-packed beef when irradiated with 2 kGy (Niemand, van der Linde, and Holzapfel 1981). During storage of this beef at 4°C, radiation-resistant lactobacilli grew and became dominant. The majority of strains isolated from minced beef radurized with 5 kGy were *Lactobacillus sake* (Hastings and Holzapfel 1987). After storage for 15 days at 3°C, spoilage odors were detected in ground beef treated with 1.03 kGy but not in that irradiated with 1.54 kGy (Dempster et al. 1985).

The treatment of fishery products with 1 to 6 kGy can increase the refrigerated storage life (Angel et al. 1986; Licciardello et al. 1984; Venugopal et al. 1982). The effectiveness of radurization varies with the type and initial quality of the product and the associated microbial flora. One problem is the presence of *C. botulinum* type E and their potential radia-

tion resistance in fishery products. However, according to Rowley, Firstenberg-Eden, and Shattuck (1983), any type E spores surviving 1 to 4 kGy are injured and do not produce toxin at 10°C or less.

The foods that rank highest in merit for radurization are shrimp, fin fish (haddock, flounder, cod), blue crab, strawberries, and mushrooms. These foods are perishable and have high spoilage rates during marketing. They are of relatively high value and, at some marketing stage, they have large volume concentrations.

Radicidation. This is a low-level irradiation treatment to destroy viable nonsporeforming pathogenic organisms to reduce or eliminate the problem of foodborne illness. Spores of *C. botulinum* or *C. perfringens* are not destroyed at levels normally used for radicidation. This term was suggested to describe the ionizing radiation treatment needed for elimination of salmonellae from food and feed.

The D values in ground beef ranged from 0.55 to 0.78 kGy for four serotypes of salmonellae, from 0.10 to 0.21 kGy for three strains of *Y. enterocolitica,* and from 0.14 to 0.16 kGy for three strains of *C. jejuni* (Tarkowski et al. 1984). In frog legs, the D values for three serotypes of salmonellae ranged from 0.18 to 0.30 kGy (Nerkar and Lewis 1982), which is considerably lower than the D value reported for salmonellae in ground meat.

Tarkowski, Beumer, and Kampelmacher (1984) stated that a dose of 1 kGy eliminates or reduces some pathogens in raw ground meat to acceptable levels. Mulder (1984) suggested that to decrease potentially pathogenic organisms and to increase shelf life of poultry, doses ranging from 2 to 9 kGy are needed, but doses over 5 kGy may cause undesirable side effects in this food. In a review by Dempster (1985), a level of 7 kGy is recommended to eliminate salmonellae.

Besides eliminating pathogens, this low-dose irradiation will destroy parasites such as tapeworms and trichinae.

Heat treatment can be used to eliminate salmonellae from dried feed such as meat, bone, fish, and blood meal, but there is a potential for recontamination prior to packaging, and heating decreases the available protein. A dose of 8 to 10 kGy can reduce salmonellae in packaged dry feeds to below detectable levels. Irradiation of animal feeds to eliminate salmonellae would be an important step in breaking the salmonellae cycle and might reduce the involvement of animal foods in human salmonellosis.

Dose levels of 2 kGy are sufficient to destroy all expected levels of *Staphylococcus aureus* in most food products. The problem with *S. aureus* is the recontamination by people handling the food during preparation for serving. Radicidation cannot prevent this from occurring.

Paster, Barkai-Golan, and Padova (1985) found enhanced production

of ochratoxin by *Aspergillus ochraceus* after irradiation with 0.5 to 2.0 kGy. No growth occurred when a dose of 3 kGy was used. These researchers suggested that care should be taken when agricultural products on which mycotoxigenic fungi can grow are irradiated with low doses.

These low levels are not expected to control sporeforming bacteria. Hence refrigeration of foods so treated is needed to prevent germination, outgrowth, and toxin production by organisms such as *C. botulinum.*

Since both radurization and radicidation utilize essentially the same intermediate levels of radiation, some authors prefer to use the term *radiation* pasteurization for both processes. Although this level of irradiation is included in the international level of usage of 10 kGy, the present FDA level of 1 kGy is not sufficient to allow the treatment of foods to eliminate salmonellae and many other pathogens.

Thermoradiation. This is a combination treatment of heat and radiation. Enzymes and spores of *C. botulinum* are relatively resistant to radiation but are more sensitive to heat. Conversely, thermophilic spores are very heat resistant, while relatively more sensitive to radiation. A combination of heat and radiation, either in sequence or simultaneously, may be of value in food preservation.

Preheating to sublethal temperatures can sensitize vegetative cells but not spores to irradiation. Preirradiation sensitizes spores to subsequent heat treatments. In some cases, simultaneous heating and irradiation are more effective than applying the treatments in sequence. Heat and irradiation are synergistic when used together for killing spores of anaerobes. Thus, it is possible that this synergism can be used to sterilize food so that the quality of the food is less affected than with the level needed for either treatment applied alone. The synergistic effect is evident only in certain temperature and radiation dose ranges, and these depend upon the organism being treated.

Shrimps subjected to heat at 121°C for 8 min and irradiated with 1.0 kGy can be stored up to two months at ambient temperatures (Savagaon et al. 1972). Canned products with excellent quality can be obtained by thermoradiation.

Synergistic effects of temperature and radiation have been determined for several organisms, including *B. stearothermophilus.* For this organism, the D of 2.10 kGy at 27°C was reduced to 1.57 kGy at 60°C (Tulis, Fogarty, and Sliger 1973).

INACTIVATION OF TOXINS. Microbial toxins reportedly are inactivated at radiation levels comparable to bacterial spores. However, the amount of radiation needed to inactivate botulinum toxin is greater than that needed to destroy the organisms (see Table 12.6).

The purer the toxin, the more sensitive it is to irradiation (Miura et al. 1970). This indicates that impurities protect the toxin in much the same way that cells are protected. Miura and researchers (1970) found that serum albumin, casein, DNA, and RNA protected purified and activated toxin against radiation inactivation. Sugars or ascorbic acid showed little or no protection.

Staphylococcal enterotoxin is more heat stable than botulinum toxin. This higher stability tends to be true for inactivation by ionizing radiation. With the D values for inactivation of enterotoxin determined by Read and Bradshaw (1967), it would be impossible to rid a food of significant amounts of this toxin without destroying the quality of the food. Low doses of irradiation would have little or no effect on either botulinum or staphylococcal toxins. A dose exceeding 10 kGy was needed for total destruction of aflatoxin B_1 (Van Dyck et al. 1982).

INACTIVATION OF ENZYMES. The complete, or at least partial, inactivation of enzymes has been considered essential for preservation of foods. The inactivation of enzymes requires higher doses of radiation than that for microbial destruction.

To determine if milk has been heat pasteurized, researchers test for the presence of phosphatase enzyme. During irradiation of milk, phosphatase is not affected.

Heat treatment of meats to inactivate the enzymes prior to irradiation improves the storage stability of these products. This preheating treatment applies not only to red meats, but also to poultry and seafoods. Even 45 to 52 kGy of radiation will inactivate only about 75 percent of the proteolytic activity of beef (Losty, Roth, and Shults 1973).

The heat treatment needed to inactivate enzymes in fruits is sufficient for preservation, so the use of ionizing radiation is unnecessary. There is little advantage in using radiation for the preservation of vegetables.

Since enzymes are more resistant than microorganisms to irradiation, it is possible to use this process to sterilize or at least significantly reduce the microbial level in commercial enzyme preparations.

Quality of Irradiated Foods

At some level of ionizing radiation, microorganisms, toxins, and even some enzymes can be destroyed or inactivated. However, at the doses necessary to destroy viruses, spores, or even some vegetative cells, the quality of the food may suffer.

The effect of ionizing radiation varies with the type of food. Some foods, such as milk, are very susceptible to even low doses of radiation. Other foods, such as cured meats, can be treated with relatively high

doses of radiation without the quality being seriously affected. According to a review by Reineccius (1979), odor changes in meats result from the formation of carbonyl compounds due to reactions of components of fats and from side chains of proteins, as well as from carbohydrates.

Ionizing radiation can cause undesirable effects in food; however, other systems of preservation, especially heating, can change the organoleptic and nutrient factors in food. In many cases, the effect of radiation is less drastic than is that of heating.

Undesirable effects resulting from radiation are generally due to free radicals produced when ionizing radiations react with water in the food. Thus, the effects on quality can be reduced by irradiation in the frozen or the dried state, by the use of free radical acceptors, or by irradiation in inert atmosphere or in vacuum. Although these procedures will help reduce, they do not necessarily prevent changes in food.

The irradiation of wheat with low levels may increase the resultant bread volume and overall quality (Paredes-López and Covarrubias-Alvarez 1984). However, they found that the overall bread quality was reduced at medium doses of radiation. MacArthur and D'Appolonia (1983) found a reduced quality of bread at 1.0 kGy treatment of wheat.

Wholesomeness

In the United States, as well as many other countries, statutes require that proof of safety for consumption (wholesomeness) must be adequate to convince the appropriate health-regulating officials before irradiation of foods will be approved. The factors considered are nutritional adequacy, microbiological safety, absence of induced radioactivity, and absence of carcinogenic or other toxic substances that may be formed by exposure of the food to ionizing radiations.

Under provisions of the Food, Drug and Cosmetic Act, the FDA must approve the use of ionizing radiation for each food given this treatment. The USDA has legal responsibility for meat and poultry and their products.

The wholesomeness of irradiated food is determined by laboratory analysis of constituents in the food, as well as by using animal feeding studies. The many wholesomeness studies have revealed that irradiated foods are as wholesome and nutritious as thermally processed foods (Brynjolfsson 1985). Hospital patients taking immunosuppressant drugs, people with immune system disorders, and astronauts have consumed irradiated foods with no apparent bad effects.

NUTRITIONAL ADEQUACY. There is concern about the effect of radiation on the proteins, energy sources, and vitamins. In general, the loss

of any nutrient is dependent on the dose of radiation, as well as the environment during irradiation. The low doses of radiation used for sprout and insect control would be expected to cause insignificant nutritional damage. At high doses (45–55 kGy) needed for sterilization, the nutritional adequacy of irradiated foods is comparable to conventional heat-processed foods.

In some cases there is an apparent increase in the nutritional value of food due to irradiation (Jami, Pubols, and McGinnis 1980; Reddy, Pubols, and McGinnis 1979). One concern is the loss of thiamin due to irradiation (Skala, McGown, and Waring 1987).

MICROBIOLOGICAL SAFETY. With radappertized foods that are shelf stable at ambient temperatures, it is essential that all cells and spores of *C. botulinum* be destroyed. This is also true of thermally processed canned foods. In either case, residual, viable spores can result in a toxic food.

There is a possibility that food sterilized by radiation may contain or develop toxin after processing. The amount of radiation required to inactivate the toxin is greater than that needed to destroy the organisms.

Inasmuch as radiation sterilization does not have the many years of widespread distribution and consumer use as does thermally processed food, it is difficult to compare the potential hazards of the two treatments. In general, it is agreed that with proper controls throughout processing and distribution, irradiated foods will present no microbiological hazards to the consumer.

By virtue of eliminating organisms such as salmonellae, *S. aureus, Vibrio parahaemolyticus,* and shigellae, irradiation can be used to reduce the number of foodborne illnesses.

INDUCED RADIOACTIVITY. No radioactivity is induced when suitable radiation sources are used. Electron accelerators with energy levels above 10 million electron volts (Mev) may induce small, but significant, amounts of radioactivity. Therefore, this treatment is considered unsuitable for irradiation of food. Below 10 Mev no induced radioactivity can be detected. Great Britain limits the accelerator level to 5 Mev.

Irradiation of food with gamma rays from cobalt^{60}, even at high dose rates, induces less radioactivity than can be detected with the most sensitive instruments available.

CARCINOGENS AND OTHER TOXIC FACTORS. The effect of ionizing radiation on the chemical components in food and an estimation of the concentration of radiolytic products has been reported (Diehl and Scherz 1975; Merritt 1980; Nawar 1978). Numerous animal feeding stud-

ies have revealed no toxic effects of foods when irradiated at doses below 68 kGy (Renner et al. 1982; Thayer et al 1987).

Most of the radiolytic products that have been identified also occur naturally in a wide variety of foods. The concentration of radiolytic products is such that they are generally not considered to be of toxicological significance.

Animals fed irradiated foods have developed tumors. However, the incidence of tumors has been no higher with a diet of irradiated than unirradiated foods. Proof of the absence of carcinogens or other toxic substances in food due to irradiation is extremely difficult. No demonstrable carcinogenicity has been shown in the extensive tests that have been done.

Economic Aspects

The irradition cost per unit depends to a large extent upon the annual volume of an irradiation plant, the type of irradiation source (including its efficiency and price), the capital investment, and the proximity of the processing plant to the food source.

The centralization of a food industry is necessary before irradiation can be accomplished at a reasonable cost per unit. On the other hand, the equipment needed for thermal processing or freezing is available for various sized operations. Only certain foods are accumulated in large volume in one location.

According to Swientek (1985), cost estimates have ranged from less than 2 cents per kilogram to more than 20 cents per kilogram of food. Beattie and Wiblin (1984) reported that, although the cost of fruit and vegetable treatment could be quite low, there is a lack of demand for the process, and an irradiation facility could not be operated economically in Australia.

Less energy is needed for irradiation of foods than for heat treatment, refrigeration or freezing (Diehl 1983). However, radiation-pasteurized foods must be refrigerated, and radappertized foods must be heated to inactivate enzymes and then frozen to prevent undesirable reactions during irradiation, which must be considered part of the overall energy requirements for this treatment.

There are some benefits of radiation treatments. No other processing system can eliminate salmonellae from frozen meat or poultry products. No one system will eliminate salmonellosis completely, but if 50 percent of the cases of foodborne illness could be eliminated by irradiation of certain foods, a considerable savings could be envisioned.

Some benefits that have been suggested for irradiation of food include the money saved by reducing food spoilage, the extension of stor-

age life, a saving in transportation and storage costs, and a reduction in the number of cases of foodborne illness. It has been suggested that, with irradiation, there would be a larger variety of foods essentially free of pathogens and parasites. The losses due to insects and sprouting would be reduced. There would be an increased availability of foods to undernourished peoples, expanded export potential, and market stabilization. These advantages would be for both the processor and the consumer.

The energy requirements for radappertized and radurized foods should be compared with thermal processing, freezing, drying, or other preservation methods. If radiation treatments save energy, they may be necessary in the future if we hope to maintain some semblance of our present standard of living. When all of the benefits are considered, for some foods they outweigh radiation costs by a wide margin. However, for other foods, irradiation is not practical due to quality alterations or the cost involved.

Summary

Preservation methods, including irradiation, will not improve the quality of a food, but they will tend to retain the attributes of high-quality food. Irradiation has the potential to reduce the enormous loss of food so that more food will be available to the ever-increasing population.

Radiation is not expected to have any substantial effect on the current systems of processing and distribution. However, radiation has the potential for reducing food loss due to sprouting, insects, and microorganisms, of decreasing hazards, of increasing food quality and convenience, and of decreasing costs of storage, distribution, and marketing.

Radiation will be of considerable importance in eliminating salmonellae from meats and poultry products, as well as from bulk animal feeds. No other process we now have can accomplish this task. In 1976, the World Health Organization approved the radiation of poultry with doses up to 700 Krad. In Canada, levels up to 750 Krad are approved for controlling salmonellae on frozen poultry. The Netherlands has approved irradiated poultry with doses up to 300 Krad. It is expected that the FDA and the USDA will approve the use of irradiation to eliminate salmonellae from poultry and red meats. Presently these foods are contaminating kitchens throughout the United States with salmonellae.

In the United States, we have highly acceptable frozen, dried, and thermally processed foods. However, in less developed areas in which these or other systems are not as adequate, radappertized foods have a greater importance. With the continued problem of energy sources, irradiation of food might become an important preservation method in the future of the human race, regardless of the country.

ULTRAVIOLET RADIATION

The antimicrobial effect of ultraviolet (UV) light has been known for about 100 years. The lethal action of UV light varies with the intensity of the light and the time of exposure. The temperature, pH, relative humidity, and amount of microbial contamination influence the effectiveness of UV light. According to Severin, Suidan, and Engelbrecht (1983), UV disinfection is relatively insensitive to temperature changes.

Useful Aspects

The energy of UV light is much lower than that of ionizing radiations. This means that UV light has low penetrating power. Ultraviolet light is effective against microorganisms in air, liquids that are clear or in thin films, and on surfaces that do not produce shadows. Microorganisms are protected by dust in the air, as well as by dust on UV light bulbs. Food particles or other substances that UV cannot penetrate protect the microorganisms. Clumping of microbial cells allows those in the central part of the clump to escape the biocidal effect of UV light. Over 50 percent of the radiation energy is lost at a depth of 5 cm in clear water. Penetration into milk is only about 1 or 2 millimeters.

The suggested or actual uses of UV light include the control of microbial growth in bulk-stored maple sap, tenderizing, aging, and retailing of meat, the removal of ethylene from banana storage areas, sterilization of packaging materials used in aseptic packaging, control of surface growth on bakery products, treatment of water and wastewater, destruction of thermophiles in refined sugars intended for canned foods, sanitation of equipment, and air purification (Brown and Russo 1979; Gilpin et al. 1985; Huang and Toledo 1982; Mans 1987; Qualls and Johnson 1983; Robinson and Adams 1978; Severin 1980; Stannard, Abbiss, and Wood 1985; Stermer, Lasater-Smith, and Brasington 1987). The production of ozone by certain wavelengths of UV light gives an added germicidal effect.

Relative Microbial Resistance

With some exceptions, UV irradiation is about equally effective against yeasts and either Gram-positive or Gram-negative bacteria. Certain species of *Deinococcus,* such as *D. radiodurans,* have exceptionally high resistance to UV as well as ionizing radiations, apparently due to a very efficient repair system. *E. coli* is more resistant to UV than either *Y. enterocolitica* or *C. jejuni* (Butler, Lund, and Carlson 1987). Of the pseudomonads tested, *P. aeruginosa* was the most sensitive to UV radiation (Abshire

and Dunton 1981). They also reported increasing resistance from *S. aureus* through *E. coli, S. faecalis, Candida albicans,* to *D. radiodurans.* However, *C. albicans* was only about 25 percent as resistant as *D. radiodurans.* Researchers found that UV inactivation of *E. coli* was comparable to that for *S. typhi, Shigella sonnei,* and *S. aureus* (Chang et al. 1985). Poliovirus and rotavirus were three to four times as resistant, and spores of *B. subtilis* about nine times as resistant as *E. coli.*

Light rays can cause chemical changes and biological damage only if they are absorbed. Light that passes through a cell has no effect. Due to the high absorption of light with wavelengths between 210 and 300 nm, there is a strong biocidal effect. Several "most effective" wavelengths have been listed in the literature. The wavelength most often suggested is 253.7 nm; however, the wavelength with optimum biocidal effect varies with different microorganisms.

Ultraviolet radiations between 210 and 300 nm are absorbed by proteins and nucleic acids. As a result of chemical reactions, there is chromosome breakage, genetic mutation, and enzyme inactivation which can result in cellular death. There is a variety of genetic alterations including base-pair substitution and frameshift mutations, deletions, mitotic crossing over, and mitotic gene conversion induced by UV light (Hoffman and Morgan 1976).

Ultraviolet light produces potentially lethal photoproducts in the cellular DNA. A large portion of the bactericidal activity of UV results from nucleotide dimer formation. These dimers are the most abundant and stable of the photoproducts. They inhibit DNA synthesis and, to a lesser extent, RNA and protein synthesis.

Repair of Lesions

The irradiation of bacteria with ultraviolet light produces a variety of pyrimidine dimers in the DNA. The repair of these lesions can be accomplished by photoreactivation, excision (dark) repair, or postreplication repair. Both photoreactivation and dark repair mechanisms have been described in a variety of microorganisms.

Photoreactivation is the reversal of short-wavelength UV damage by postirradiation exposure to long-wavelength light. Photoreactivation is enzymatic and functions by splitting the dimers in situ. The photoreactivation of damage caused by UV light clearly establishes a major role of pyrimidine dimers in the inactivation of various cells, especially radiation-sensitized mutants. The enzymes involved in this repair are called the photoreactivating enzymes.

Excision repair of pyrimidine dimers is considered to be a sequence of enzymatic steps. This involves incision breaks, removal of the dimers,

resynthesis of the excised regions, and bonding between the newly synthesized and preexisting DNA to restore the continuity of the repaired DNA strand. Strains deficient in excision repair are more sensitive to UV radiation than are repair-proficient strains.

At high fluences, lethality may be enhanced by damage to the excision and recombination repair systems (Webb and Brown 1976). Certain enzymes can be inactivated by light. The inactivation of photoreactivating enzymes by UV light has biological significance. One of the causes of death of cells is the disappearance of the enzyme responsible for the final step in the repair of DNA.

The rate of inhibition and death depends, at least partially, on the rate of repair. With large doses of radiant energy, the repair cannot maintain a balance and photoinhibition occurs. The occurrence of sigmoidal inactivation curves could be related to such effects.

The repair of DNA damaged by UV light has been discussed in several reports (Adkins and Allen 1982; Friedberg 1975; Pollard and Fugate 1978; Prakash 1976; Sutherland 1977).

Protective Agents

Plasmids may confer partial protection against the bactericidal effects of UV irradiation. Siccardi (1969) reported that of thirty-one plasmids, fifteen protected, eleven had little or no effect, and five increased the susceptibility of *E. coli* K-12 to UV irradiation. How these plasmids afford protection is not known. However, the authors offered suggestions as to the possible mechanisms for this occurrence.

E. coli was protected from mutagenesis and inactivation by broad spectrum near UV light when catalase was incorporated into the plating medium (Sammartano and Tuveson 1984). Glycerol protects bacterial cells against near UV (365 nm) but not against 254 nm UV light (Peak and Peak 1980; Peak, Peak, and Foote 1982).

Interactions

The lethal interaction of UV radiation and mild heat on *E. coli* was studied by Tyrrell (1976). Irradiation with most wavelengths sensitized the cells to temperatures of 45° or 48°C. He proposed that, in addition to DNA damage, both mild heating and near UV treatment interfere with DNA repair mechanisms. Hence, the combination of the two treatments leads to a strong positive interaction.

The simultaneous use of UV and ozone produced synergistic effects on molds but not on bacteria (Kaess and Weidemann 1973).

The interaction of UV light and ionizing radiation was studied by

Schneider and Kiefer (1976). Submitting *Saccharomyces cerevisiae* to UV light followed by X-rays, and also in the reverse order, revealed differing reactions. With UV, then X-ray treatment, synergism was observed with all strains tested. With the reverse treatment, differences in synergism were obtained with different strains.

The simultaneous treatment of hydrogen peroxide and UV light has a synergistic killing effect on both sporeforming and nonsporeforming bacteria (Bayliss and Waites 1980, 1982).

Other studies have revealed an X-ray-ultraviolet synergism for diploid yeast, certain strains of *E. coli, Aspergillus nidulans, Clostridum botulinum,* and *Serratia marcescens.*

OTHER SYSTEMS

There are other systems that can inactivate microorganisms, but they have not been developed for commercial usage. This is probably because, at present, there are few, if any, practical applications for these processes.

Visible Light

The effects of visible light on microorganisms were reviewed by Harrison (1967). He listed the types of damage induced by light as loss of colony-forming ability, photolysis, lengthening of lag period, loss of fermentative ability, and reduction of ability to support phage growth. The presence of carotenoid pigments in the bacteria sometimes enhances the survival of bacteria when exposed to light.

Strains of *E. coli* inactivated by near UV light also are sensitive to inactivation by visible light (Tuveson and March 1980). Visible light kills *Bacillus* spores exponentially (Abad-Lozano and Rodriguez-Valera 1984) as well as prevents or retards the outgrowth of surviving spores (Quesnel and Spencer 1985).

The photodynamic effect on bacteria can be enhanced by simultaneously exposing the organism to light, a photosensitizing compound (usually a dye) and to oxygen (Banks, Board, and Paton 1985).

Laser Energy

Pulsed laser energy effectively inhibited the growth of *E. coli* (Takahashi et al. 1975). Other research has indicated the requirement of a photosensitizing dye for inhibition of microorganisms to occur. In the pulsed laser energy study, however, researchers found no difference in inhibition with or without added toluidine blue, and less inhibition when

acridine orange was present (Takahashi et al. 1975). They stated that technology was just beginning to produce tunable laser irradiation powerful enough for microbiological research.

Ultrasonic Energy

Ultrasonics is the transmission of vibrational energy above the level of audible sound. When alternating current is applied to a crystal, the shape of the crystal changes with the electrical field. These continuous changes in shape cause pulsations or waves which travel through a liquid. These waves have alternate compressions and rarefactions. If the amplitude is high enough, a phenomenon known as cavitation is produced. This is the making and breaking of microscopic bubbles. These bubbles tend to become larger and when they grow to what is known as resonant size, they collapse instantly and violently, producing high local pressures. This pressure is the force that alters the microbial cells.

When microorganisms are subjected to ultrasonics, the overall result may be molecular bond breakage, physical damage by shock waves, denaturation, reactions with free radicals, or a combination of these effects.

Microbial cells vary in their ease of ultrasonic disruption. Cells that are relatively easy to disrupt include *Pseudomonas, E. coli,* salmonellae, and vegetative cells of *B. cereus.* Cells that are considered difficult to disrupt include *Staphylococcus aureus,* streptococci, lactobacilli, *Serratia marcescens, B. cereus* spores. *Candida albicans* spores, and poliovirus. Generally, the smaller and rounder the cell, the more difficult it is to disrupt. Spores are not greatly affected by ultrasonic treatment.

In a laboratory, suspensions of cells can be inactivated after long exposure to ultrasonics. However, commercially, this process seldom provides a 100 percent kill of microorganisms. Reportedly, another potential use of ultrasonics is the inactivation of toxins.

The combination of ultrasonics and UV radiation acts synergistically to destroy microbial cells. Synergism has been noted with chemicals, such as chlorine or hydrogen peroxide and ultrasonic treatments (Ahmed and Russell 1975). The time required for the fumigant gases ethylene oxide or propylene oxide to kill microorganisms is reduced by simultaneous treatment with ultrasonic waves.

Ultrasonic treatment lowers the resistance of *D. radiodurans* and *S. faecalis* to ionizing radiation. According to Burgos, Ordonez, and Sala (1972), the heat resistance of *B. cereus* and *B. licheniformis* spores decreased markedly after ultrasound treatment. Ordonez and Burgos (1976) reported that ultrasonic waves do not affect the heat resistance of some strains of *Bacillus* species.

Electricity

Stersky, Heldman, and Hedrick (1970) used various voltages to inactivate microorganisms aerosolized into a chamber. As the voltage increased from 6,000 volts to 20,000 volts, the reduction of the bacterial population increased. The highest mean reduction they reported was 59.2 percent.

Edebo (1969) found the microbicidal effect of transient electric arcs on *E. coli* in tap water was similar to that of UV radiation, both biologically and chemically.

Gram-positive bacteria and yeasts were less sensitive than were Gram-negative bacteria (Hülsheger, Potel, and Niemann 1983). Cells in the logarithmic phase of growth were more sensitive than were those in the stationary phase. They found there was selective damage to the inner cell membranes.

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13

Regulations and Standards

NEED FOR REGULATIONS

When the United States began to change from a rural society to an urban one, food science was nonexistent, microbiology was in its infancy, and little was known about the chemistry of foods. Workers in the food business were trained in the need for sanitation. The adulteration of food was common.

As people became aware of the problems in their food supply, they demanded some action. In 1906, Congress passed the Pure Food and Drugs Act, as well as the Meat Inspection Act.

Some regulations are needed to eliminate unfair competition. For example, a process such as pasteurization will help control a health hazard, but increases the cost of production. If every processor is not required to pasteurize a particular product, it is difficult for those willing to do it to compete. The inspection of meat and poultry, with elimination of diseased animals, is a cost that an unscrupulous person would not be willing to pay without government regulations.

PURPOSE OF REGULATIONS

The basic purpose of food laws and food regulatory agencies is to ensure that all foods reaching the consumer are safe, wholesome, and truthfully labeled. The production and marketing of wholesome foods should be important to everyone affiliated with the food industry.

There is general agreement that we, as consumers, want clean, wholesome food that is produced and processed in a sound, sanitary manner and is free of microbial pathogens, toxins, or harmful chemicals. Some consumer advocates demand 100 percent assurance of safety of all consumer goods. However, there is no way that absolute safety of food or any other commodity can be proven. The safety of food involves the evaluation of a variety of risks. Everything we do in life involves some risk, including eating food.

Since we have an abundant supply of food in the United States, the present demands for food safety overshadow the hunger in many other

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countries. A hungry person is more interested in satisfying an appetite than in the esthetic aspects of the food that is offered.

FOOD LAWS

The enforcement of the Pure Food and Drugs Act and Meat Inspection Act of 1906 was delegated to the Secretary of Agriculture in the United States Department of Agriculture.

The Pure Food and Drugs Act made illegal the adulteration of food entering into interstate commerce. The individual states maintained the sole authority to regulate intrastate production and marketing of food.

The enforcement of the law improved the quality of the food supply. However, as advances were made in technology, it became apparent that a major revision was needed. In 1933, a bill was introduced in Congress and, after many revisions, was passed in 1938. It became the Food, Drug, and Cosmetic Act. This act is still the basic legislation, although it has been amended several times. The law applies to exports, imports, and commerce between states, in the District of Columbia, and in the territories. The act prohibits adulteration of food and requires certain information on the food label. It authorizes definitions or standards for food. The specific standards are the standards of identity, standards of quality, and standards of fill of container.

Among the amendments to the act, the Miller Pesticide Amendment of 1954 allows the establishment of tolerances of residues of pesticides on raw agricultural commodities shipped in interstate commerce.

The Hale Amendment of 1954 simplifies procedures for establishing food standards. The Factory-Inspection Amendment of 1954 provides that inspections of processing plants and other establishments are authorized upon presentation of credentials and a written notice to the owner, operator, or agent in charge.

The Food Additives Amendment of 1958 requires proof of safety before a chemical additive can be used in food. It allows the use of substances that are safe at the levels of intended use. This amendment put the burden of proof of the safety of additives on the chemical and food industries rather than on the regulatory agencies. The Delaney Clause in this amendment forbids the use of any substance in any amount if it is found to induce cancer in humans or animals. This clause has resulted in much controversy.

The Fair Packaging and Labeling Act of 1966 requires the use of retail packages that are not deceptive. The label must contain certain meaning-

ful information about the value of the food and the ingredients. The name and address of the manufacturer or distributor are required.

The Meat Inspection Act of 1906 requires the inspection of meat-processing plants that ship products in interstate commerce. The Wholesome Meat Act of 1967 requires that state inspection services for intrastate meat processors have regulations and requirements comparable to those of the federal regulatory agency.

Prior to 1959, poultry was inspected on a voluntary basis, with the poultry industry paying for the service. The Poultry Products Inspection Act of 1959 made poultry inspection mandatory for any processor that ships products in interstate commerce. The Wholesome Poultry Act of 1968 requires comparable regulations for intrastate processors.

In the past, foods not in interstate commerce were not subject to federal laws and regulations. However, this was changed with the Wholesome Meat Act and the Wholesome Poultry Act. The food services on ships, trains, airplanes, and buses, and on interstate highways are subject to federal inspection. Many foodservice facilities are national chains, some using foods prepared in central kitchens and shipped across state lines.

Besides the laws directly aimed at the food industry, there are many other federal laws and agencies that regulate and affect our food supply. There is no limit to the number of laws and agencies that can be created to regulate every segment of our life, including foods.

Besides all of the laws and regulations imposed at the national level, the food processor must comply with city, county, and state laws and regulations. For those engaged in exporting and importing, there are international considerations. Through organizations such as the Codex Alimentarius Commission and the International Commission of Microbiological Specifications for Foods, foods are being standardized at the international level.

The purpose of the Codex Alimentarius Commission is to protect the health of consumers and ensure fair practices. The primary means of accomplishing this is through establishment of food standards and codes of practice that we hope will be adopted by cooperating countries.

THE ENFORCERS

A number of government agencies are involved in the enforcement of laws and regulations that affect the food industry either directly or indirectly. According to Sloan (1976), there were fifty regulatory agencies

in the federal government, many of which exert influence upon or control the food industry.

Food and Drug Administration (FDA)

This is the agency generally recognized as the main enforcer of the regulations concerning food. The FDA is part of the Public Health Service located in the Department of Health and Human Services.

The FDA enforces the Food, Drug, and Cosmetic (FD&C) Act, as amended. It is responsible for regulating the production, storage, and labeling of all foods, except red meats and poultry. These foods are regulated by the USDA. The FDA has the responsibility to enforce tolerances for pesticide residues.

The FDA also administers the Fair Packaging and Labeling Act as related to foods and household products.

The FDA devotes part of its effort to the sanitation of milk processing, shellfish, restaurants, and interstate travel facilities included in the Public Health Service Act.

FDA ACTIVITIES. The food processor has the prime responsibility for the safety, wholesomeness, and nutritional quality of his food. The role of the FDA is to see that industry is meeting its responsibility.

About 40 percent of the FDA budget is allocated to food programs. In these food activities, most of the effort is devoted to food safety and the remainder to economic aspects. The food safety programs include food sanitation control, chemical contaminants, mycotoxins, food and color additives, and nutrition. The economic aspects include food standards and controlling unfair competition due to economic cheating.

In food sanitation control, there are efforts to reduce and control the incidence of foodborne illness due to salmonellae, the toxins of *S. aureus* and *C. botulinum,* as well as other pathogenic and toxic agents. The food sanitation and control program is responsible for the development and enforcement of the good manufacturing practice regulations.

The FDA periodically inspects processing and storage plants to determine if they are sanitary. The inspectors check the wholesomeness of ingredients and finished food products and the legality of packages and labels.

It is impossible for the FDA to regulate the entire food industry as well as imported foods. Hence, the FDA depends upon the integrity of responsible people in the food industry as well as upon quality-assurance programs as described by Majorack (1982).

Another program of the FDA is to maintain surveillance on commodities that have demonstrated a potential for mycotoxin contamination.

Products that are contaminated above acceptable levels are removed from the market.

The FDA enforces safe levels of use for chemical additives. Standards are established for classes of foods, and all such products must then meet the minimum requirements.

PRODUCTS NOT IN COMPLIANCE. When a product that is not in compliance with the various laws and regulations is discovered, the FDA has certain courses of action. Among the possible actions it might take are recall, seizure, injunction, and prosecution.

Recall. Some recalls begin when the firm finds a problem. Others are conducted at the FDA's request. Recalls may involve the physical removal of products from the market or correction of the problem where the product is located.

Recalling a product from the market has become the most effective method of removing all units found to be adulterated, misbranded, dangerous to health, grossly fraudulent or deceptive, or materially misleading to the detriment of consumer health and welfare. The food industry can recall a product more efficiently and effectively than the FDA. If the firm refuses to recall the product, or the recall is not effective, seizure actions and FDA publicity are alternative considerations.

There are three classes of recalls. For Class I recalls, there is a reasonable probability that the use of the product will cause a serious adverse health consequence. The Class II recall involves products which, when used, will cause a temporary or medically reversible adverse health consequence or present only a remote possibility that there will be serious adverse health effects. Class III recalls are those in which the food is not likely to cause an adverse health effect.

Seizure. This is an action taken to remove a product from commerce because it is in violation of the law. The FDA initiates a seizure by filing a complaint with the U.S. District Court where the goods are located. A U.S. marshal is then directed by the court to take possession of the goods until the matter is resolved.

Injunction. This is a civil action filed by the FDA against an individual or company seeking, in most cases, to stop a company from continuing to manufacture or distribute products that are in violation of the law.

Prosecution. This is a criminal action filed by the FDA against a company or individual charging violation of the law.

NATURAL OR UNAVOIDABLE DEFECTS. Growing crops in an open field can result in natural or unavoidable defects. It has never been possi-

ble, nor is it now possible, to prevent certain levels of contamination. With this realization, the FDA has developed food defect action levels. The alternative to allowing certain defects would be to use increased levels of pesticides. This could result in higher residues of pesticides in our foods. Growing one's own food organically does not ensure the absence of insect or other contamination.

United States Department of Agriculture (USDA)

The USDA has many functions in food production and processing. The agency most directly associated with the food industry is Food Safety and Inspection Service (FSIS). The FSIS is charged with the inspection of domestic and imported meat, poultry, and their products, as well as establishing ingredient standards and the approval of recipes and labels for meat and poultry products.

Other USDA agencies involved in food include the Packers and Stockyards Administration, Animal and Plant Health Inspection Service, Agricultural Research Service, Economic Research Service, and the Agricultural Marketing Service.

National Marine Fisheries Service (NMFS)

This organization is under the National Oceanic and Atmospheric Administration (NOAA) in the Department of Commerce. Through its Fishery Product Inspection and Safety Program (FPISP), it offers a voluntary fee for service inspection to processors of fishery products. This service includes inspection for plant sanitation, inspection of the product for safety and wholesomeness, laboratory analyses, product grading, lot inspections, and consulting services. The NMFS has developed memos of understanding with the FDA and the USDA regarding the areas of jurisdiction which each organization has with fishery products.

Occupational Safety and Health Administration (OSHA)

This organization was created in the Department of Labor to enforce the Occupational Safety and Health Act. OSHA develops and promulgates safety and health standards and regulations, investigates to determine compliance with these standards and regulations, and issues citations with proposed penalties for noncompliance. Every employer who is engaged in interstate commerce is subject to the act and to the regulations of OSHA. The law requires detailed reports from the employers concerning every category of safety, noise, and other working conditions.

Environmental Protection Agency (EPA)

This agency was established in 1970 to coordinate programs of the federal government concerning the environment. This includes air, water, solid wastes, pesticides, radiation, and noise. As a regulatory agency, it establishes and enforces environmental standards. Two of the acts it enforces are the Insecticide, Pesticide, and Rodenticide Act and the Water Pollution Control Act. These are important to food processors.

Federal Trade Commission (FTC)

The FTC is an independent agency that enforces various laws. It includes the Bureau of Competition, Bureau of Consumer Protection, and the Bureau of Economics. In the consumer protection area, the food industry is affected by rules concerning marketing practices and national advertising.

FOOD STANDARDS

Section 401 of the FD&C Act provides that whenever, in the judgment of the Secretary, such action will promote honesty and fair dealing in the interest of consumers, he shall promulgate regulations fixing and establishing for any food, under its common or usual name, a reasonable definition and standard of identity, standard of quality and/or standard fill of container. Fresh and dried fruits and vegetables are exempt, except for avocados, cantaloupes, melons, and citrus fruits. Also, butter is exempt since it was defined by an Act of Congress in 1923. Meat and poultry products are in the domain of the USDA.

The Hale Amendment of 1954 simplified the procedures for establishing food standards. A standard constitutes the official specification for that food. Supposedly, these standards are important to the consumers and food manufacturers, as well as the enforcers. Standards prevent certain types of debasement and promote honesty and fairness to the consumer, and serve to prevent unfair competition.

The Codex Alimentarius Commission is concerned with establishing food standards on a worldwide basis rather than having many regional standards. Establishing international standards for foods is complicated. Over the years many countries have developed their own standards and codes of practice. The differences in these standards must be reconciled by participating countries with those being developed by the Commission.

The FDA has stated that it will be its policy to accept the standards

recommended by the Codex Alimentarius Commission insofar as the requirements of those standards can be shown to be reasonable and calculated to promote honesty and fair dealing in the interest of the American consumers.

Although these standards should promote better international trade, they, like any other standards, must be somewhat flexible and subject to change when new technology requires a change.

The regulatory agencies attempt to improve sanitary standards through the development of ordinances, codes, manuals, and guides to safe practices. This is done at the local, state, and federal level. Although the FD&C Act does not specifically mention sanitary standards, these are written on the premise of increasing food safety. Reportedly, the sanitary standards are used to prevent foodborne illness in the United States.

The GMP's of the FDA and the Regulations of the USDA for operating meat, poultry, and egg products plants can be considered sanitary standards for food processors. Some food processors have in-house standards that exceed the requirements of the federal agencies.

The sanitary standards usually include observation of the overall plant and facilities, raw material, plant operations, clean-up operations, and finished products. Many also include the handling and storage of the finished products, especially if the product is perishable.

MICROBIOLOGICAL CRITERIA

A criterion is a standard that can be used to evaluate the correctness of a judgment. When the microbial load of a food is determined, we then need to make a judgment as to whether the food is satisfactory or not. Hence, microbial criteria are established to help make a valid judgment concerning the safety and keeping quality of a food.

Another reason for adoption of microbial standards for foods is to compel people in the food industry to adopt quality-control procedures to protect the microbial quality of food from producer to consumer.

Most microbiological criteria are developed to protect the safety of the food supply and reduce the risk to the consumer. Since the FD&C Act states that harmful or deleterious substances are prohibited in food, the position of the FDA is that standards are needed to evaluate quality. In previous chapters, several potential pathogens and toxins were discussed. Of all of these, only salmonellae are determined in foods on a regular basis.

It is well known that salmonellae are present on much of our raw red meat and poultry. Although these products are not under the jurisdiction

of the FDA, they are an example of foods that are known to contain pathogens and, at present, they are tolerated.

Since the local health authorities and the FDA cannot adequately inspect all of the food processing establishments in the United States, the establishment of microbiological criteria for the finished product is the only method to estimate the microbial quality of food.

The functions of microbiological criteria should be to

1. Control the risk to the consumer from pathogenic organisms or their toxins.
2. Reveal if gross contamination has occurred.
3. Assure that a reasonable shelf life of the food is attainable.

There are various criteria that can be established for foods. A microbiological criterion of a food should include the microorganisms and/or their toxins of concern, the methods for analyzing the food for detection and quantification, a sampling plan that includes the number of samples to be obtained and the size of the sample unit, the limits for the microorganisms and/or toxins that are appropriate for the specific food, and the proportion of sample units that should conform to these limits.

The microbiological load of any food is influenced by various production and processing procedures. These variations should be considered in establishing microbial limits. The limits should be attainable by using good manufacturing practices. There should be tolerances due to the inherent errors in sampling and microbial analyses.

Types of Criteria

To evaluate the microbiological condition of foods, the criteria that have been used are guidelines, recommended limits, specifications, and standards.

GUIDELINES. Microbiological guidelines have been developed for many foods. A guideline is used when no official standard exists for a particular food. Guidelines might be considered as administrative standards.

A guideline usually is an estimate of the microbial load that can be attained readily in a satisfactory plant using satisfactory methods of processing and materials.

Although not an official standard, a guideline can be useful for quality-control personnel to make judgments about the acceptability of raw materials, finished products, or operations during processing.

Both processors and regulators can establish microbiological guidelines. The FDA has used guidelines in conjunction with processing-plant inspections. When the microbial load exceeds the guidelines, it supports the observations of poor sanitation and failure to follow good manufacturing practices (Read and Baer 1974). These guidelines have been useful at the processing level but less useful at the retail level. The analysis of food at the retail level has merit if the consumer is considered. Abuses that might occur between processing and retailing could influence the consumer risk.

The defect action levels for molds or mold fragments established by the FDA for certain fruits and various other foods can be considered microbiological guidelines. Some defect action levels for microorganisms are listed in Table 13.1.

TABLE 13.1. MICROBIAL DEFECT ACTION LEVELS IN SELECTED HUMAN FOODS

Product	Defect Action Level
Potato chips	6% of the chips by weight contain rot.
Dried whole eggs	Decomposed as determined by direct microscopic count of 100,000,000 bacteria per gram.
Frozen eggs	Two cans contain decomposed eggs; and subsamples examined from cans classed as decomposed have counts of 5,000,000 bacteria per gram.
Citrus fruit juices (canned)	Microscopic mold count average of 10%.
Pineapple (canned, crushed)	Microscopic mold count average of 30%.
Plums (canned)	Average of 5% by count of plums with rot spots larger than the area of a circle 12 mm in diameter.
Prunes (dried and dehydrated low moisture)	Average of 5% by count insect infested and/or showing mold and/or dirty fruits or pieces of fruit.
Raisins	Average of 5% by count of natural raisins showing mold.
Strawberries (frozen) (whole or sliced)	Microscopic mold count average of 45% and mold count of 55% in one-half of the subsamples.
Apple butter	Microscopic mold count average of 12%.
Cherry jam	Microscopic mold count average of 30%.
Black currant jam	Microscopic mold count average of 75%.
Allspice, bay leaves, capsicum, cassia, or cinnamon	Average of 5% moldy berries by weight.
Beets (canned)	Average of 5% by weight of pieces with dry rot.
Greens (canned)	Average of 10% of leaves by count or weight showing mildew $\frac{1}{2}''$ in diameter.
Spinach	Average of 10% leaves by count or weight show mildew or other type of decomposition $\frac{1}{2}''$ in diameter.

SOURCE: FDA (1973).

RECOMMENDED LIMITS. These are the suggested maximum acceptable number of microorganisms or of specific types of microorganisms, as determined by prescribed methods in a food (NAS 1964). They are similar to guidelines.

There are many recommended microbiological limits. Most of these recommendations are based on too few samples or other considerations to be of much value. The ranges of recommended limits for various foods as found in the literature are listed in Table 13.2. Due to the wide variation, it is evident that most of these are not valid.

SPECIFICATIONS. A specification is a document that provides a detailed description of a material. Included in a specification of a food may be acceptable microbiological quality. A microbiological specification has been defined as the maximum acceptable number of microorganisms or of specific types of microorganisms, as determined by prescribed methods, in a food being purchased by a firm or agency for its own use (NAS 1964).

The acceptable microbiological limit that is written into a specification is not always easy to derive. A microbiological specification should consider both the buyer and the capabilities of the supplier. If the specification is too strict, there may not be enough acceptable material to meet the need of the buyer. If too lenient, the buyer may obtain rather poor quality material. By analyzing many samples, or using information obtained through guidelines, a compromise may be made between the microbial quality available and what is considered to be acceptable. To obtain a high-quality product, the purchaser may expect to pay a premium price for the material.

A practical specification should have tolerance limits set wide enough to allow permissible deviations. However, the limits should be narrow enough to ensure that the material will be acceptable consistently.

TABLE 13.2. RANGES OF SUGGESTED LIMITS FOR TOTAL VIABLE AEROBES IN VARIOUS FOODS

Food Product	Microbial Range (per g)
Frozen precooked foods	2,000 to 100,000
Precooked meats	10 to 10,000
Raw ground meat	100,000 to 50,000,000
Frozen whole eggs	200,000 to 10,000,000
Fish	100,000 to 10,000,000
Shellfish	50,000 to 1,000,000
Vegetables	50,000 to 500,000
Frozen desserts	5,000 to 1,000,000

SOURCE: NAS (1964).

The specification should include sampling and analytical procedures so that there is no question of the system that is to be used.

The microbiological limits that appear in various military and federal specifications for food products are listed in Tables 13.3 through 13.5.

STANDARDS. Of these criteria, only microbiological standards have a regulatory function. The National Academy of Sciences (NAS 1964) defined a microbiological standard as that part of a law or administrative regulation designating the maximum acceptable number of microorganisms or of specific types of microorganisms, as determined by prescribed methods, in a food produced, packed, stored, or imported into the area of jurisdiction of an enforcement agency.

A microbiological standard should not be established haphazardly. A standard must be meaningful and attainable by what is considered to be good manufacturing practice. It should be subject to reevaluation as new technologies are developed. Above all, to be of value, a microbiological standard must be enforceable. When enforced, a standard should reduce the public health hazard of a food.

When the need for microbial standards for food is discussed, an example often used is how the establishment of standards eliminated or reduced the involvement of milk in foodborne diseases. Microbiological standards, per se, do nothing to increase the safety or quality of the food. It is the acceptance of the standards by all segments of industry and the enforcement by the regulators that is necessary before a standard has value.

TABLE 13.3. MICROBIOLOGICAL SPECIFICATIONS FOR MILITARY AND FEDERAL PURCHASES—BEEF AND PORK PRODUCTS

Product	Aerobic Plate Count	Coliform	*E. coli*	Salmonellae (25-g sample)
Beef, cooked, dried	150,000	60	—	—
Beef stew, cooked, dried	75,000	—	neg.	—
Beef with rice, cooked, dried	75,000	—	neg.	—
Swiss steak w/gravy, frozen	100,000	100	neg.	neg.
Pork slices, cooked, dried	110,000	20	—	—
Pork sausage, cooked, dried	200,000	40	—	—
Pork loin, sliced, w/gravy, frozen	100,000	100	neg.	neg.
Pork and beef chop suey, frozen	100,000	100	neg.	neg.
Meat balls and meat ball products, cooked, dried	150,000	40	—	—
Spaghetti w/meat sauce, dried	75,000	—	neg.	—
Escalloped potatoes w/pork, cooked, dried	75,000	—	neg.	—
Beef hash, cooked, dried	75,000	—	neg.	—

TABLE 13.4. MICROBIOLOGICAL SPECIFICATIONS FOR MILITARY AND FEDERAL PURCHASES—MILK AND DAIRY PRODUCTS

Product	Maximum Count per Gram				
	Aerobic Plate Count	Coliform	Yeast & Mold	Salmonellae	Direct Microscopic Count
Milk and milk products, fresh, fluid, concentrated, and frozen	20,000	10	—	—	—
Milk, whole, dry	—	10	—	—	40,000,000
(premium)					
(extra grade)	—	10	—	—	75,000,000
Milk, nonfat, dry	50,000	—	—	neg.	—
Processed American cheese, dehydrated	50,000	90	—	—	—
Milk, fat	5,000	10	20	—	—
Frozen fudge bar	50,000	20	—	—	—
Flavored dairy drink, dry chocolate or coffee flavored	20,000	10	—	neg.	—
Cream substitute	20,000	10	—	—	—
Cottage cheese	—	10	10	—	—
Cream, sour, cultured, or acidified	—	10	10	—	—
Ice cream, ice milk:					
Vanilla	50,000	10	—	—	—
Chocolate, nuts, fruits	50,000	20	—	—	—
Malted milk	30,000	10	—		—
Buttermilk	—	10	—	—	—

TABLE 13.5. MICROBIOLOGICAL SPECIFICATIONS FOR MILITARY AND FEDERAL PURCHASES—POULTRY PRODUCTS

Product	Maximum Count per Gram				
	Aerobic Plate Count	Coliforms	*E. coli*	Salmonellae 25-g sample)	Yeasts & Molds
Frozen chicken à la king	100,000	100	neg.	neg.	—
Frozen chicken cacciatore	100,000	100	neg.	neg.	—
Frozen turkey with gravy	100,000	100	neg.	neg.	—
Cooked turkey, dried	200,000	40	—	—	—
Cooked chicken with rice, dried	75,000	—	neg.	—	—
Cooked chicken and cooked chicken products, dried	75,000	—	neg.	—	—
Frozen eggs and egg products, whole egg, egg white, yolk	50,000	—	—	neg.	50
Table grade	20,000	—	—	neg.	50
Egg mix, dried	25,000	10	—	neg.	—

Microbial Groups and Species

Microbiological criteria could include the determination of many groups or species of microorganisms. Keeping the number of required tests to a minimum has advantages. There is less expense and time involved in the analysis, and the interpretation of the results is less complex when fewer tests are needed.

The microbial limits might include the aerobic plate count for mesophiles, as well as coliforms, fecal coliforms, *E. coli,* salmonellae, *Staphylococcus aureus, Clostridium botulinum, C. perfringens,* group D streptococci, Enterobacteriaceae, yeasts, and molds.

The standard aerobic plate count of mesophilic bacteria is a rough measure of the bacterial content and the sanitary conditions during processing or temperature abuses of a food. The aerobic plate count cannot be related unequivocally either to a health hazard or to spoilage.

Some foods normally have high microbial loads. Depending on the raw material and the processing conditions, a particular number of organisms may represent a good or bad product. Precooked or other heat-treated foods would be expected to have lower counts than raw foods. High microbial numbers in a heated food may be due to poor processing control; contamination after heating; poor holding temperature, allowing multiplication; or a combination of these factors. The viable microorganisms tend to decrease in frozen or dried foods and to increase in refrigerated foods. Hence, the microbial number that is obtained will depend on the time of analysis after processing.

The coliform group is determined in food mainly because this group is important in water microbiology, and researchers once believed that this importance carried over to food products. The presence of coliforms cannot be related unequivocally to fecal contamination. In a pasteurized or precooked food, the presence of coliforms indicates poor processing controls, postprocess contamination, or both.

The presence of *E. coli* reportedly indicates either direct or indirect fecal contamination. However, this organism is part of the normal flora of many raw animal products, even when prepared with acceptable processing techniques, as well as plant products that contact the soil. *E. coli* is heat sensitive. When found in heat-processed food, it indicates contamination after heating by fecal material that also could conceivably include enteric pathogens. Certain strains of *E. coli* are enteropathogenic or enterotoxigenic. Distinguishing these types from any other *E. coli* would be too time consuming and costly for practical purposes regarding microbial standards.

The determination of *E. coli* reportedly can be used to estimate the plant sanitation, postheating contamination, poor handling practices,

cross-contamination from raw products, or a combination of these factors. Being pathogenic, salmonellae should not be associated with food.

Staphylococcus aureus has been included in some microbiological criteria. In certain situations, *S. aureus* may be a pathogen, a source of enterotoxin, an indicator of unsanitary practice, or be of little significance and concern (Bartram 1967). Foods that are handled by people are likely to contain coagulase-positive staphylococci. Standards for *S. aureus* at the processing level would have little value in preventing foodborne illness.

Clostridial spores are prevalent in the environment, especially in soil and dust. Hence, organisms, such as *C. botulinum* and *C. perfringens* may be present in foods. Unless the food is an acceptable environment and is allowed to remain at temperatures that allow growth, a few clostridia or *S. aureus* are not considered to be a health hazard. However, processing and handling should be designed to minimize contamination and prevent the growth of these organisms.

Organisms in the group D streptococci have been considered indicators related to human contamination of processed meats. They have been proposed as indicators of fecal pollution of various processed foods due to their ability to withstand certain processing and environmental conditions.

The presence of yeasts and molds is used in the microbiological criteria for some foods. A high count (above 50/g) in cocoa powder indicates poor sanitary conditions after roasting. The guidelines for product defects (Table 13.1) consider certain levels of molds unsatisfactory in foods. Tomato products contain high levels of mold due to moldy tomatoes not being properly sorted or trimmed or to contaminated processing equipment. In either case, a high level of mold indicates poor processing technique. In some products, the presence of molds may indicate the possible presence of mycotoxins that may or may not be a health hazard.

The tests that are most useful or used in microbiological criteria are the standard plate count of aerobic mesophiles, coliforms, *E. coli,* and salmonellae.

Types of Food

There is more justification for microbiological standards for some foods than for other foods. When there is a risk or hazard, a standard for a food might be needed. There are some foods for which there is no substantive need for microbiological standards.

Those foods that have demonstrated a potential health hazard should be the first ones considered for establishment of any microbial standards. Foods that contain animal products are the primary vehicles for organisms that cause foodborne illness.

Frozen and chilled precooked foods that support the multiplication of pathogenic bacteria when held at growth temperatures are considered to be the main foods that need microbiological standards. However, even though there is a potential hazard, these foods have been involved in relatively few known outbreaks of foodborne illness.

Foods under consideration for international standards by the Codex Commission include egg products, dried milk, molluscan shellfish, frozen frog legs, precooked frozen shrimps and prawns, food for infants and children, and frozen desserts. Those considered for prompt attention include chilled frozen poultry and meat, precooked frozen crabmeat, and lobster, cheese, and coconut.

Microbiological standards have been applied to milk for a number of years. Standards for bottled drinking water and local standards for a few foods have been established. Milk and water were involved in many disease outbreaks before microbial standards were established. These liquids are homogeneous and can be subjected readily to heating, filtration, or chemicals to control the microbial load. Hence, a microbial limit can be attained relatively easily and the standard can be enforced fairly easily. These problems are increased with heterogeneous solid foods.

In 1973, the FDA proposed microbial standards for food-grade gelatin and frozen ready-to-eat banana, coconut, chocolate, or lemon cream-type pies. These standards were rescinded in 1978. In 1977, the FDA planned microbiological standards for fish cakes, fish sticks, and crab cakes. Information regarding the microbial content of various foods has been reported by researchers of the FDA, apparently with the intent of establishing microbial criteria. These foods include dry milk mixes and milk-substitute infant formulas (Schwab et al. 1982), pasta products (Swartzentruber et al. 1982), cocoa powder, dry instant chocolate milk, nondairy creamer and frozen nondairy topping (Payne et al. 1983), crabmeat, clams, and oysters (Wentz et al. 1983), biscuit dough, snack cakes, and soy protein meat extender (Swartzentruber et al. 1984).

Sampling Plan

Microbiological criteria should include a sampling plan, as well as methods for analysis. Since the microbial contamination in most foods is heterogeneous, the testing of as many samples as possible is necessary to satisfy statistical requirements. Statistical sampling plans can ensure that the tests that are done are sufficient to give a good assessment of the microbial condition of the food. It is important to minimize the chances of rejecting an acceptable lot of food (producer risk) or accepting a substandard lot (consumer risk).

Sampling plans for food have been amply described by Thatcher and

Clark (1968) and ICMSF (1974). Sampling of food is discussed briefly in Chapter 2. The sample units should be representative of the lot and be obtained in an unbiased manner.

The number and size of samples to be analyzed should be included in the microbiological standard or other criterion.

Methods for Analysis

Standard methods for analysis must be selected to determine if foods meet the requirements of microbiological standards. Methods published by AOAC (1985), APHA (1984), the FDA (1978), or Thatcher and Clark (1968) could be acceptable. The AOAC methods are standardized and are accepted in courts of law in the United States.

On an international basis, the standardization of methods is more difficult. There are different media manufacturers in different countries. The organisms that are prevalent vary from continent to continent. Food from a tropical country would undoubtedly contain more thermophiles than would the food from colder climates. However, standardized methods would help facilitate international trade in foods.

Whatever methods are designated, they should not require the use of unusual and expensive equipment not found in most laboratories. The procedure should be easy to follow by a technician and not be so complex that only a research scientist can master the techniques. However, the method must give reproducible results so that the findings from different laboratories will agree within reasonable limits.

Methods that are applicable to many commodities or groups of commodities are preferred to methods that apply to only one or a few foods.

Microbial Limits and Variables

There are various ways in which the limits of microbiological criteria can be developed. One method is to make an extensive survey of the microbiological condition of the finished product. The samples should be obtained from several processors that operate with what are considered acceptable sanitation controls. With these data, the expected level of microorganisms in a particular product can be estimated.

A system can be used in which an inspection team visits the processing plant and evaluates the apparent sanitary conditions on various days. Samples of the finished product are obtained during these visits and analyzed for the microbial condition. In this way, it may be determined if there is a correlation between sanitation and microbial condition.

The microbial limits must be achievable when good manufacturing practices are used in processing the food. In establishing limits, there

should be a consideration of the risk associated with the presence of the organism(s) and the number of organisms that would affect the acceptability of the food.

Any microbiological limit that is established by a regulatory agency may be challenged in court. It may be difficult to prove that a certain number of bacteria above a certain limit is a health hazard or an indication of filth and decomposition.

It is well recognized that there are variations in the microbial content between sample units of food. These variations must be considered when microbial limits are applied to a food product.

The good manufacturing practices that are applied today might be considered poor or unacceptable in a few years. Also, the factors that are considered as health hazards are continually changing. To be effective, microbiological criteria must be reevaluated and altered as experience is gained and as technology is changed.

DECISION CRITERIA. These are part of the overall sampling plan. However, they are used in the application of the data obtained from analyzing sample units to the microbial limits that have been established.

We can assume that the limit for the aerobic plate count (APC) is 100,000/g. This limit is designated as *m*. In sampling by variables plans, the lot is accepted if the mean or average of the APC of the sample units is *m* or less. If the mean is greater than *m*, the lot is substandard. Variable sampling requires some knowledge of the distribution of microorganisms within the product (Thatcher and Clark 1968).

The attribute sampling plan can be used with any food that is tested. With this system, either the APC of a sample unit exceeds *m* or is equal to or less than *m*, the microbial limit. If ten sample units are analyzed and the APC of eight are equal to or less than *m*, and two are greater than *m*, what decision can be made regarding the acceptability of the lot?

The Two-Class Plan. In a two-class plan, several sample units are analyzed. The number of sample units that must be examined from a lot of food to satisfy the requirements of a particular sampling plan is designated as *n*. If ten units are needed, then $n = 10$. With a two-class sample plan, a certain number of nonacceptable or defective sample units are allowed. The maximum allowable number of defective sample units is designated as *c*. If two defective sample units are allowed of the ten that are examined, then the entire lot of a food product that has eight acceptable and two defective units would be considered as acceptable. If there are three defective units, the lot would be rejected.

The Three-Class Plan. In the two-class system, the sample unit is either acceptable or nonacceptable. With a three-class system, a marginal area is

included. If we keep m as 100,000, then any sample units of m or less are still considered as acceptable. An upper limit is established which we can assume to be 5,000,000, which is designated as M. In this example, sample units with APC above m (100,000) and equal to or less than M (5,000,000) are considered marginal. A product with any sample unit with an APC over M is unacceptable. With a three-class system, of ten sample units, two could be between m and M, with no sample units greater than M. Our plan can be formulated as: $n = 10$, $c = 2$, $m = 10^5$, $M = 5 \times 10^6$. For salmonellae, which are not tolerated in food, the plan would be $n = 10$, $c = 0$, $m = 0$. There would be other values for m, M, or c if *E. coli*, coliforms, or other types of microorganisms were in the standard.

Different foods might have different types of organisms listed in the standards, as well as different values for n, c, m, and M; these values depend on the type of food and the microorganisms or group of microorganisms. The value of M should be so high that it constitutes a definite hazard or evidence of overt spoilage. Hence, no food could be tolerated with sample units above this level. With high potential health hazards, the value of n would be high and c would be low. For cases in which there is essentially no hazard and leniency is designated, n may be low and c relatively high, or n might be 5 with a c of 3. In other words, three of five samples could exceed the limit m without rejection of the lot.

With any sampling plan, there is a probability that acceptable products will be rejected and products not in compliance will be accepted. The probability of these occurrences with various sampling plans is discussed by ICMSF (1974), Read and Baer (1974), Thatcher and Clark (1968), Clark (1982), and Jarvis and Malcolm (1986). The three-class plan has been proposed for various foods such as chocolate (Collins-Thompson et al. 1981), infant cereal and formula (Collins-Thompson et al. 1980), pasta products (Rayman et al. 1981), and frozen cream-type pies (Todd et al. 1983).

Disposition of Substandard Food

Foods that exceed the acceptable microbiological limits in a standard are not necessarily wasted. The standards as proposed by the FDA were for quality determination. Foods that did not qualify for the standards were to be labeled "Below Standard in Quality—Contains Excessive Bacteria." It is evident that this statement on a food will decrease sales. If there are excessive bacteria in a food, it may be subject to seizure on the basis that it is adulterated by having been "prepared, packed, or held under unsanitary conditions."

According to Elliot (1970), when poultry and red meat products exceed the microbiological criteria, production of further products is

halted to determine the cause of the contamination. The disposition of the product depends upon the extent of contamination, the existence of a hazard, presence of decomposition or adulteration, and the nature of the microorganisms. After considering these factors, the product may be released, sorted, reprocessed, or destroyed. If the food is not acceptable for human consumption, depending upon the contamination, it might be fed to animals. Food should be not destroyed unnecessarily. However, if the hazard is such that it cannot be remedied, destruction may be the only solution.

Miscellaneous Standards

Although specifications or guidelines have been established for most foods and food ingredients, there are very few official standards. Many of the so-called standards in the literature are specifications, guidelines, or suggested standards.

WATER. In the United States, the Environmental Protection Agency develops regulations for drinking water according to the Safe Drinking Water Act. The FDA has proposed microbiological standards for bottled water used for drinking.

Tap Water. Standards have been established for the standard plate count (SPC) and coliforms. The SPC limit is 500 organisms per milliliter, based on the arithmetic average of samples collected in a calendar month.

Either the membrane filter method or the fermentation tube method can be used to determine coliforms. For the membrane filter method, a 100-ml sample is used. The arithmetic mean coliform density of all samples examined per month shall not exceed 1/100 ml. Per sample, the coliform colonies shall not exceed 4/ml in more than one sample if fewer than twenty samples are examined per month, or more than 5 percent of the samples if more than twenty samples are examined.

For the fermentation tube method, five portions of either 10 ml or 100 ml are tested. When 10-ml portions are tested, not more than ten percent in any one month shall show the presence of the coliform group. The presence of coliforms in three or more 10-ml portions of any one sample are not allowed if this occurs in either more than one sample per month if fewer than twenty samples are examined, or more than 5 percent of the samples if over twenty samples are examined during the month.

When 100-ml portions are tested by the fermentation tube method, not more than 60 percent in any one month shall show the presence of coliforms. The presence of coliforms in all five of the 100-ml portions in

a sample shall not be allowed if this occurs in either more than one sample per month when fewer than five samples are examined per month, or more than twenty percent of the samples when five or more are examined per month.

There are provisions for analyzing more samples if the level of organisms is above the standards. Besides microorganisms, limits are set for various chemicals. These standards are used as guidelines by many states and cities in setting up limits for potable water.

The World Health Organization standard for drinking water includes coliform limits. In 90 percent of the samples, there should be fewer than one count/100 ml, and for all samples, there should be fewer than ten counts/100 ml.

Bottled Water. The standard for bottled water includes the coliform index. The sample used is ten containers (sample units) of water selected from the lot.

The coliforms can be determined by either the membrane or fermentation method. For the membrane filter system, not more than one of the sample units shall have more than four or more coliforms/100 ml, and the arithmetic mean of the coliform density shall not exceed one coliform/100 ml.

For the fermentation tube method, not more than one of the sample units shall have an MPN of 2.2 or more coliforms/100 ml, and no analytical unit shall have an MPN of 9.2 coliforms/100 ml.

MILK. Due to its widespread consumption by infants and children, and its potential for containing health hazards, milk was the first food for which microbiological standards were established.

The microbiological standards for milk and other dairy products are listed in Table 13.6. The SPC of pasteurized grade A milk must not ex-

TABLE 13.6. MICROBIOLOGICAL STANDARDS FOR SOME GRADE A DAIRY PRODUCTS

	Maximum Numbr per Gram	
Product	Standard Plate Count	Coliform Count
Raw milk, at pick-up	100,000	—
Raw milk, commingled	300,000	—
Pasteurized milk	20,000	10
Pasteurized, condensed milk	30,000	10
Pasteurized, condensed whey	30,000	10
Nonfat dry milk	30,000	10
Dry whey	30,000	10

ceed 20,000 organisms per milliliter and more than ten coliforms per milliliter. Restrictions imposed by individual dairy plants are frequently much more rigorous than those imposed by public health requirements. No limit is established for the SPC of fermented products, due to the addition of a starter culture. However, there is a limit established for coliforms.

For milk used in manufacturing, such as in making cheese, the USDA recommended standards for a standard plate count are as follows:

Grade	Organisms/ml
1	not over 500,000
2	not over 3,000,000
3	over 3,000,000

The bacterial level in manufacturing milk is much higher than allowed in grade A milk.

Other countries have other standards. Quite often, a reductase test using resazurin or methylene blue is used in place of the SPC to estimate the bacterial quality of the milk or milk product (Cox 1970).

OYSTERS. The National Shellfish Sanitation Program established a microbiological criterion at the wholesale level. A satisfactory sample should not have a fecal coliform density of more than 78/100 g, with occasional values to 230/100 g (MPN) and a standard plate count of no more than 100,000/g, with occasional samples at 500,000/g.

Ayres (1975) stated that, for bivalve shellfish, there is no justification for a standard based on total plate counts that often exceed 1,000,000/g. He suggested that the relation of nonspecific bacteria growing at 20° and 37°C might be useful in assessing the potential health risk of raw shellfish.

SUGAR. This ingredient may contain spores of thermophilic bacteria that are important in the spoilage of low-acid canned foods. In 1931, the National Canners Association (NCA) formulated and published standards for sugar for use in canning of low-acid canned foods. These standards have been applied by industry, as well as federal and state laboratories. By definition, these microbiological criteria for sugar are specifications, since there are no official standards for sugar. However, due to the importance of these limits, they are included in this section.

Limits have been established for total thermophilic spores, flat sour spores, thermophilic anaerobic spores, and sulfide spoilage spores. The sample from a lot or shipment consists of five sample units of above 225g each taken from each of five bags or containers.

For the five sample units, there shall be a maximum of 150 total thermophilic spores and an average of not more than 125 spores/10 g of sugar. There shall be a maximum of seventy-five flat sour spores and an average of not more than fifty spores/10 g of sugar. Thermophilic anaerobic spores shall be present in not more than three of the five sample units and in any one sample unit to the extent of not more than four of six inoculated tubes. For this test, an amount of solution equivalent to 4g of sugar is distributed equally among six tubes containing liver infusion broth.

The sulfide spoilage spores shall be present in not more than two of the five sample units and not more than five spores in 10 g of any one-sample unit.

STARCH. This ingredient can cause spoilage when added to low-acid canned foods due to the presence of thermophilic spores. The microbial limits for starch are the same as for sugar.

PEANUTS. There are no standards for the number of microorganisms that can be present in peanuts or peanut products. However, there is a tolerance of aflatoxin of less than 20 μg/g of peanuts.

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